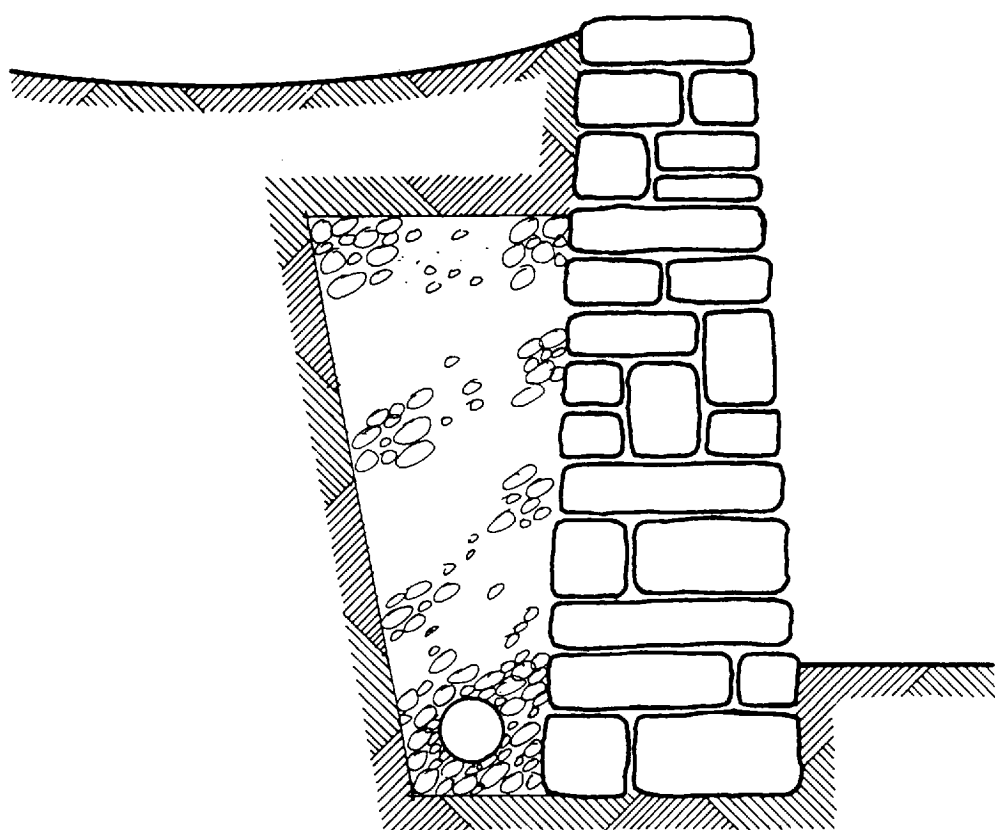




THIRD EDITION
MICHAEL LITTLEWOOD

LANDSCAPE DETAILING



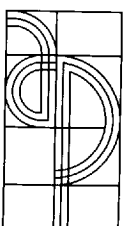
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ENCLOSURES

LANDSCAPE DETAILING

Volume 1 ENCLOSURE

Third Edition Michael Littlewood

Architectural Press



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WILL PAY FOR BTCV TO PLANT AND CARE FOR A TREE.

CONTENTS

Volume 1 ENCLOSURE

Foreword	vii	Appendix A References	185
Acknowledgements	ix	Appendix B Bibliography	186
Introduction	xi	Appendix C British standards	187
Copings, cappings and bonds	1	Appendix D Manufacturers and Suppliers	189
Free-standing walls	14	Appendix E Institutions and Associations	191
Screen walls	46	Appendix F Graphic symbols	192
Retaining walls	53	Appendix G Conversion tables	193
Fences	90		
Gates and stiles	143		
Barriers	165		

FOREWORD

The success of both editions of *Landscape Detailing* has resulted in a review of material and data for the third edition. In view of the many more details that have been produced since the second edition it was felt that the user would prefer to have them in two volumes for ease of use. This book covers details relating to enclosures and the second relates to surfaces.

Many landscape architects, architects, other professionals and students responsible for the production of drawn details and specifications for landscape construction works have a need for ready reference. This book has been produced to meet that need and it can be extended by additional sheets. It has been arranged for ease of copying of sheets and it is sufficiently flexible for designers to use the details for their specific requirements.

The range of materials for external works and their possible combinations for enclosures would make it impossible to provide a definitive book of details.

It is not the intention of this book to supplant the landscape designer's own skill and experience, which is vital to the success of any project. This is still essential in evaluating the site conditions, assessing the character of the environment and creating sensitive design solutions.

It is hoped that the book, if used correctly, will allow the designer to spend more time on design details, avoiding the need to produce repetitive drawings for basic construction elements. It has been found that the details can be very useful for costing purposes and to support the preliminary design when presented to a client. To assist the designer and to save further time in writing specifications, check lists for these have been included in this edition along with technical guidance notes and tables.

Design information has been excluded; many other publications deal with this subject much more adequately than could be achieved in this book. General comments on appearance have been given only where it was felt appropriate.

ACKNOWLEDGEMENTS

I must give particular thanks to many people who have supported me in some way – no matter how small – and who have encouraged me to complete this third edition, which has been greatly enlarged.

My particular thanks must go to Caroline Mallinder and Paddy Baker of Butterworth-Heinemann – my publishers – both of whom have supported my work and put up with so many frustrating delayed publishing dates. Thank you for being so patient. Also to landscape architects Andrew Clegg, Melissa Bowers, Naila Parveen, Donna Young, Peter Dean and Craig Schofield, all of whom have succumbed to my persistence in drawing the details and reading the text.

My appreciation must also go to Colin MacGregor of NBS for his ready and willing assistance on specification matters as well as Alistair Smythe of *Specification* and Barrie Evans of the *Architect's Journal*. A very special thanks to Doris Evans for typing the text and correcting it so many times.

I am also very grateful to civil engineers John Williamson and Alan Taylor for their advice on retaining walls and to Peter Morrison of Ibstock Building Products Ltd for his kind assistance on brick walls in general.

All of the above have contributed to this book to ensure that it eventually reaches the publishers, after such a long time.

INTRODUCTION

The landscape detail sheets have been produced in an effort to eliminate needless repetition in detailing landscape works covering hard elements. It is possible to use them without alteration, but in some cases minor modifications and additions to dimensions or specifications may be necessary. Lettering has been standardised by the use of a stencil (italic 3.5 mm). When a detail is required which is not available on a detail sheet, the new detail can be drawn by the designer using the standardised format, which will enable it to be added to the original collection of details and to be easily re-used on other projects. Readers are invited to send the publishers copies of their own details which they think would merit inclusion in future editions of this book. Appropriate acknowledgement will be made.

Each sheet portrays a detail without reference to its surroundings. This approach has been adopted because it affords to each detail the maximum number of possibilities for re-use. No attempt has been made to recommend a particular detail for a particular situation. This remains the responsibility of the landscape architect, architect or designer.

There are, of course, a great many other details which might be included on specific projects or in specific situations. In some cases, the detailing of site elements and site structures can be coordinated very carefully with the architect or building designer in order to ensure a uniformity of form and material. In yet other instances, various agencies and organisations may have standard details which must be used on their particular projects.

Notes

The notes which precede each section are intended to give only the briefest outline of main points. For more detailed guidance, the publications listed in Appendices A and B should be consulted.

Specifications

Specifications should not be written without a knowledge of the content of the relevant British Standards in Appendix C. Some British Standards contain alternative specifications which may prove more suitable in a particular case.

The task of writing specifications has now been made very much easier by the use of the word processor. Nevertheless, if a specification is to serve its purpose efficiently it must be concise and accurate, otherwise it could be misunderstood by all the people involved in the project.

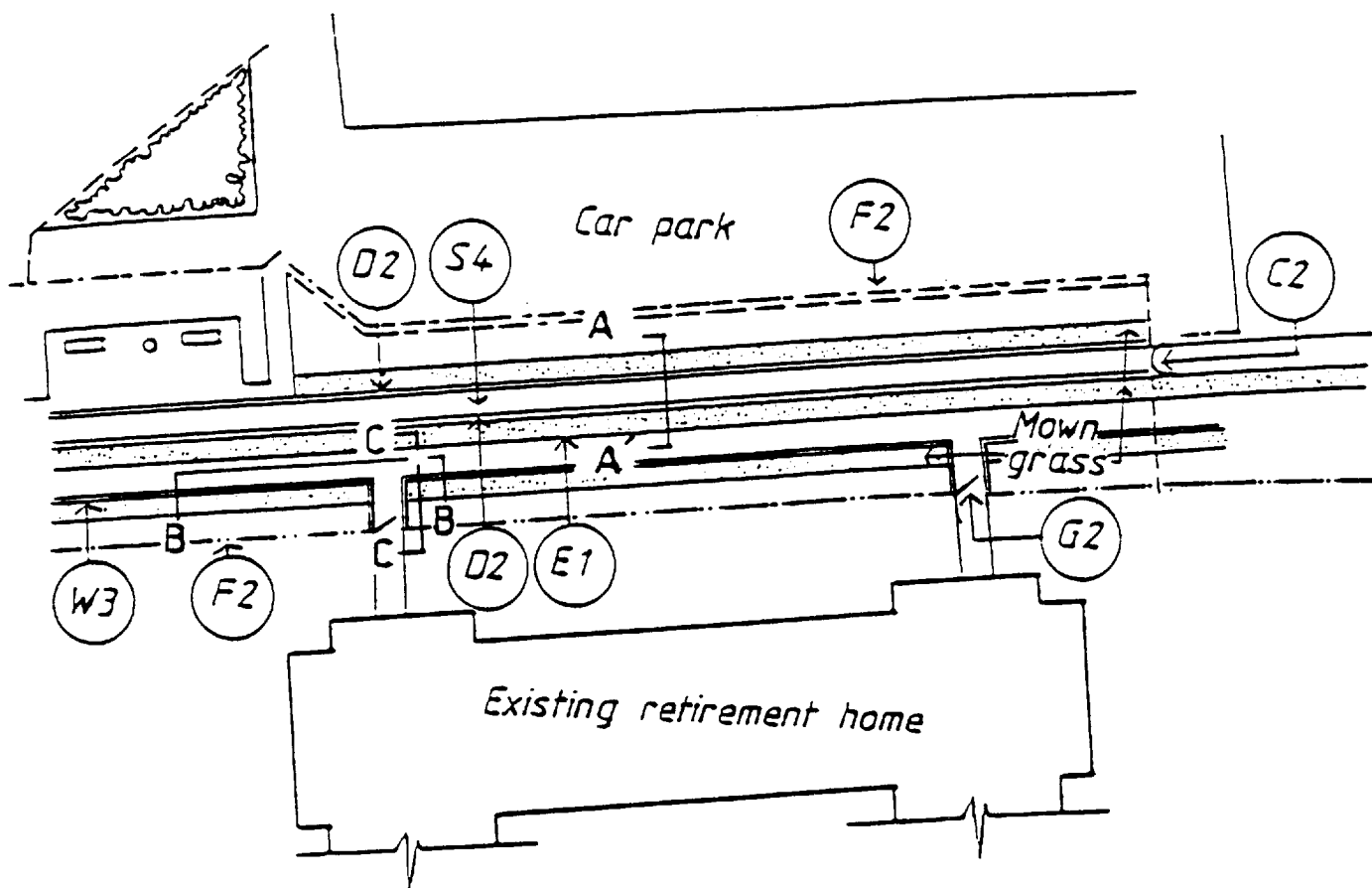
To assist the designer and to ensure that he or she makes the minimum of omissions, a check list has been provided after the notes for each chapter or section. Ease of access to a particular section will encourage a contractor to read the specification and conform with its requirements. So many contractors ignore the specifications and use only the bills of quantities. Probably the best way to ensure that the completed specification is satisfactory is for the designer to read it as if he or she were the contractor and could complete the project accordingly.

Reference should be made to two main sources for specifications, namely the NBS of Newcastle-upon-Tyne and the publication *Specification*. Full details of their services are given in the Appendices.

Use of the detail sheets

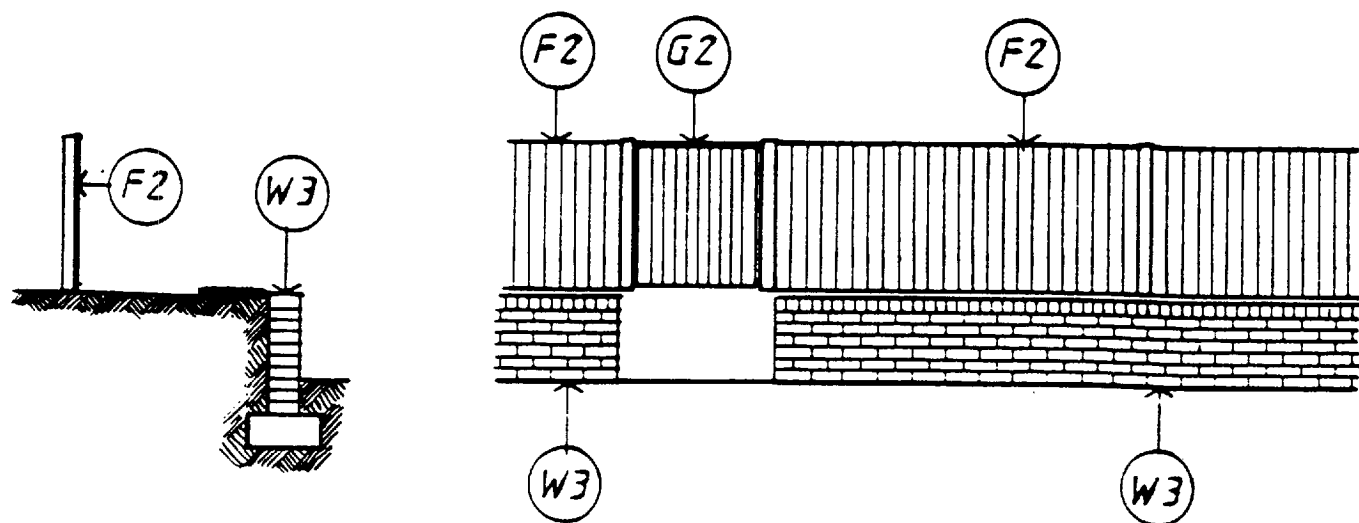
The collection of detail sheets, as purchased, may if users wish be photocopied, punched and stored in a ring binder. The detail sheets have been laid out in such a way as to facilitate this operation. In the form of individual leaves the details can easily be traced or copy negatives can be made.

The sheets must be used in conjunction with a site layout drawing, preferably at 1:200. These may be more than one sheet,



PLAN

Figure 1



SECTION C C

ELEVATION B B

N.T.S

Figure 2

depending upon the size of the project. The layout drawings will convey all information on levels, directions of falls and setting-out dimensions. They also indicate the location of the elected details and the deployment of surface finishes. (See Figure 1.) Simple conjunction of details (for example, a fence, a wall and a gate) can be indicated on section and elevation drawings quite easily. (See Figure 2.)

Standards

British Standards and Codes of Practice are referred to where necessary. Users of this book living in countries where British Standards are not used should delete the reference to the British Standard and, if they feel it necessary, either insert a reference to an equivalent national standard or describe what is required in empirical terms.

Production of new detail sheets

Where the use of a detail not included in the original collection of detail sheets is required, the new detail can be produced on A4 tracing paper using a standard format. This will enable it to be added to the original collection and to be easily re-used. New details will be assigned a reference number by the design office, using their own reference system. The title of the new detail, as shown in the centre label at the foot of the drawing, can then be added to the contents list prefacing each section.

Issue of detail sheets

Detail sheets can be used in two ways. A set of photocopies can be issued to the contractor of the selected details, after completion of the title panel reference, and number-stamping each detail with the office stamp. The second method is to trace or

copy a batch of details, grouped according to type and identified with key numbers, onto an A1 sheet of tracing paper and include the drawing with the contract set in the normal way.

Design detailing

The creation of good design can only come from the designer, and no amount of drawn details can be a substitute for this fact. The principles must be followed as Fraser Reekie has stated in his book *Design in the Built Environment*:

To make an objective assessment of a design, or to set about the process of designing, consideration has to be given to the three aspects which may be summarised as:

1. Function: The satisfying of requirements of use or purpose;
 2. Structure: The physical implementation of function by the best available material(s), construction, manufacture and skills as conditions permit;
 3. Appearance: The obtaining of satisfactory visual effects (sometimes referred to as 'aesthetic values').
- Other words can be used to describe these three aspects but, on analysis, whatever words are used it will be found that almost every writer on building design, which may be extended to cover the built environment, is dealing with the same three fundamentals.

These three constituent parts of design are closely interrelated and each, to a greater or lesser extent, according to the nature of the subject, influences the others. An urban composition or a building or a detail that is truly well designed is one in the creation of which all three aspects have been fully considered and integrated. Integration may well be the key-word in good design. Not only does it mean the correct combining of parts into a whole but it implies, by association with integrity, soundness and honesty.

COPINGS, CAPPINGS AND BONDS

GUIDANCE NOTES

Copings and cappings

Their primary purpose is to prevent water seepage into the wall from the top, and second, to shed water clear of the face of the wall, both as effectively as possible. Reference should be made to BS 3798: 1964 for copings of clayware, concrete or stone. A minimum weight of 1.5 kN/m² is preferred for copings with concrete or stone units. The overhang, if any, of any coping or capping should include a throating recess or a drip not less than 13 mm wide, with the outer drip edge at least 40 mm from the face of the brickwork.

The practice of using a brick on edge coping is acceptable because it is the damp proof course incorporated under the coping that functions as a coping and not the brick on edge. The presence of a damp proof course causes the brick copings to become saturated and susceptible to frost damage. Only bricks that are resistant to such damage should therefore be used. Engineering or concrete bricks are usually preferred. End bricks should be held in position by galvanised metal cramps. Cramps can also be built into the top of the wall at a spacing of 900 mm to secure the coping bricks along the length of the wall. Concrete block walls are often capped with concrete slabs or bricks. Concrete copings may need to be dowelled together for strength, especially in areas susceptible to vandalism. Tile, slate or metal could also be used for copings. A coping for a wall needs to be simple, bold and effective, although random rubble walls look far more convincing when finished with the traditional

method for the locality. It should be remembered that the choice of suitable coping for the situation may decide the choice of brick in the wall and limit the possibilities available to the designer.

Bonds for brickwork

The way bricks are arranged within a wall is known as the bond, and while they do have marginally differing strengths their main virtue is aesthetic. The bond will to some extent depend upon the thickness of the wall. For a one-brick-thick wall the bond is usually of stretchers only and the height is normally under one metre. For higher walls a Flemish or English bond is used which has a double thickness of brick with headers (bricks laid end-on) laid to bind the wall.

DETAIL SHEETS

Copings and cappings

Brick (2)

Concrete

Mixed

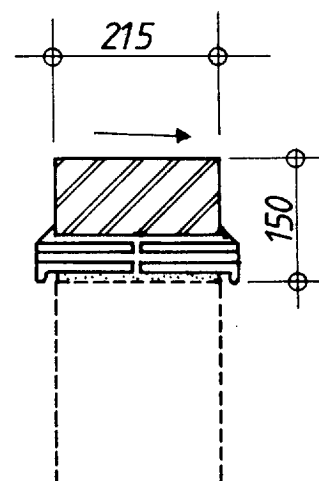
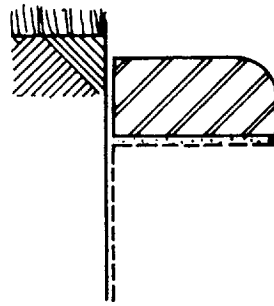
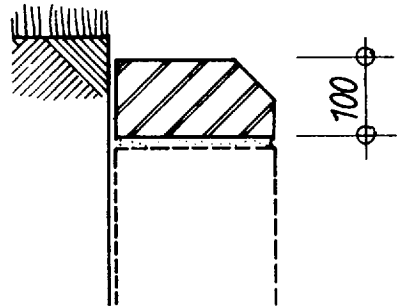
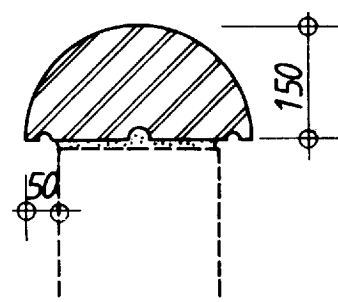
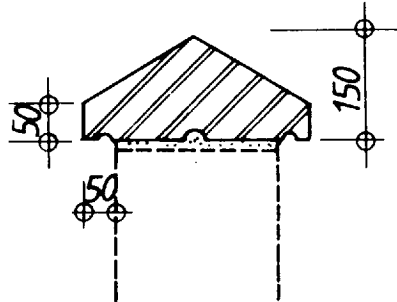
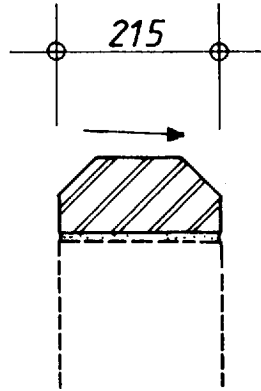
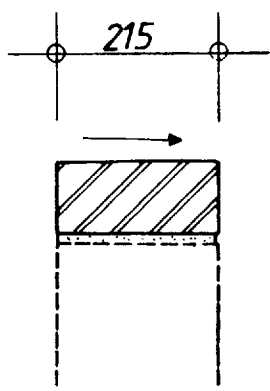
Bonds

Brick bonds (5)

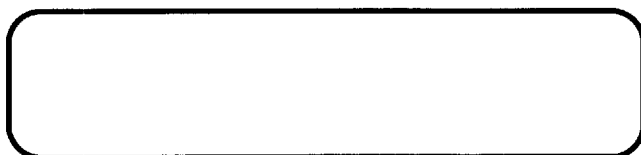
Stone

Random rubble masonry patterns

Ashlar stone masonry patterns

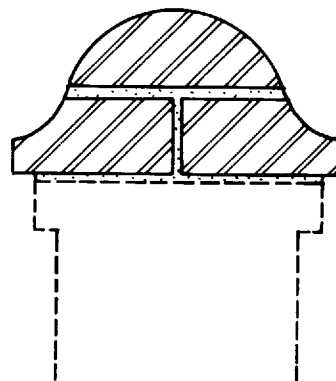
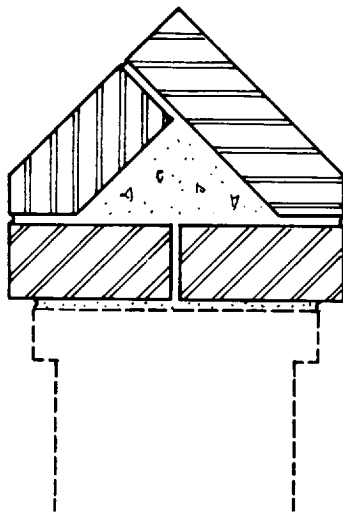
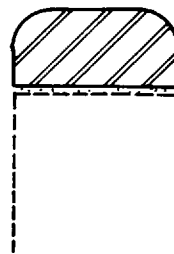
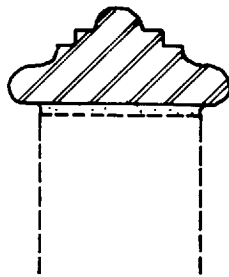
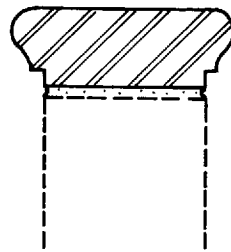
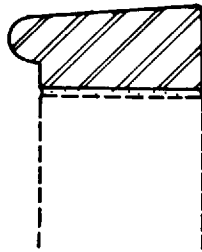
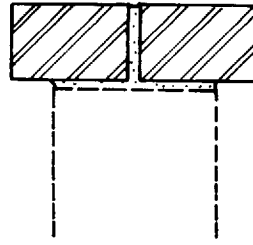
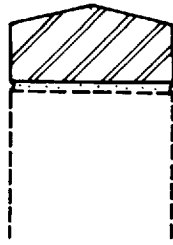


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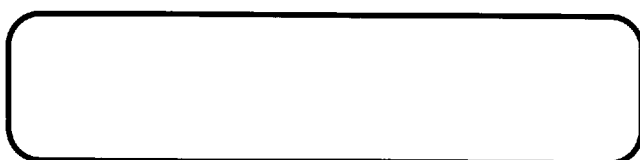


COPINGS &
CAPPINGS, brick



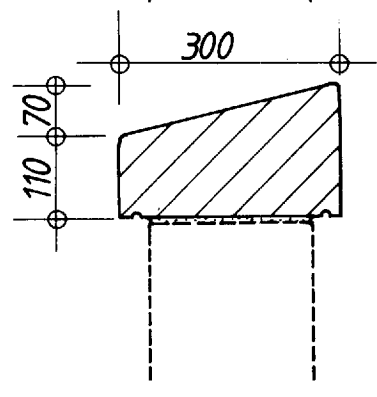
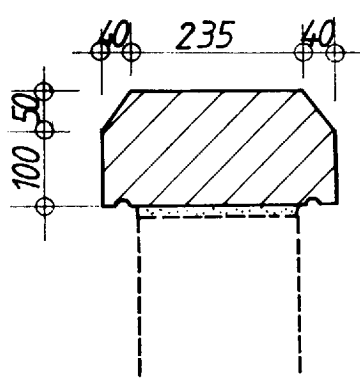
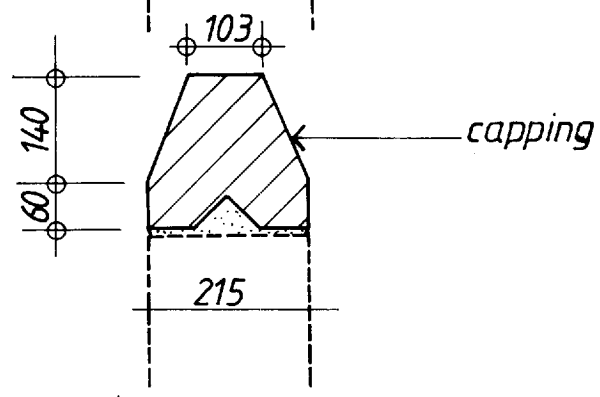
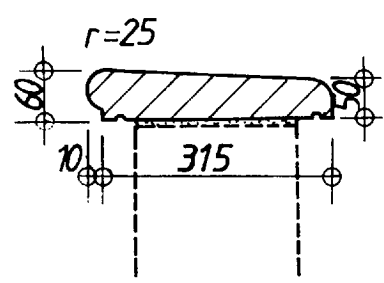
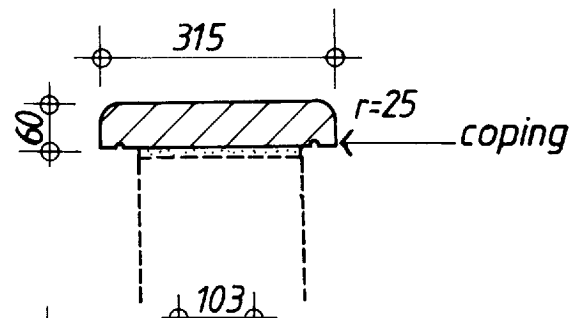
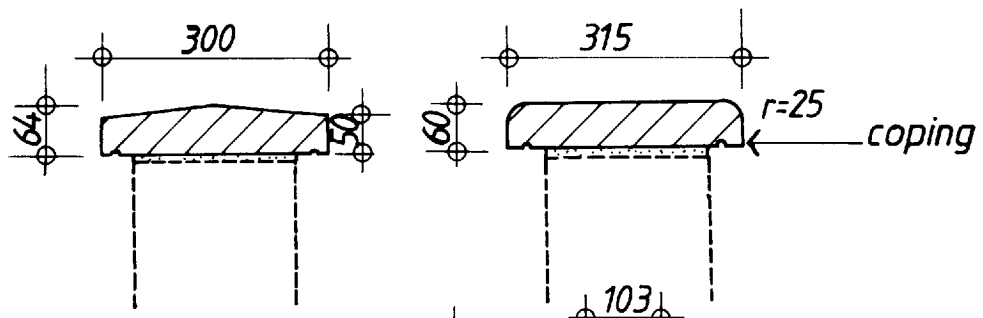


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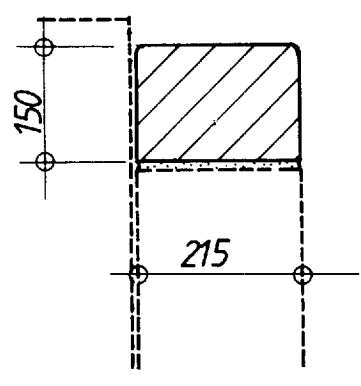
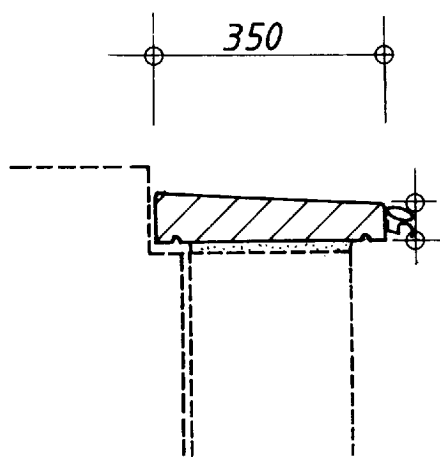


COPINGS &
CAPPINGS, brick

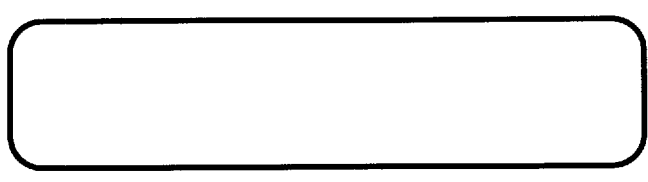




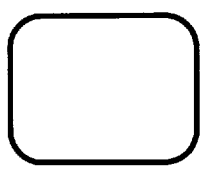
Note dpc is required
between coping/capping
and wall

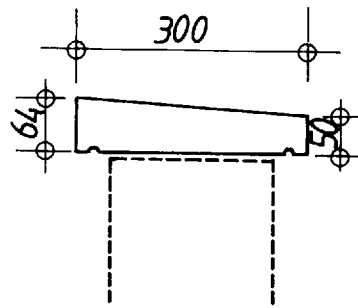
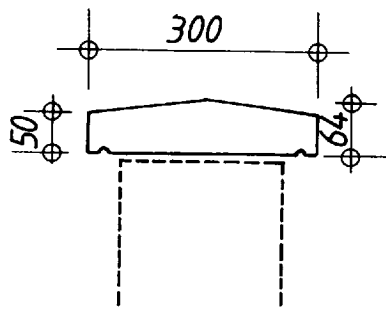


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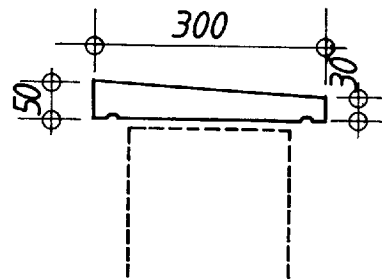
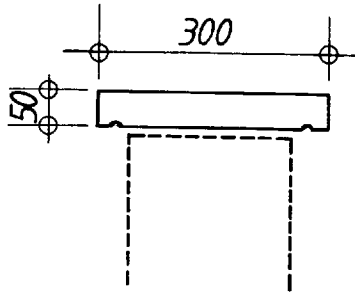


COPINGS &
CAPPINGS, concrete

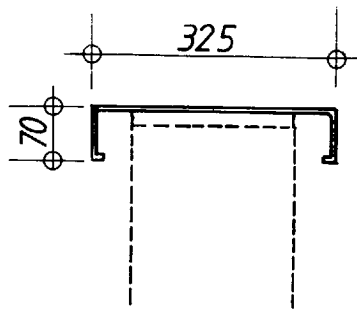




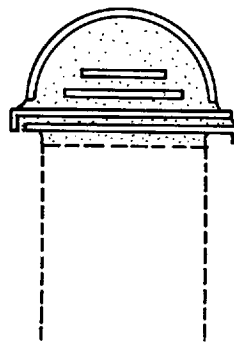
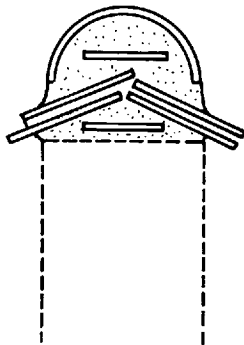
Clay copings



Slate copings

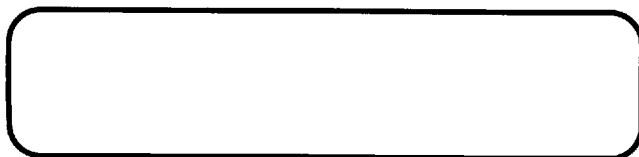


Metal coping



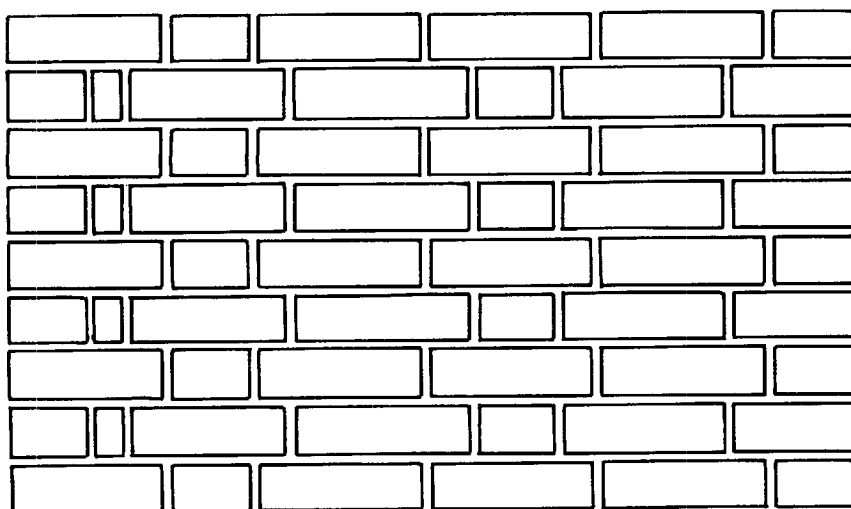
Tile copings

Scale 1:10

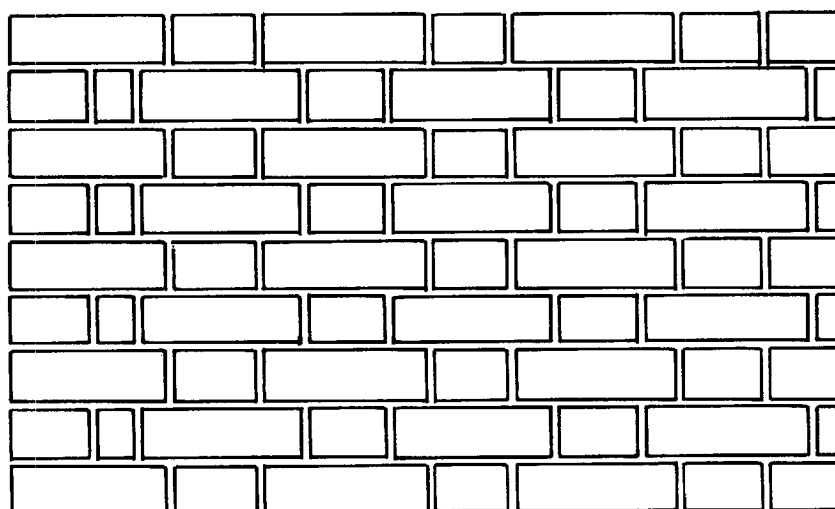


COPINGS
mixed



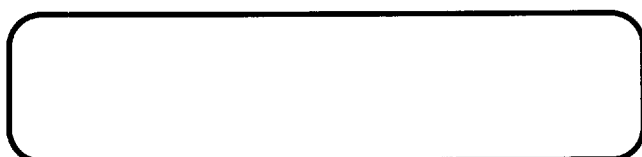


Garden wall

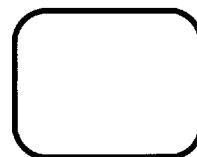


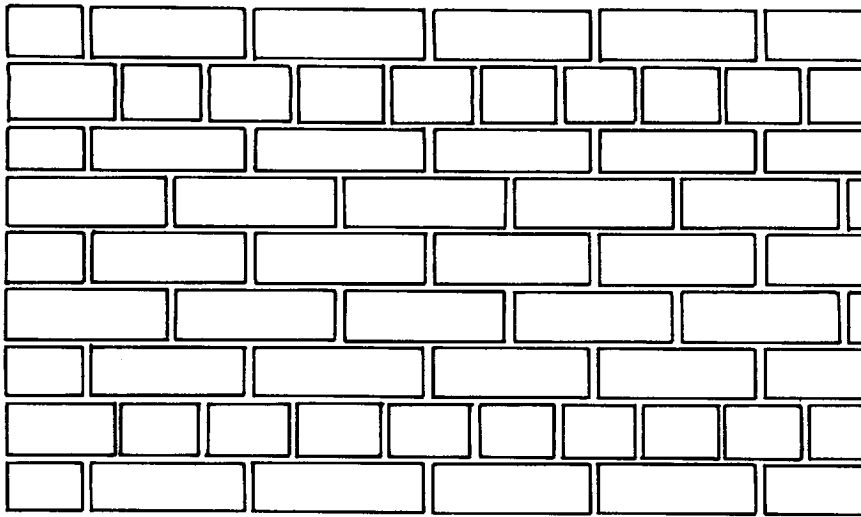
Flemish double bond

Scale 1:10

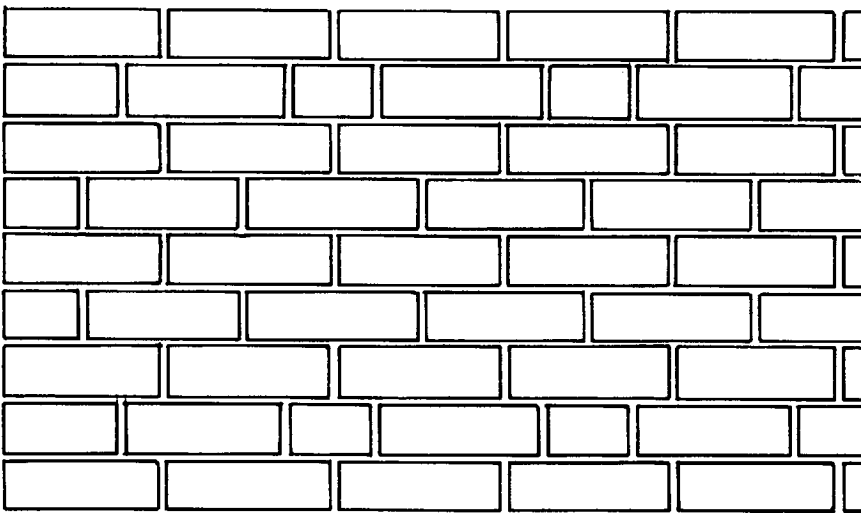


WALLS
brick bonds





6th course headers
Common



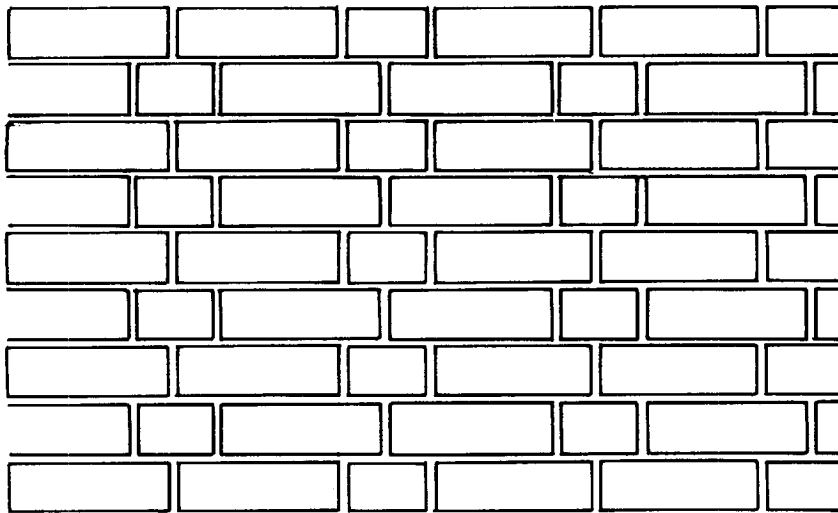
6th course flemish headers
Common

Scale 1:10

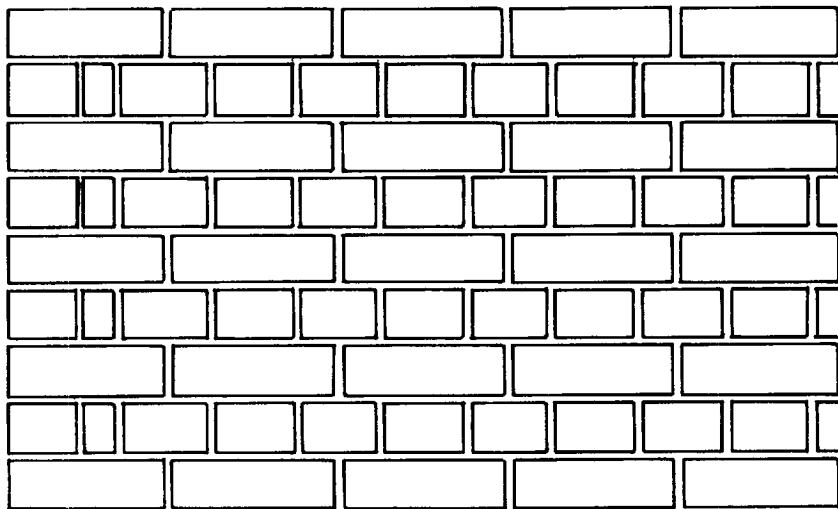


WALLS
brick bonds





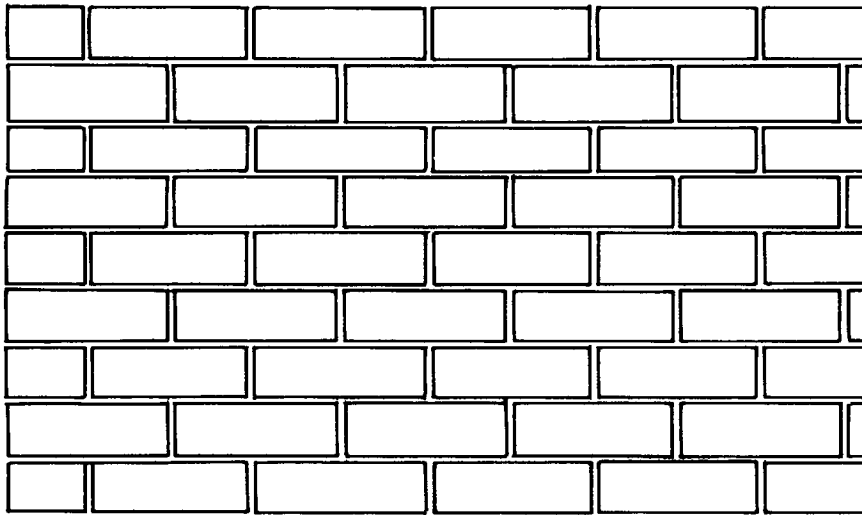
Flemish (double stretcher)



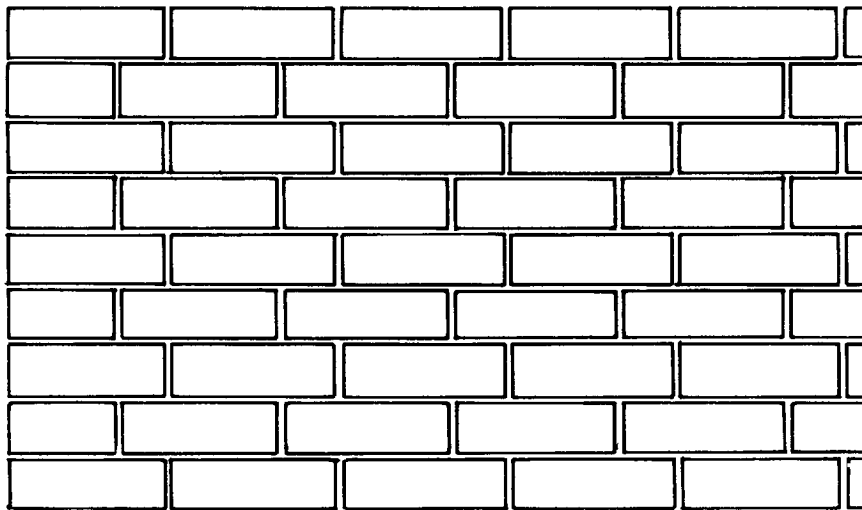
English cross or dutch

Scale 1:10

WALLS
brick bonds



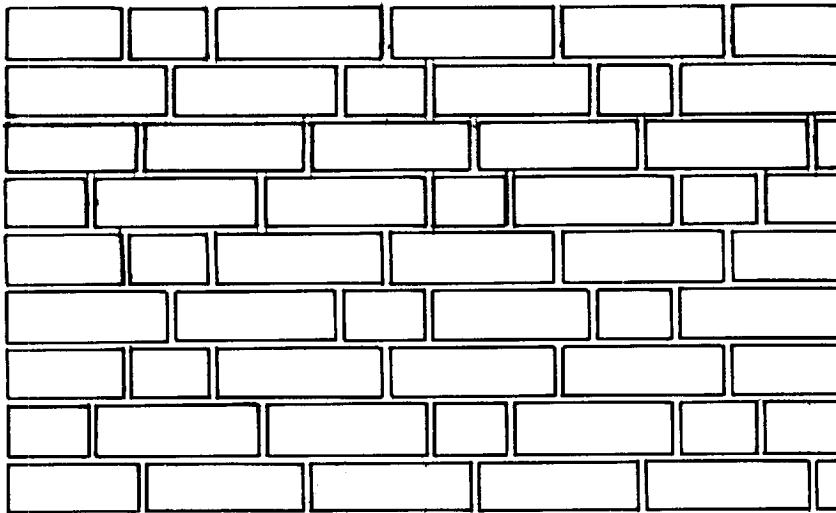
Stretcher bond



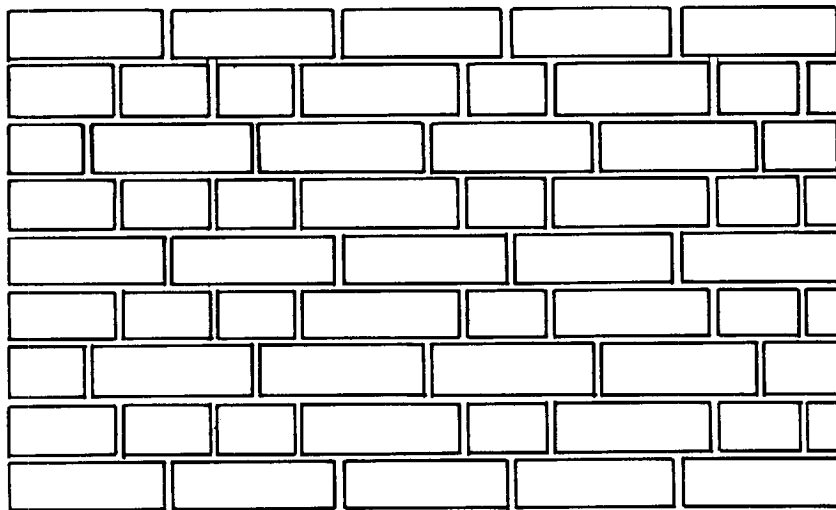
1/3 running

Scale 1:10

WALLS
brick bonds



Flemish (diagonal)



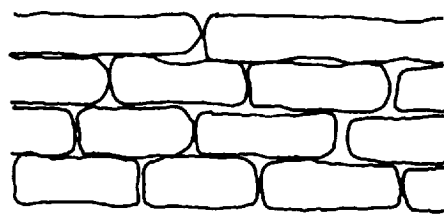
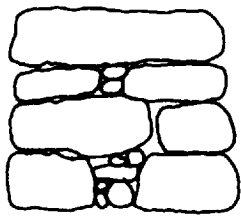
Flemish (cross)

Scale 1:10

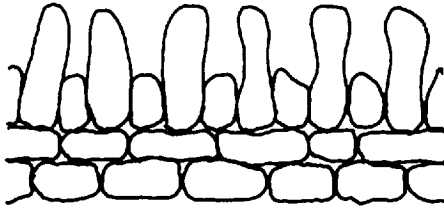
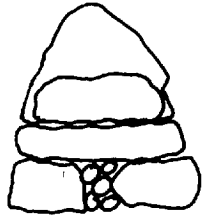


WALLS
brick bonds

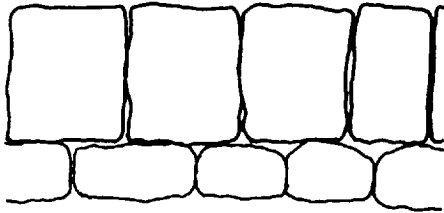
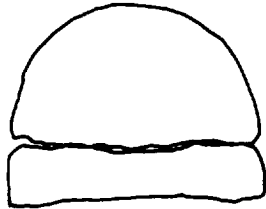




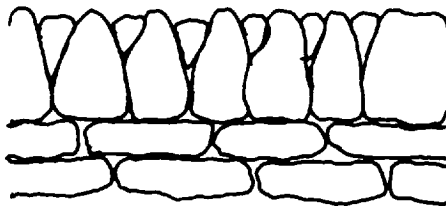
Flat



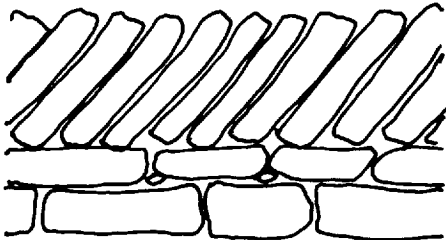
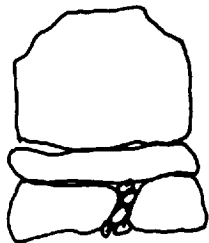
Buck and doe



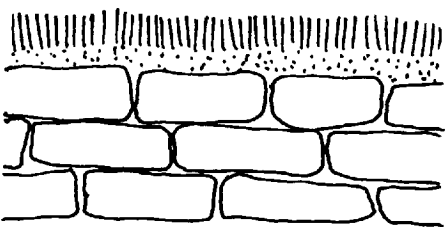
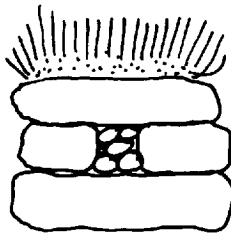
Dressed



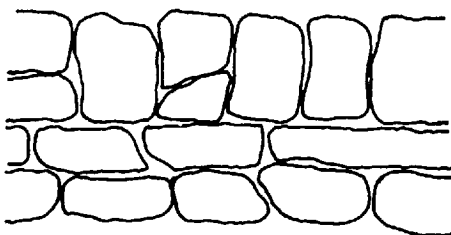
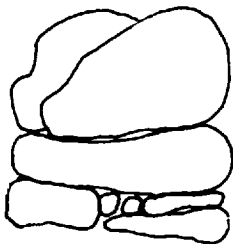
Locked



Tilted

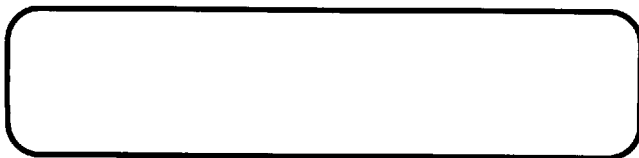


Turf

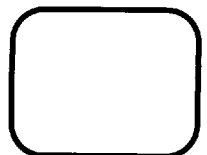


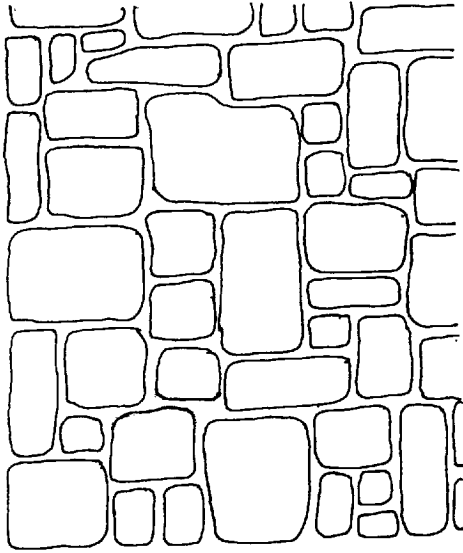
Rubble

NTS

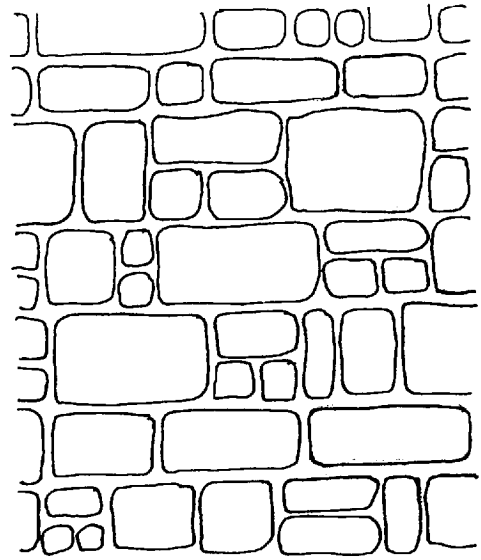


WALL COPINGS
stone

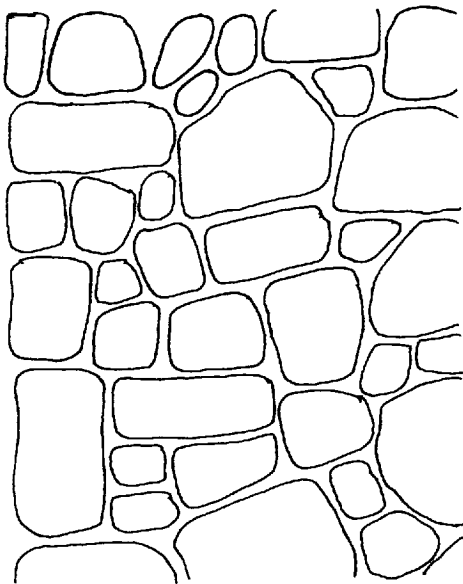




*Uncoursed random with
roughly squared rubble*



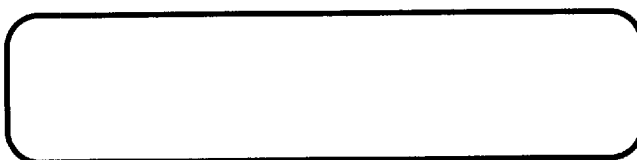
*Coursed random with roughly
squared rubble*



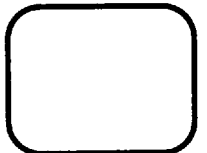
*Uncoursed random with
rough rubble*

ELEVATIONS

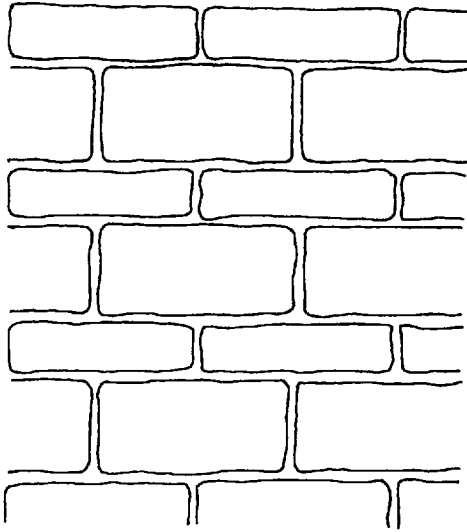
Scale 1:10



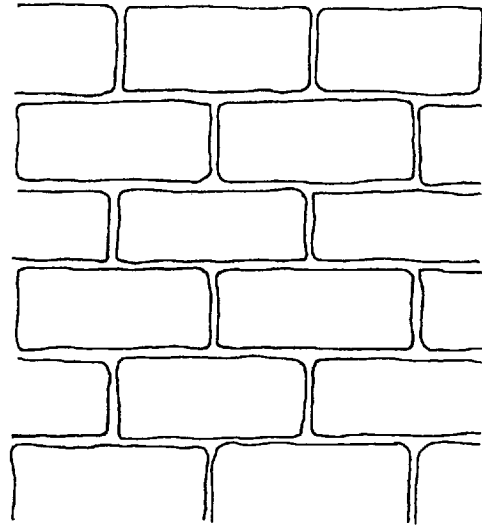
WALL
Random rubble
masonry patterns



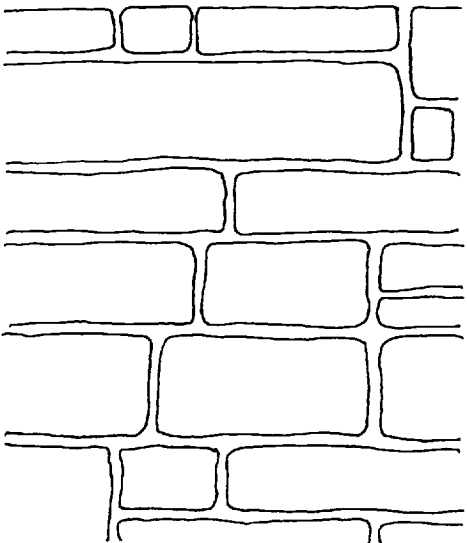
Stone can be up to 600 long
depth 150/200/250/300
height 75/100/125/150



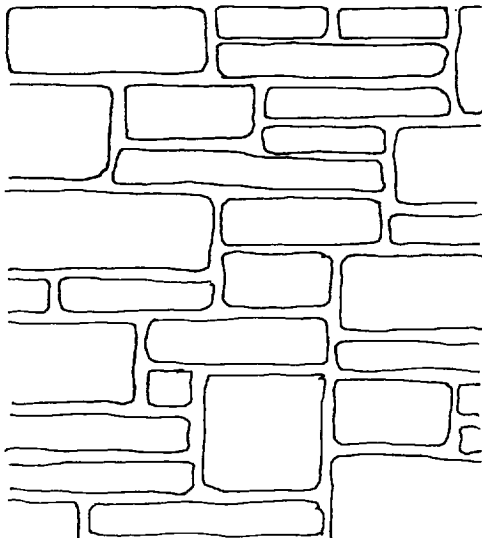
Coursed Ashlar with irregular courses



Coursed Ashlar with regular courses



Coursed Ashlar with broken



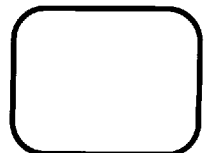
Random coursed random bond

Note: Units are precast and dressed before delivery to site

Scale 1:10



WALL
Ashlar stone
masonry patterns



FREE-STANDING WALLS

GUIDANCE NOTES

Appearance

The choice of materials available today is quite considerable and great care must be taken to ensure that the appearance of any wall is in harmony with the local vernacular of style. Too often, materials are used which bear no relation to either the place where the walls are being built or to the surrounding landscape. Natural stone is generally much more appropriate in rural areas, whereas concrete blocks are better used in urban areas, unless they are rendered or painted. Bricks can be used in almost any locality, provided the selected colour is appropriate to the area.

Natural stone walling should never be used from another part of the country and the local materials and techniques will provide the catalyst for the design. Walls in the open landscape are usually used only as alternatives to fences and hedges where stone is plentiful.

Walls higher than eye level are usually used to form a complete physical and visual barrier and are often more associated with the architecture. Walls below eye level provide partial enclosure and are useful where a view can still be obtained. Dwarf walls are often used as a physical barrier where strength is required by retaining all the visual qualities. Brick walls cannot 'flow' with the landscape in the same way as stone walls, because they cannot take up changes of level without the top of the wall being stepped.

With such a wide choice of bricks available today great care must be taken in their selection, especially over the use of colour and finish. Minor details such as the raking out of joints, for example, will be of less importance in the open landscape scene than in the urban one.

Design factors

Special considerations in the design of external free-standing walls include:

- *Rain exposure*: rain may penetrate the wall from both sides and above, affecting both durability and appearance unless material quality and design detailing take account of the special requirements.
- *Thermal exposure*: damp proof courses in free-standing walls are subject to extremes of temperature, and therefore to high thermal movement.
- *Wind exposure*: in a building, a wall of similar height would be restrained both at the top and bottom by floors and roof and laterally by cross walls. The free-standing wall, in contrast, is an unrestrained vertical cantilever, and is often designed without any structural calculations having been made.
- *Appearance*: there will be two exposed sides to fair-face, unless an applied finish is specified.
- *Foundation design*: because free-standing walls do not carry applied loads there is a tendency to economise on foundation design. This often leads to expensive remedial work.
- *Building in cold weather*: precautions are essential when walls are exposed to wind, rain and frost on both faces.
- *Damp proof courses and movement joints*: detailing may differ from the norm in walls of buildings.
- *Associated planting*: adjacent planting may affect detailing.

Stability

Although in many cases the ratio of height to width (slenderness ratio) is the major factor affecting stability, the list of design criteria should include:

Wall plan shape
Type of bond
Use of reinforcement
Weight of brick and block
Slenderness ratio
Wind load
Movement joints

Foundations
Special strength requirements

Appearance

The stability of a free-standing wall can be improved by adding piers, staggering the wall in bays, or increasing the thickness of the wall. The thickness of a free-standing wall can be estimated by reference to appropriate publications given in Appendix B.

All walls over 2,000 mm high should be referred to a structural engineer for checking. Stability calculations for such walls may also be required by the district surveyor or building inspector. Approval is usually required from a local authority when the height above ground level exceeds 1,800 mm.

Brick

Brick is the most commonly used material for building walls. There are three types that are suitable for external use:

- Commons, which are for general-purpose building, have no finish and are particularly subject to weathering.
- Facings, used where appearance is of prime importance, have either a hand finish or a wire-cut weather-resistant finish on the side facings only.
- Engineering bricks are the hardest and most impervious brick available.

Bricks used in free-standing external walls, being exposed on both sides, are more prone to deterioration through frost actions than the same bricks used, for example, in the external walls of a building.

Frost-resistant bricks are generally those with low water absorbencies. However, certain types of stock bricks with absorbencies of up to 20 per cent are frost-resistant because their highly porous structure allows ice to expand within the brick without causing damage. Users should contact the manufacturers to confirm that their bricks are suitable for the degree of exposure intended.

External free-standing brick walls are also

subject to deterioration following the reaction of soluble sulphates present in the bricks with the tricalcium aluminate constituent in ordinary Portland cement. The reaction is accompanied by expansion of the mortar and subsequent cracking. Sulphate attack can be avoided by the use of a 'special' quality brick as defined by BS 3921: 1974. The BS limits the soluble salt content of bricks of 'special' quality, but not those of 'ordinary' quality. Alternatively, use a richer mortar mix, e.g. 1:3, or a sulphate-resisting cement.

Thickness

The thickness and height of a brick wall determine the frequency of piers or buttresses used to provide support. None is required if the wall is staggered in a zig-zag way or by bringing the line of the wall forward and backward by the thickness of the brick. Curving or serpentine walls will also support themselves, although they need more space. Single-brick walls look unstable, whereas a double-brick wall looks more safe and solid. In all but exceptionally exposed positions a double-brick wall, unbuttressed, should be satisfactory for heights up to 2.0 metres.

Concrete blocks

These are very strong and, because they are also much larger than bricks, a wall can be constructed more quickly and therefore will be cheaper. The large size of cavity blocks used for walling makes piers unnecessary for a standard wall, but solid concrete blocks are half the thickness and will require the support of piers used with the same frequency as for brick walls.

Similarly, concrete foundations are required as well as copings. Concrete blocks are now available in a wide range of finishes and colours to match local brick or stone. The blocks may be utilised in a variety of ways to produce either a decorative or a plain finish. They can easily be combined with other materials, such as brick, to relieve monotony in a long length of wall. Concrete blocks used in free-standing external walls should be type A, as defined in BS 2028,

1364: 1968. A rough surface texture helps reduce weather staining. Special pierced concrete blocks are also available. Calcium silicate bricks should be Class 3 or higher as defined in BS 187: Part 2: 1978.

Stone

Natural stone

Most kinds of stone can be used for wall construction – granite, limestone, slate or sandstone. The two main types of stone-walling are ashlar and rubble walling, of which the latter comprises three groups:

- Random rubble, which is of uncut stone, is laid coursed or uncoursed.
- Squared rubble, which is of roughly dressed stone, is laid regularly coursed, irregularly coursed, or uncoursed.
- Miscellaneous rubble walling uses traditional materials and construction methods.

Most stone walls are built in random rubble. No cutting is done to fit the stones together and when they are placed within the wall there is an adequate distribution of pressure over a maximum area, with no continuous vertical joints. To stabilise the wall, header stones are used approximately every square metre and should ideally run right through the wall. Whether this type of wall is coursed or uncoursed, all joints should be well filled and flushed with mortar. Variation can be achieved by filling some of the joints with soil and growing plants in them. To construct a wall in stone of which one side only is visible, a concrete block wall can be faced with stone, tying the two together with galvanised wall ties. The elements of a natural stone wall are usually thick enough to be self-supporting and do not require any piers or buttresses. Natural stone walls usually have the coping or cappings traditionally used in the area.

Reconstituted stone

Different areas produce different rock, which can be crushed and used as the aggregate in the composition of

reconstituted stone blocks. Reconstituted stone block is easier to transport and lay than natural stone. Walls of these materials must be laid in the same way as for bricks.

Mortars

It is essential that there be a permanent tensile bond between the bricks and the mortar to ensure that free-standing walls resist lateral pressure. In sulphate-free conditions use a relatively weak mortar, e.g. 1:1:6, with clay bricks. If sulphates are present in the bricks a sulphate-resisting cement or a stronger mix should be used, as recommended above. Calcium silicate bricks are generally laid in a weaker mix than clay bricks. Brickwork below ground level and up to 150 mm above ground level should be laid in a 1:3 cement/mortar mix. See Tables 1 and 2.

Damp proof courses

A damp proof course, 150 mm from ground level, is required. The use of any damp proof course near ground level must provide the necessary adhesion across the joint. Only two methods meet this requirement:

1. Two courses of (engineering) bricks, to BS 3921: 1974 or BS 743: 1970, having a water absorption of not more than 7 per cent laid in 1:(1/4):3 mortar.
2. Two courses of 5 mm slates fully half-lapped and bedded in 1:(1/4):3 mortar to BS 743: 1970.

In certain situations where walls are constructed of frost-resistant bricks a damp proof course may not be necessary. A damp proof course is also required under the coping if the coping is of a porous material such as brick on edge. If the coping is of an impermeable material, such as metal, PVC or engineering bricks, no damp proof course is needed. Although natural slate or clay tile is the best damp proof course in this position, a lead core bituminous damp proof course can also be used.

If a damp and mossy appearance to the lower parts of a wall is not objected to, the

damp proof course may be omitted provided that the bricks be of 'special' quality and a strong mortar mix (e.g. 1:3) is used. The same applies to the damp proof course under the coping.

Expansion joints

Expansion or movement joints are necessary to accommodate thermal and moisture expansion. They must be constructed in such a way as to ensure a complete separation over the full height of the wall, including the damp proof course and coping/capping, but not the foundations. A 10 mm joint should be provided every 12 m for clay, bricks and concrete blocks and every 6 m for calcium silicate (sand lime) bricks. Expansion joints may be left open but are more commonly filled with a proprietary flexible mastic jointing compound.

Foundations

The depth and width of a foundation will vary according to the type of subsoil. On average subsoils a depth of 450–600 mm is usually sufficient; on shrinkable clay the depth may have to be 900 mm or greater. A maximum width of 525 mm for unreinforced concrete foundations for 225 mm wide walls is advised, but site conditions will determine a final size after consultation with an engineer. A concrete mix of 1:6 is used in all the details, although other mixes may be equally suitable. Where a wall is below 500 mm in height, foundations can be omitted altogether. The topsoil should be removed and the brickwork built off the top of the subsoil, which should be thoroughly compacted. Where more than 150 mm of topsoil has to be removed, the level of the excavation can be made up with well-consolidated granular material. The depth of facing brickwork below ground should not exceed 200 mm.

TABLE 1. MORTAR MIX GROUPS FOR BRICKWORK AND BLOCKWORK

	Group 1	Group 2	Group 3	Group 4
Cement:lime:sand	1:0–¼:3	1:½:4–4½	1:1:5–6	1:2:8–9
Cement:premixed lime and sand (proportions of lime to sand given in parentheses)	1:3 (1:12)	1:4–4½ (1:9)	1:5–6 (1:6)	1:8–9 (1:4½)
Cement:sand and air entrainer	–	1:3–4	1:5–6	1:7–8
Masonry cement:sand	–	1:2½–3½	1:4–5	1:5½–6½

Note

Specify the mortar by group or by type and proportions, including any special requirements, e.g.

Group 3 as clause 460 with sulphate-resisting cement.

1:1:6 cement:lime:sand.

1:5–6 white cement:selected white sand.

TABLE 2. MINIMUM QUALITY OF MORTARS FOR BRICKWORK AND BLOCKWORK IN VARIOUS SITUATIONS

The mortar groups given are the weakest of those recommended in BS 5628: Part 3, table 13. Higher-strength mortars may be required for environmental (see notes below) or structural reasons. Also, for work below ground, if the soil or ground water contains sulphates it may be

necessary to use a stronger mortar mix (i.e. containing more cement) and/or sulphate-resisting cement. If mortar is susceptible to sulphate attack then so also may be concrete bricks and blocks used in the same location.

Masonry condition or situation	Clay silicate	Calcium Brick	Block	Concrete
Work below or near external ground level				
• low risk of saturation with or without freezing	3	3	3	3
• high risk of saturation without freezing	2*	3	3	2
• high risk of saturation with freezing	2*	2	3	2
Brick dpcs	1	NS	NS	NS
Unrendered external walls (other than chimneys, cappings, copings, parapets, sills)				
• low risk of saturation	3	4†	3	4†
• high risk of saturation	2**	3	3	3
Rendered external walls (other than chimneys, cappings, copings, parapets, sills)	3*	4†	3*	4†
Internal walls and inner leaves of cavity walls	4†	4†	4†	4†
Unrendered parapets (other than copings or cappings)				
• low risk of saturation, e.g. low parapets on some single-storey buildings	3	3	3	3
• high risk of saturation	2*	3	3	2
Rendered parapets (other than copings or cappings)	3**	3	3	3
Copings, cappings and sills	1	2	2	2
Free-standing boundary and screen walls (other than copings or cappings)				
• with coping	3*	3	3	3
• with capping	2	3 SRC	3 SRC	2
Earth-retaining walls (other than copings or cappings)				
• with waterproofed retaining face and coping	2*	3	2	2
• with coping or capping but no waterproofing on retaining face	1*	2	2	2
Manholes, inspection chambers, culverts				
• surface water	1 SRC	3	3	2
• foul drainage (continuous contact with walling)	1 SRC	2	2	NS
• foul drainage (occasional contact with walling)	1 SRC	2	2	NS

NOTES

† Where group 4 mortar is used it is essential that the walling under construction is fully protected from saturation and freezing. Alternatively, specify group 3.

* If clay bricks with a normal soluble salt content (as defined in BS 3921) are specified, sulphate-resisting cement should be used.

** If clay bricks with a normal soluble salt content (as defined in BS 3921) are specified, sulphate-resisting cement should be used or the mortar changed to the next stronger group.

NS not suitable.

SRC sulphate-resisting cement.

TABLE 3. FREE-STANDING BRICK WALLS: CONSTRUCTIONAL DETAILS

Thickness (mm)	Max. height (m)	Piers		Expansion joints		Mass concrete foundations		Notes
		Size (mm)	Spacing (m)	Preferred spacing (m)	Max. spacing (m)	Min. width (mm)	Min. depth (mm)	
112	1.2	225 × 112	4.5	9	9	337	150	Expansion joint at every second pier position: 12 mm joint between two No. 225 × 112 mm piers
112	1.8	337 × 225	4	8	9	337	150	Expansion joint at every second pier position: 12 mm joint between two No. 225 × 225 mm piers
112	2.4	337 × 225	2.6	7.9	9	337	150	Expansion joint at every third pier position: 12 mm joint between two No. 225 × 225 mm piers
225	1.8	—	—	6	9	675	225	
337	2.75	—	—	6	9	1,012	337	

SPECIFICATION CHECK LIST

Brick/block walling

General

Specification may be by reference to a standard by proprietary name(s) or both. Clause headings should describe the use and/or location of the brick/blockwork in a helpful way, e.g.

FACING BRICKWORK ABOVE DPC
FACING BLOCKWORK FOR WALLS
ENGINEERING BRICKWORK FOR
MANHOLES

The clause heading need not be absolutely precise but more a confirmation that the clause is the correct one for the situation identified and/or cross-referenced on the drawings or in the schedule of work.

For foundations see separate section.

Type of walling

Select type of walling required, e.g.

Facing brickwork
Second-hand facing brickwork
Facing blockwork
Reconstituted stonework
Common brickwork
Common blockwork
Engineering brickwork

Drawings

State drawing references

Materials

Clay bricks

BS 3921 designates clay bricks according to frost resistance and soluble salt content.

Calcium silicate bricks

BS 187 classes 3 to 7 can be used in any situation with the exceptions of:

Cappings, copings and sills	Minimum class 4
Manholes with foul effluent in continuous contact with the brickwork	Minimum class 7
Damp proof courses	Not suitable

See BS 5628: Part 3, table 13.

Concrete bricks and blocks

BS 6073: Part 1 does not specify functional properties such as thermal conductivity, nor does it give recommendations as to suitability for particular purposes.

Concrete bricks and blocks should not be used in contact with ground from which there is danger of sulphate attack unless they have been specifically made for this purpose, i.e. with sulphate-resisting cement and/or a certain minimum cement content – see BRE Digest 250.

For detailed guidance on durability see BS 5628: Part 3, table 13.

Reconstituted stonework

This clause is for proprietary blocks which simulate natural stone rubble walling, e.g. rough hewn brought to courses, Ashlar coursed and bonded as shown on drawings.

Provide details of make, manufacturer's reference, size, colour and finish.

Special shapes

Bricks should be to BS 4729 as shown in drawings. Specify detail of special shapes for blocks.

Mortar

State mix for mortar. Where mortar is specified by group number select any mortar in that group, using the same mortar throughout any one type of facing work.

For guidance on selection see BS 5628: Part 3, table 13 or BRE Digest 160. In general, stronger bricks and blocks and more severe exposure conditions require mortars of higher strength, i.e. with a greater cement content. When coloured mortar is required, it is better to specify coloured ready-mixed material. Site mixing using pigments is rarely satisfactory;

adequate control of colour consistency being very difficult.

Samples

Indicate if any samples of bricks or blocks are to be seen for approval prior to commencement of works.

Damp proof course

State type of damp proof course required for both wall and coping.

Workmanship

Foundation

Provide details on type and size of foundation.

Laying

State details of wall construction.

Bond

Describe the type of bond required. See illustrations of brick and block bonding. To reduce the number of types of brickwork, one type can cover more than one bond if these are in different thicknesses and the brickwork is identical in other respects.

Mortar

Give details on the use of mortar and its strength

Joints

This item applies only to facework, including fair-facework. Wall faces which are to be plastered or which will not be visible in the finished work are covered by general clauses.

Profiles may be specified as:

Flush
Weathered
Recessed square
Bucket handle
Approved

Movement joints

Describe type and material of movement joint required along with frequency.

Features

Insert details of any unusual requirements,

cross referring where appropriate to other sections.

Damp proof course

Ensure bricks are to BS 3921 Engineering Class. Give details of mortar, mix, bond and joints.

Copings

Describe the type of coping required and type of mortar as copings are vulnerable to frost action.

Control samples

The specification of control samples should be related to the size and importance of the job. It may be difficult to justify control samples on small jobs. Insert type, item(s) of walling reference, location, minimum size and any features to be included, i.e. copings, etc.

Inclement weather

State procedure for work in wet and cold weather, storage of bricks/blocks and any other materials. Describe remedial works where mortar has failed.

Appearance

State requirements to achieve uniform appearance with consistent joints cut bricks/blocks.

Cleanliness

Describe method for keeping work clean.

Jointing

Specify method of jointing and profile.

Pointing

State procedure for pointing and profile.

Protection

Give details of how work is to be protected both during and after construction.

Natural stone walling

General

Specification may be by reference to a standard or proprietary names or both. This section deals with the construction of walling using natural stone blocks of varying

sizes. The stone may be used as it comes out of the quarry or can be roughly dressed. Rubble stonework is principally used in external walls built from local granite, whinstone, limestone or sandstone. Rubble walls may be stone faced both sides, or stone faced one side and backed with bricks or blocks separated by a cavity. This section does not cover the repair, renovation or cleaning of existing stonework. If ashlar dressings are required they should be specified separately. For brickwork and blockwork used in association with rubble walling use separate specification.

Type of walling

Use this to briefly describe the type of walling required, e.g.

Random rubble uncoursed
Random rubble brought to course
Squared rubble uncoursed
Squared rubble sneaked
Squared rubble coursed
Kentish rag
Lakeland

Materials

Stone: Insert the stone required, e.g.

Portland Roach
Clipsham
Darley Dale
Springwell

Visit the quarry and select stone with regard to proven durability, weather resistance and suitability for local atmospheric conditions and the nature of the work.

For advice on selection see BS 5390, BRE Digest 269 and *Stone in Building* published by The Architectural Press. For sources and availability see the *Natural Stone Directory*.

Size of stones will vary with type and quarry, but generally allow 150–180 mm for thickness. Stones used in rubble walling are usually small enough to be manhandled. State type of finish required, e.g. 'as found' or hammered, tooled or sawn.

Type of joints

State either

Flush or
Bucket Handle

Features

Insert brief details of any unusual requirements, e.g. wall laid to batter against concrete block wall.

Mortar

Insert proportions of mix required, including any special requirements, e.g. 1:3:12 white cement:lime:crushed stone.

See BS 5390, clause 23 for mortar mixes. Consult local specialists for advice on composition and proportions of mortar mixes to suit the type of stone, method of walling and exposure conditions.

Accessories

Wall ties, fastenings, damp proof courses, etc, should be specified.

Workmanship

Operatives

State that work is to be executed by skilled masons. Request evidence of previous experience and work.

Appearance

Include clause requesting inspection of samples of dressed stone before delivery to site.

Inclement weather

State procedure for work in wet and cold weather, storage of stone and any other materials. Describe remedial works where mortar has failed.

Control samples

A panel will only be necessary to verify that the contractor can provide the standard of workmanship required and/or to check aspects of design. For simple panels, one square metre should be sufficient. List any features which are to be included in the panel.

Laying

Describe procedure for laying generally including size, shape and colour of stones, quoins, jambs and bonders.

Courses and joints

Specify how coursed work and joints are to be undertaken.

Protection

Give details of how work is to be protected during and after construction.

DETAIL SHEETS

Brick

Brick (low) (2)
Brick (medium) (2)
Brick (high) (4)
Brick with piers
Brick rat trap bond
Brick serpentine
Brick with stone plinth

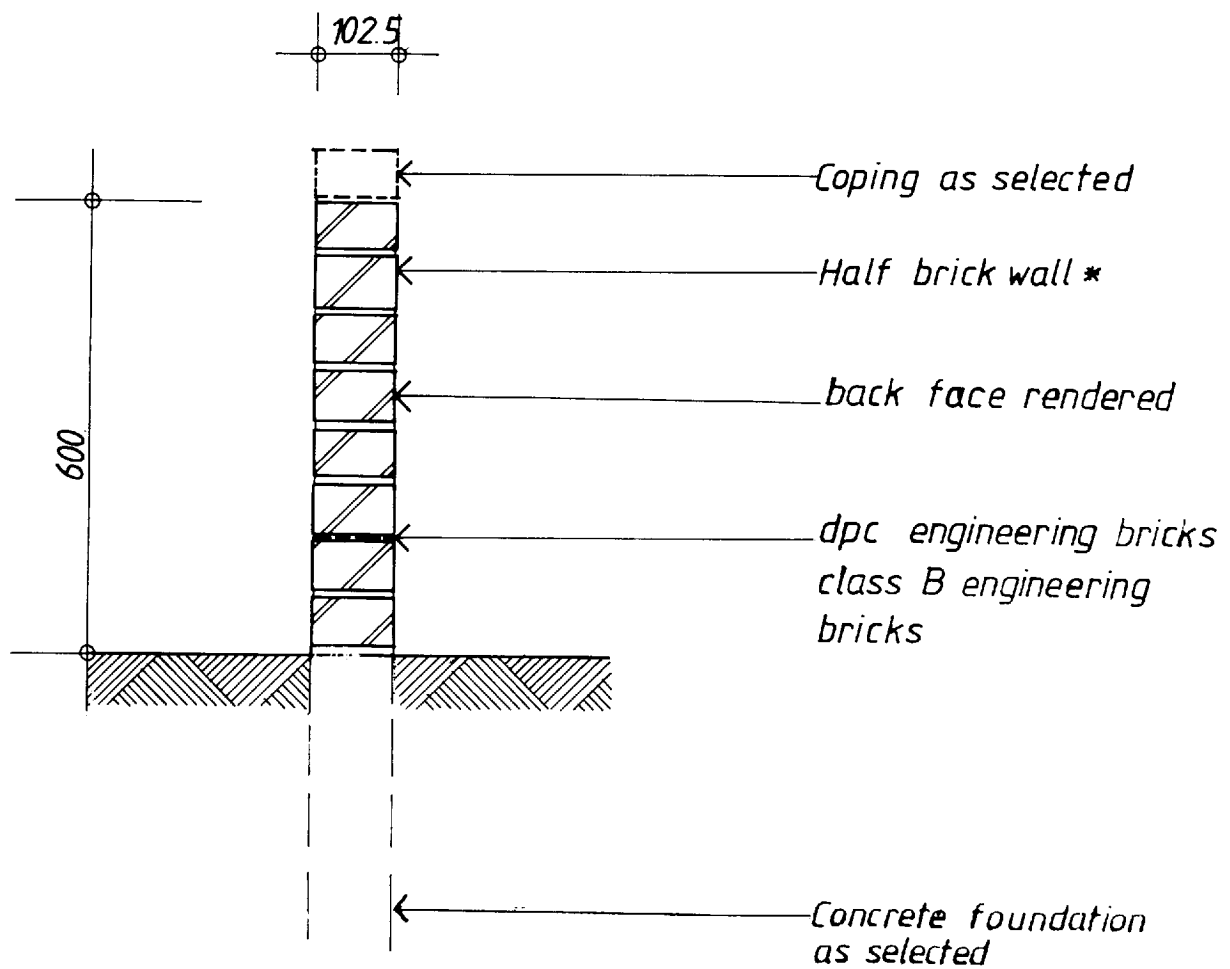
Concrete

Concrete
Concrete block (low)
Concrete block (high)
Concrete block (gravity)

Stone

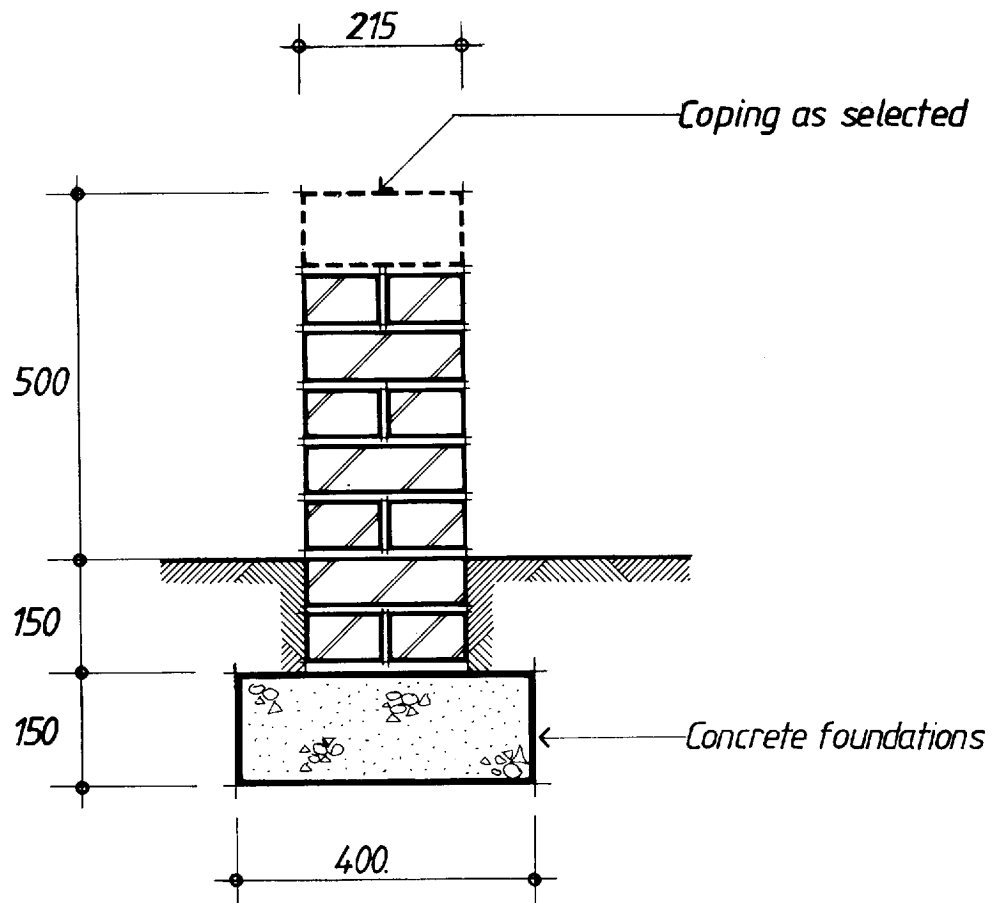
Cornish hedge (stone)
Hedge bank and fence
Stone hedge
Natural stone: mortared/coursed
Ashlar: mortared/coursed
Natural stone: mortared/uncoursed
Natural stone, dry

- * Only certain bricks have an acceptably visual finish to the back stretcher face. This could be rendered

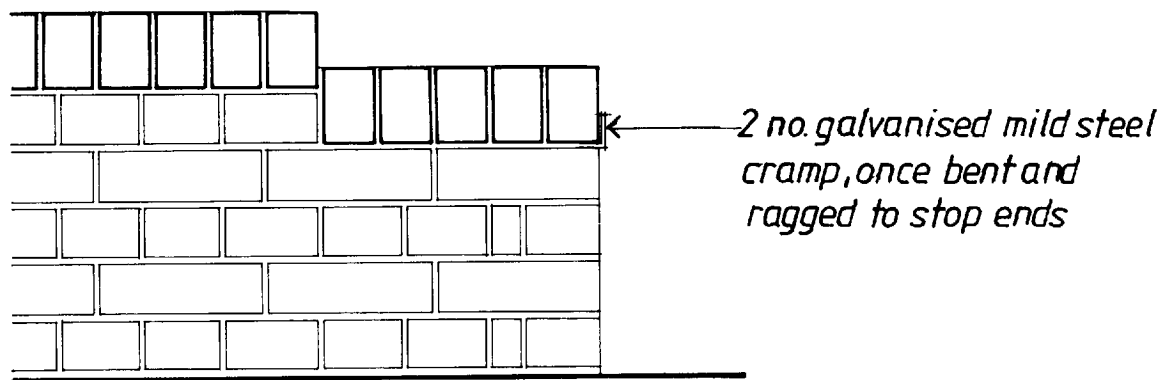


Scale 1:10

WALL
brick (low)



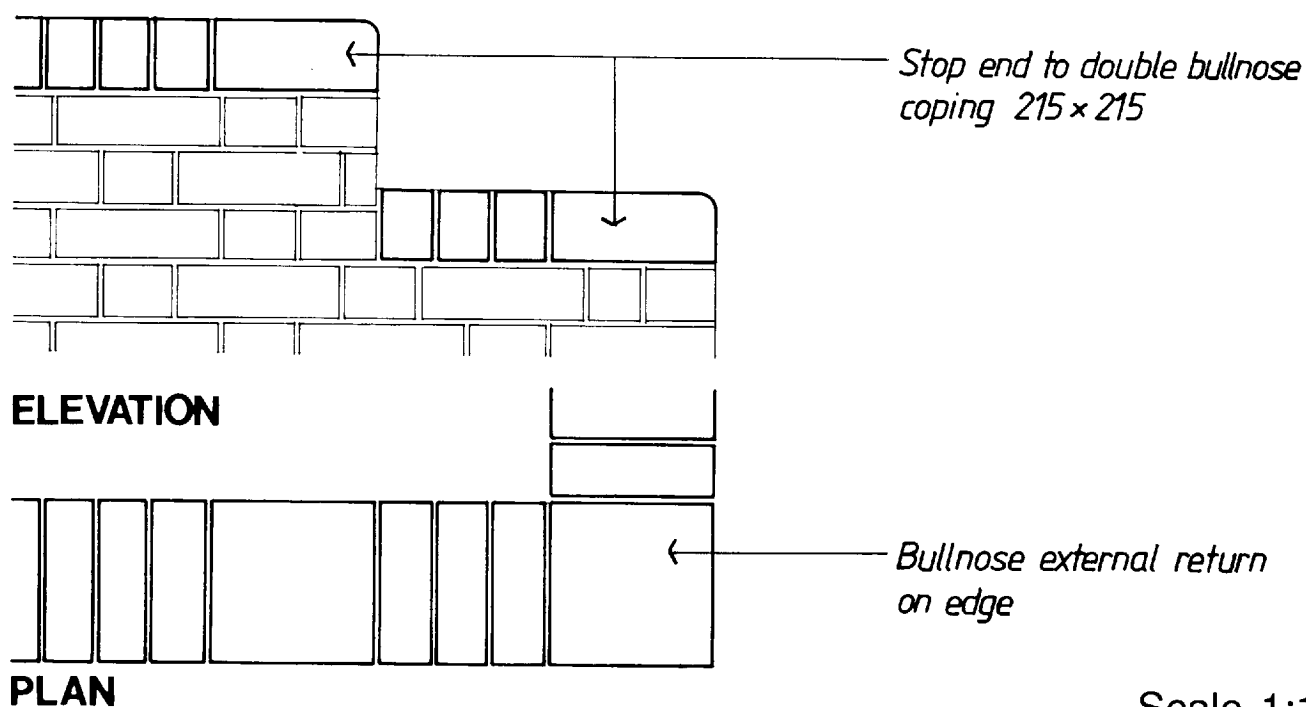
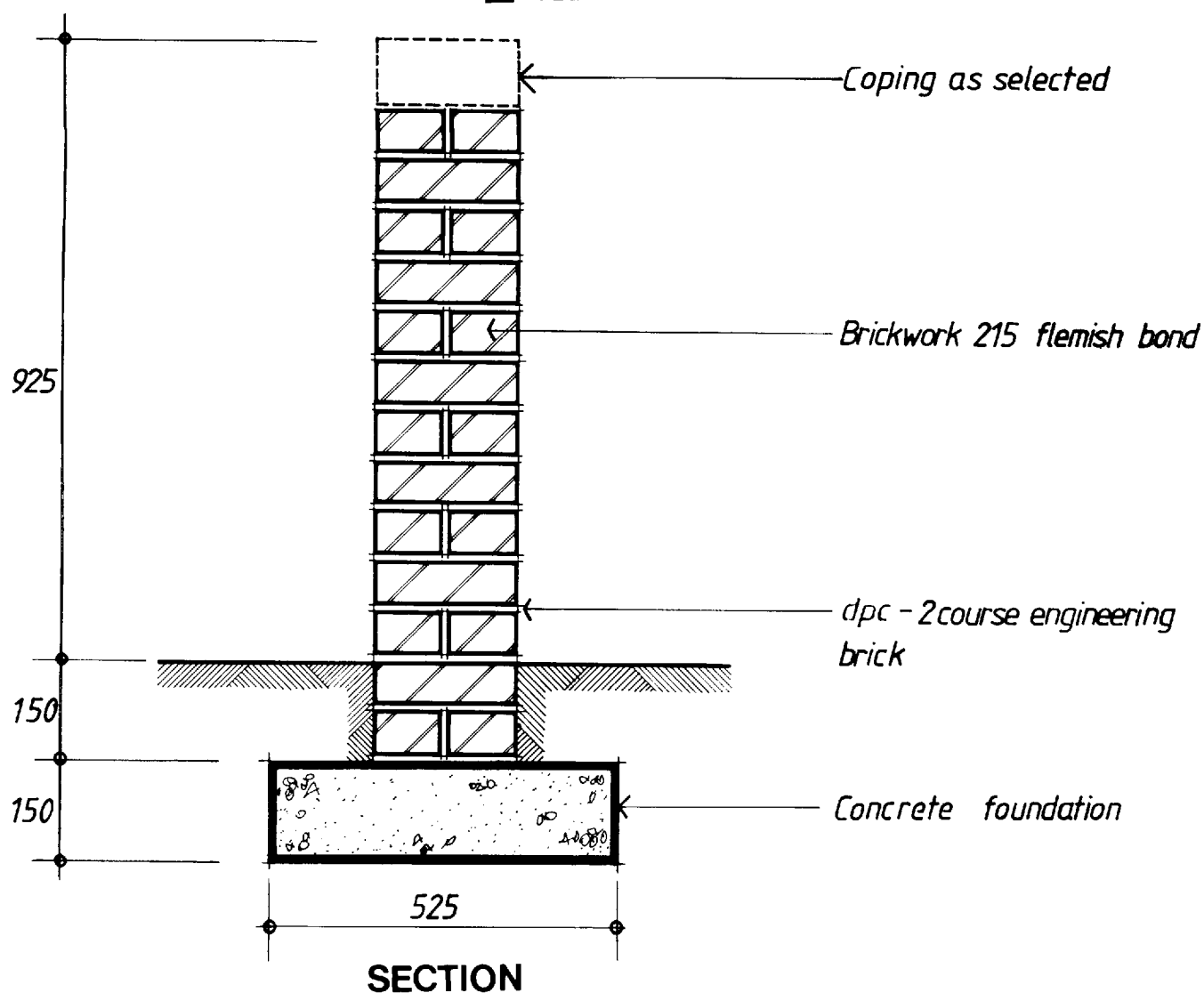
SECTION



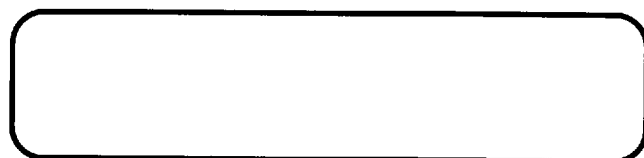
ELEVATION

Scale 1:10

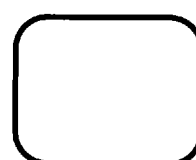
WALL
brick (low)

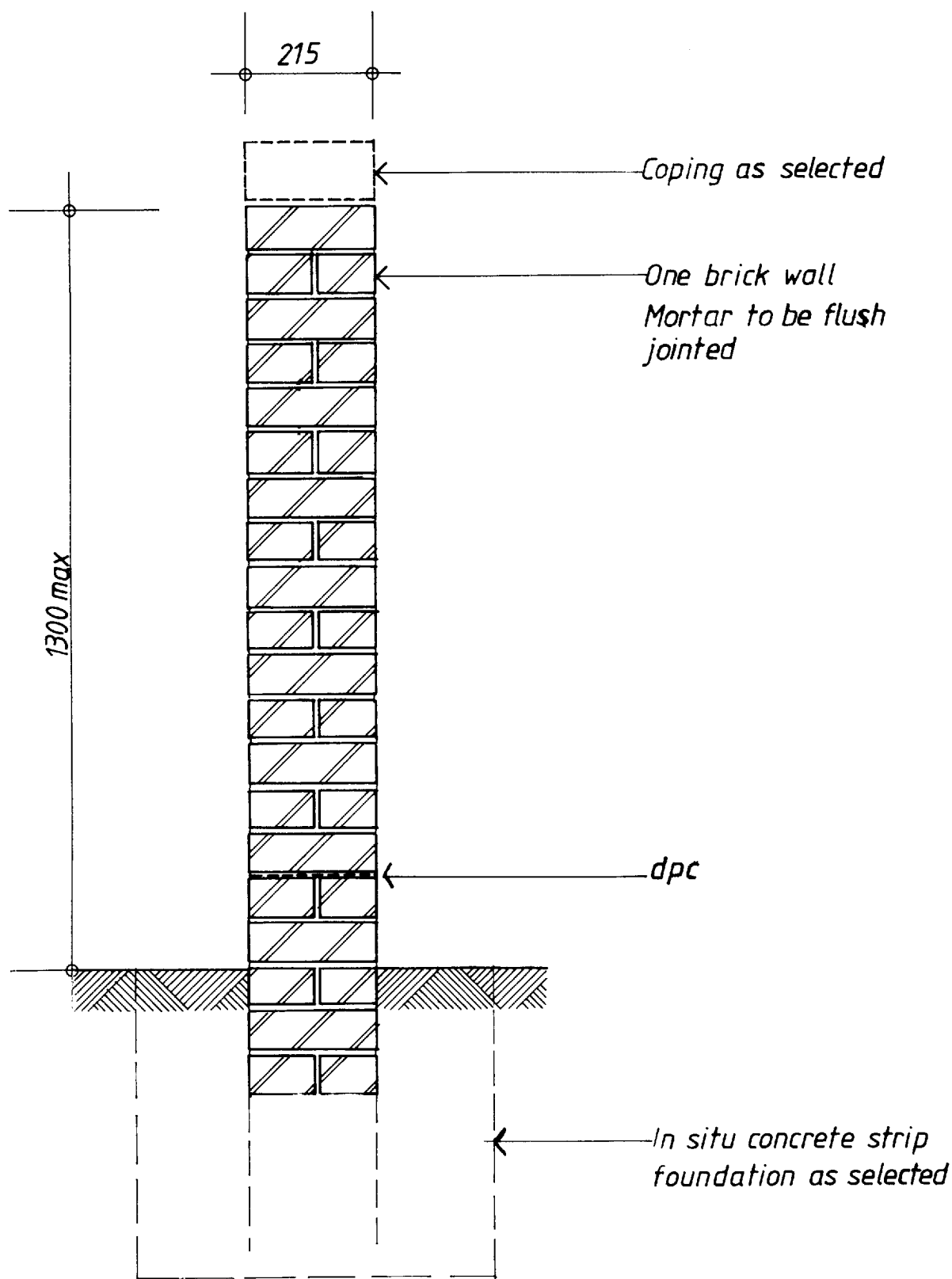


Scale 1:10

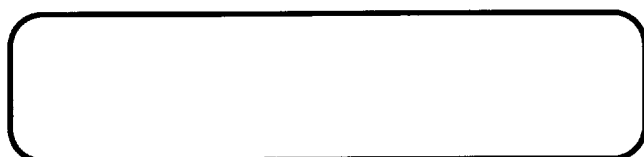


WALL
brick (medium)

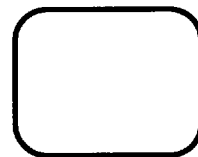


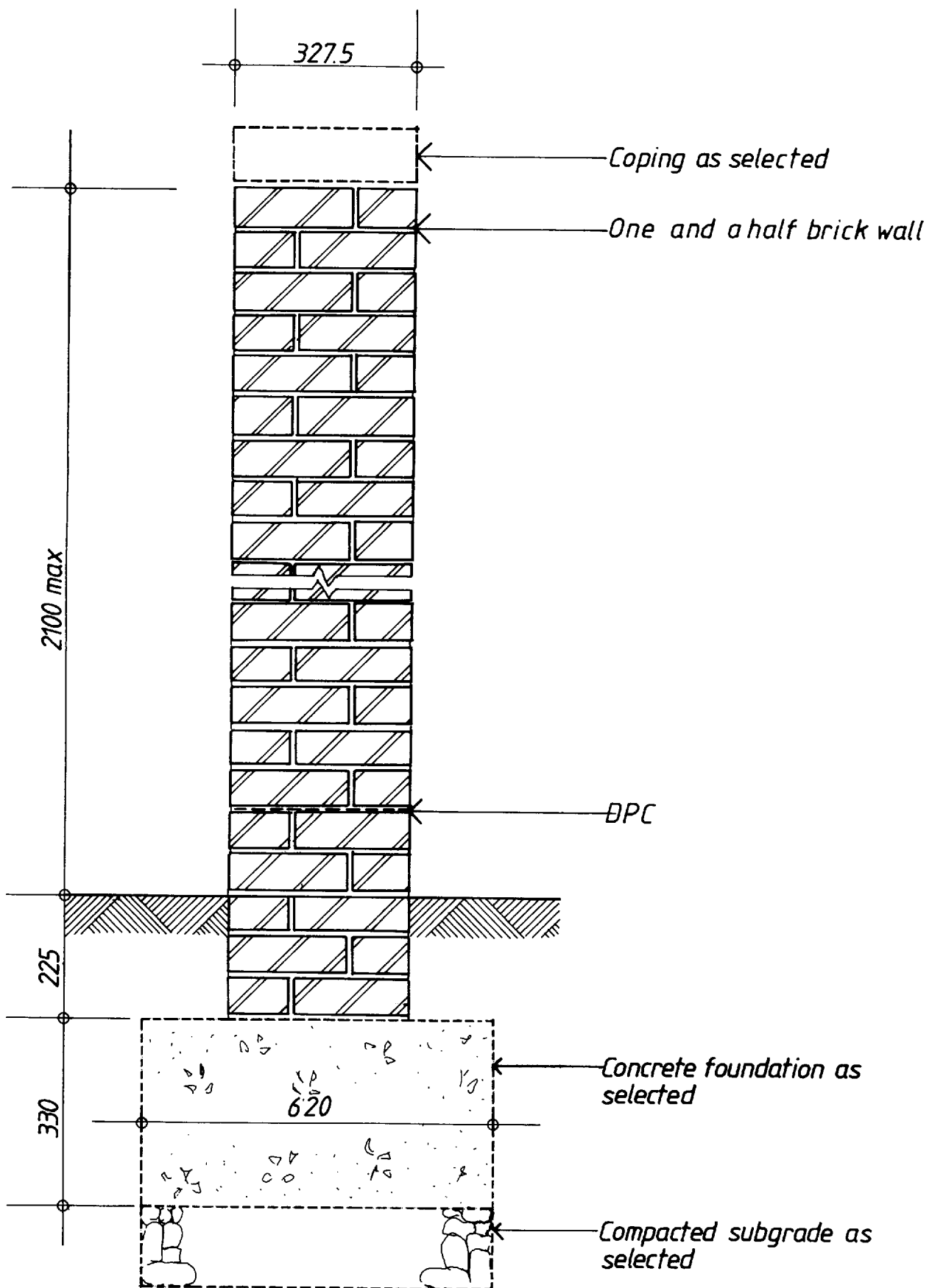


Scale 1:10

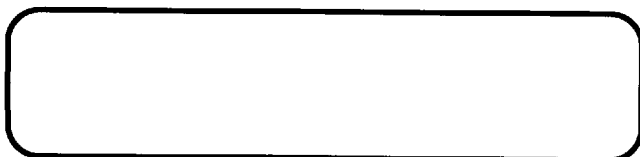


WALL
brick (medium)

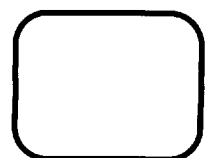


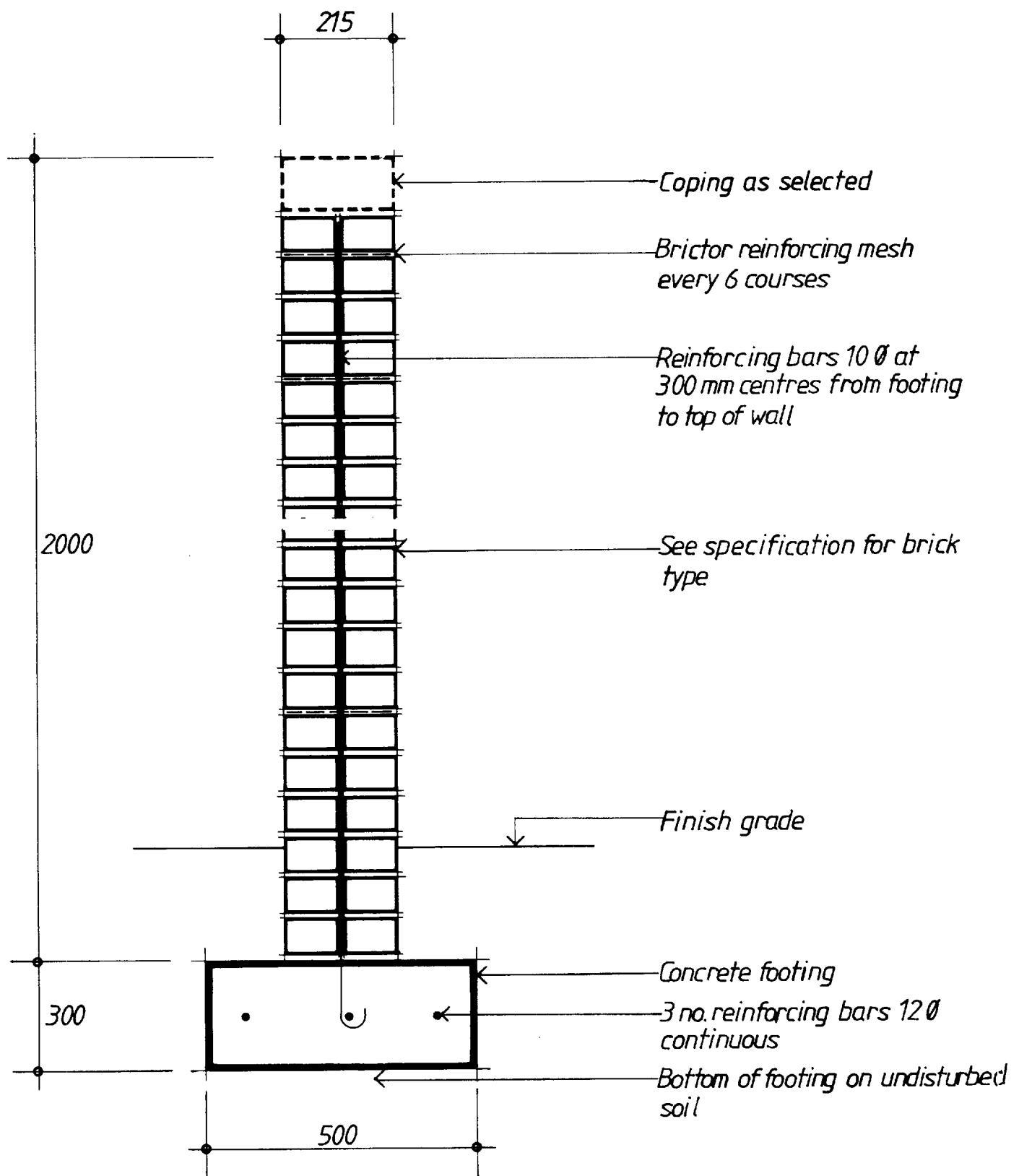


Scale 1:10



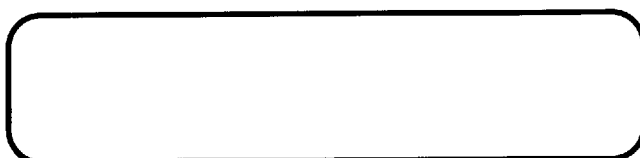
WALL
brick (high)



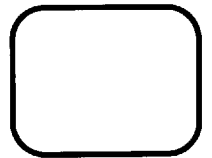


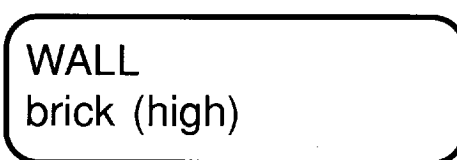
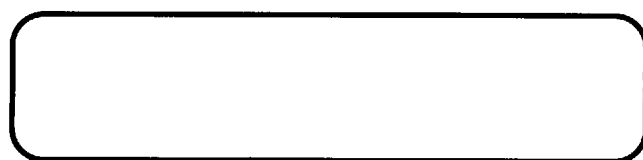
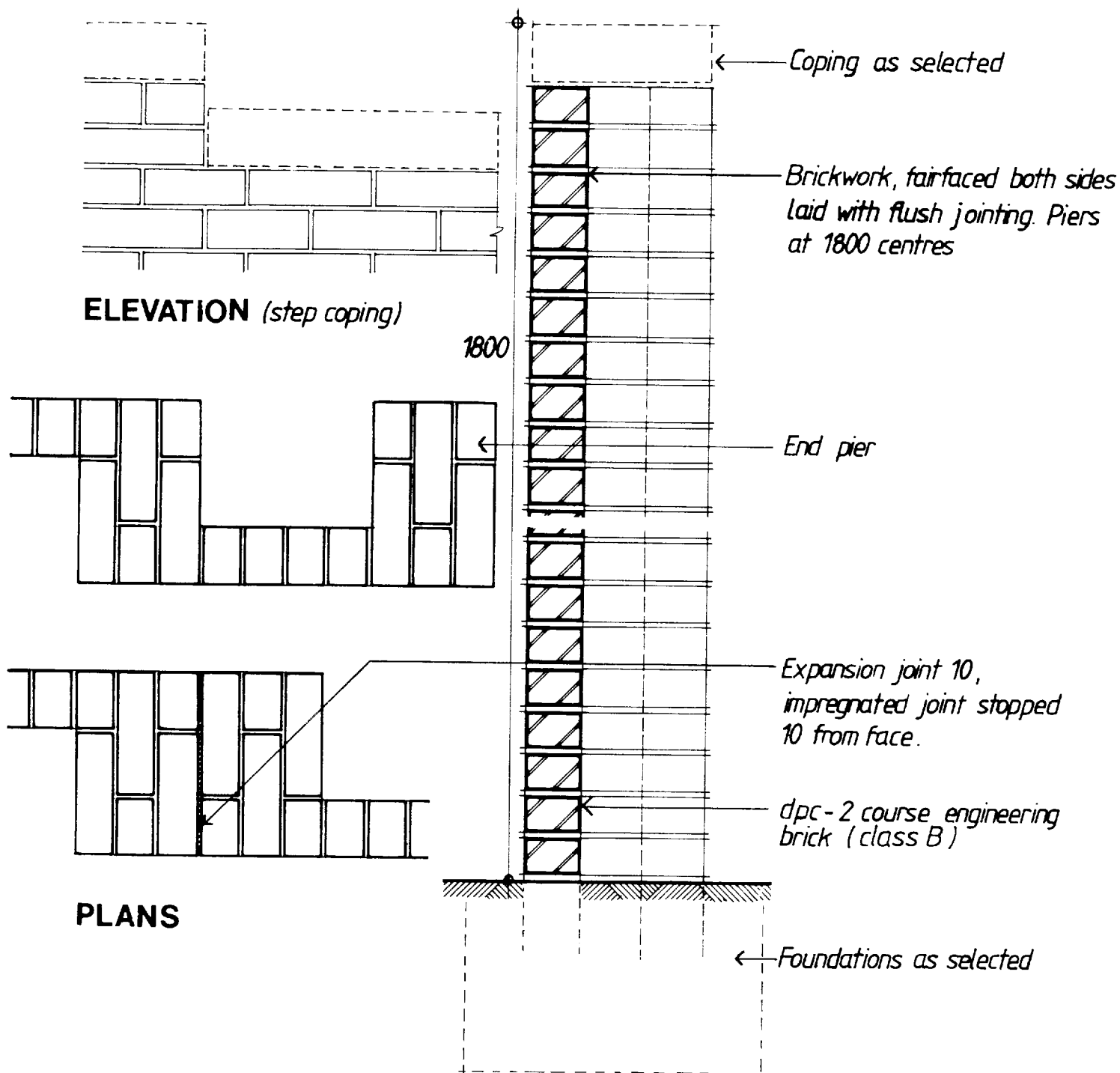
SECTION

Scale 1:10



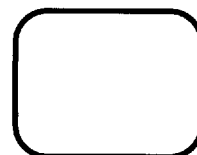
WALL
brick (high)

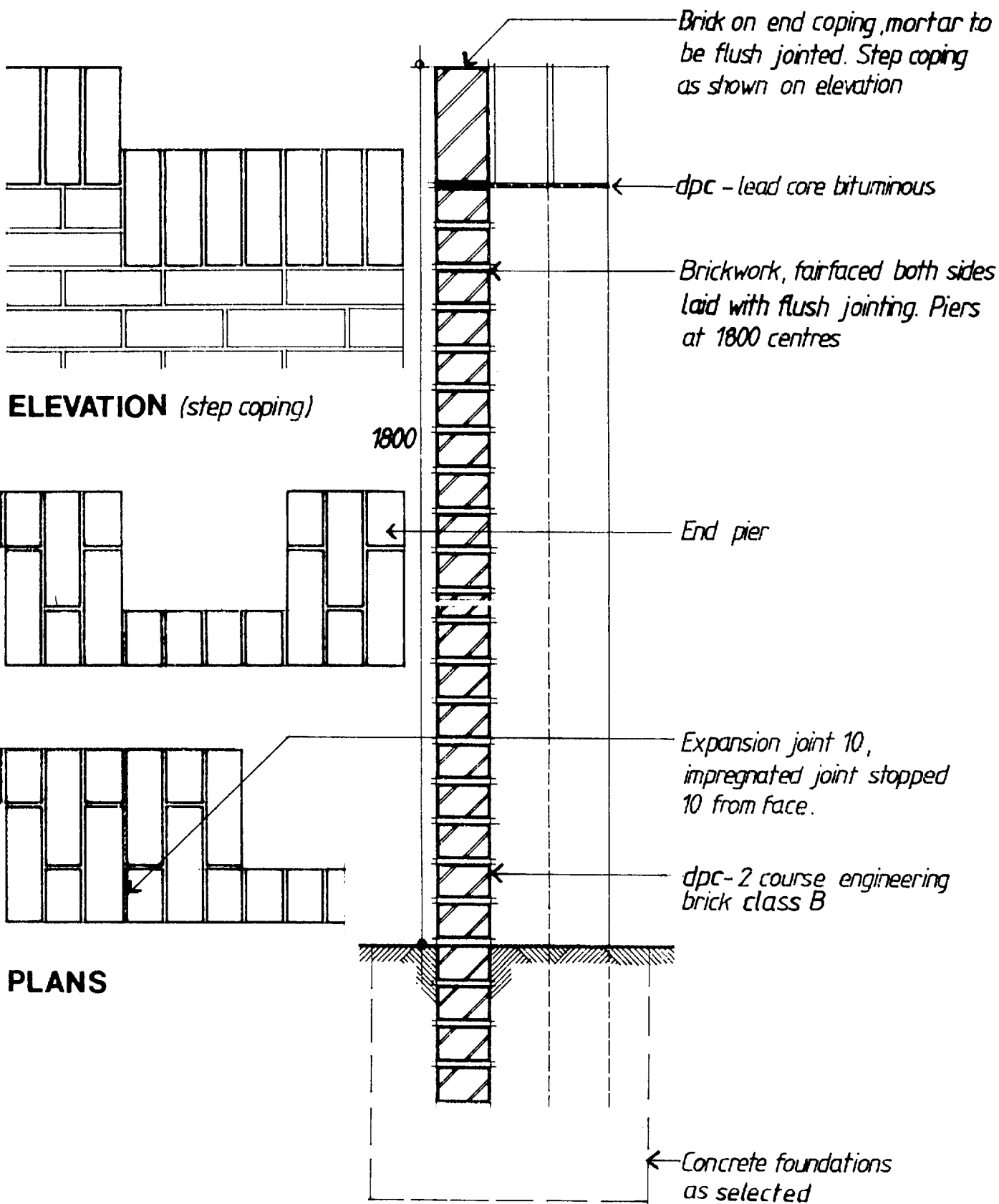




WALL
brick (high)

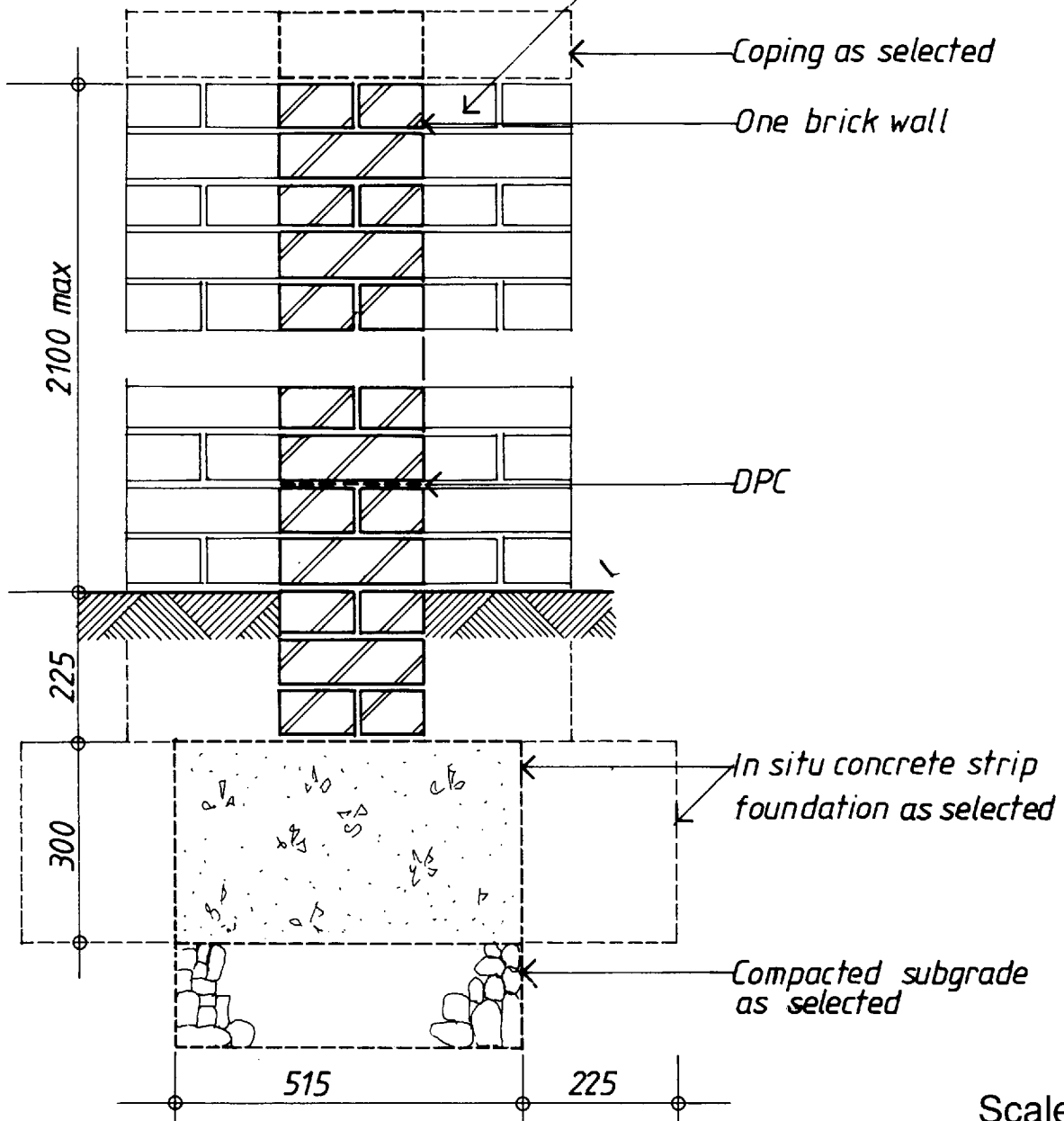
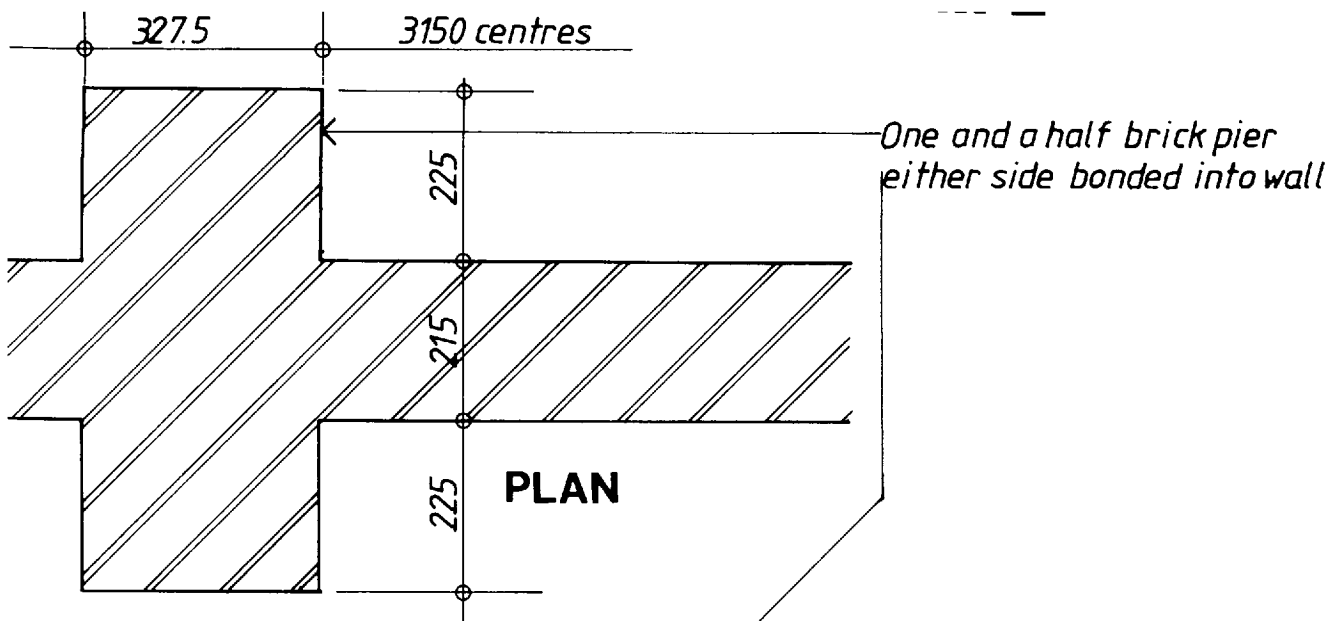
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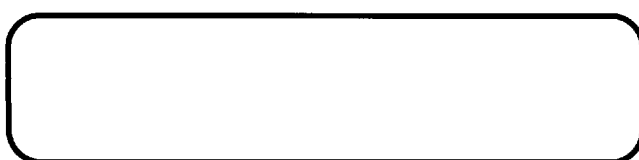


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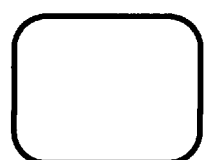
WALL
brick (high)

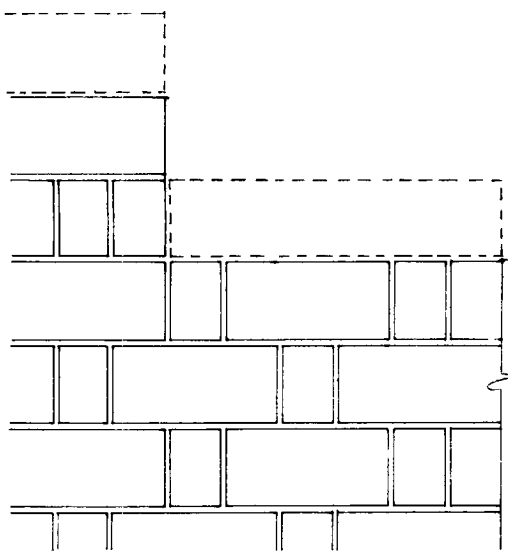


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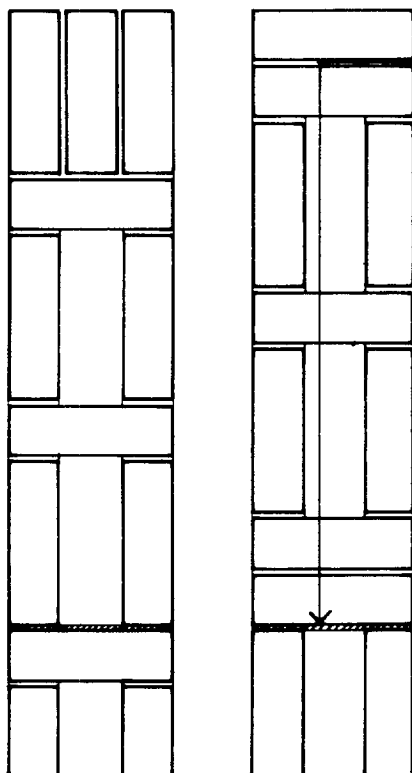


WALL
brick with piers

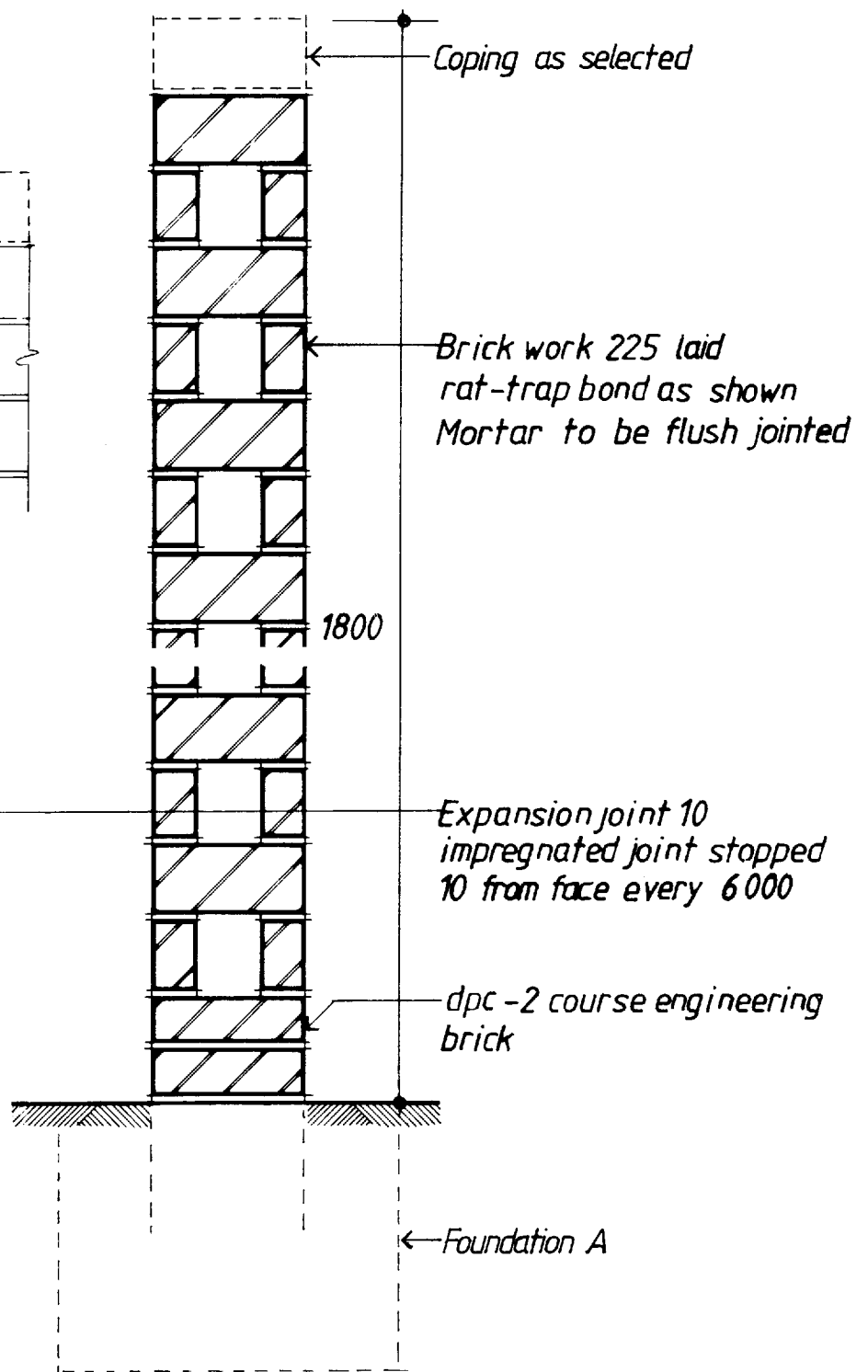




ELEVATION



PLAN



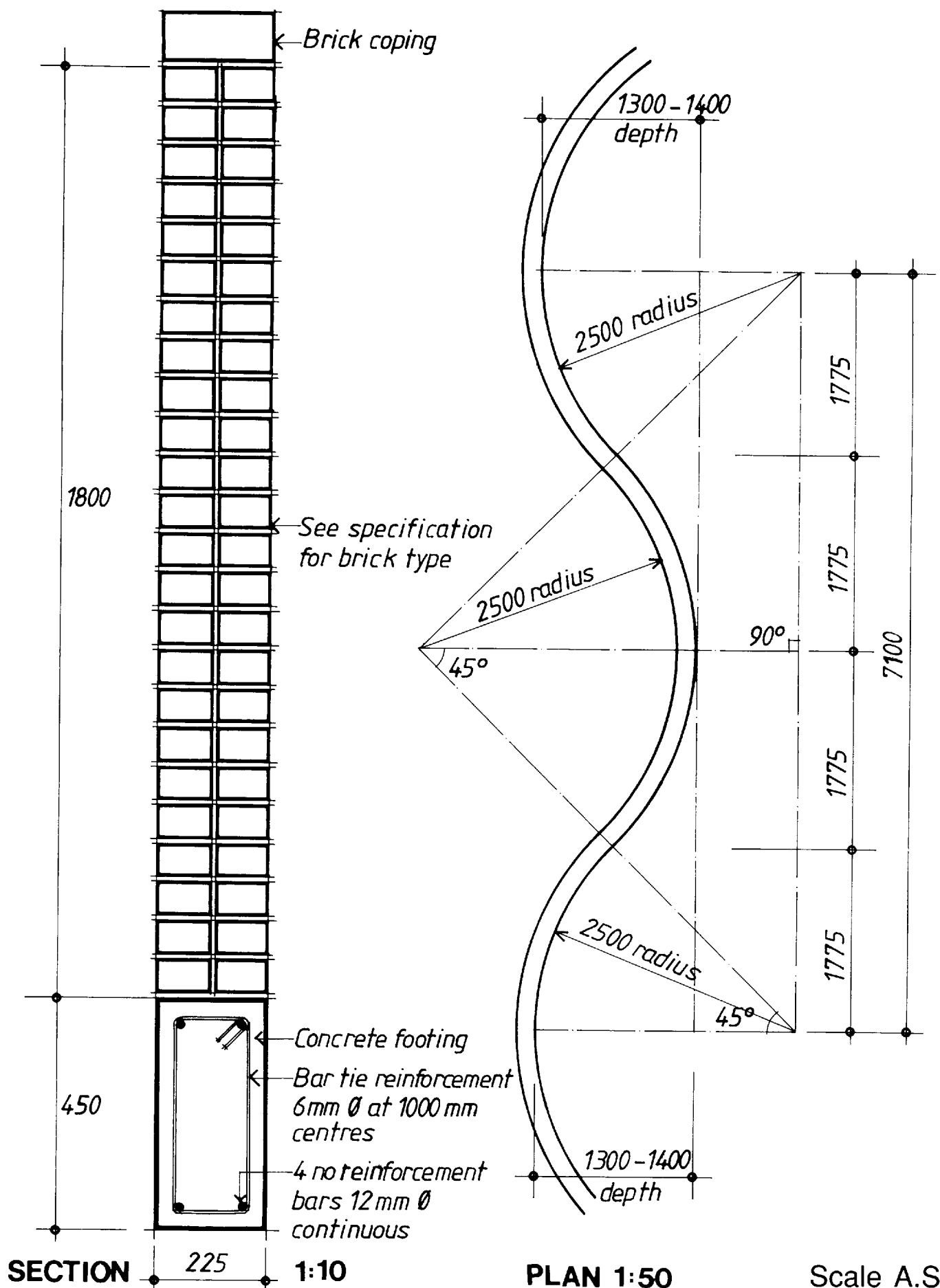
SECTION

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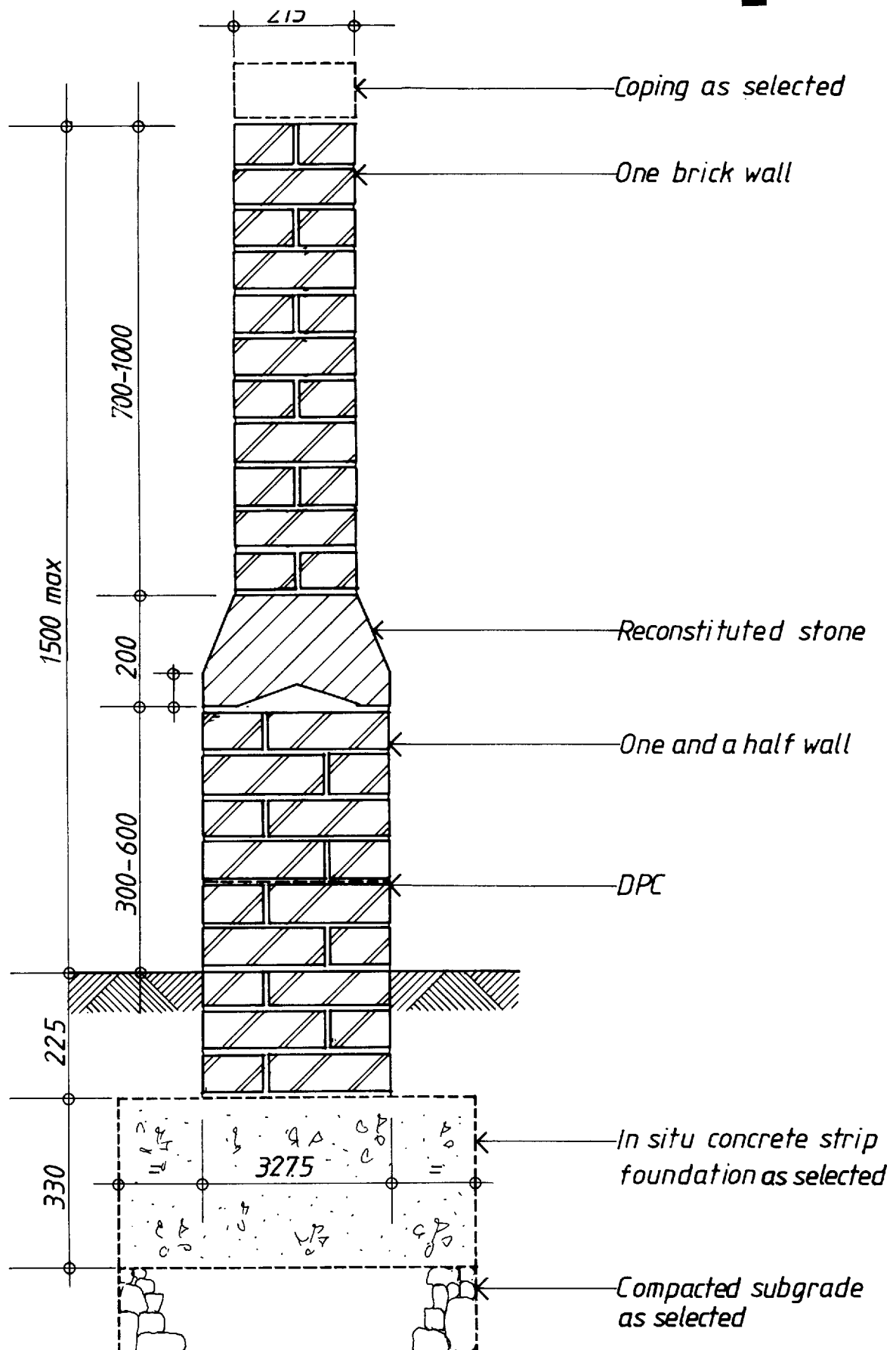


WALL
brick rat trap bond



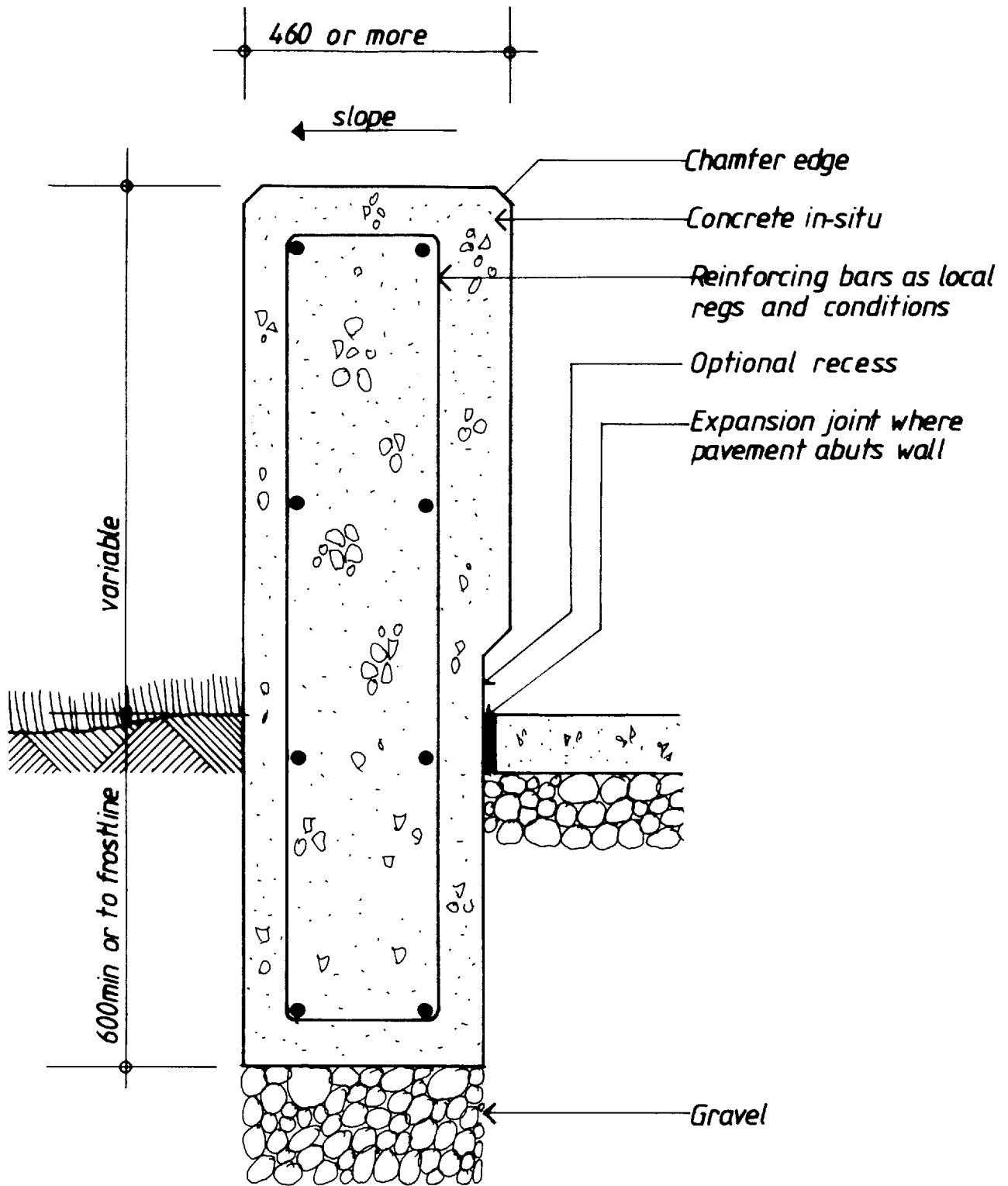


WALL
brick serpentine

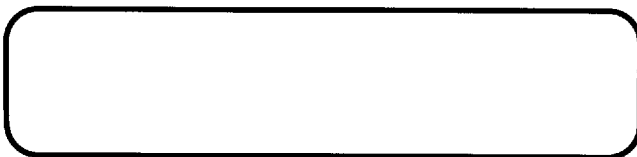


Scale 1:10

WALLS
brick with stone plinth

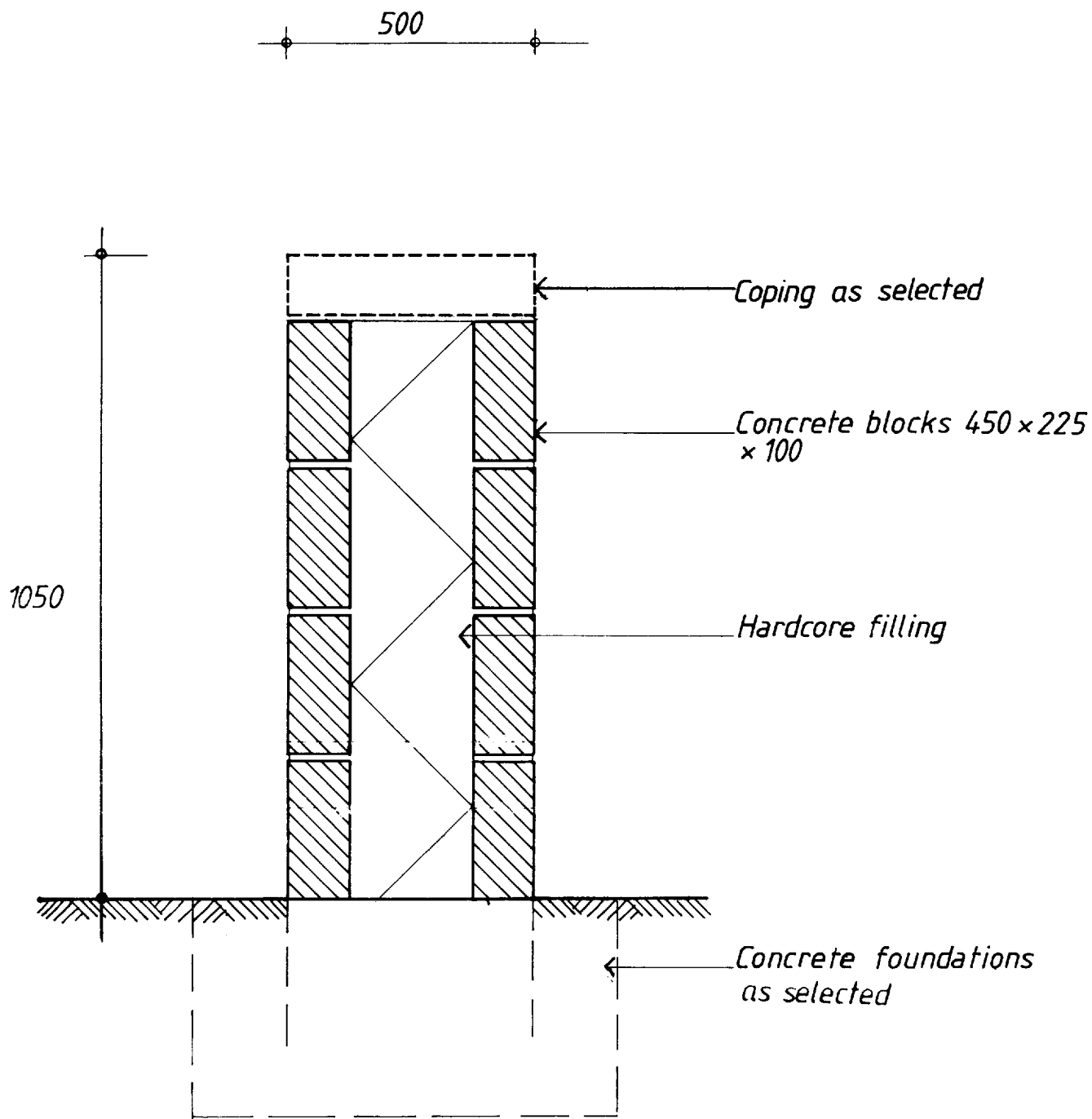


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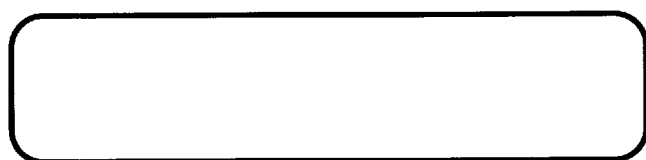


WALLS
contrete

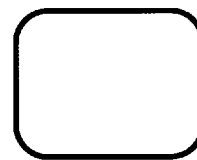


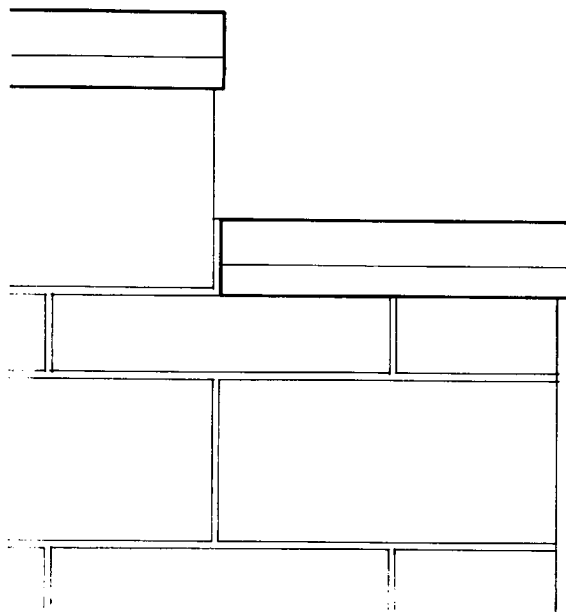


Scale 1:10

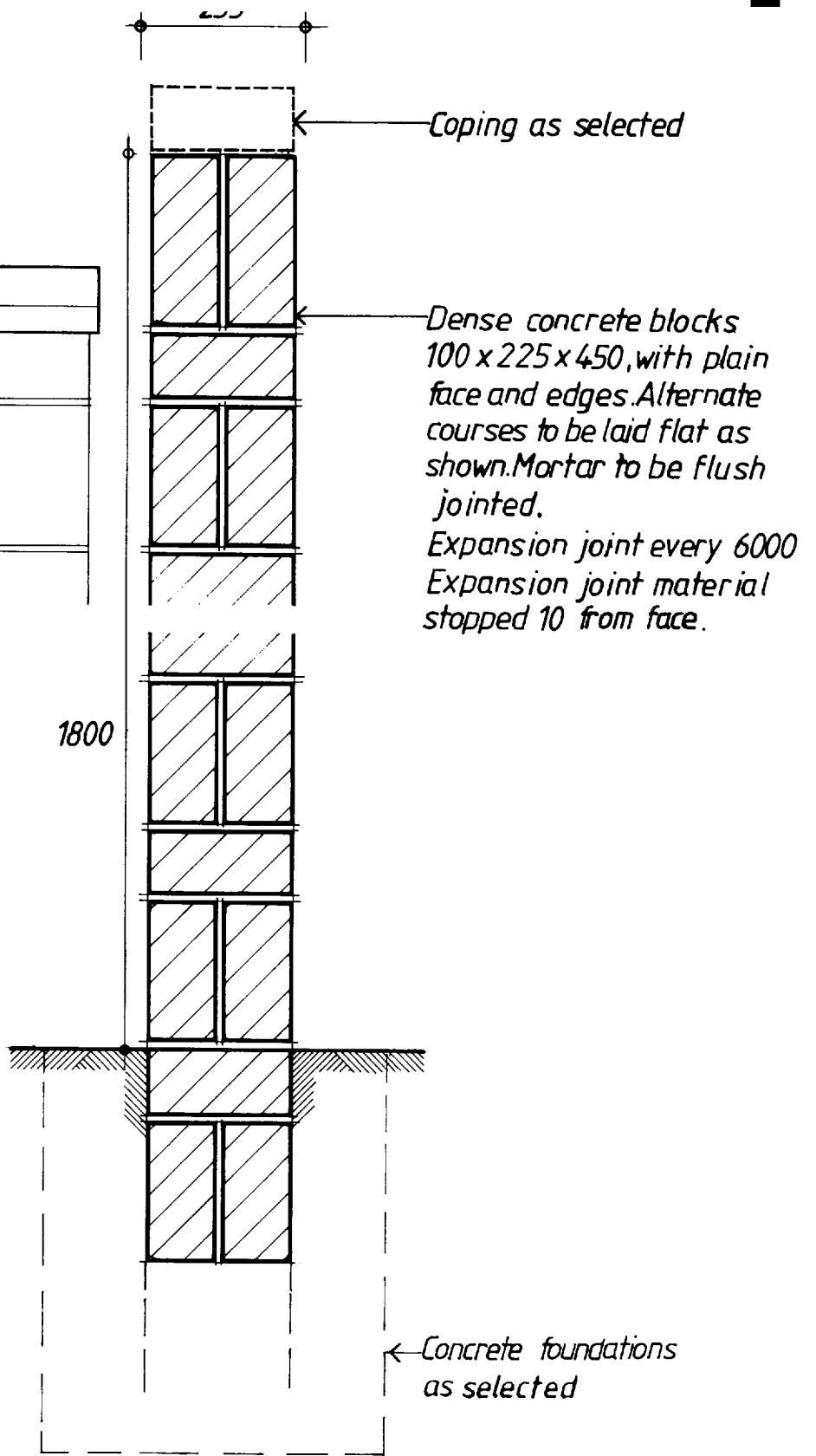


WALL
concrete block (low)



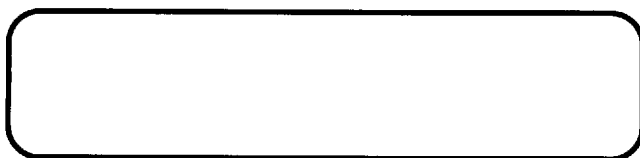


ELEVATION



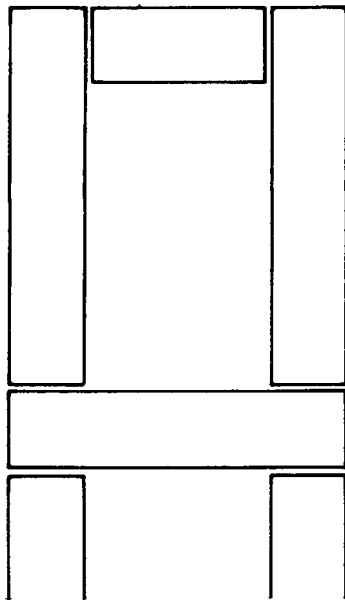
SECTION

Scale 1:10

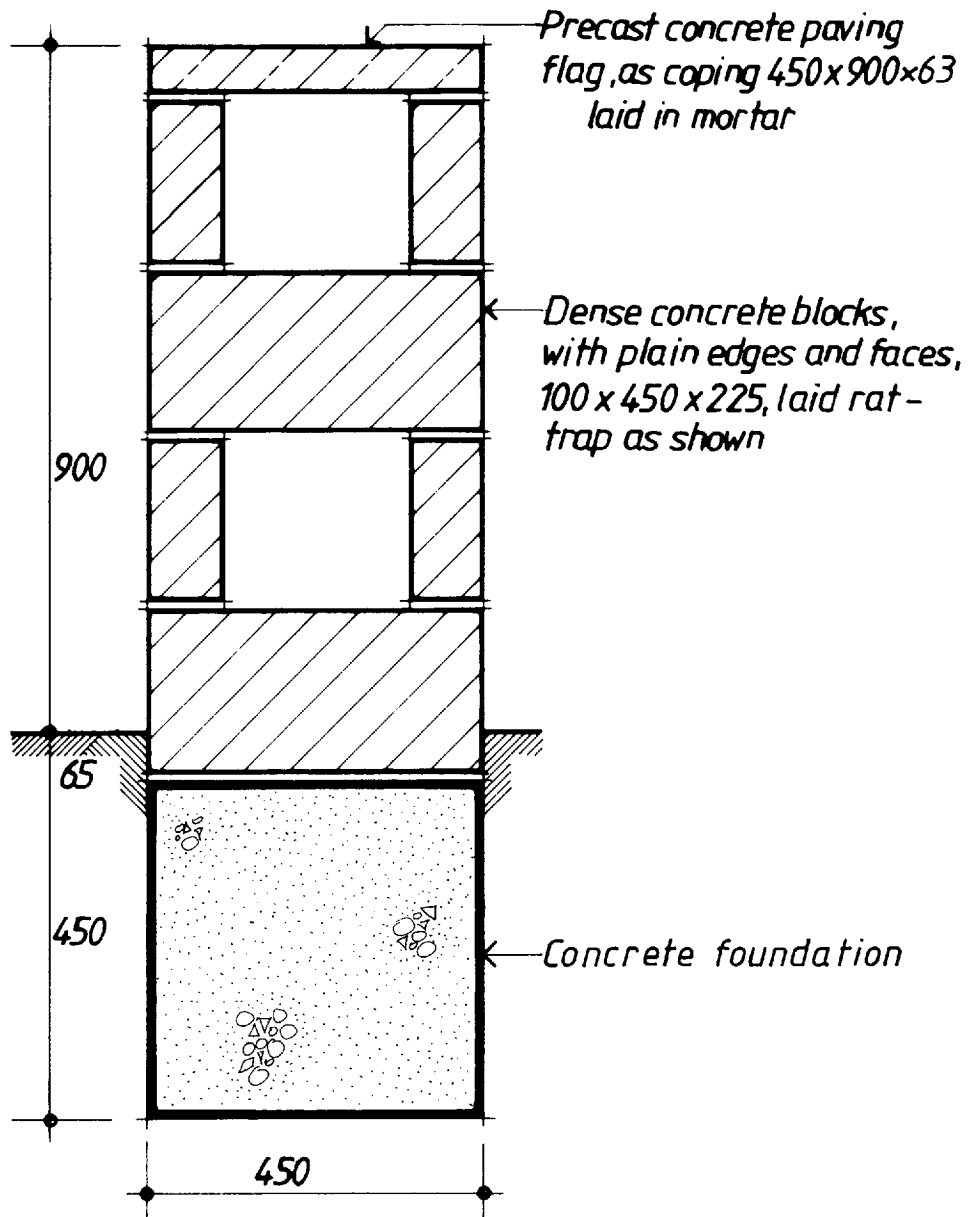


WALL
concrete block (high)

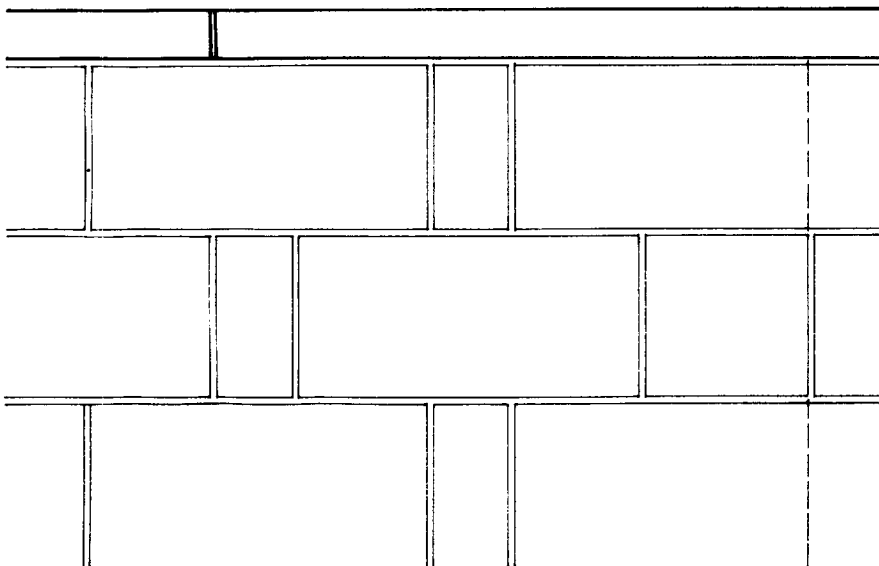




PLAN



SECTION

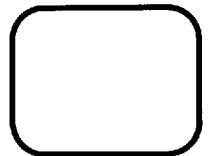


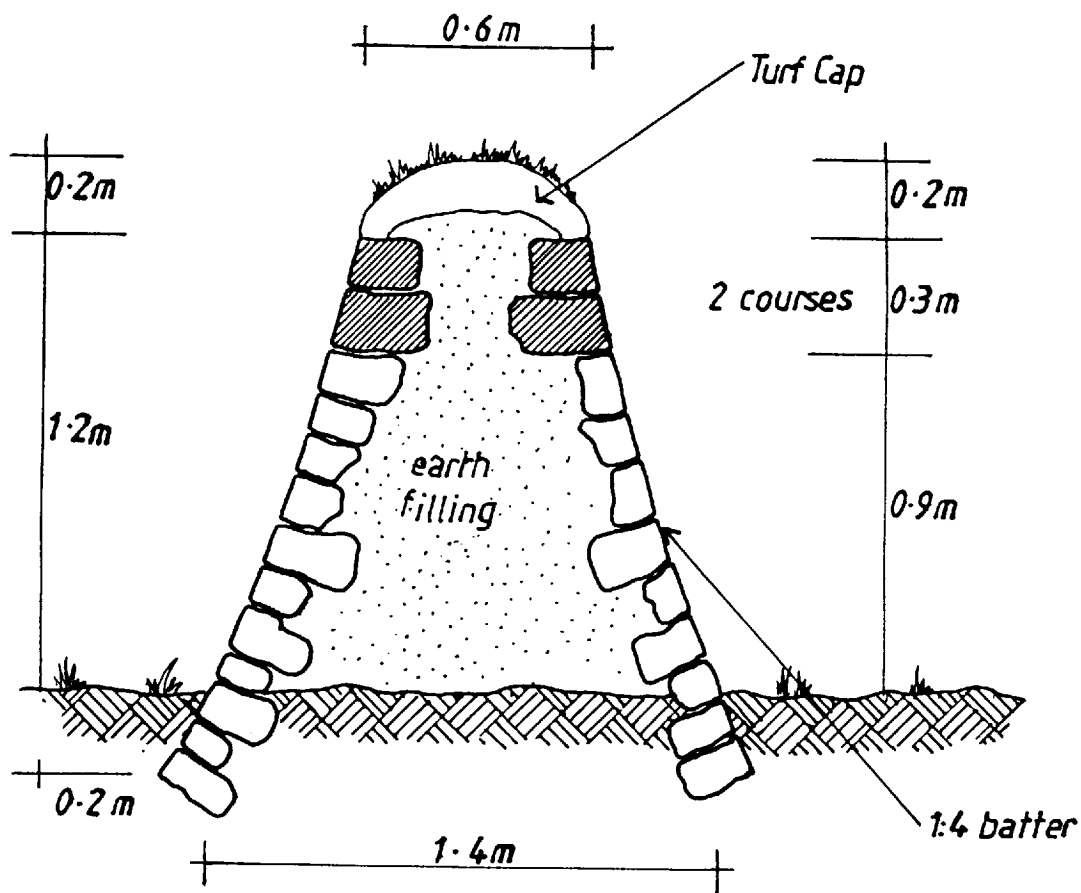
ELEVATION

Scale 1:10

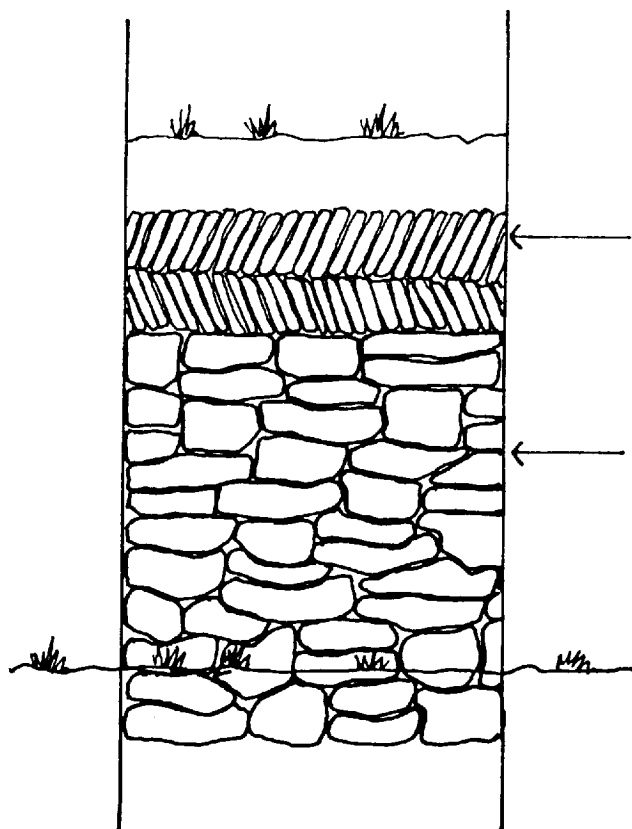


WALL
concrete block (gravity)





SECTION



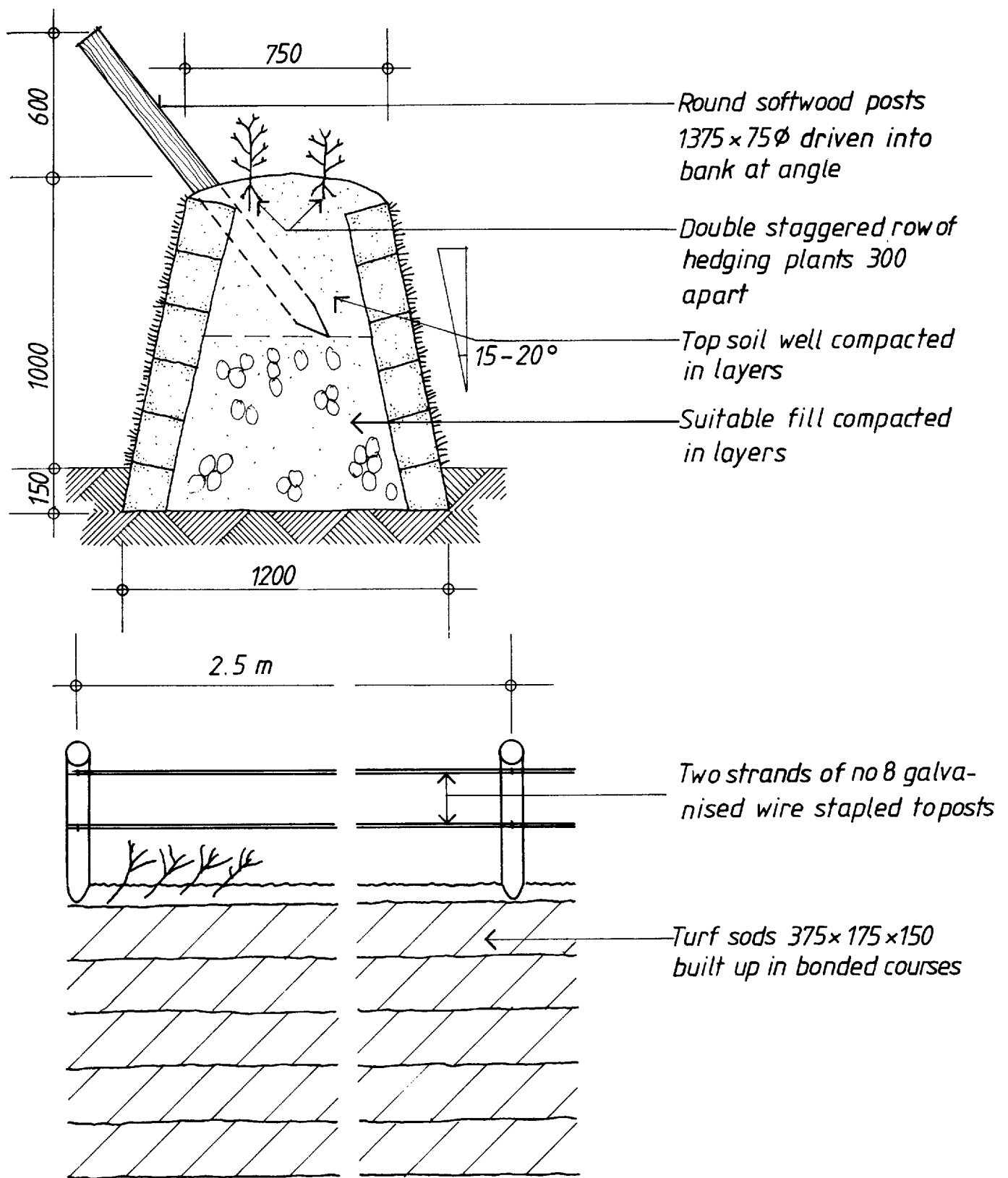
ELEVATION

Jack and Jill
200 mm x 40 mm to 150 mm x 40 mm on face
and 100 mm to 150 mm deep approx.

Stretchers
610 mm x 150 mm to 150 mm x 50 mm
and 100 mm to 250 mm deep approx.

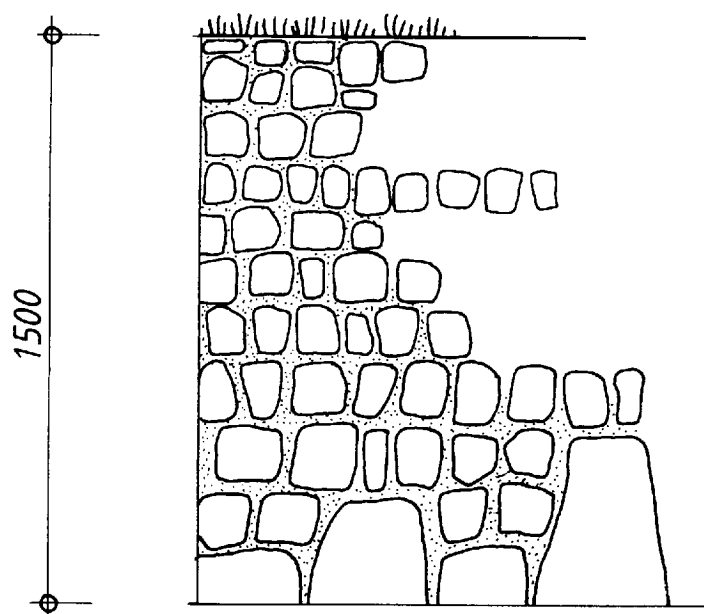
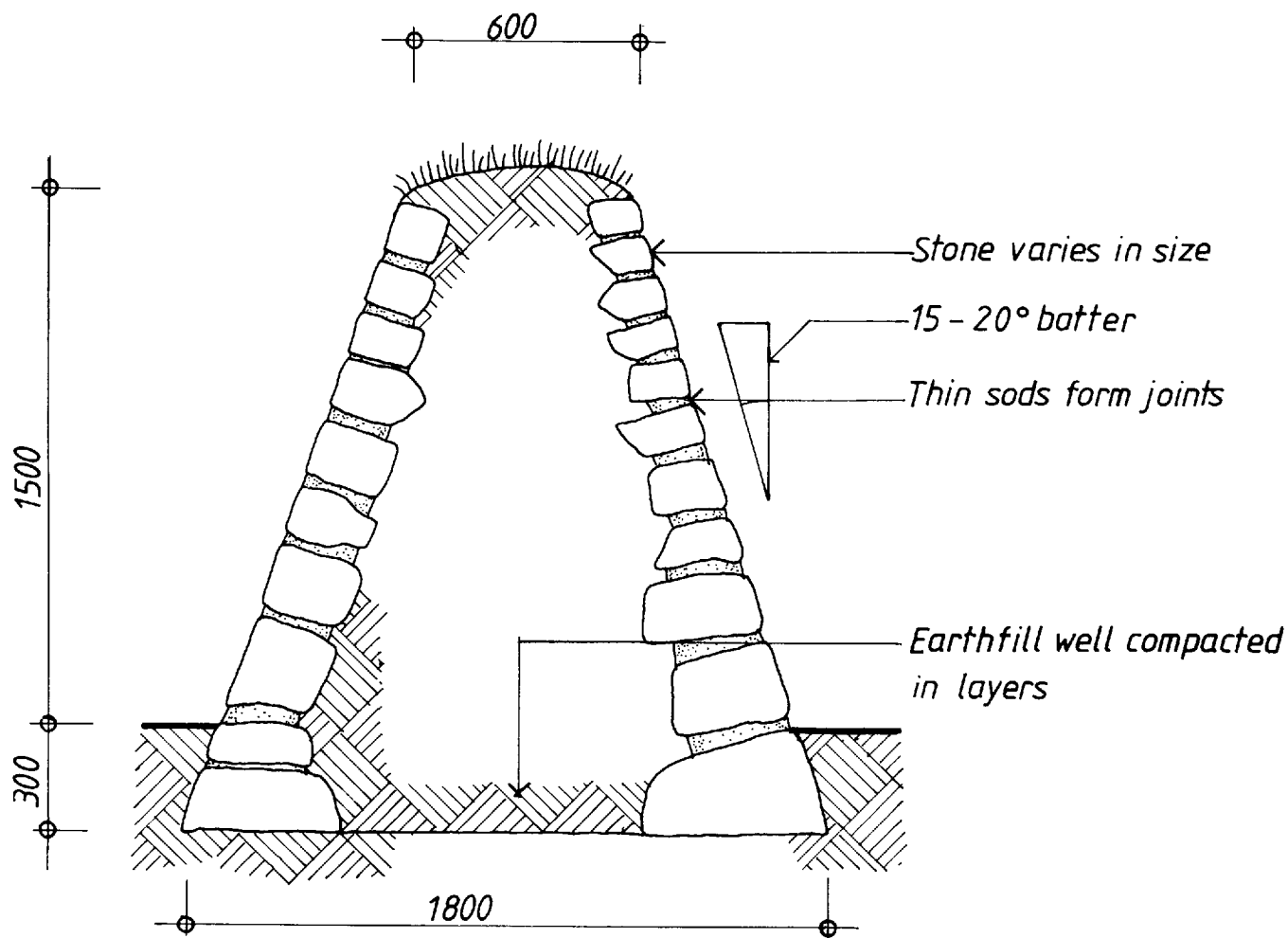
Scale 1:20

CORNISH HEDGE
natural stone



Scale 1:20

WALLS
hedge bank + fence

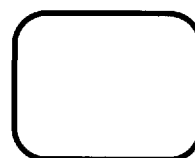


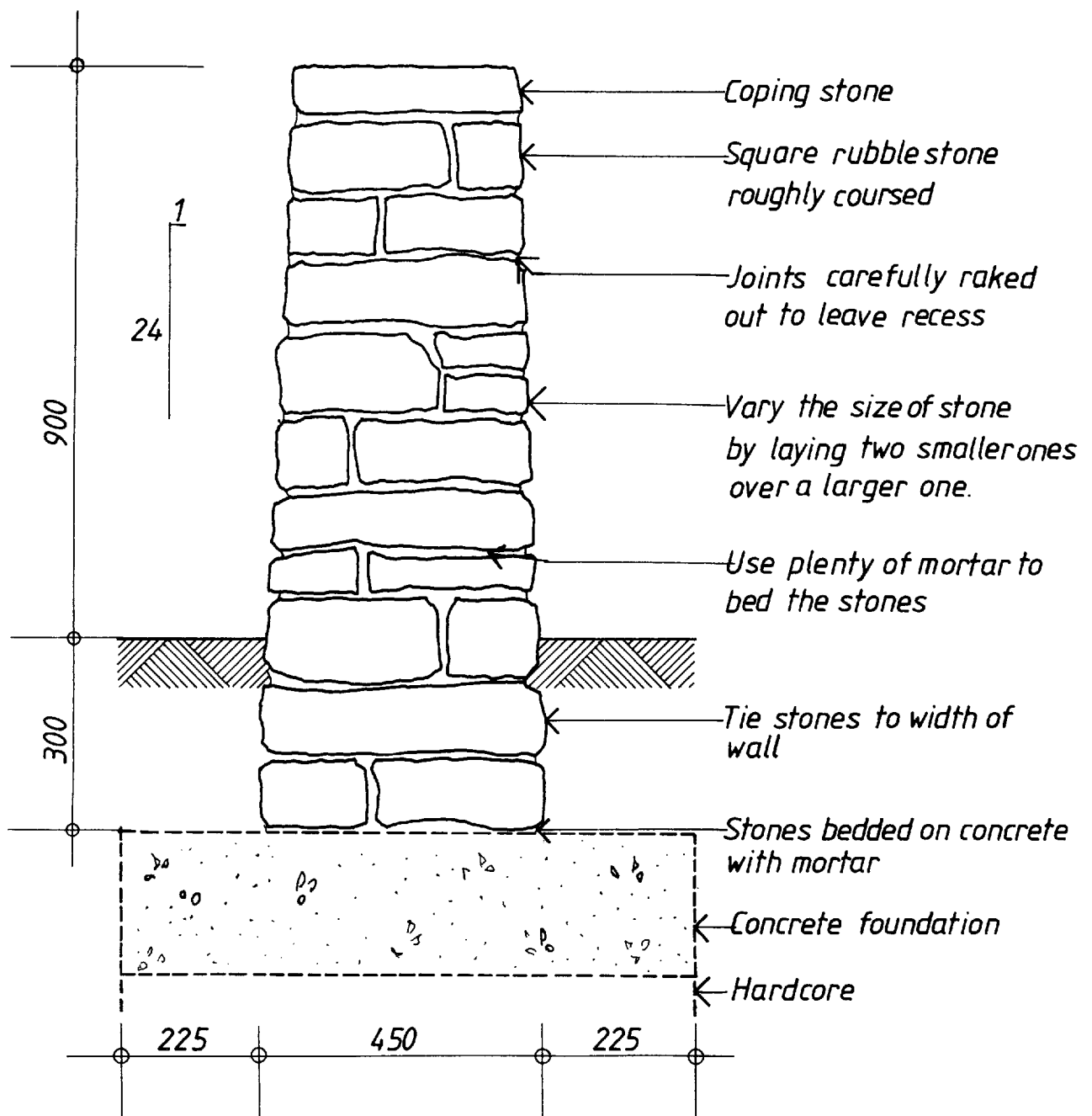
Turf in joints gradually grasses over stones

Scale 1:20

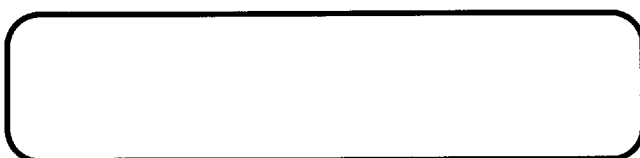


WALLS
stone hedge

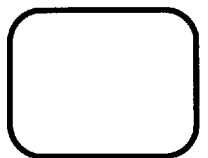


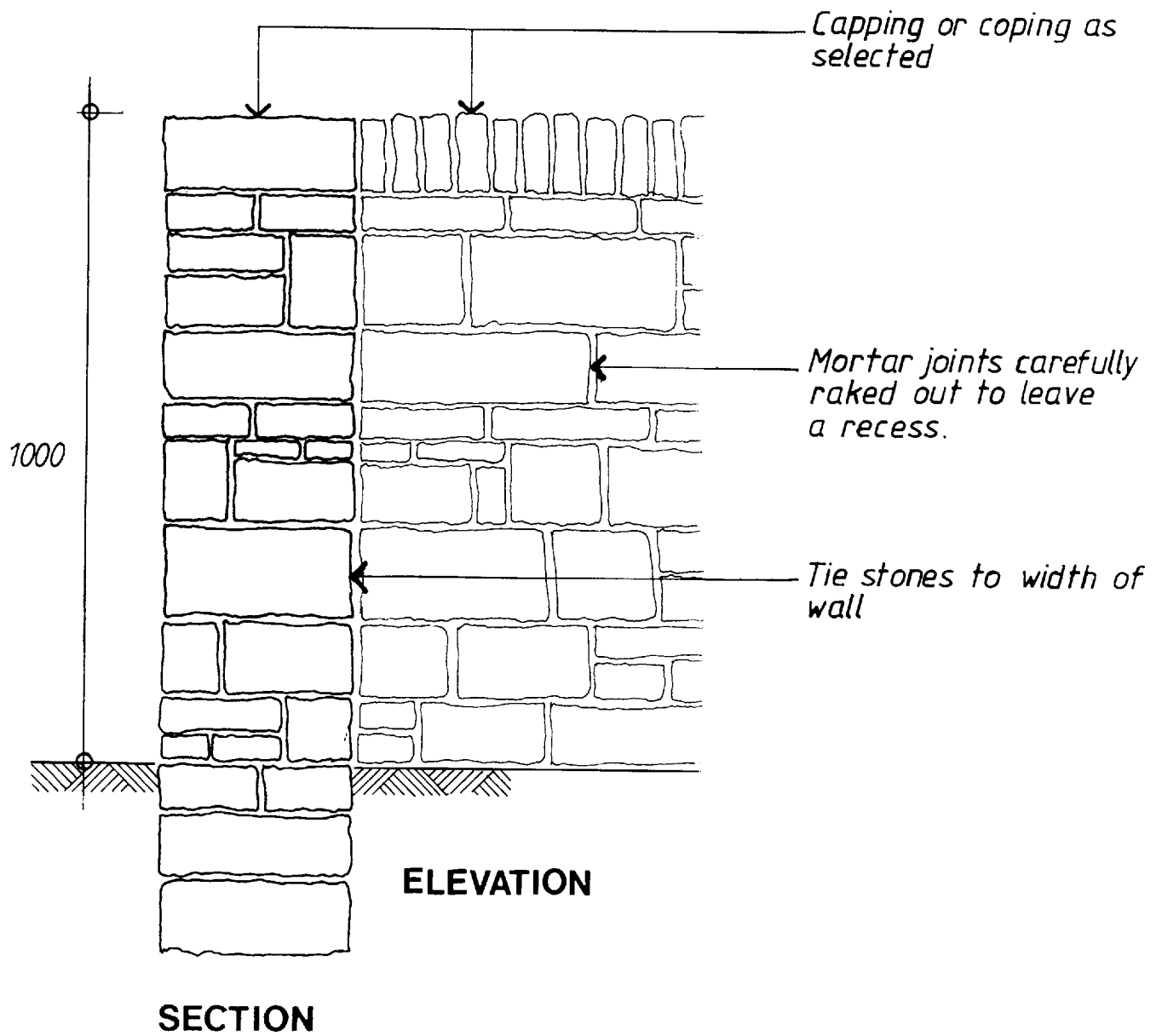


Scale 1:10



WALL
natural stone
mortared/coursed

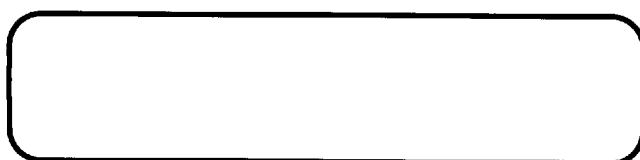




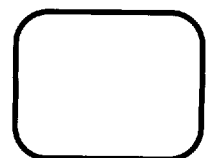
Note

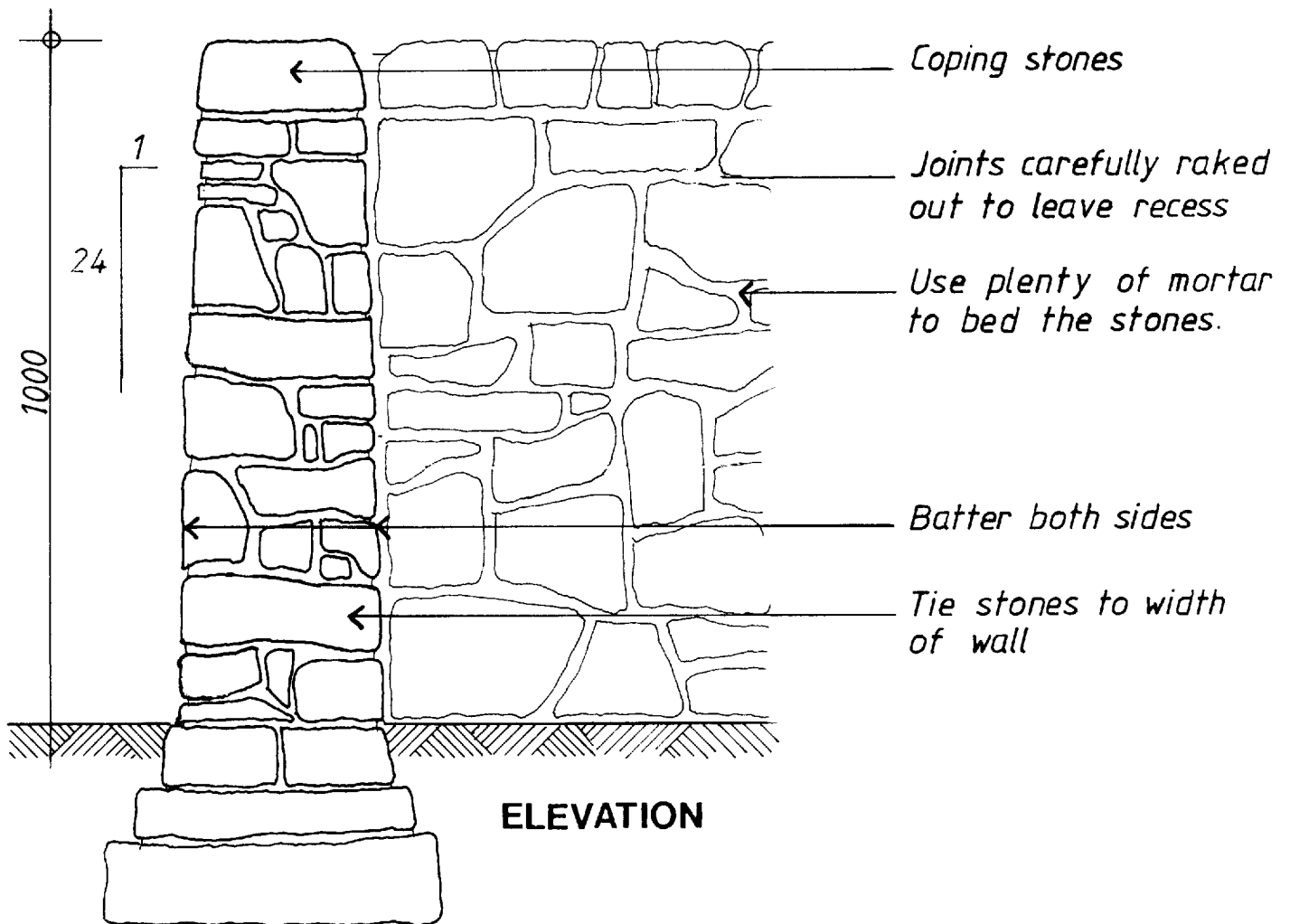
Units are precast
and dressed before
delivery to site

Scale 1:10



WALL
ashlar mortared/
coursed



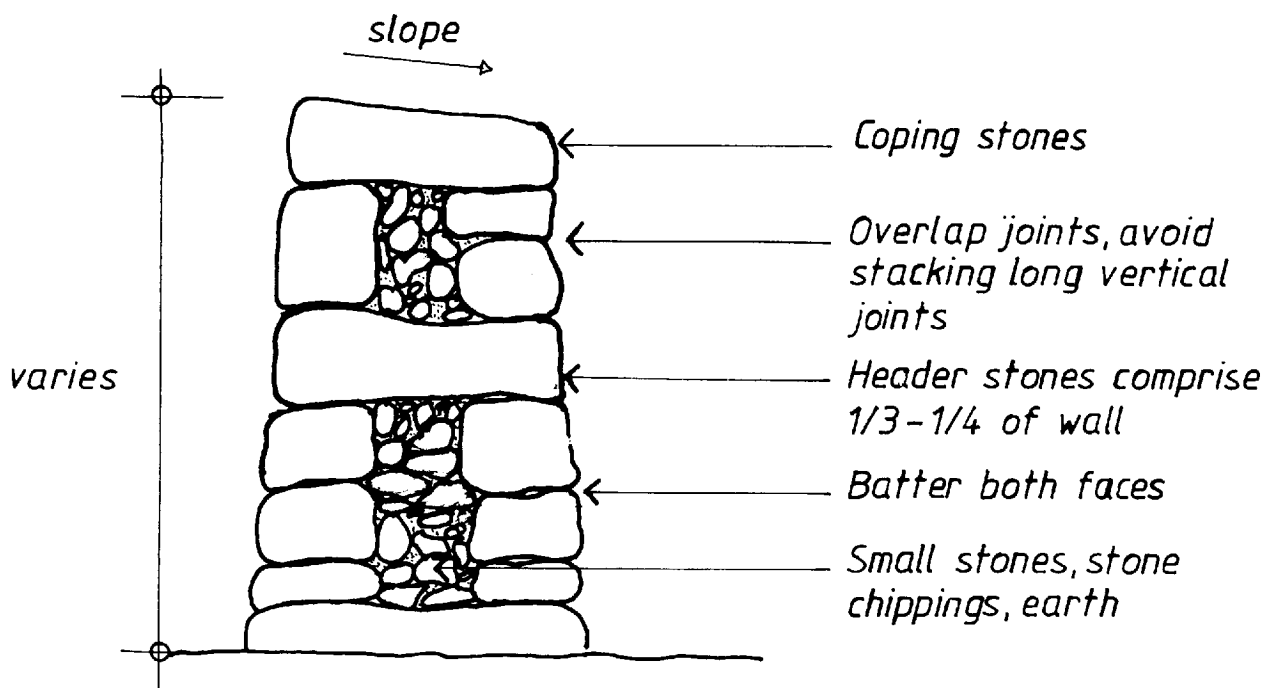


Scale 1:10

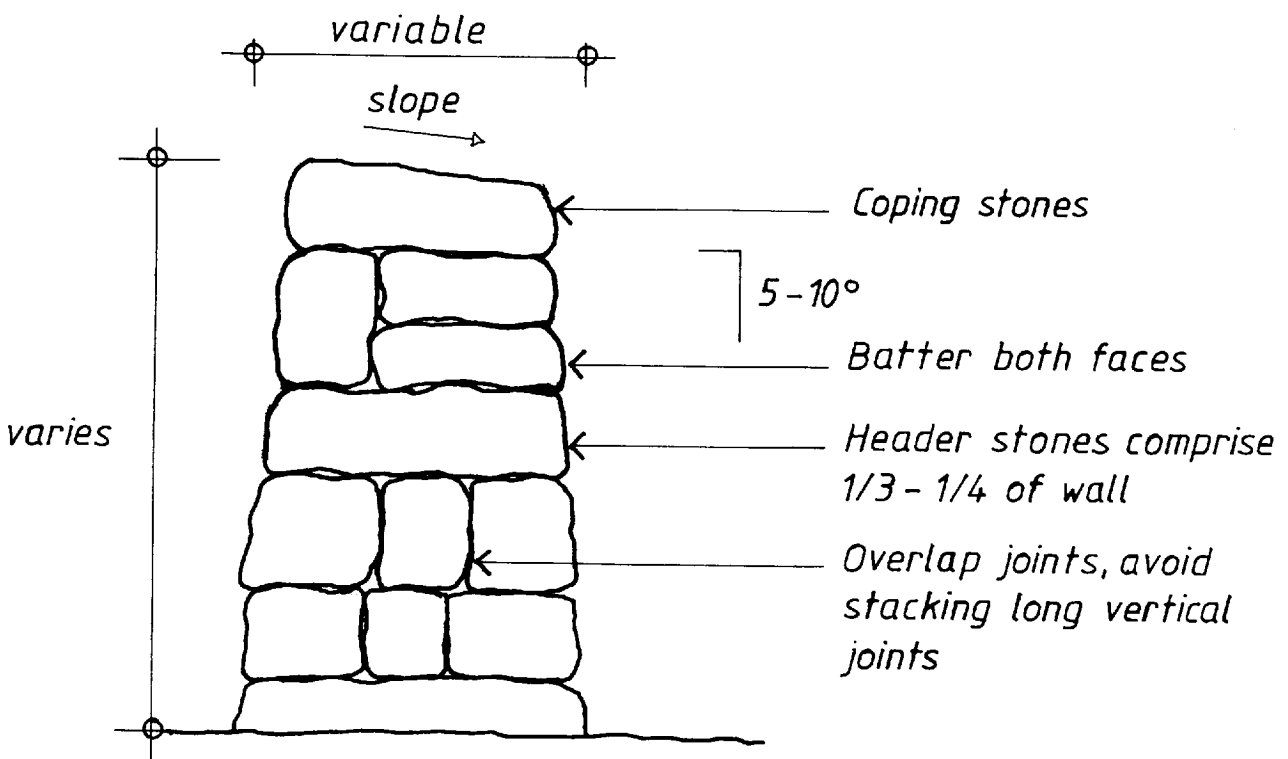


WALL
natural stone
mortared/uncoursed



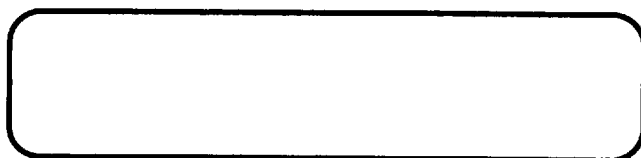


DOUBLE TIER WALL

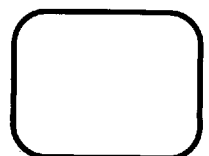


SINGLE TIER WALL

Scale 1:10



WALL
natural stone, dry



SCREEN WALLS

GUIDANCE NOTES

Appearance

Scale, texture and pattern should decide the type of open screen unit required. Since any large area would become boring, an open screen is usually more successful when used sparingly as a decorative element in a solid wall or in short lengths. These walls look their best when there is strong contrast of light and shade and especially when used with large foliage plants.

This type of walling is quite easy to erect and reinforcement is not usually necessary. Concrete blocks are available in several standard designs, but in general they are approximately 300 mm square, and 90 mm thick. Colour and finish vary according to the manufacturer. Alternatively, a screen wall can be achieved using bricks and perforated bricks.

Foundations

Screen blocks can be mortared directly onto the concrete foundations of 1:6 mix. They should be 200 mm wider than the wall (100 mm each side) and at least 200 mm deep, depending upon the wall height and ground conditions. Soft ground may necessitate thicker foundations. Alternatively, lay two courses of ordinary concrete blocks on a hardcore bed, with a flat coping stone on top.

Mortar

This should be workable, but not sloppy. Protective measures should be taken to prevent spillage onto the block faces. A 1:5 mortar mix should be used.

Joints

These should be 9 mm throughout.

Piers

Pilaster or pier blocks are needed at each end of a free-standing wall and at 300 mm centres, depending upon the height of the wall. Walls above four courses high should be reinforced with rods set into the foundations. The cavity must then be filled with mortar and capped with a special capping block.

Coping

A special coping block should be placed on top of the screen blocks to provide a suitable finish.

Design notes

Infinite combinations of texture, pattern and perforation make brick modules a unique and challenging design material for screen walls, adding visual interest. They improve microclimatic conditions as they allow breezes into a garden and discourage wind turbulence and frost pockets, which often occur on the leeward side of solid walls. Before designing a perforation or texture it is useful to review the available sizes and possible placement of bricks within a wall. Brick bonds are used primarily to strengthen the wall and tie the bricks together; they also create various patterns which provide the system for designing perforations. The simplest way to pierce a wall is to leave out one or more of the brick units in a standard bonding pattern, using loose bricks as temporary spacers. There are few materials which offer brick's colour, pattern and texture possibilities. From pattern-to-pattern placement, the variations in brick garden walls are limited only by the designer's imagination. Additional variety can be achieved by breaking from ordinary coursing and/or orientations. These variations can also be constructed much more efficiently if the voids conform to a brick module. In such cases, temporary brick fillers can be used to support the courses above, and brick cutting can be minimised.

Construction types

Because free-standing brick walls are vertical structures subject to wind loads, the resistance to wind loads is the primary design consideration. The wall height-to-thickness ratio is usually the major factor affecting lateral stability, but wall shape and/or reinforcement can make thinner walls acceptable.

Structural analysis for the specific design and wind conditions is necessary to

determine the safest and most economical thickness. But, as a rule, solid brick walls under 2.0 m high are 100 mm (one wythe, or tier of brick units), 200 (two wythe), or 300 mm (three wythe) thick. See extract from BS 5628 Part 3 attached.

They are built in one of five ways:

1. Straight with continuous footings.
2. Straight with continuous footings and pier supports.
3. Reinforced 'thin fences' with footing support only at piers.
4. Staggered.
5. Serpentine or chevron-shaped.

When designing perforated walls it is assumed that the reduction in brick weight is offset by the reduction in wind load. Consequently, the same methods used to calculate thickness for solid, free-standing walls are acceptable. All five types can be designed with perforations as long as the dimensions and reinforcing normally required are still accommodated. The required wall thickness can also influence the choice of an economical and effective perforation pattern.

Because straight walls on a continuous footing rely only on the height-to-width ratio for lateral stability, they must be thicker than the other four types.

Straight walls buttressed by piers and supported on footings are often 100 mm thick with 300 mm or 400 mm square piers. The minimum required frequency of supports is determined by wind loads, and the reduced thickness of the wall usually results in lower costs.

'Thin fence' construction is seldom used. Foundations occur only at pilasters, or piers, which support 100 mm thick, steel-reinforced brick panels. No. 2 and larger steel bars are used, their exact size and frequency determined by wind load and distance between piers. Reinforcing requirements could be difficult to meet on a wall with a continuous perforation pattern. However, perforations within a limited area – particularly at the top – usually pose no problem.

Staggered walls less than 2.0 m high can be built only 100 mm thick if they are offset

200 mm every 1800 mm. A wall higher than 2.0 m can also be only 100 mm thick, but it requires more than 300 mm offset.

The shape of serpentine and chevron walls provides additional lateral resistance to wind loads; a 100 mm thick wall is often sufficiently strong. However, the geometry of the wall should be analysed to ensure that it provides the required lateral stability.

Water resistance

Often bricks are used to cap a wall rather than a solid stone or cast top. Flashing the horizontal joint below prevents water from penetrating and freezing the mortar. Some perforation patterns can leave vertical joints similarly vulnerable, and these should be avoided when possible. If a large void has vertical joints open to the weather, it is wise to flash the horizontal joint below. Joints should be completely filled to resist water penetration; weathered, concave, and flush joints are acceptable.

Mortars and joints

The use of Designations 1 and 2 mortars (1–3 and 1–4(1/2)) will greatly increase wall strength and durability – particularly at ground level (if no damp proof course is possible) and at coping and capping levels. A movement joint may be necessary at the normal centres (max. 6 cm and max. 3 cm from a corner). This will interrupt the bond and stability and would be best placed where a stiffener or pier occurs.

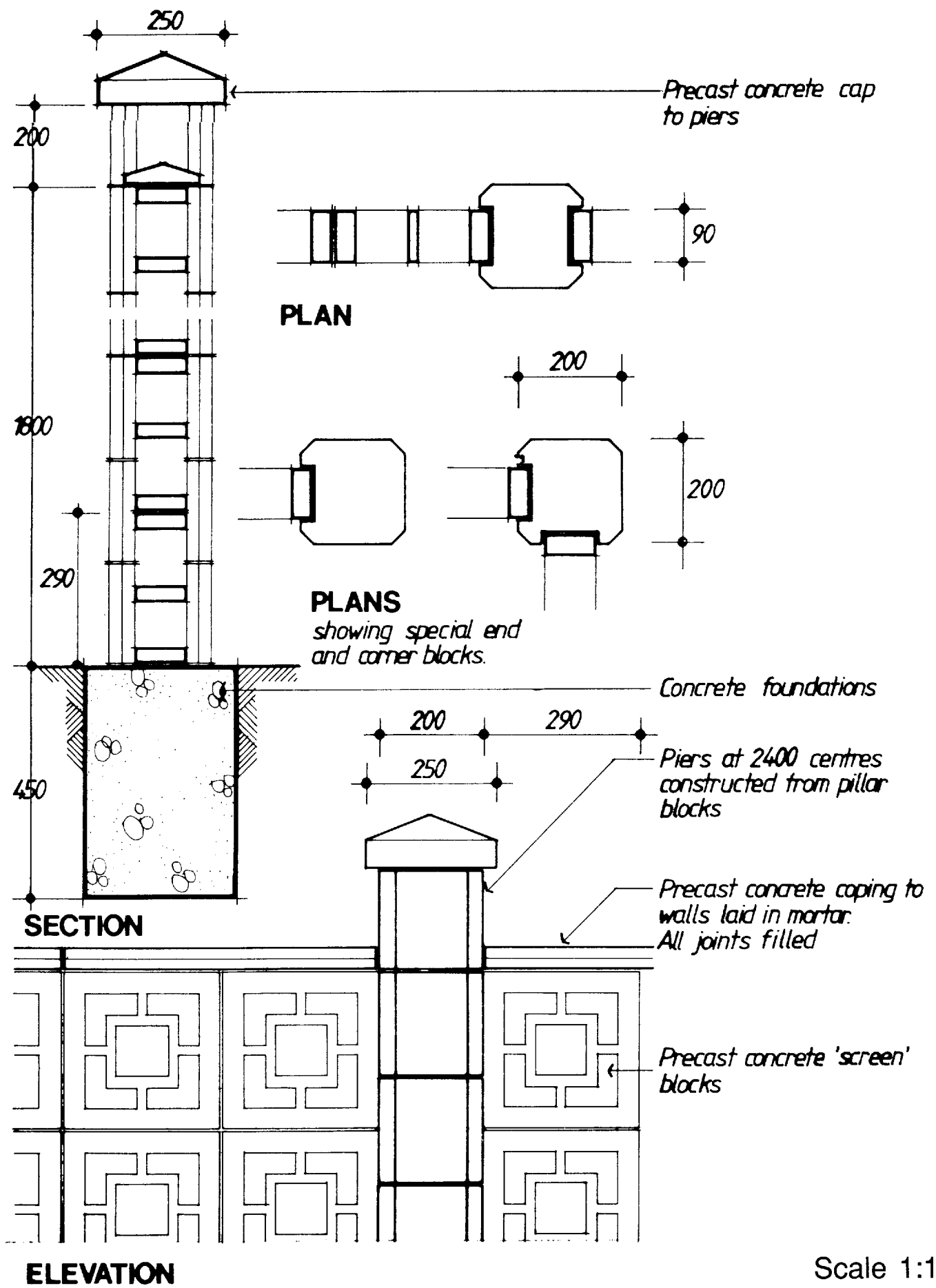
FL quality bricks should always be used for cappings and copings and preferably where they are in the ground or below a damp proof course.

SPECIFICATION CHECK LIST

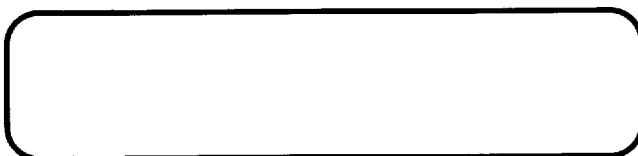
See previous chapter.

DETAIL SHEETS

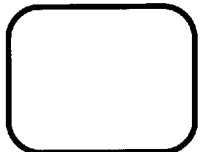
Concrete block screen (3)
Perforated screen (2)

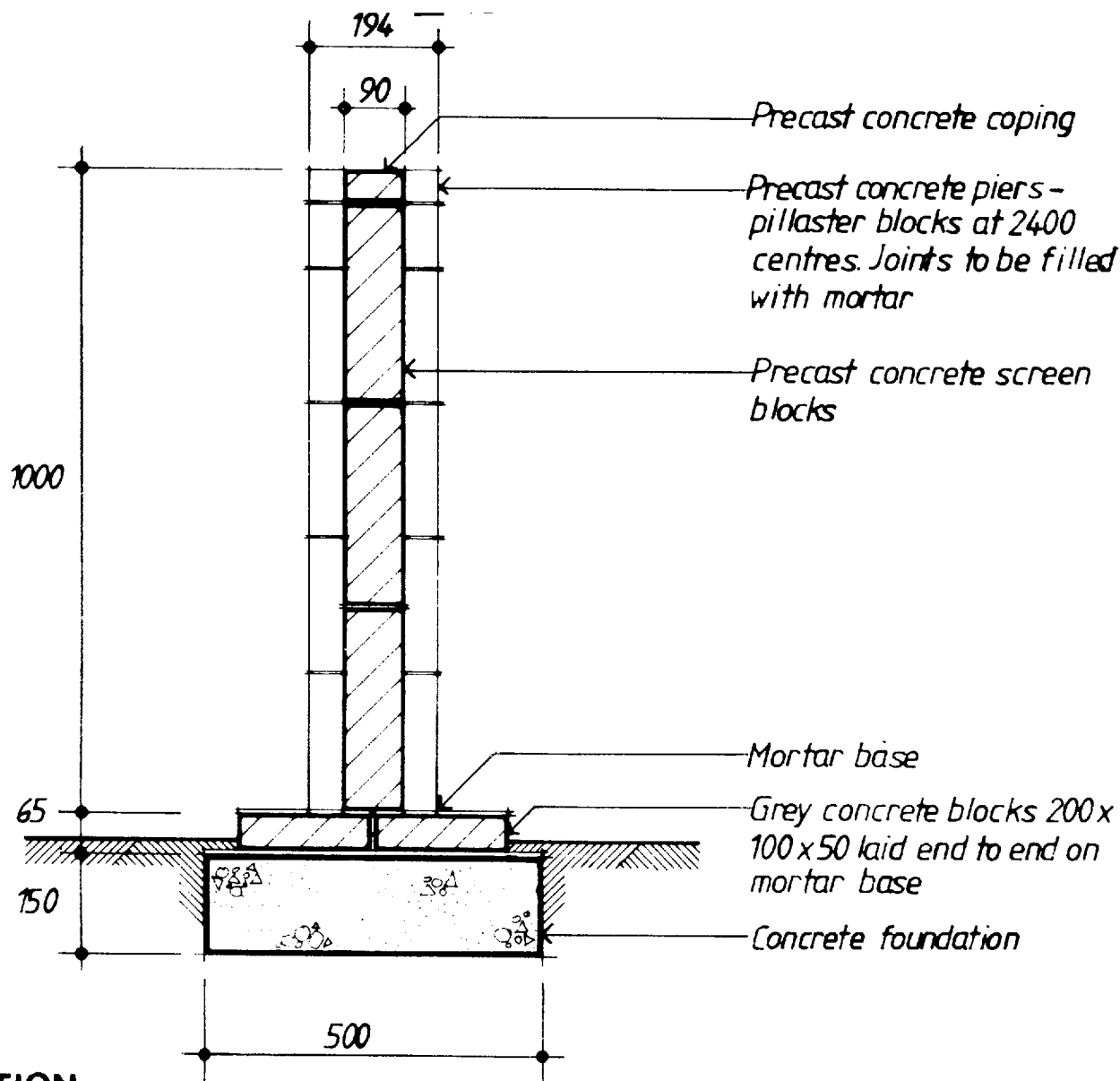


Scale 1:10

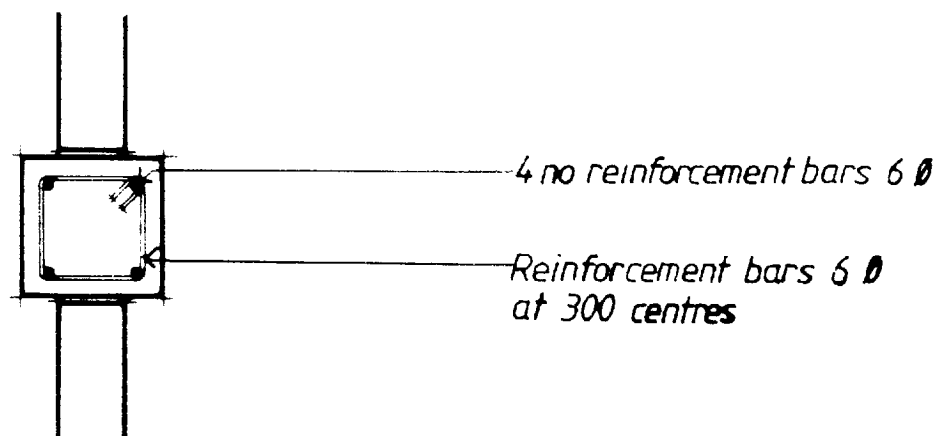


WALL
concrete block screen



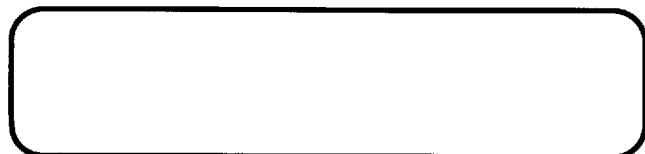


SECTION

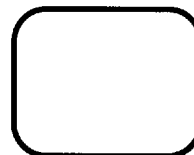


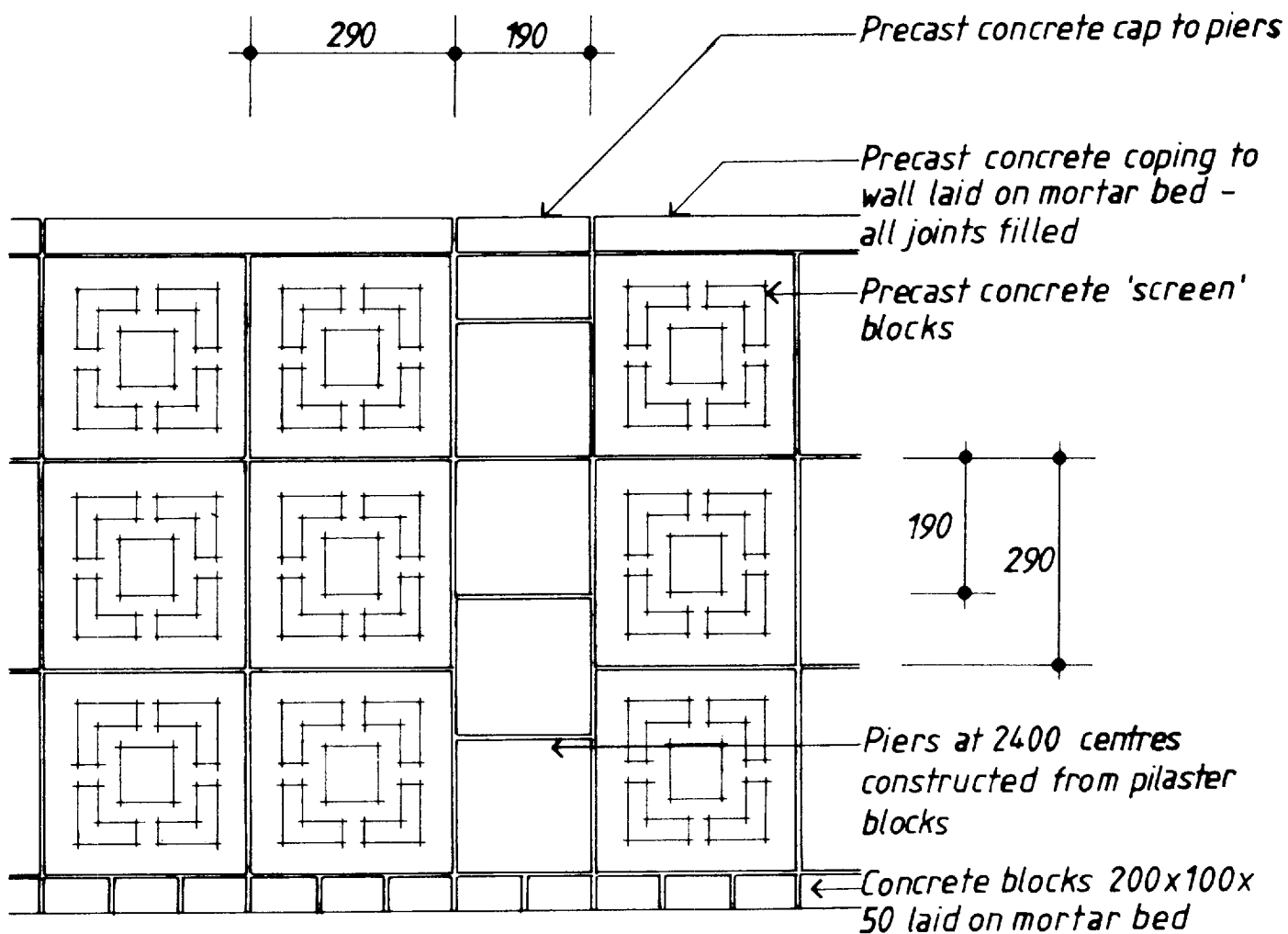
PLAN

Scale 1:10



WALL
concrete block screen



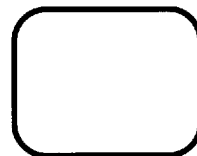


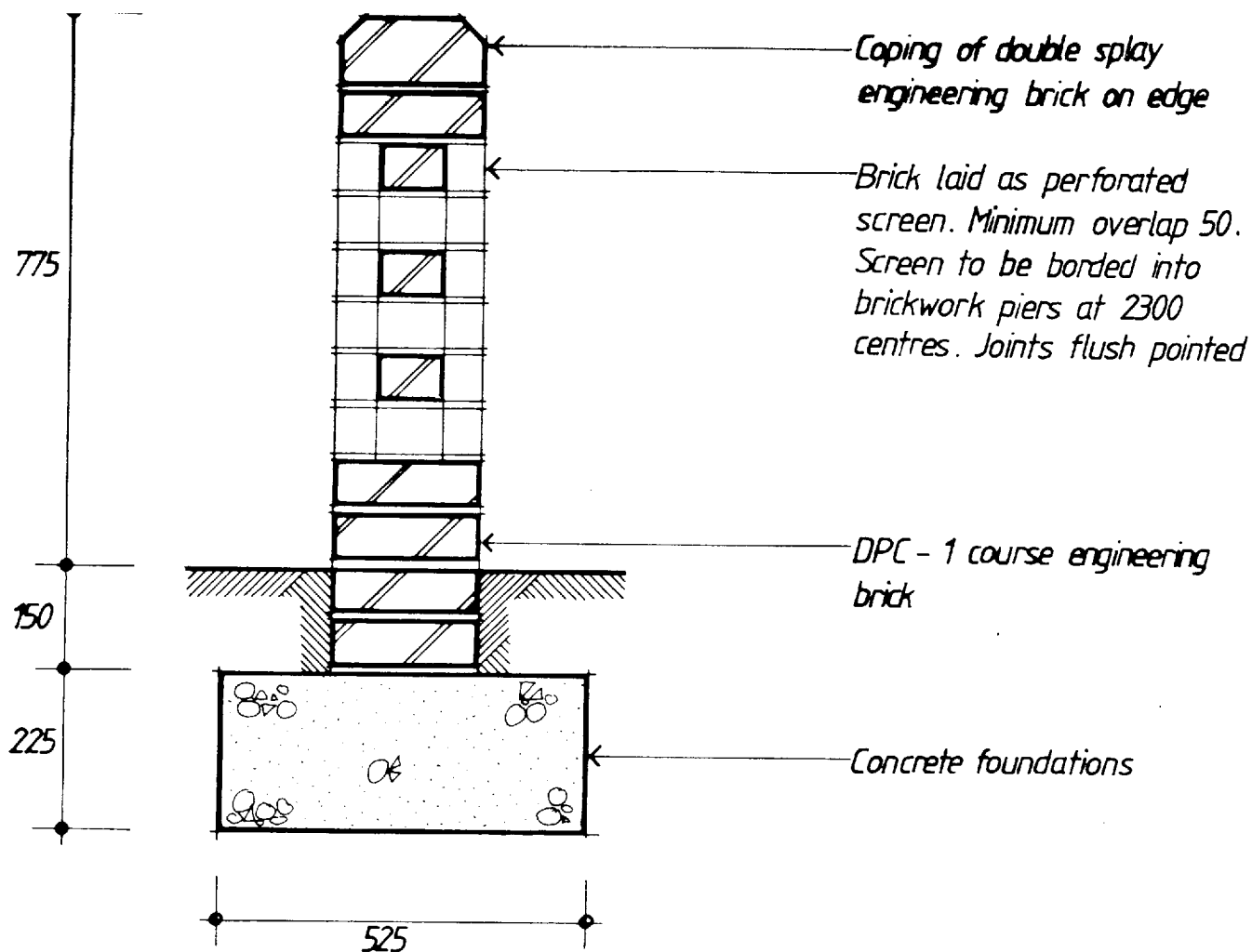
ELEVATION

Scale 1:10

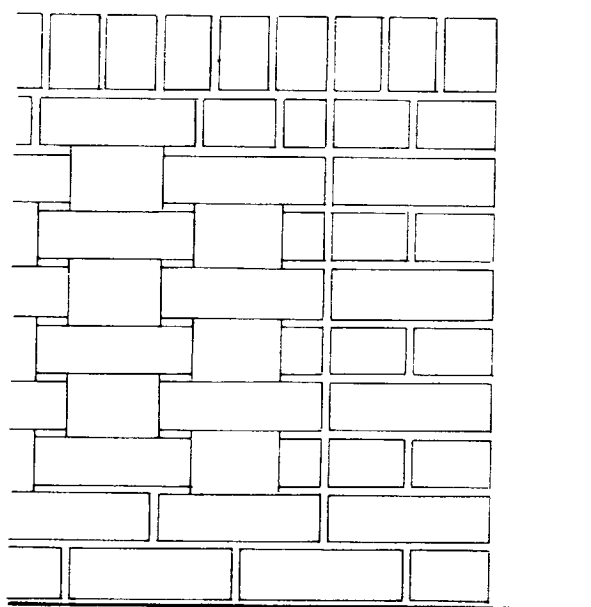


WALL
concrete block screen



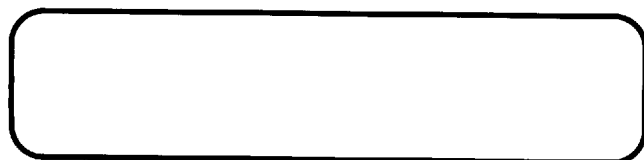


SECTION

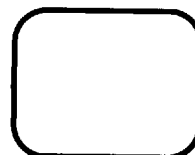


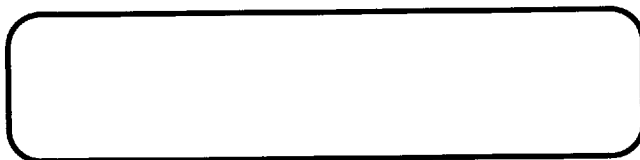
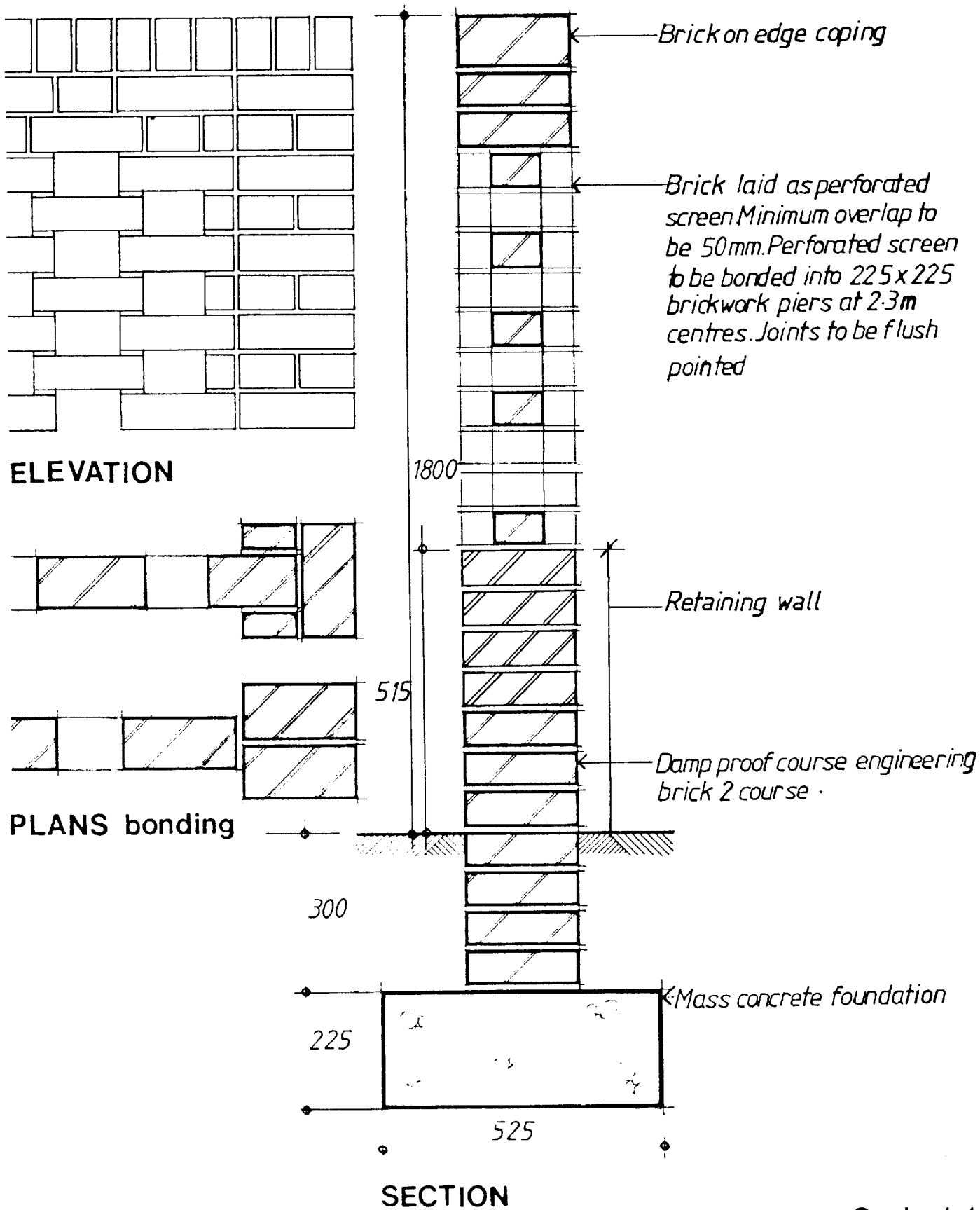
ELEVATION

Scale 1:10

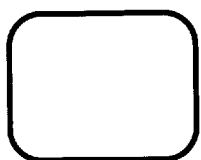


WALL
perforated screen





WALL
perforated screen



RETAINING WALLS

GUIDANCE NOTES

Design procedure

Soils cannot stand more steeply than their natural angle of repose, so that it is often necessary to construct a retaining wall. After an analysis of subgrade conditions and soil profile has been done, the normal design process is:

- Estimate the forces exerted by the material to be restrained on the back of the wall.
- Determine profile of retaining wall and base so that the structure will be stable and not fail by overturning or sliding.
- Analyse the wall itself in terms of structural stability.
- Check the bearing pressure under the base.
- Design structural elements.
- Decide drainage method in backfill.
- Allow for movement and settlement.
- Decide on finishes to wall. (The service of a structural consultant will be required on all walls over 1000 mm high.)

Basic categories of retaining structures

Retaining structures basically fall into two general categories: flexible and rigid construction.

Flexible construction

Flexible construction includes dry stone walls, crib walls, gabion walls and other types of walls that are not rigid structures. The base, or footing, for a flexible wall need not go below the frost line if the depth has a suitable bearing, i.e. undisturbed ground or a compacted sub-base. Typically, a setting bed of sand or compacted granular material is used to improve drainage and provide a level surface. The advantage of flexible construction is that it will tolerate a certain amount of differential settlement without being affected significantly.

Rigid construction

Rigid construction is used where any movement of the structure cannot be tolerated or when aesthetic considerations require it, such as in association with buildings or in formal landscape designs. Typically, rigid construction implies the use of concrete and masonry in a gravity wall or reinforced cantilevered wall type of construction.

Types of retaining walls

Three major types of structural solutions are gravity walls, cantilevered walls and crib walls. Each one has several variations that may be considered, depending upon the criteria used for selection.

Gravity walls

Gravity walls depend upon their mass (i.e. weight and volume) for stability. Regardless of their size, the ratio of the base width to height is approximately constant, varying from about 0.40 to 0.45 for a horizontally loaded wall. Gravity walls are constructed of concrete, stone masonry, or concrete faced with stone or brick. Gravity walls less than 1.5 m high are usually built vertically at the front and back or with a slight batter. In such cases, the base width should be at least 0.4 m.

Gravity walls built of stone laid without mortar, called dry stone walls, are useful in many situations where the retained heights are low (less than 3 m), the stone is easily available and is suitable for the local landscape character.

Concrete block walls

Block walls are of three main varieties:

1. *Plain blocks*, which have a comparatively smooth grey concrete surface.

2. *Profiled blocks*, in which the block is cast in a mould to give it an irregular surface, to either a geometric or a natural pattern. Their advantage over plain blocks is that marking from atmospheric pollution is less visible.
3. *Exposed aggregate blocks* composed of aggregates chosen for their colour and texture which are exposed by removing the cement skin – their advantage over the two previous types is that the greater part of the surface is of a natural material.

Cantilevered walls

Reinforced concrete

A cantilevered wall consists of a base and a stem, which are securely tied together by reinforcing rods extending from the base up through the stem. Reinforcing rods running laterally through the stem provide longitudinal reinforcement in the wall. The weight of the backfill on the base counteracts the forward pressure of the retained height and helps to keep the wall from overturning and sliding. This only applies when the back of the wall can be constructed under the retained material. In most instances this type of wall can be replaced by an embankment. Reinforced concrete cantilevered construction is particularly useful for long lengths of wall where standard metal forms can be re-used with great economy in construction. Liners can be used to achieve a variety of finishes and join patterns in the face of the wall. A veneer of brick or stone can be applied to the wall by using standard metal fasteners.

Reinforced masonry and concrete

A two-brick retaining wall is especially suitable for low-wall construction. It has the finished appearance of brick but the structural stability of a cantilevered wall. It can be laid up without any formwork, using only brick and grout.

The vertical reinforcement bars must be placed within about 10 mm of the brick on the tension side (i.e. the retained earth side) so that the bars will be bonded to the

grout and must extend the full height of the wall. The bricks should be laid with no headers projecting into the grout and in successive tiers of about 200 mm. The space should then be filled each time with grout, followed by puddling with a grout stick. All joints behind the wall should be fully grouted. Solid concrete units or stone units may also be used besides brick.

Reinforced concrete block

A concrete block retaining wall using a cantilever and grout design can be laid up without any framework. The vertical reinforcement bars must be placed within about 10 mm of the inside face of the blocks on the tension side (i.e. the retained earth side) so that the bars will be bonded to the grout. The cells are filled with grout, which should be thoroughly puddled during pouring.

Crib walls

Concrete

A concrete crib wall is constructed of precast reinforced concrete units laid up in interlocking stretchers and headers to form vertical bins which are filled with crushed stone or other granular material. They are a particularly utilitarian solution for retaining fills in situations where excavation is not necessary. Reinforced projecting lugs on the headers are typically used to lock the headers and stretchers together. Backfilling should follow closely the erection of units, and the cribbing should not be laid up higher than 1 m above the backfilled portion.

Timber

Crib walls may be built of timber when the appearance of wood is desired in the design. All timber units should be pressure-treated with a suitable preservative. New or used railway sleeper ties were commonly used in early crib wall construction and continue to be used for low walls. However, timber units cut to size and pressure-treated with copper salts or other non-bleeding preservatives are now widely available.

Other wall types

Timber

Since the timber wall's resistance to overturning depends upon one-half of its height being below finished grade, it is often not economical or practical to use this timber design for retained heights greater than 1.5 m. The timber units in the horizontal wall can be of variable length but should be at least 1.5 m long. Low timber walls are especially useful for raised planting beds and boxes.

The retained height should never be greater than eight to ten times the thickness of the timbers and at least 50 per cent of their overall length should be buried below the lower ground level in average soil; for poor soil 70 per cent and very soft soil almost 100 per cent. For example, 150 mm thick timber retaining wall could comfortably retain a 1200 mm high vertical bank provided that the timbers had an overall length of at least 2000 mm.

Gabions

Gabions are rectangular baskets in standard sizes made of galvanised steel wire or polyvinyl-coated (PVC) wire hexagonal mesh which are filled with stone and tied together to form a wall. Each gabion has a lid and is sub-divided into 1 m cells. After being filled with stone, the lid is closed and laced to the top edges of the gabion. Each gabion is then laced to the adjacent gabions.

Gabion walls, being flexible, can adapt to ground settlement. Their permeability allows water to drain through, making gabions especially suitable along stream and river banks where variations in water depths occur between flood and dry weather conditions. Volunteer vegetation establishes itself quickly in gabions, softening the structure's appearance in the landscape while also adding durability.

Materials

Retaining walls can be constructed of a wide choice of materials, such as brick, block, concrete, stone or timber. Combinations of these materials are used

in some instances, but in the interests of simplicity these have not been included.

Brick

Choice of bricks

Retaining walls can be constructed of ordinary clay bricks (BS 3921: 1974). If there is any likelihood of the brickwork becoming frozen after being saturated by high rainfall, frost-resistant clay bricks (special quality) should be used. Where ordinary clay bricks are proposed, the manufacturer's advice should be sought.

Copings/cappings

The comments in the chapter on free-standing walls apply equally to brick retaining walls. The coping selected should be effective in throwing the water clear of the exposed surfaces of the wall.

Mortars

Sulphate-resisting cement in a mortar of 1:(1/4):3 (cement:lime:sand) should be used up to and including the ground damp proof course, and above if the retaining face is not effectively waterproofed where aggressive sulphate ground conditions are found. When clay bricks of ordinary quality are used in high-rainfall areas, mortar with sulphate-resisting cement should be used for the whole wall. It is advisable to use the same mortar for brickwork below the damp proof course.

Expansion joints

Retaining walls are less susceptible to horizontal movement than free-standing walls, and although CP 121 suggests that vertical joints be provided at 12 m centres in long runs of clay brickwork walling, and at 7.5–9.0 m centres in calcium brickwork, these could be extended to 15 m and 12 m, respectively. A water bar included in the joint will prevent staining of the face. When brick on edge cappings are used, expansion or movement joints should be provided at about half the normal spacing.

Precast concrete

Concrete blocks used for the construction of retaining walls are normally reinforced and laid on foundations as for bricks.

Foundations

To simplify the shuttering of the wall, the bottom 75 mm should be constructed with the foundations. Reinforcement using starter rods should be placed in the foundations and run up into the wall. A damp proof course will not be necessary.

Expansion joints

A 12 mm open joint should be provided at 5.0 m intervals in unreinforced walls and at 10 m intervals in reinforced walls. Except in battered walls, joints should form an integral pattern in unreinforced walls to avoid cracks caused by shrinkage. Joints should be filled with approved material and pointed with non-hardening polysulphide sealant.

Natural stone

This makes a good solid retaining wall, especially when a batter is used. Construction is similar to that described in the chapter on free-standing walls. The type of stone available and the local style should determine the overall appearance. Alternatively, stone can be used to face a concrete block wall or an *in-situ* concrete wall. It is most important to ensure that soil is level with the top of the wall, not only to allow vegetation to grow over and down but also to provide a better appearance. Copings should be selected to conform to the local character and style.

Timber

Retaining walls of timber should be built of hardwood or naturally resistant wood (redwood or cedar). All timbers should still be treated with fungicidal and insect-repellent preservatives. Softwood when used must be pressure-impregnated with preservative by a vacuum process. Consideration should be given to the shrinkage of timbers, especially where joints are used.

For horizontal timbers, vertical supports tied back into the bank with metal rods set into concrete should be used. All exposed timbers should be stained with a proprietary coloured stain.

All edges of timber should be chamfered and end grains protected by cappings which should slope 10–20° to shed water. Fixings must be of galvanised mild steel or stainless steel.

Concrete

In-situ concrete

All reinforced concrete should be grade C30 with 20 mm maximum aggregate size and the back face of the wall should be lined with a bituminous damp-proof membrane. All reinforcement overlaps should be a nominal 40 diameter and the minimum concrete cover should be 40 mm. The bearing capacity of the soil must also be taken into account. As a general guide, in a wall approximately 1,700–2,000 mm high the thickness of a reinforced wall will not be less than 150 mm and mass concrete 300 mm.

If above 1.0 m in height, *in-situ* reinforced concrete retaining walls should generally be designed by a structural engineer. *In-situ* mass concrete walls without reinforcement are double the thickness of reinforced walls. Only in exceptional circumstances will it pay not to use reinforcement, but there are occasions when a heavy structure is required.

Drainage

To prevent the build-up of water pressure behind a retaining wall, a hydraulic drain should be provided near the base on the retaining side. The size will depend on ground conditions, but an average of 100–150 mm diameter is generally used. The drain must be surrounded by granular sulphate-free material, extending to the surface where possible. Weep-holes should discharge at least 150 mm above ground level and at intermediate levels if necessary. Care should be taken to ensure that the drainage is not allowed to become blocked through build-up of silt.

Reinforcement

In grouted cavity retaining walls reinforcement will vary according to the size of the wall.

TABLE 4: SOIL TYPES AND THEIR PROPERTIES

Division	Soil description	Value as a foundation material*	Frost action	Drainage
Gravel and gravelly soils	Well-graded gravel, or gravel–sand mixture, little or no fines	Excellent	None	Excellent
	Poorly graded gravel, or gravel–sand mixtures, little or no fines	Good	None	Excellent
	Silty gravel, gravel–sand–silt mixtures	Good	Slight	Poor
	Clayey gravels, gravel–clay–sand mixtures	Good	Slight	Poor
Sand and sandy soils	Well-graded sands, or gravelly sands, little or no fines	Good	None	Excellent
	Poorly graded sands, or gravelly sands, little or no fines	Fair	None	Excellent
	Silty sands, sand–silt mixtures	Fair	Slight	Fair
	Clayey sands, sand–clay mixtures	Fair	Medium	Poor
Silts and clays LL <50†	Inorganic silts, rock flour, silty or clayey fine sands, or clayey silts with slight plasticity	Fair	Very high	Poor
	Inorganic clays of low to medium plasticity, gravelly clays, silty clays, lean clays	Fair	Medium	Impervious
	Organic silt–clays of low plasticity	Poor	High	Impervious
Silts and clays LL <50	Inorganic silts, micaceous or diatomaceous fine sandy or silty soils, elastic silts	Poor	Very high	Poor
	Inorganic clays of high plasticity fat clays	Very poor	Medium	Impervious
	Organic clays of medium to high plasticity organic silts	Very poor	Medium	Impervious
Highly organic soils	Peat and other highly organic soils	Not suitable	Slight	Poor

*Consult soil engineers and local building codes for allowable soil-bearing capacities.

†LL indicates liquid limit.

Source: Douglas S. Way, *Terrain Analysis: A Guide to Site Selection Using Photographic Interpretation*, Douglas S. Way, Columbus, Ohio, 1972.

Cover to reinforcement

Grouted cavity walls: minimum cover to all reinforcement = 10 mm. Pocket Walls: minimum cover to all reinforcement = 40 mm. Quetta bond: minimum cover to all reinforcement = 10 mm. Buttressed wall or other wall with reinforcement positioned in the joints: minimum cover to main reinforcement = 25 mm.

The above recommendations are in accordance with SP 91(8), to which reference should be made for further information.

SPECIFICATION CHECK LIST

Check with consultant engineer all walls over one metre high regarding ground conditions, size of wall, drainage, etc.

Brick/block/stone faced

Use the check list for free-standing walls where retaining walls are to be faced with brick, block or stone, or built entirely of these materials.

Add details of reinforcing, waterproofing rear of unit walls, drainage and weep holes. Provide details of:

- Foundation type and size.
- Wall construction.
- Waterproofing and drainage.
- Coping type.

Concrete

Where the material is to be concrete *in-situ* or concrete block, both may have to be reinforced and therefore it is advisable to check with consultant engineer regarding type and method of reinforcing.

Timber

Where timber is to be used check the type of hardwood is suitable without any preservation. Where softwood is specified ensure that it is treated and to the appropriate grade. Ensure all hardware is galvanised. State details of finished appearance and where timber is to be stained specify manufacturer, colour and number of coats.

DETAIL SHEETS

General

Reinforcement

Foundations

Joints

Brick

Brick (low) (3)

Brick (high)

Concrete

Battered concrete

Concrete block (2)

Cobble blocks

Concrete crib (2)

Stone

Stone faced

Mortared/coursed

Uncoursed/dry

Gabions – stepped

Gabions – batter

Gabions and plants

Timber

Board (low)

Board (medium)

Angled – horizontal (2)

Stepped

Crib – square

Crib – round

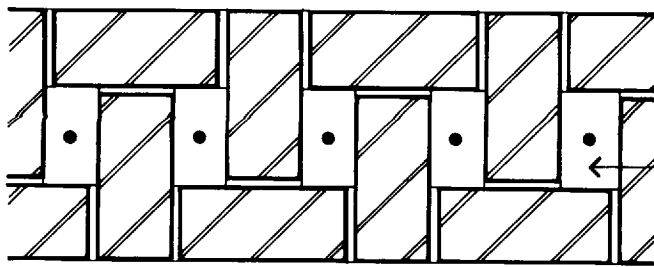
Crib – round (live)

Angled – vertical

Rail sleeper and steel

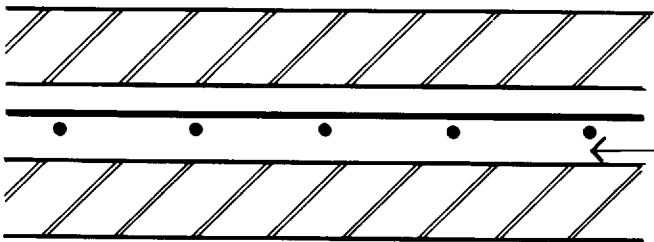
Timber and steel

Timber and mesh



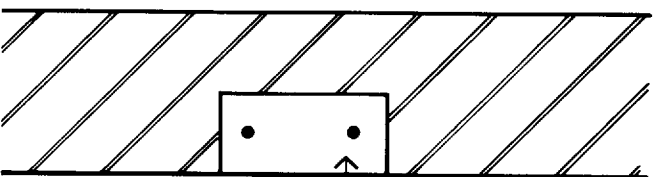
F Quetta bond

minimum 10 cover all round



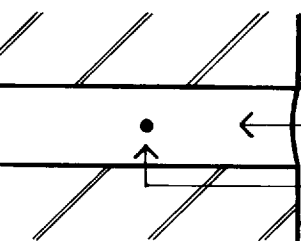
G Grouted cavity

minimum 10 cover



H Pocket wall

minimum 40 cover all round

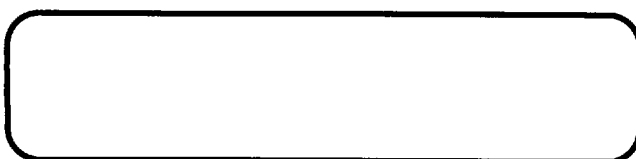


J Bed joint

minimum 25 cover to wall face

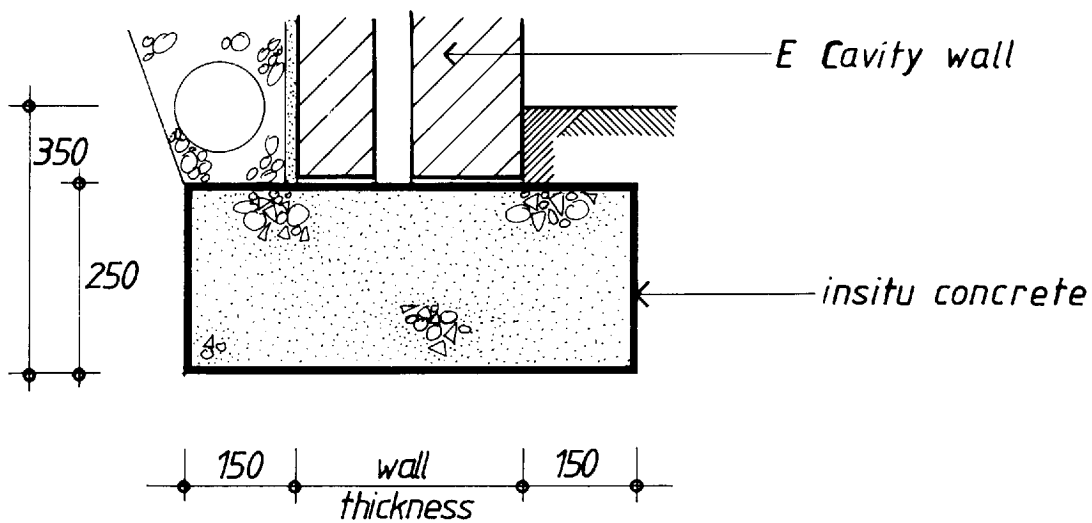
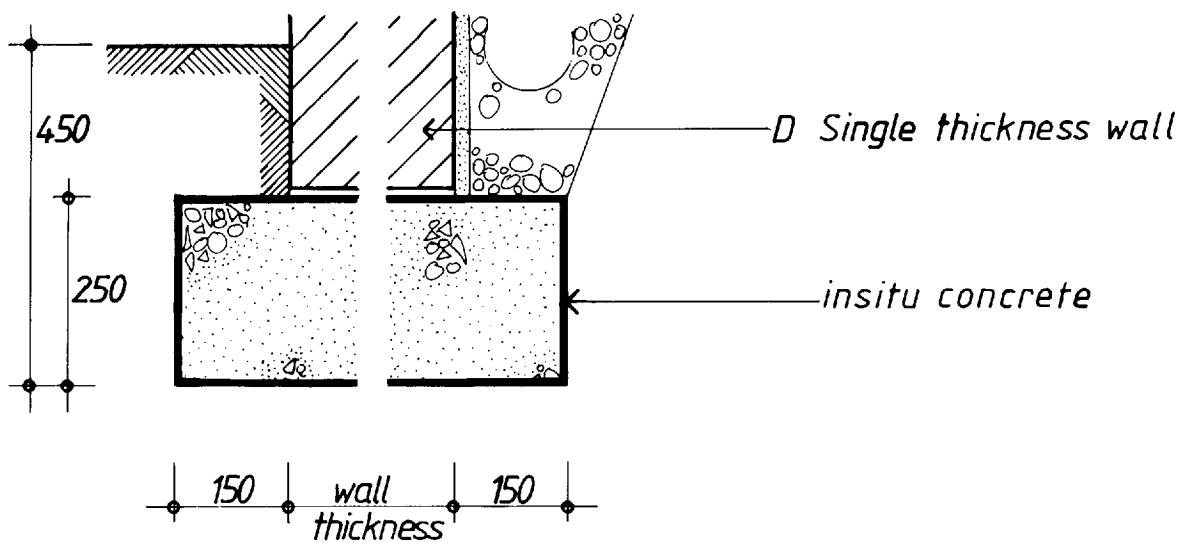
minimum 3 cover above and below bar

N.T.S



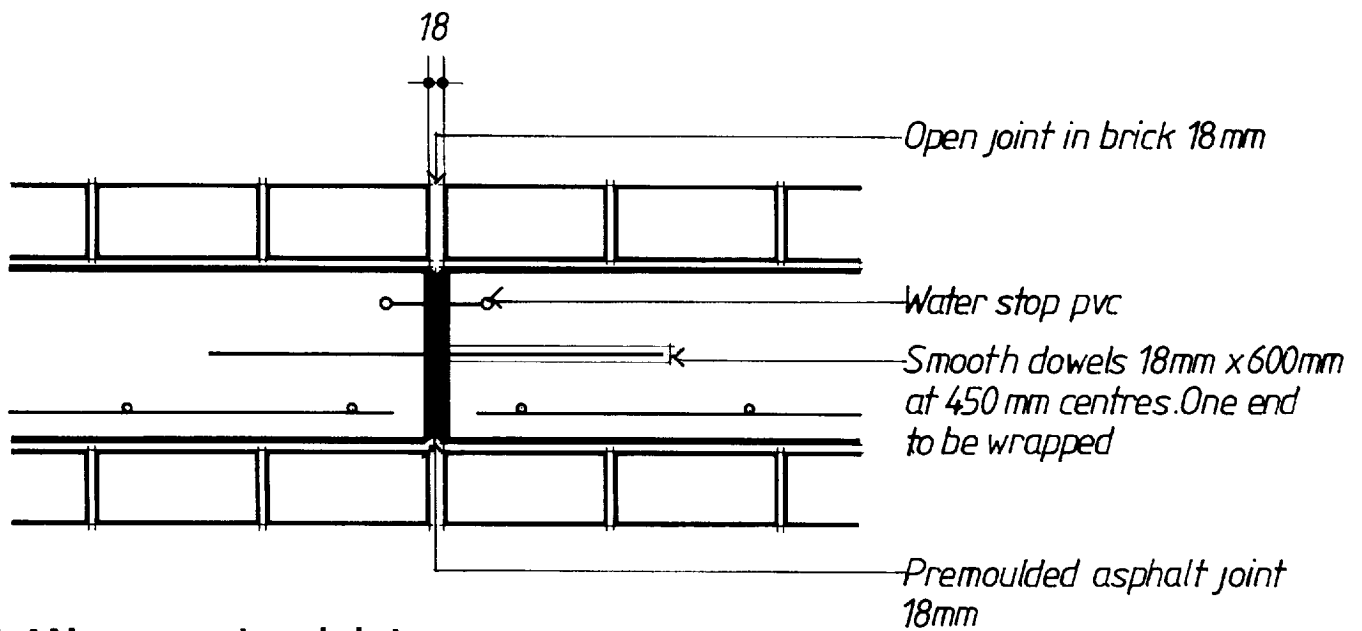
RETAINING WALLS
reinforcement



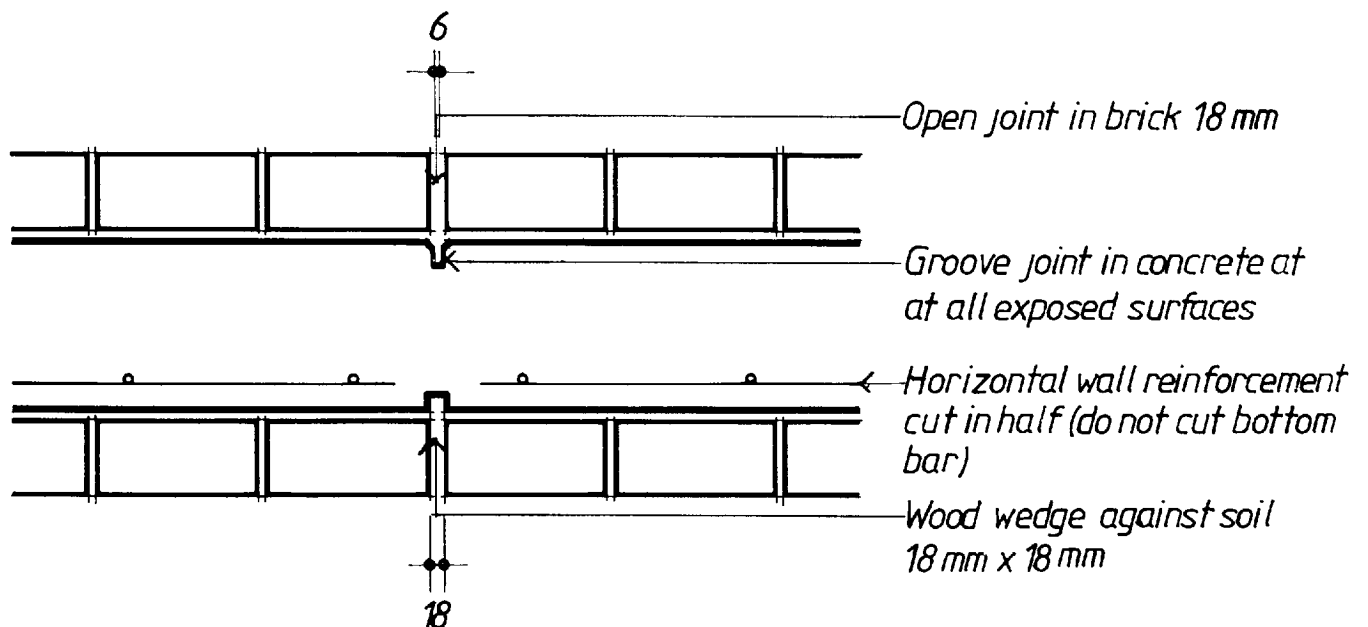


N.T.S.

RETAINING WALLS
foundations



PLAN expansion joint

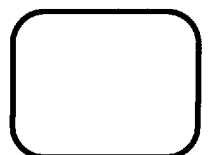


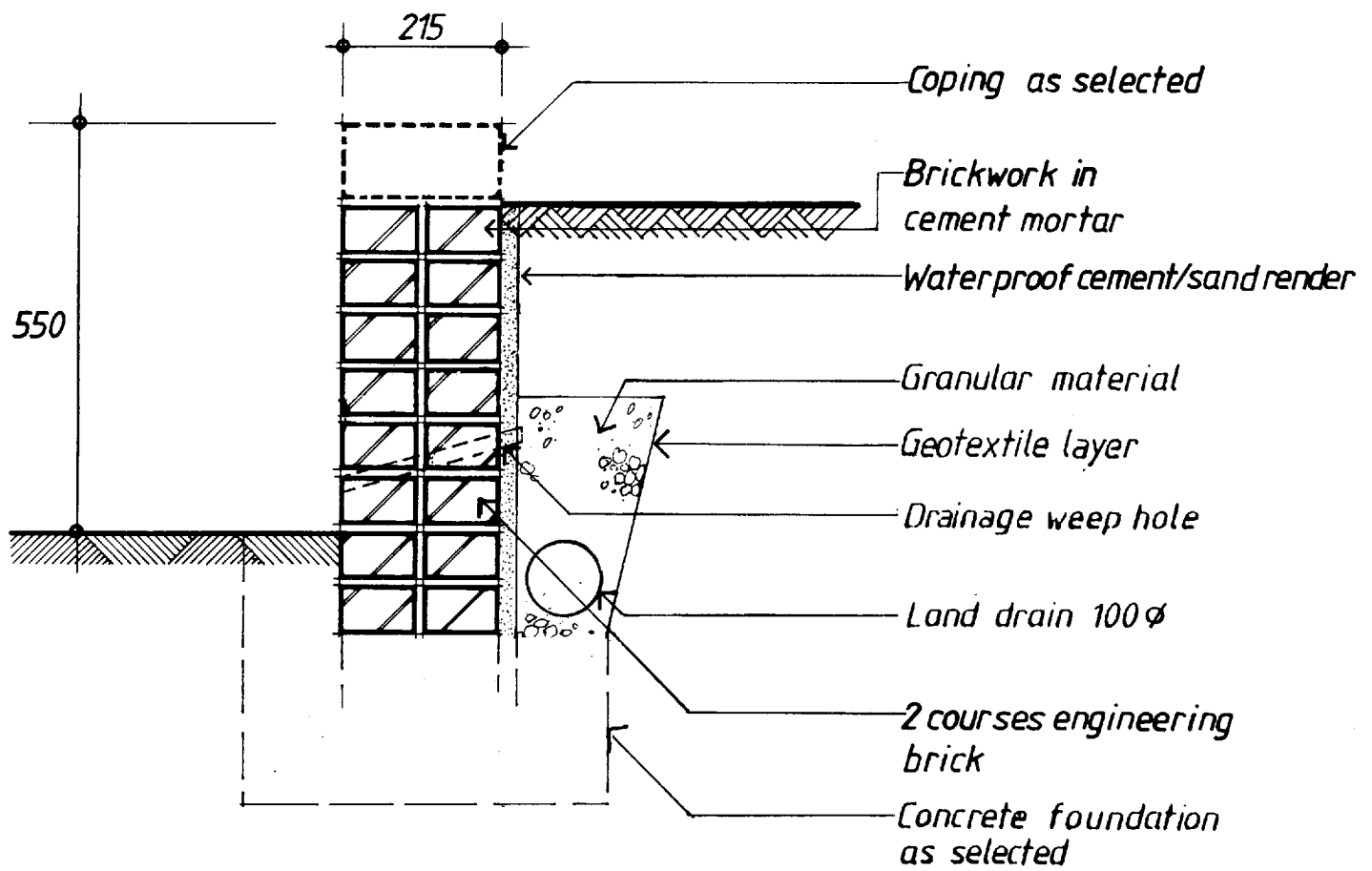
PLAN control joint

Scale 1:10



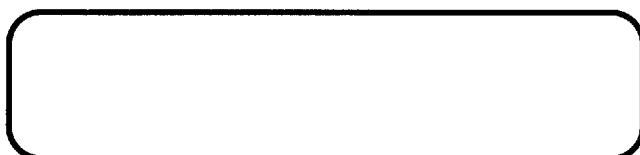
RETAINING WALL
joints





SECTION

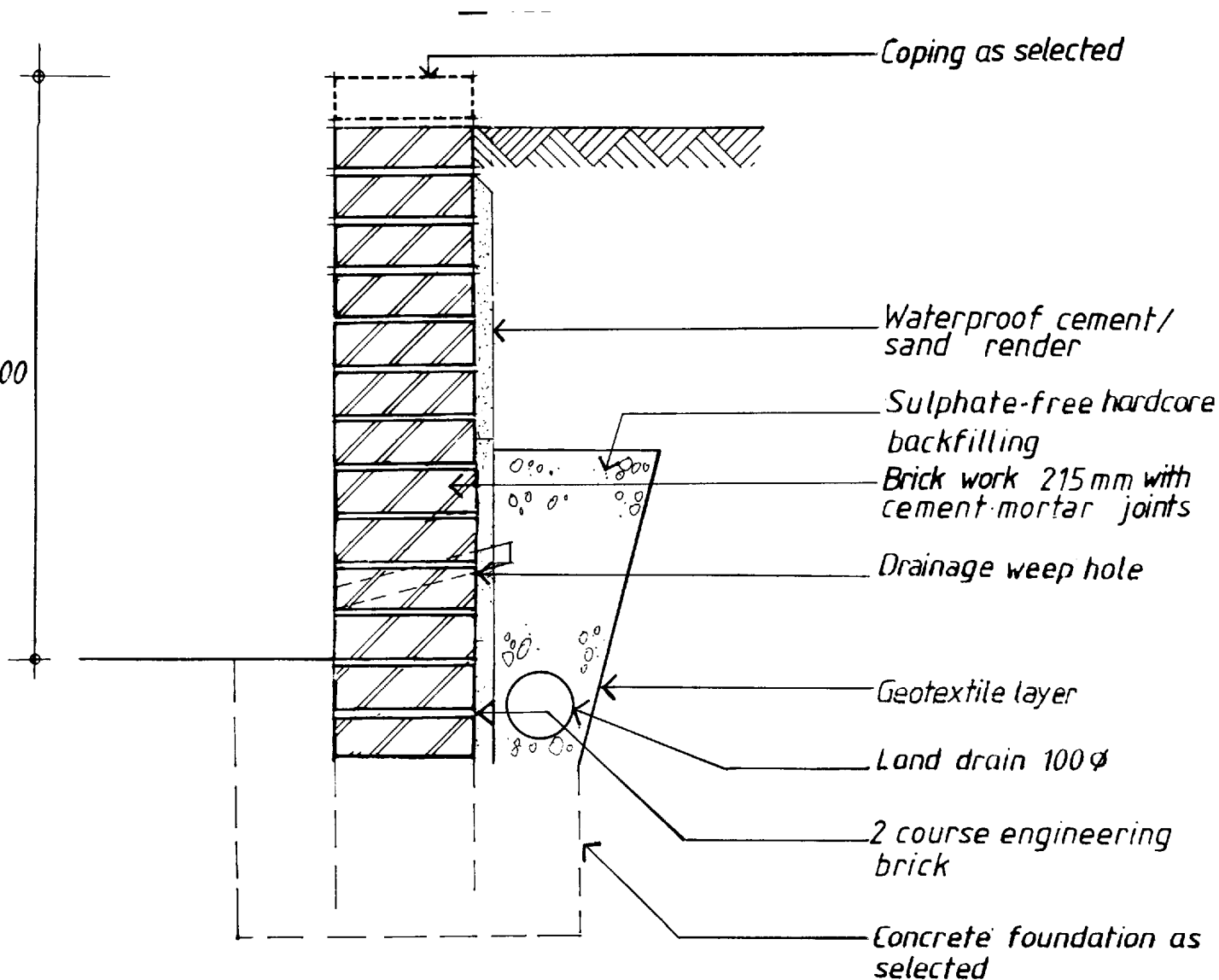
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RETAINING WALL
brick (low)



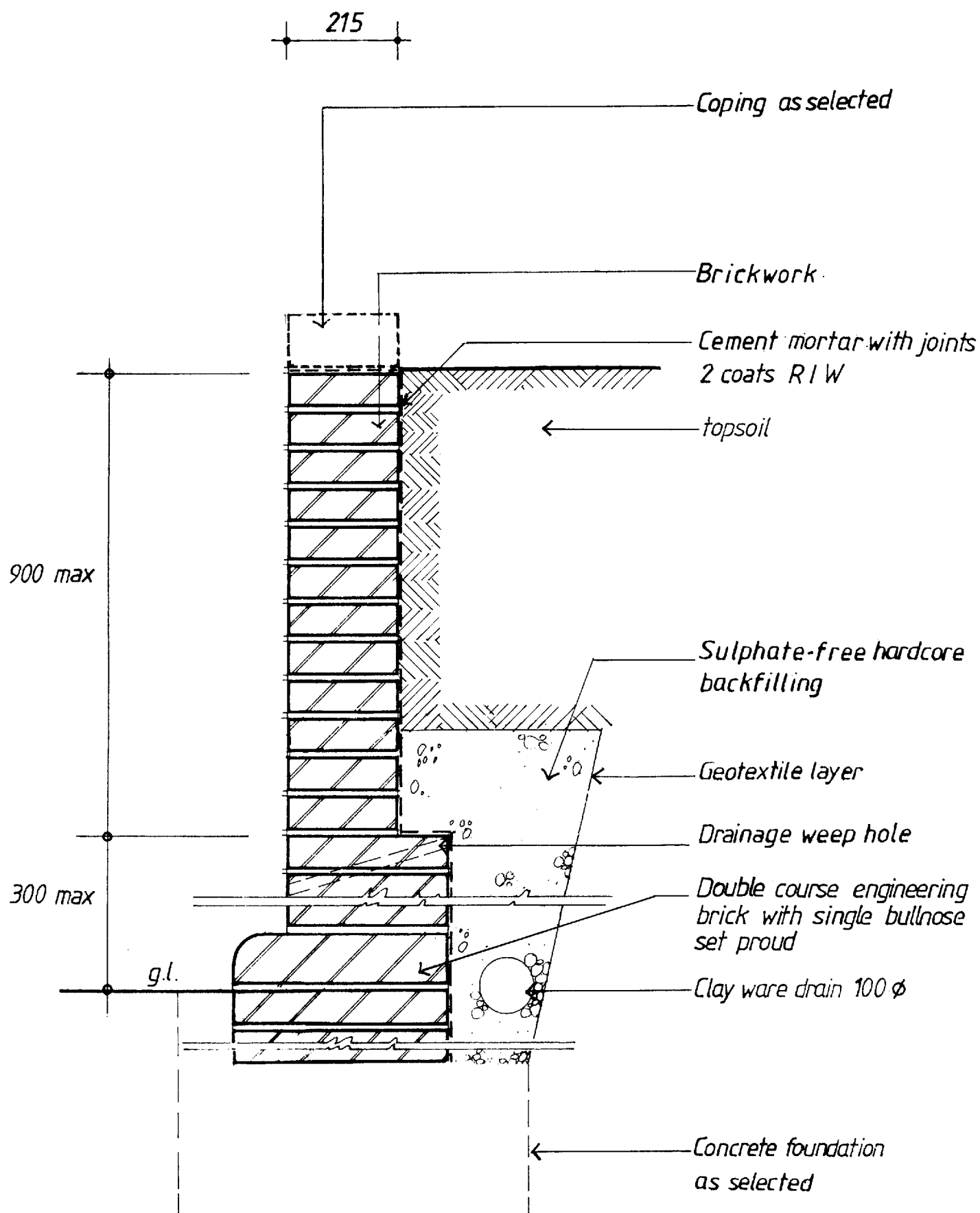
900



SECTION

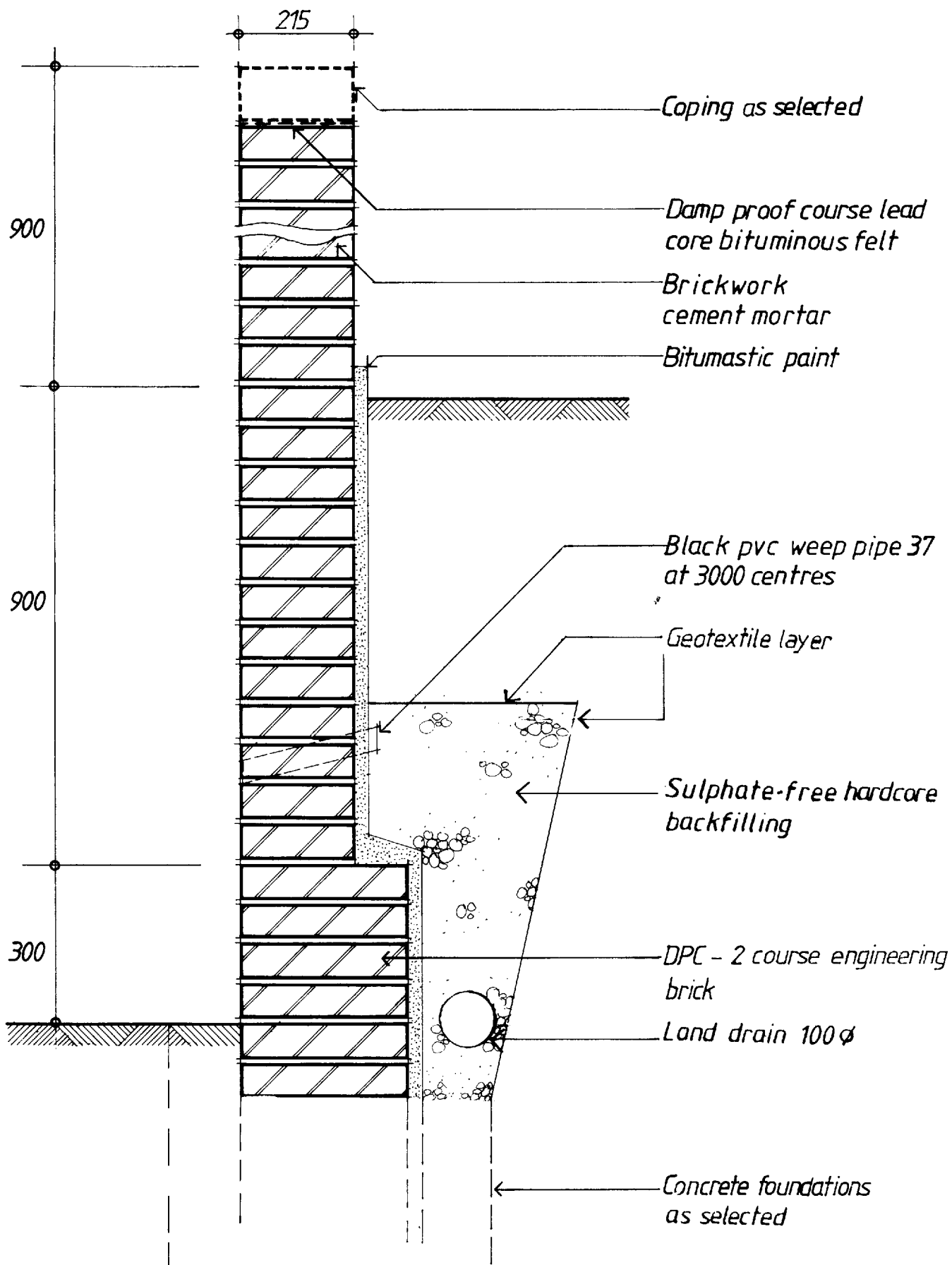
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RETAINING WALL
brick (low)



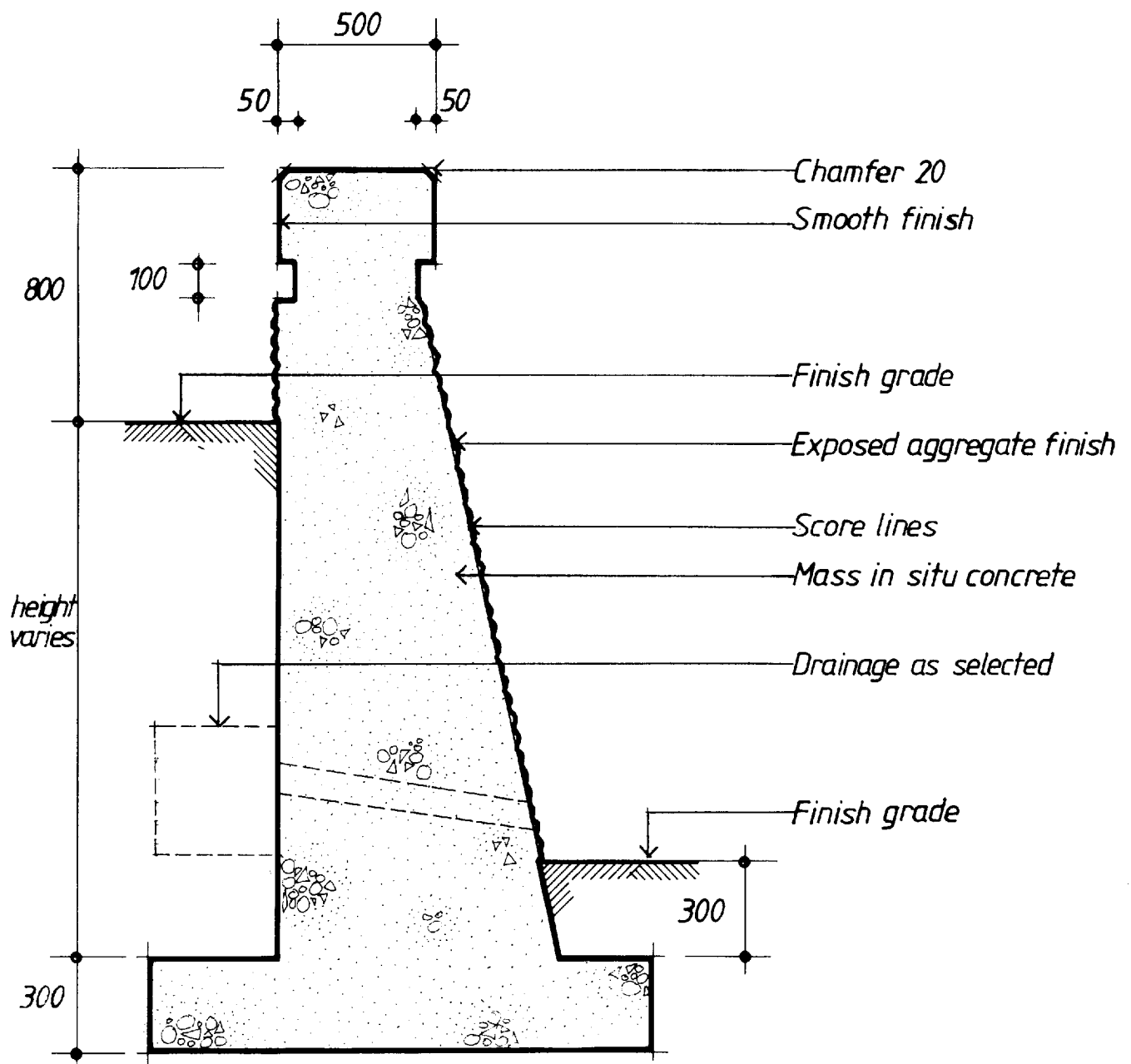
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RETAINING WALL
brick (low)



Scale 1:10

RETAINING WALL
brick (high)



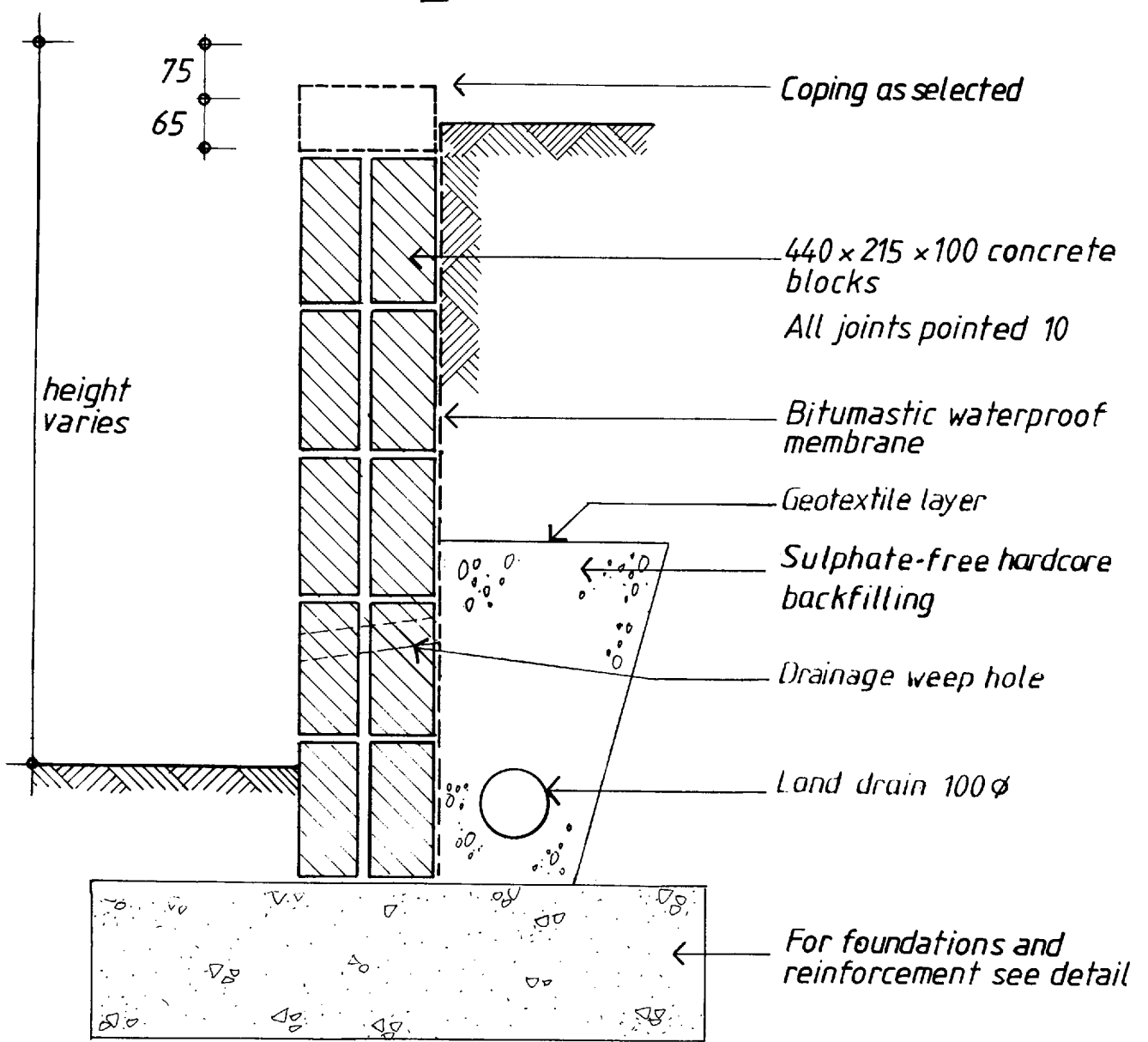
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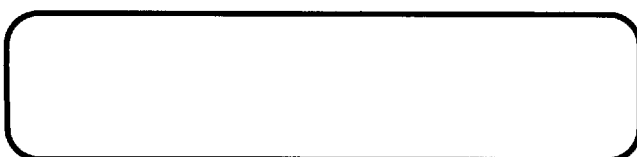
RETAINING WALL
battered concrete





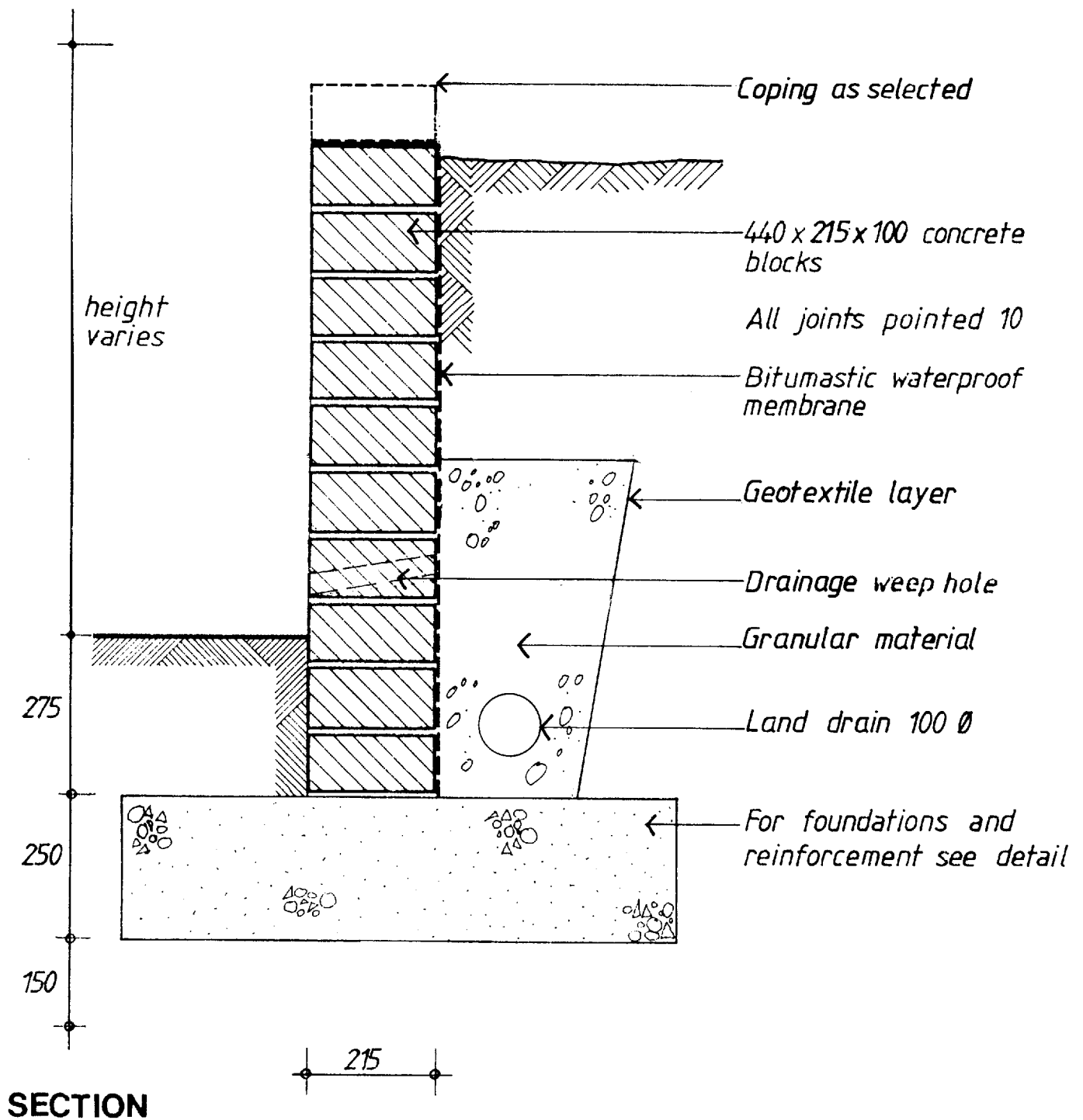
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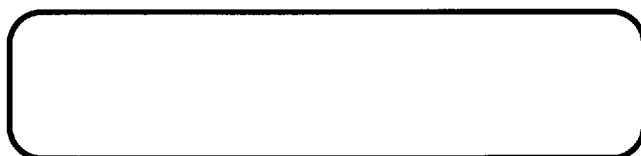


RETAINING WALL
concrete block

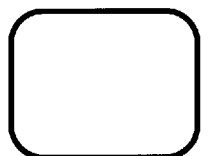


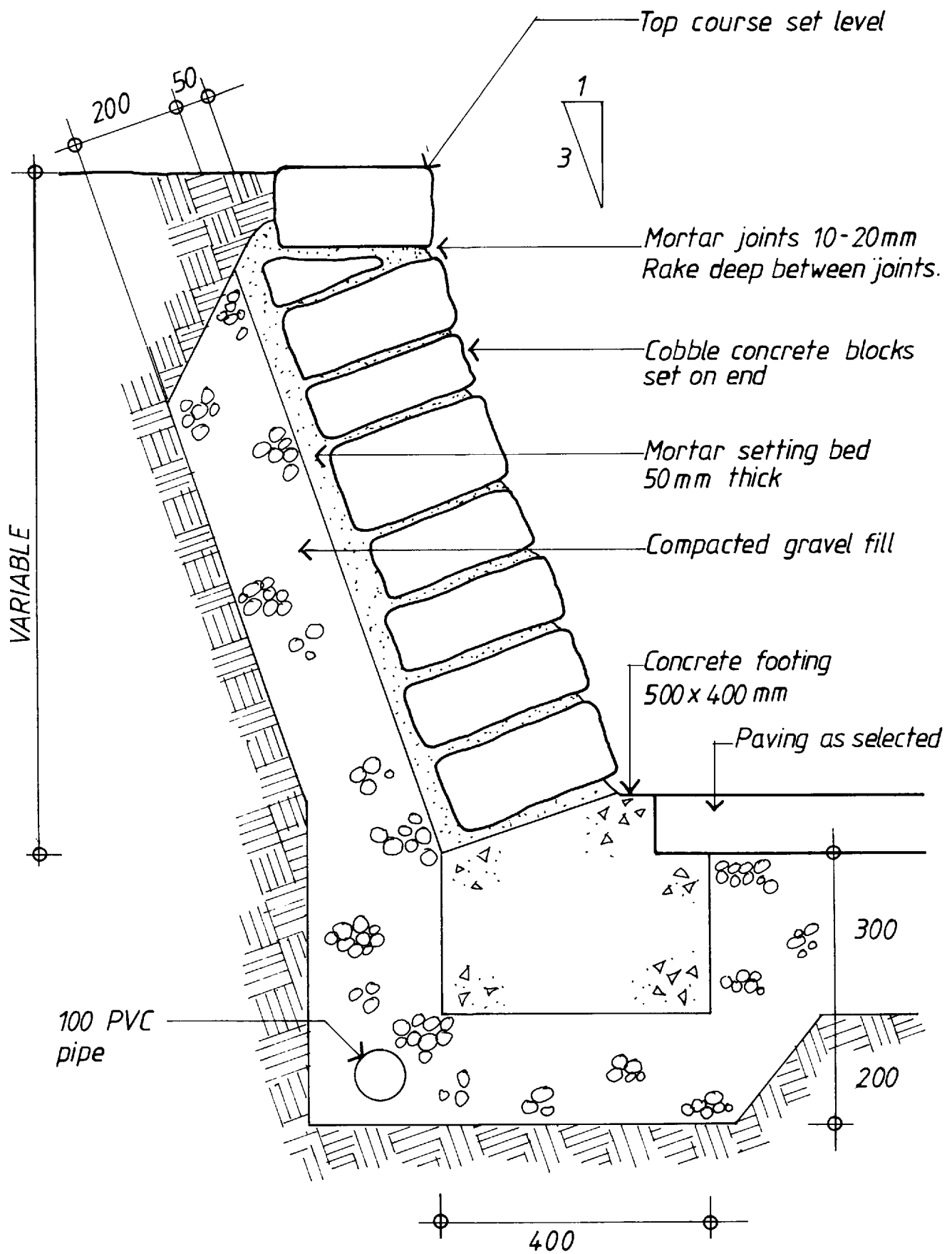


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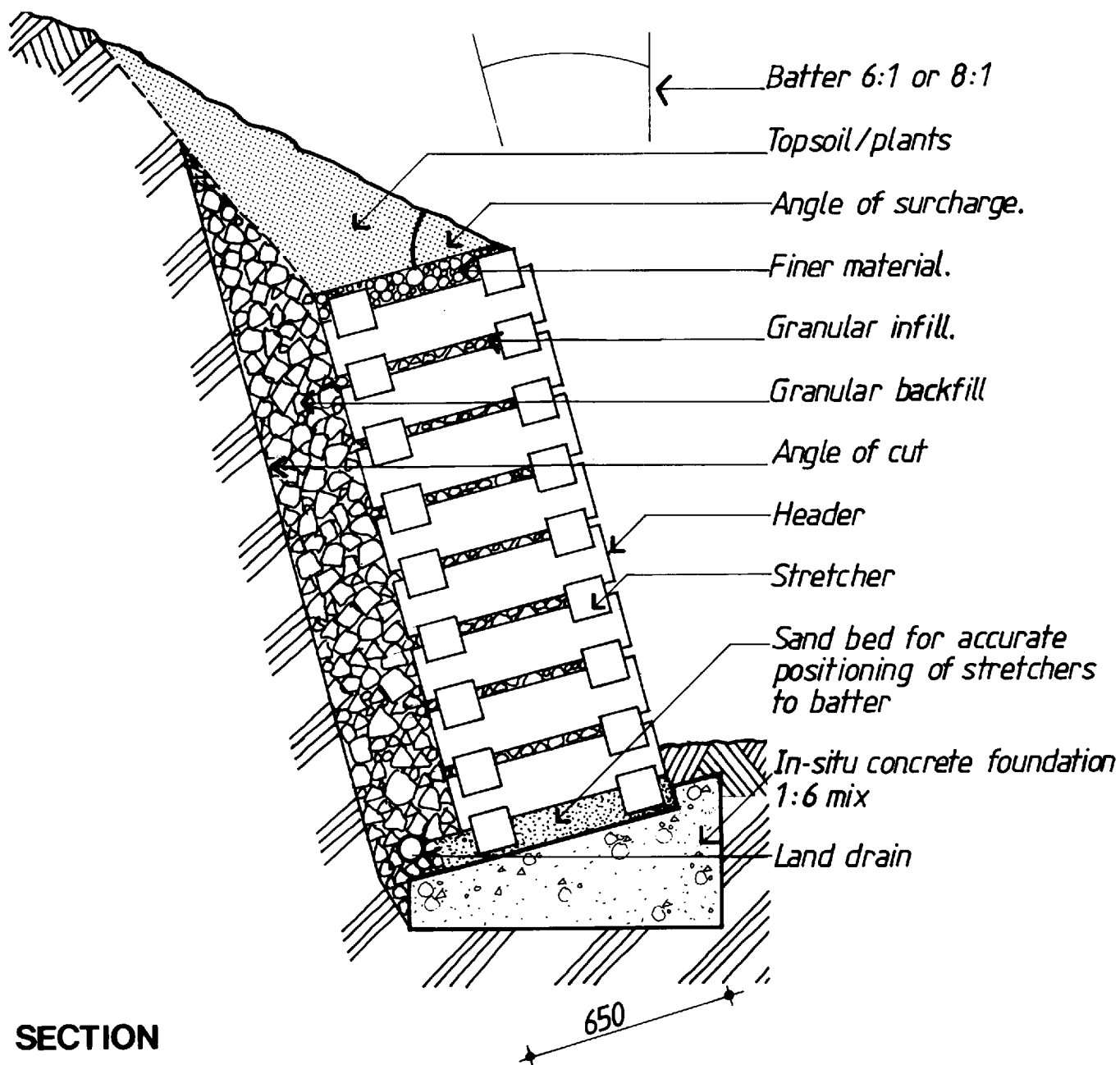
RETAINING WALL
concrete block



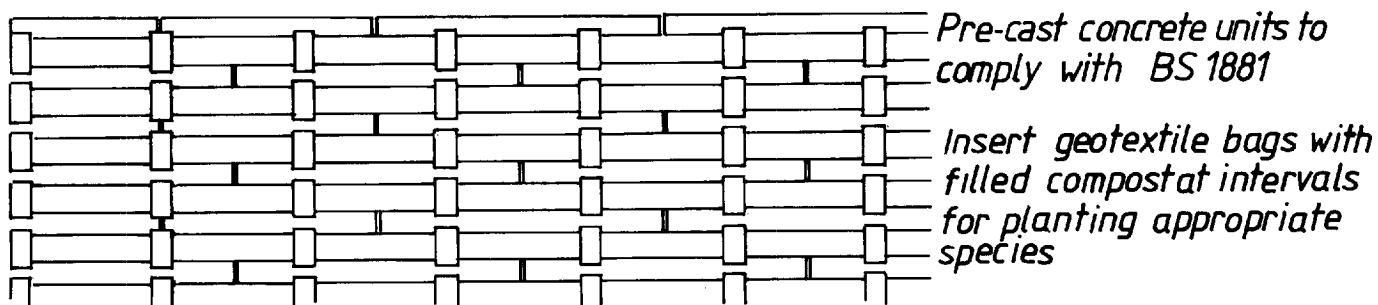


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RETAINING WALL
cobble blocks

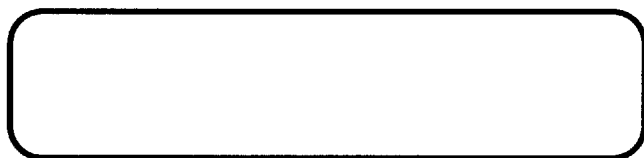


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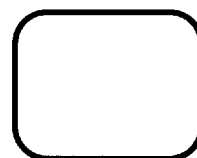


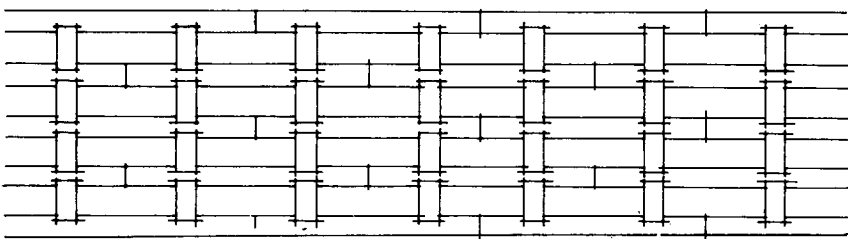
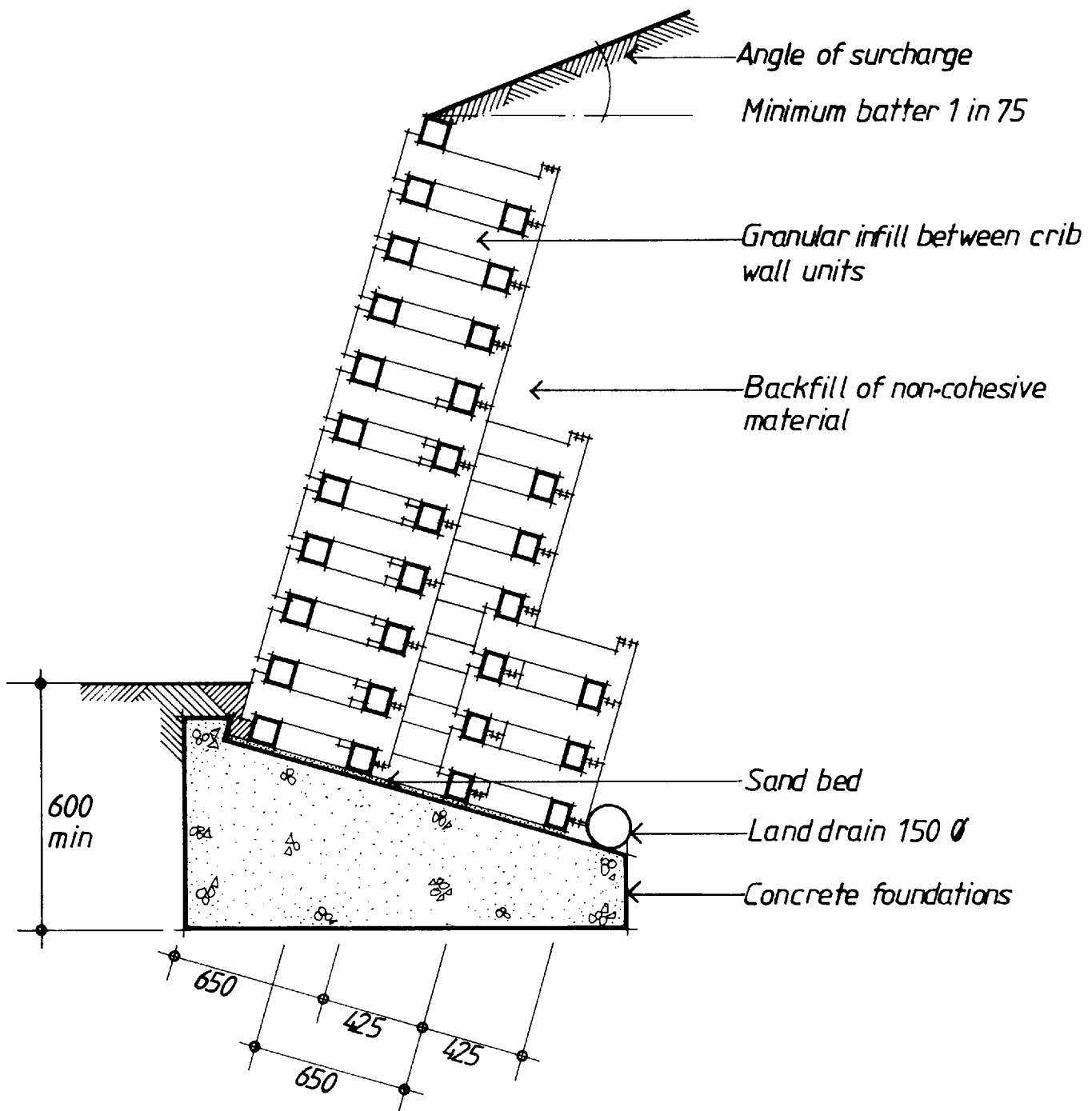
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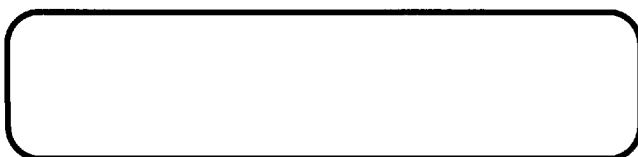
RETAINING WALL
concrete crib





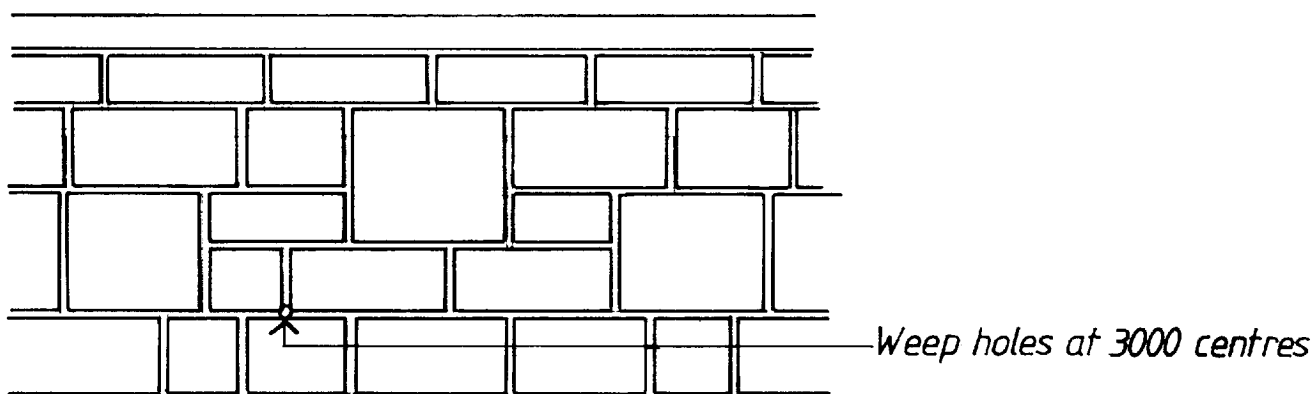
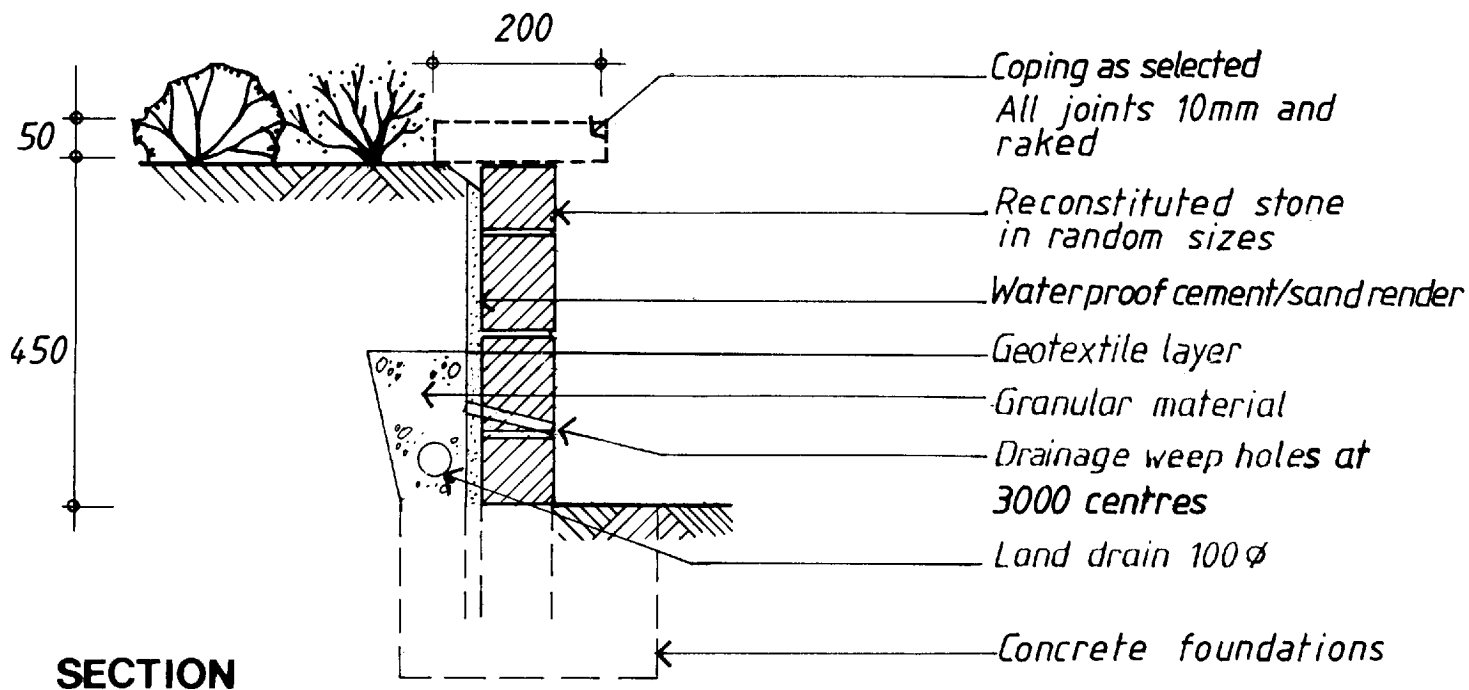
ELEVATION

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RETAINING WALL
concrete crib



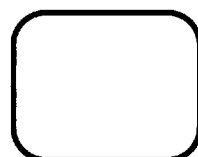


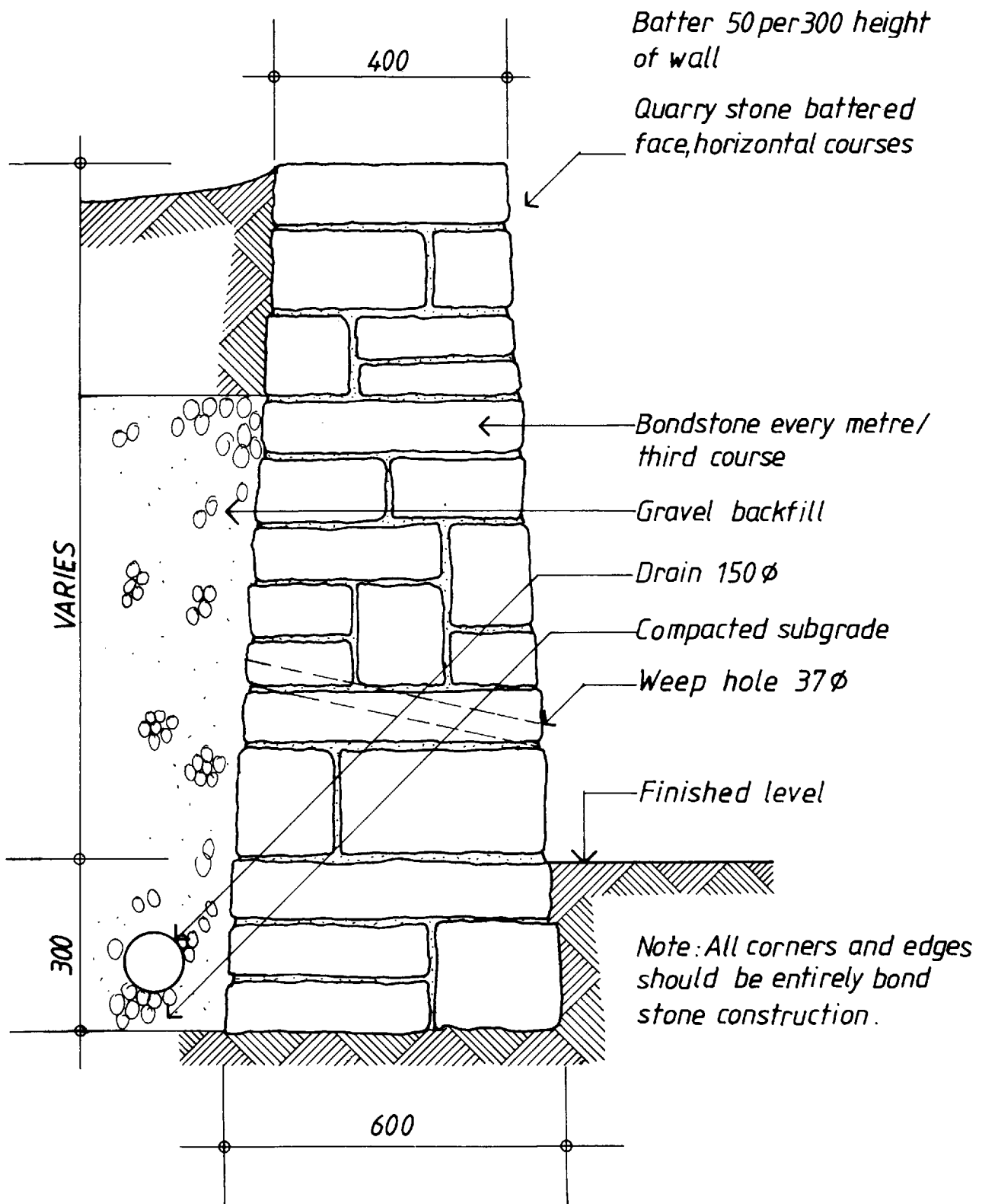
ELEVATION

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RETAINING WALL
stone faced



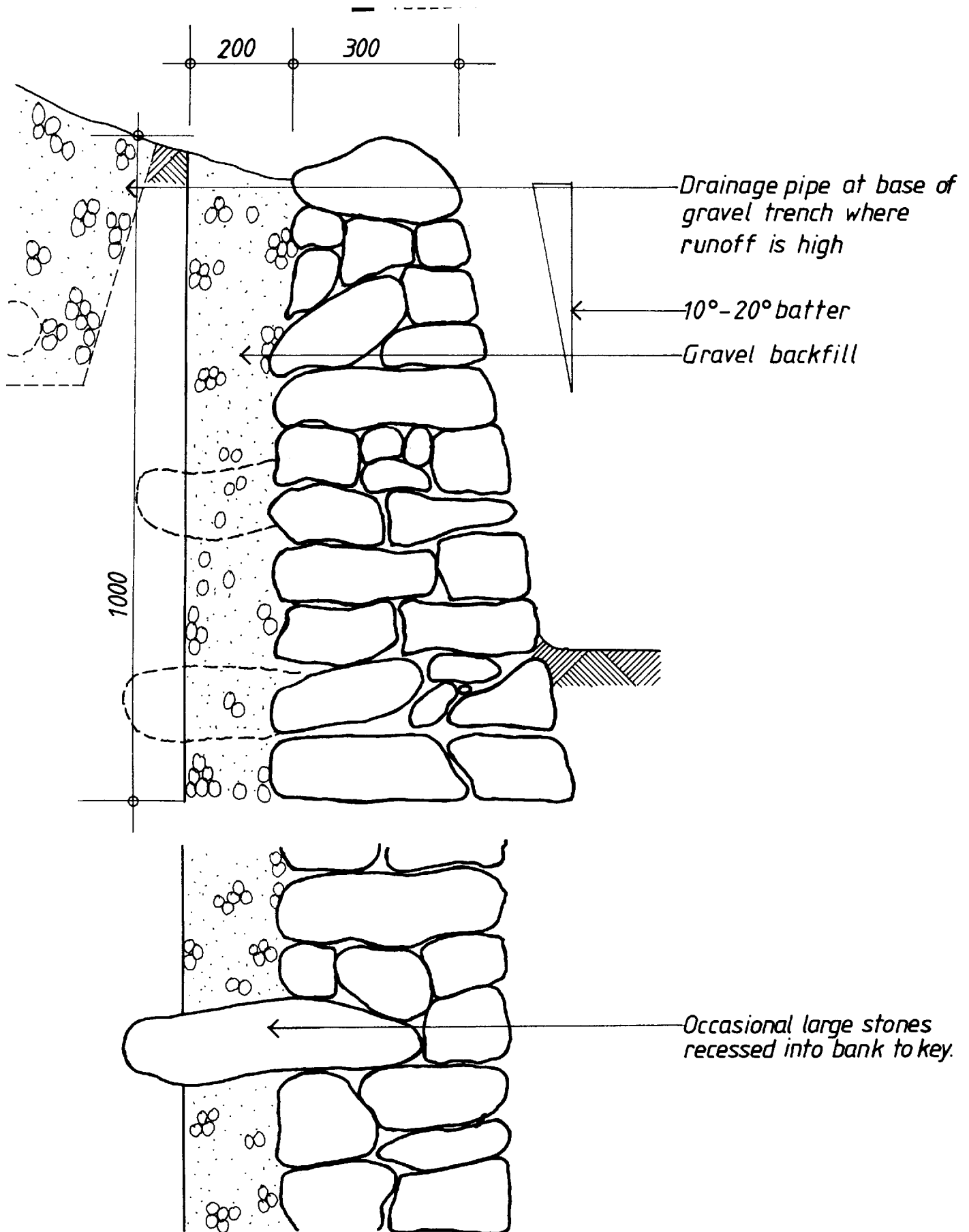


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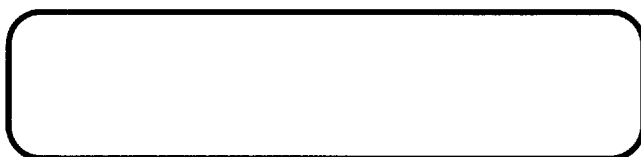


RETAINING WALL
natural stone
mortared/coursed



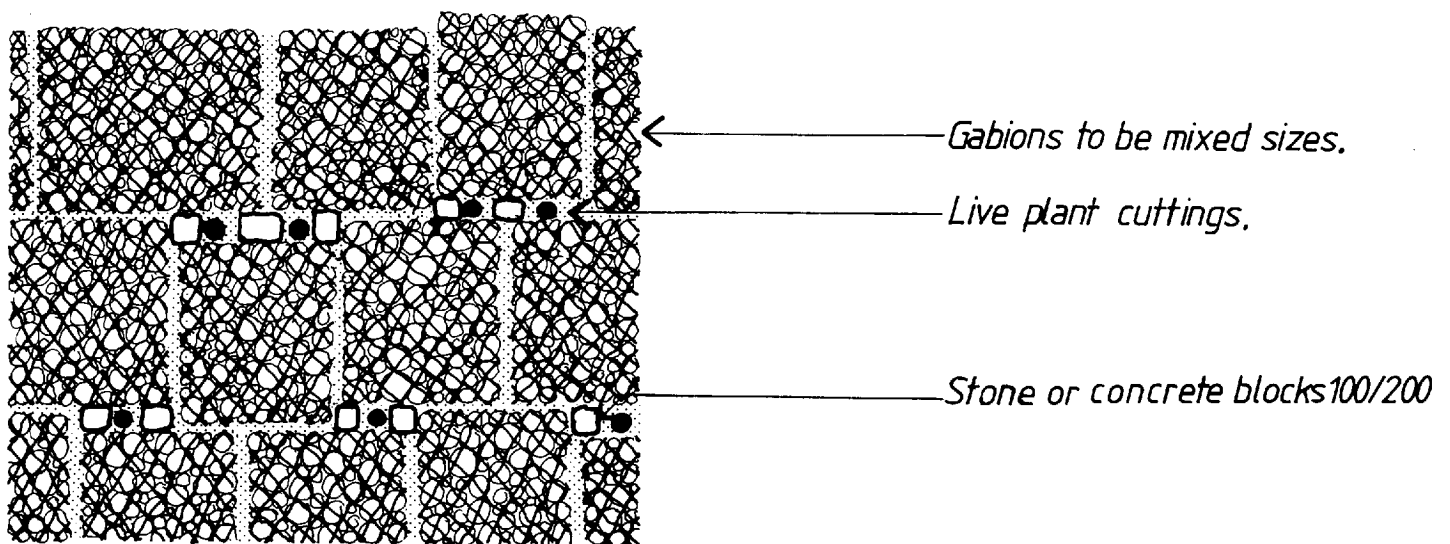
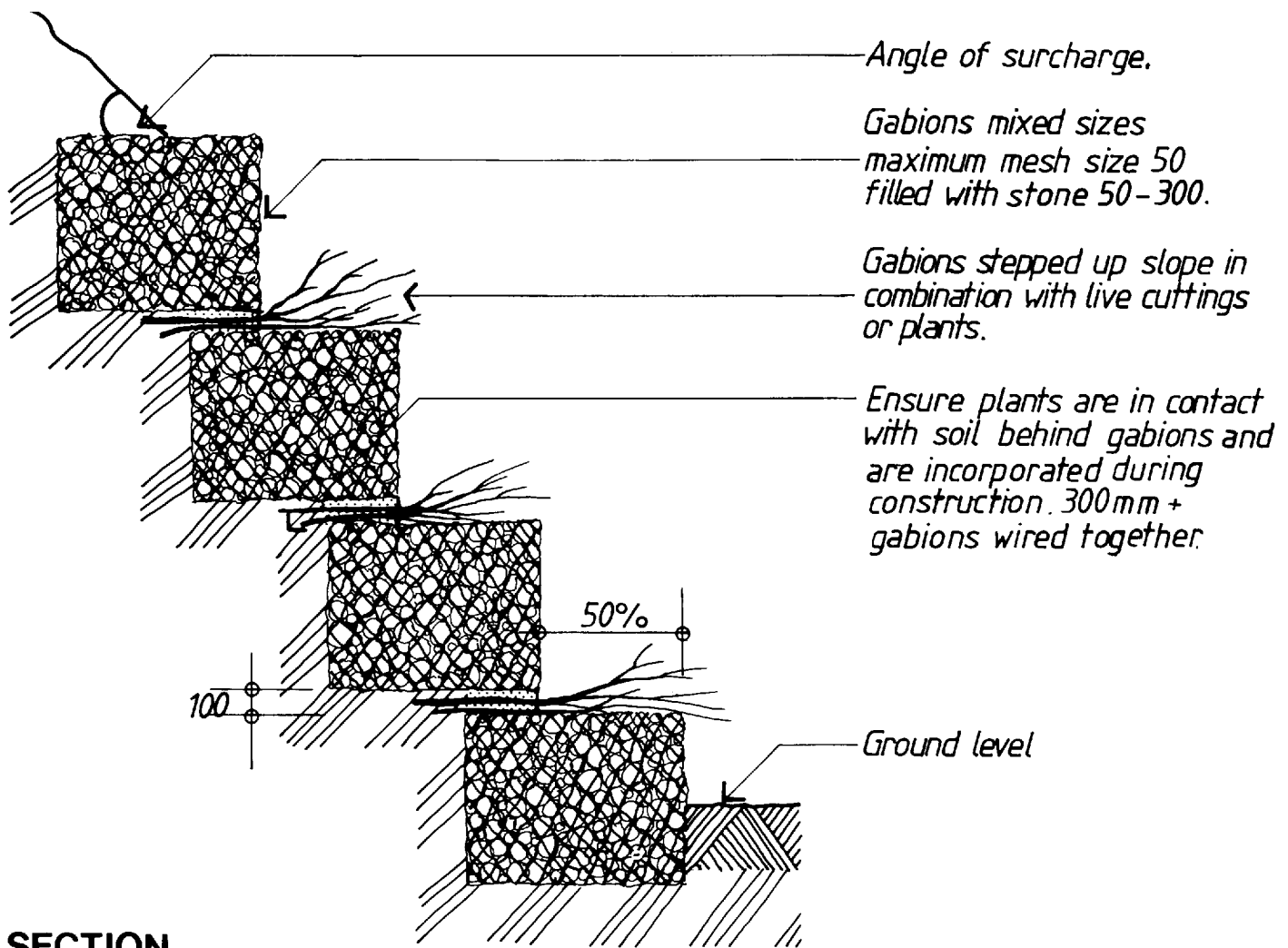


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RETAINING WALL
natural stone
uncoursed/dry

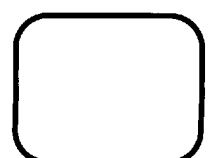


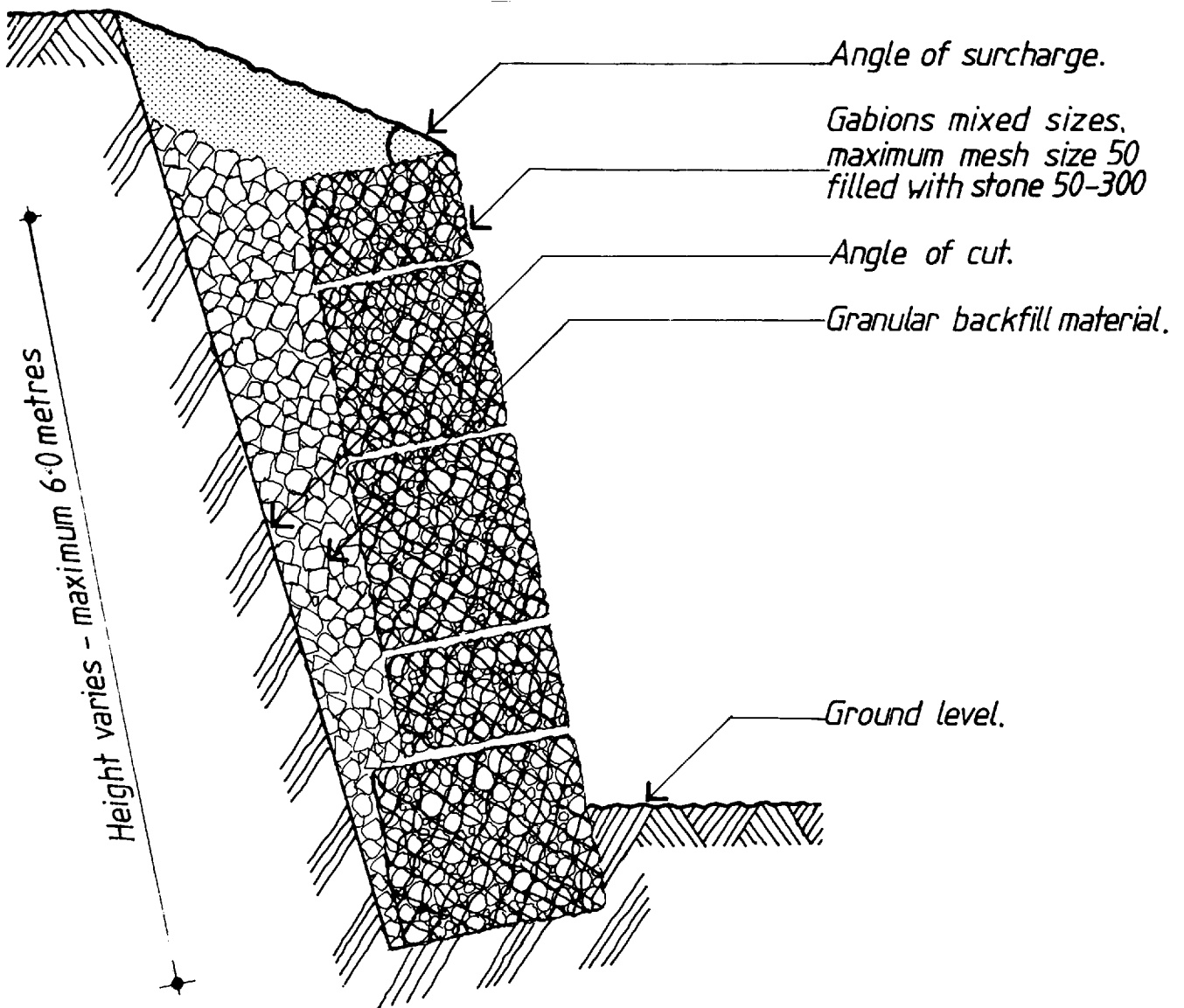


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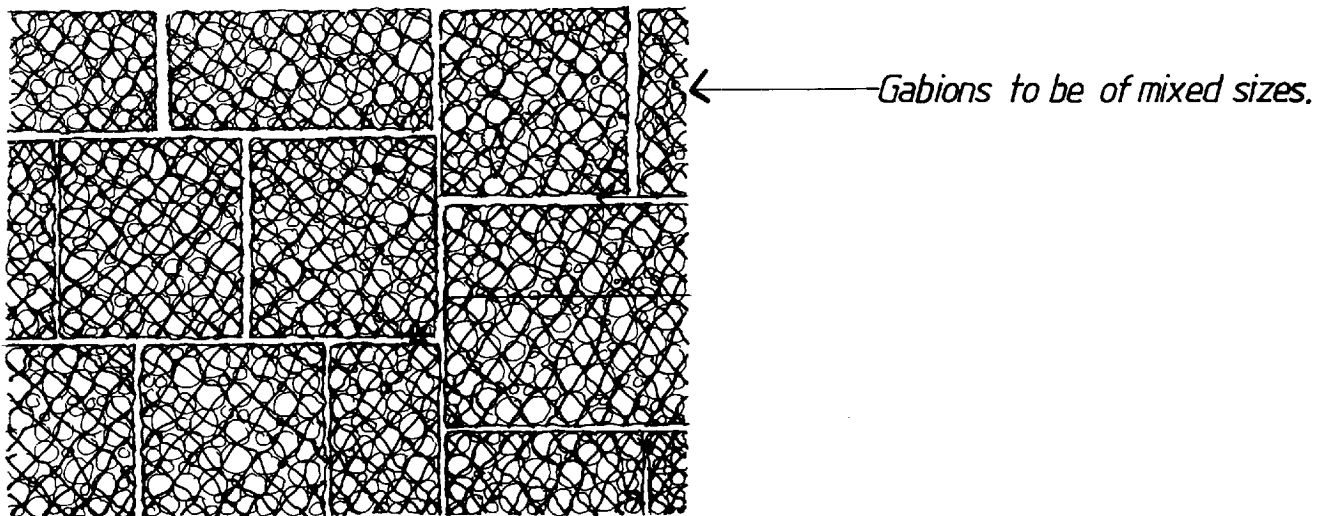


RETAINING WALL
Gabions – stepped



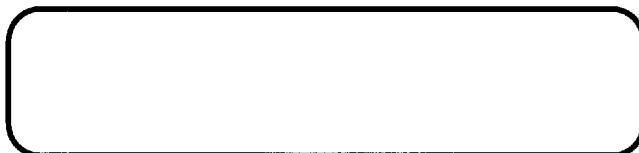


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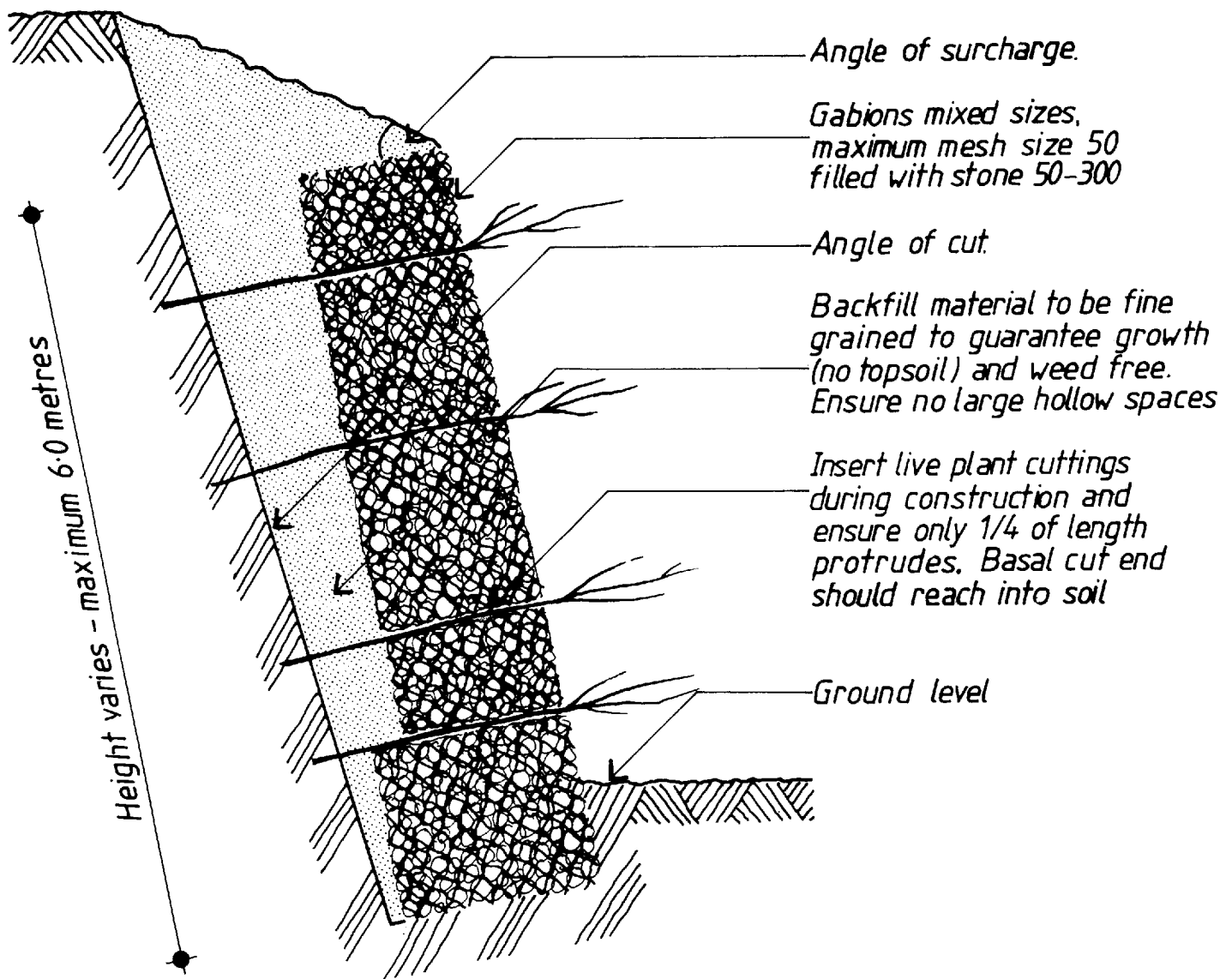
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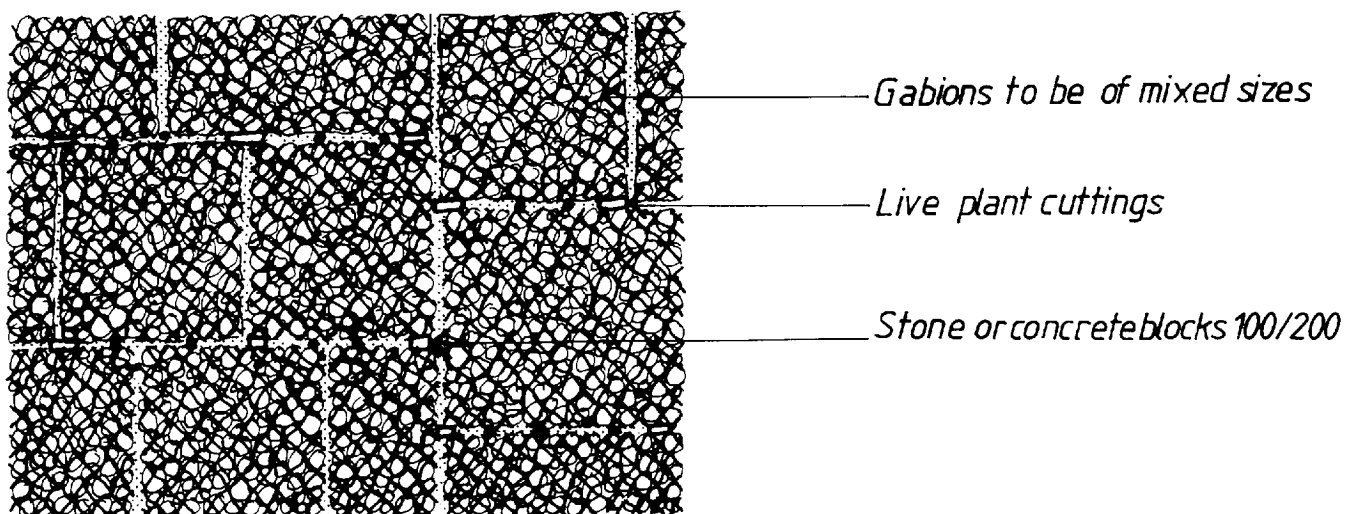


RETAINING WALL
Gabions – batter





SECTION



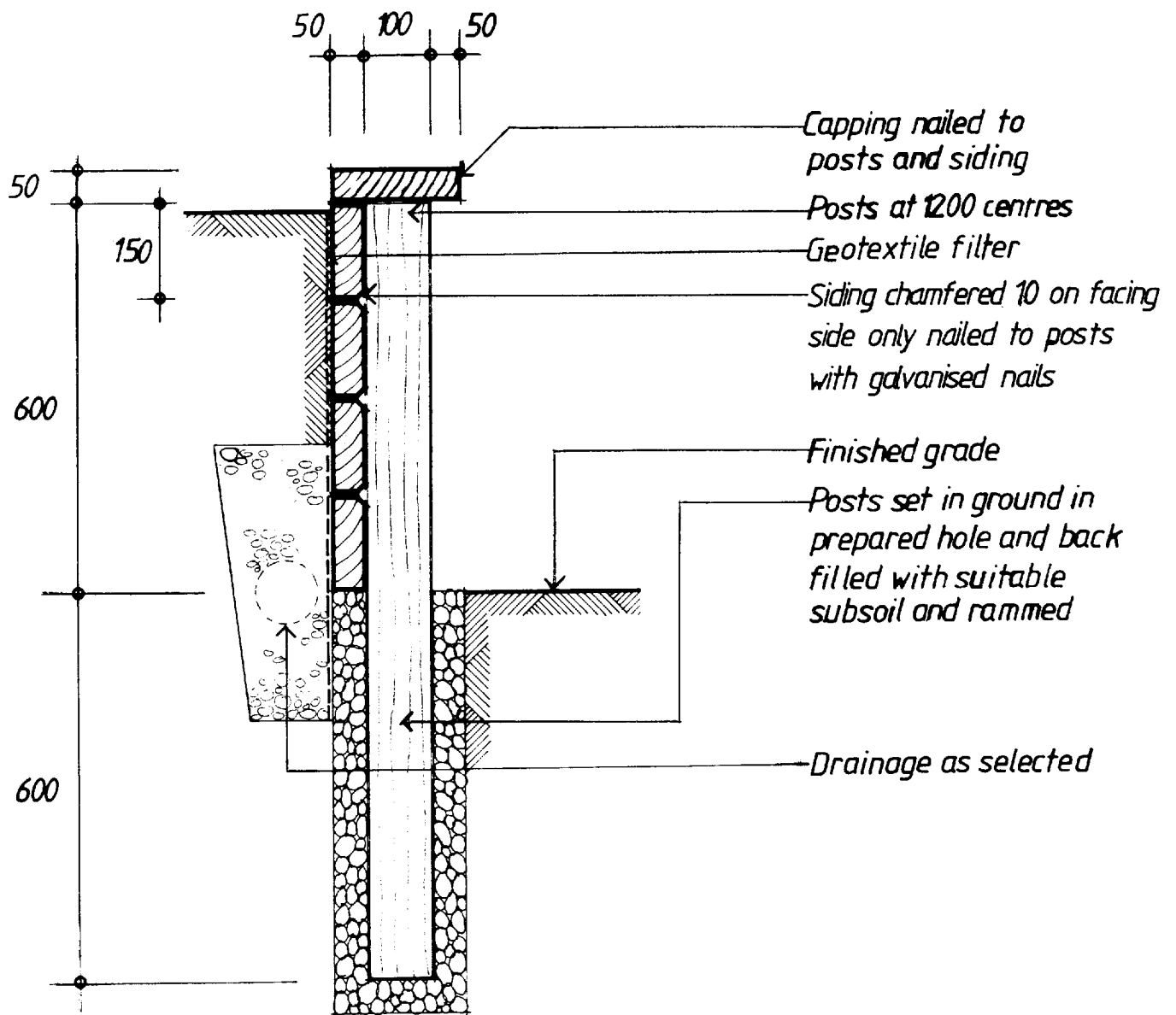
ELEVATION

Scale 1:20



RETAINING WALL
Gabions and plants

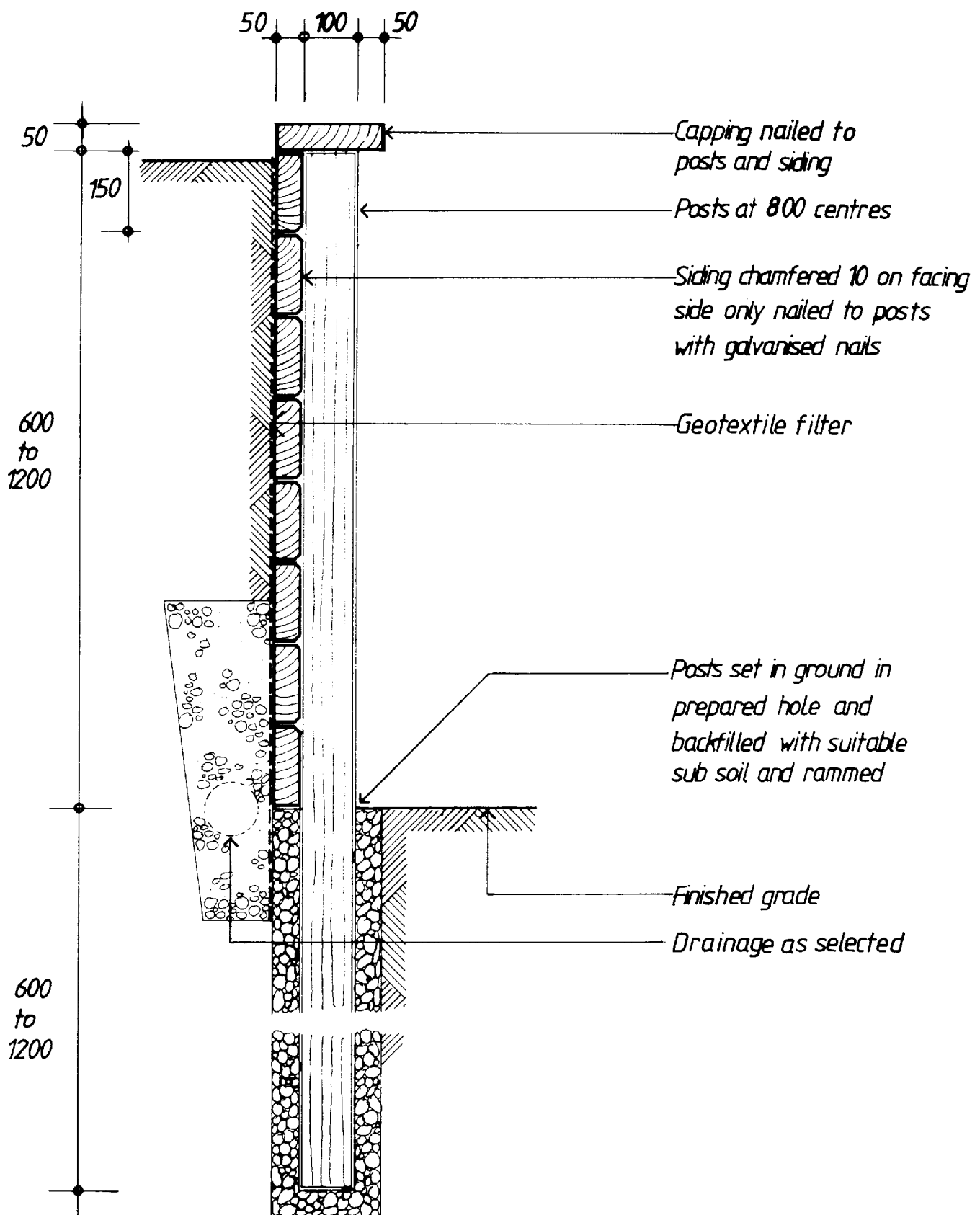




SECTION

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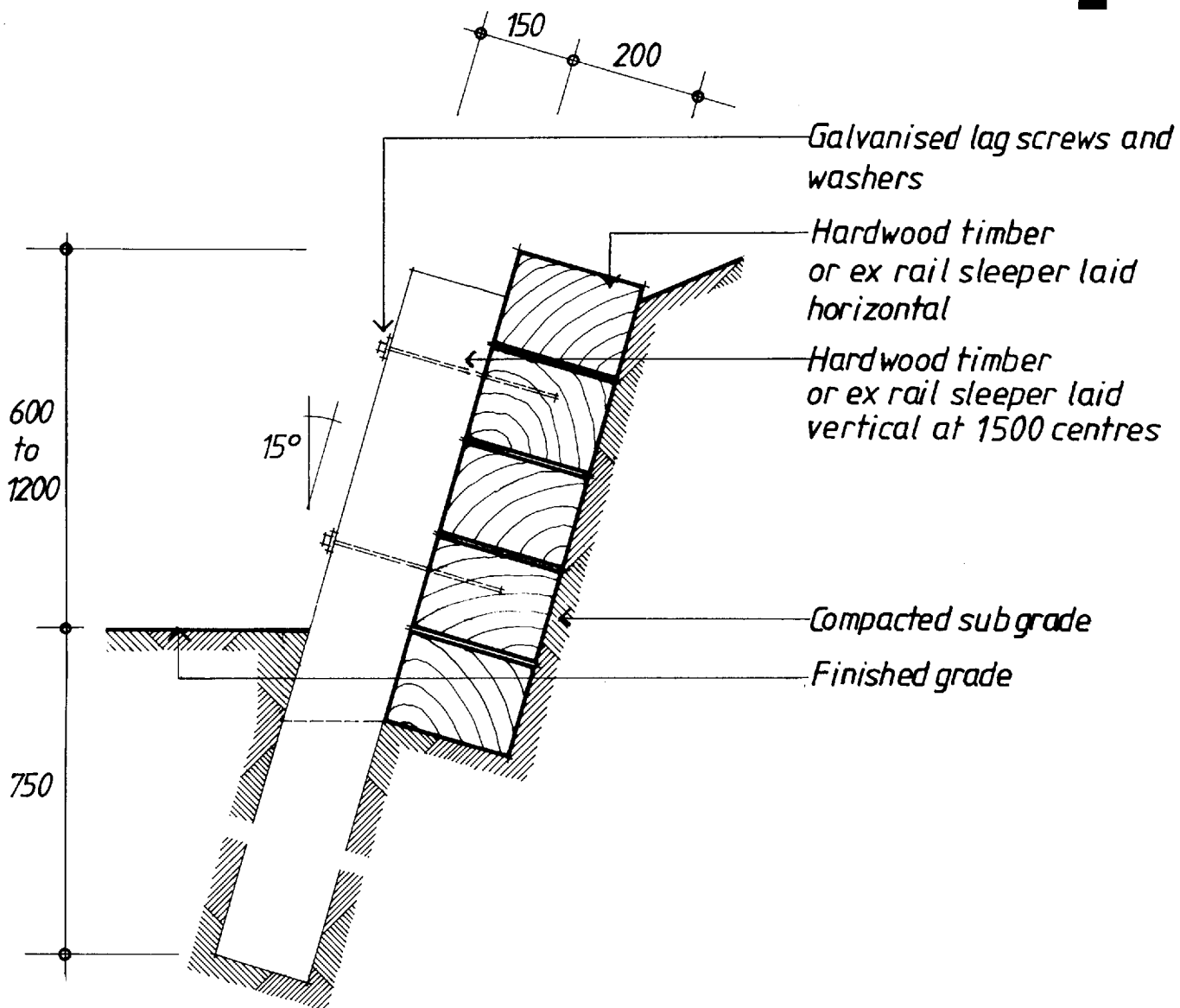
RETAINING WALL
timber (low)



SECTION

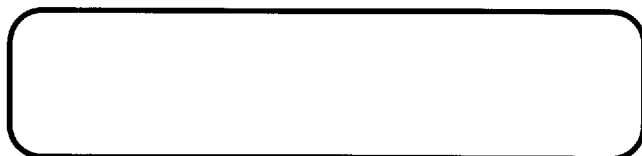
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RETAINING WALL
timber (medium)

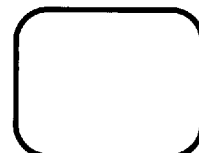


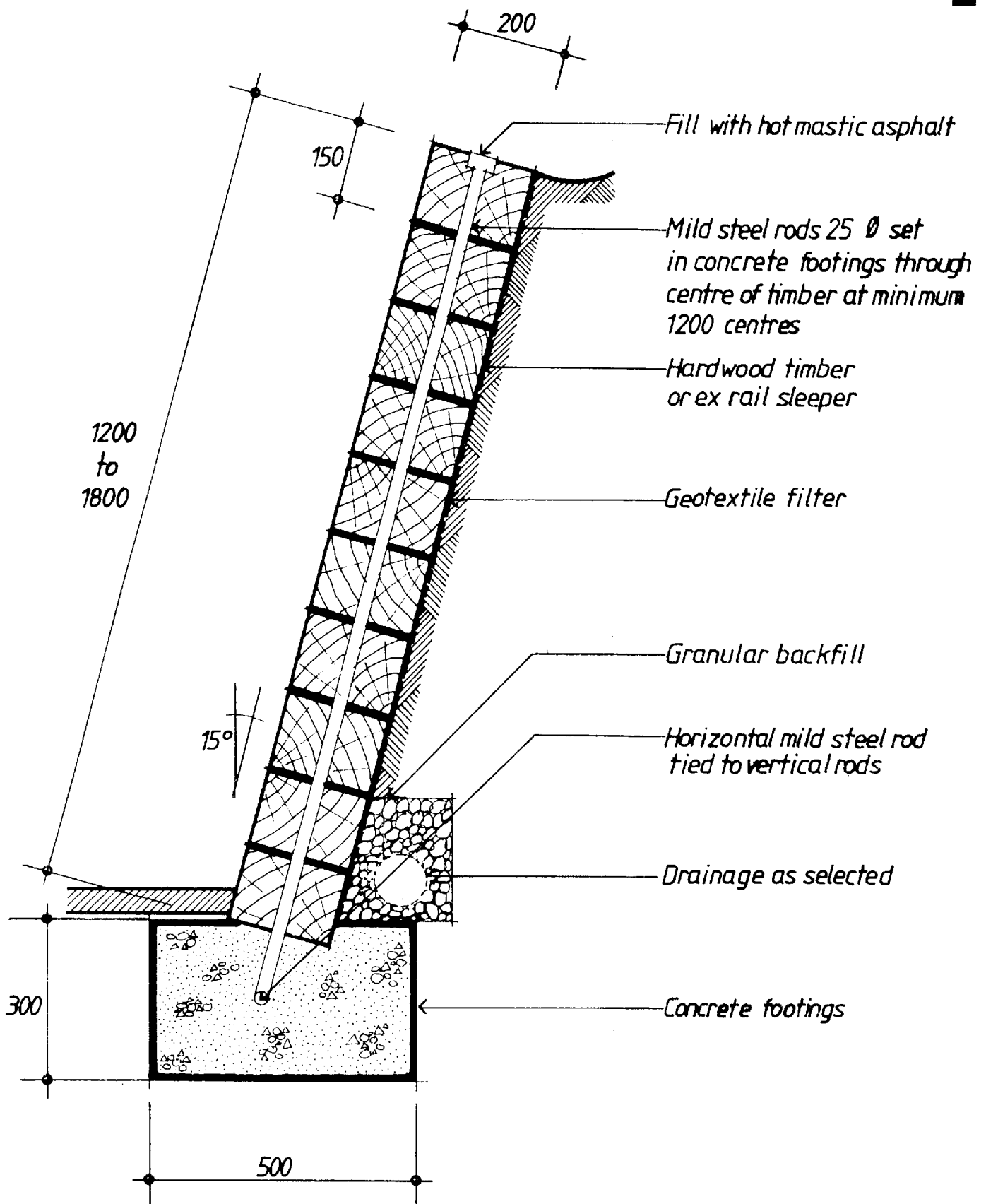
SECTION

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RETAINING WALL
timber angled –
horizontal





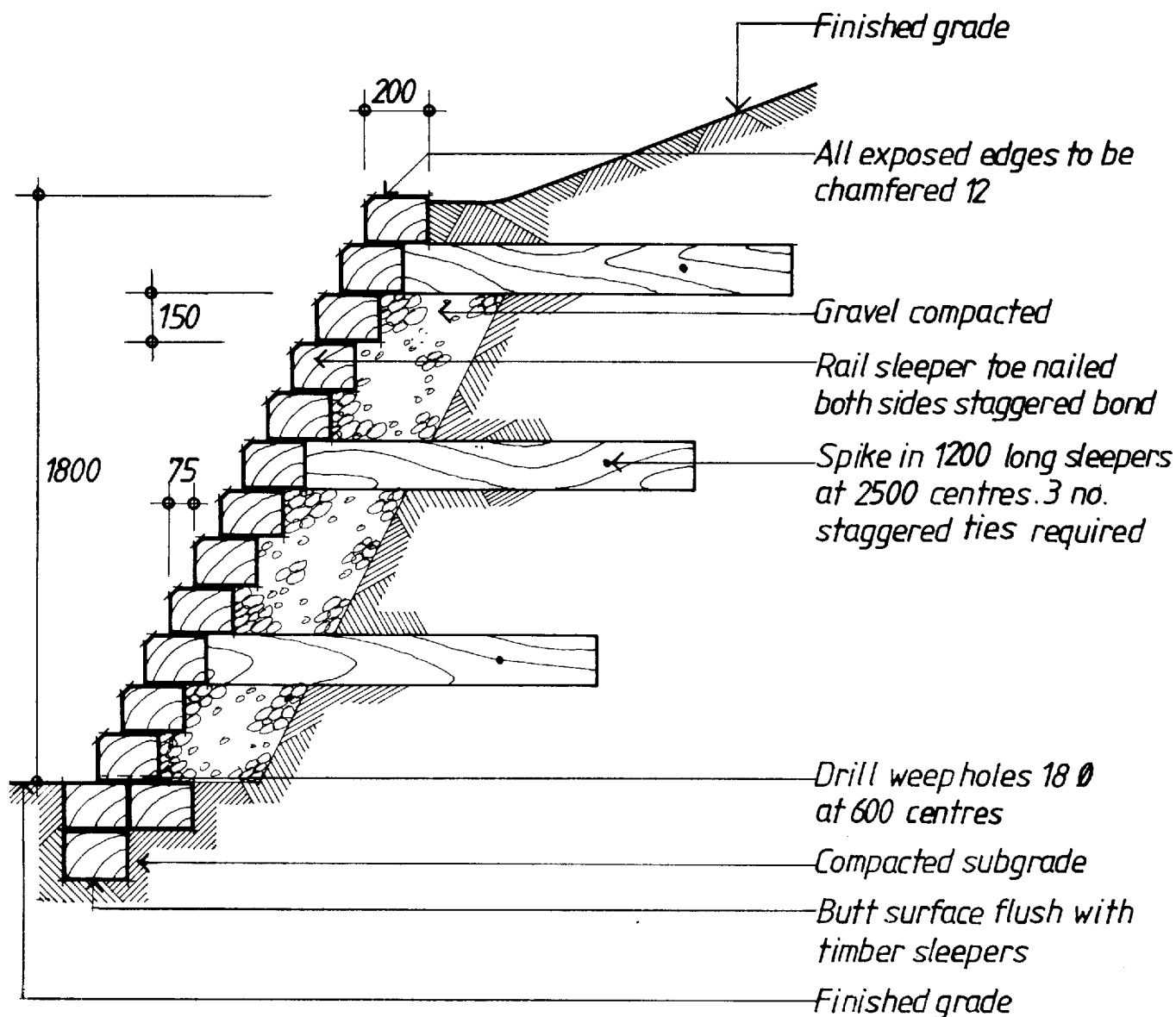
SECTION

Scale 1:10



RETAINING WALL
timber angled





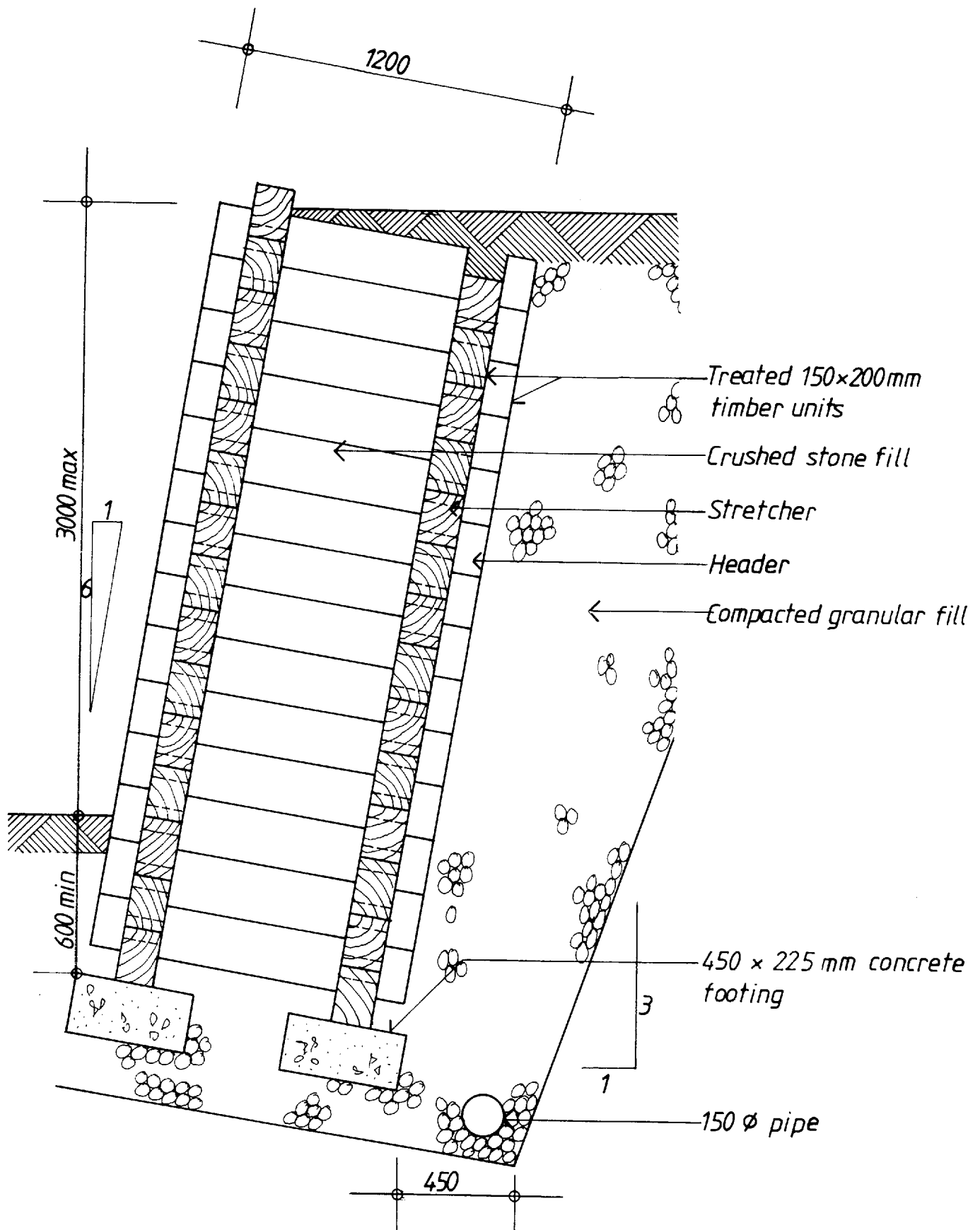
SECTION

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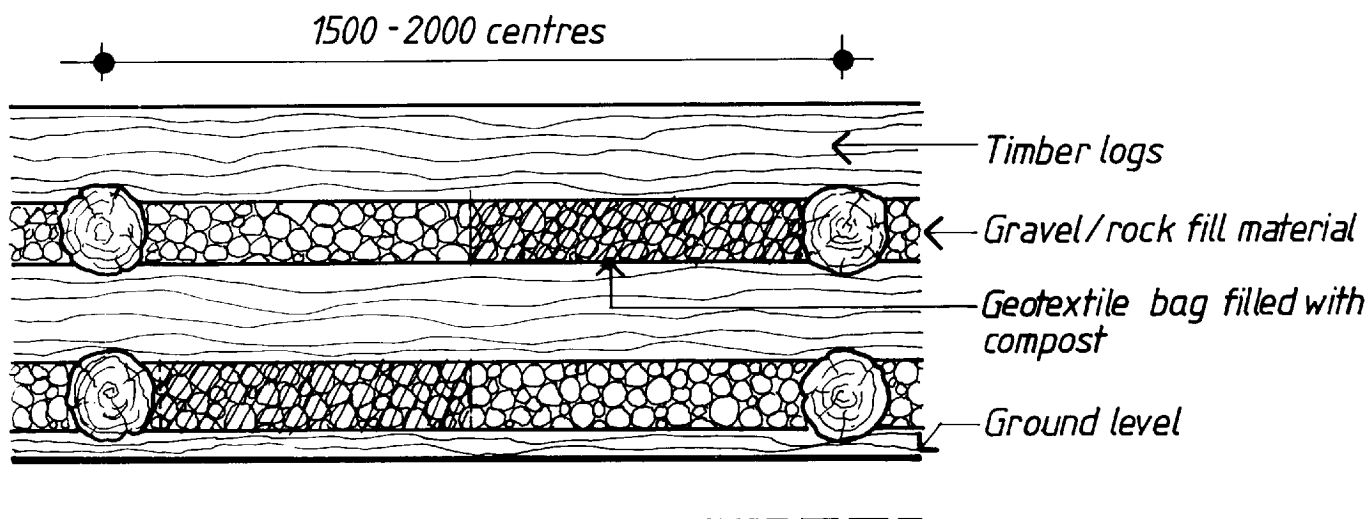
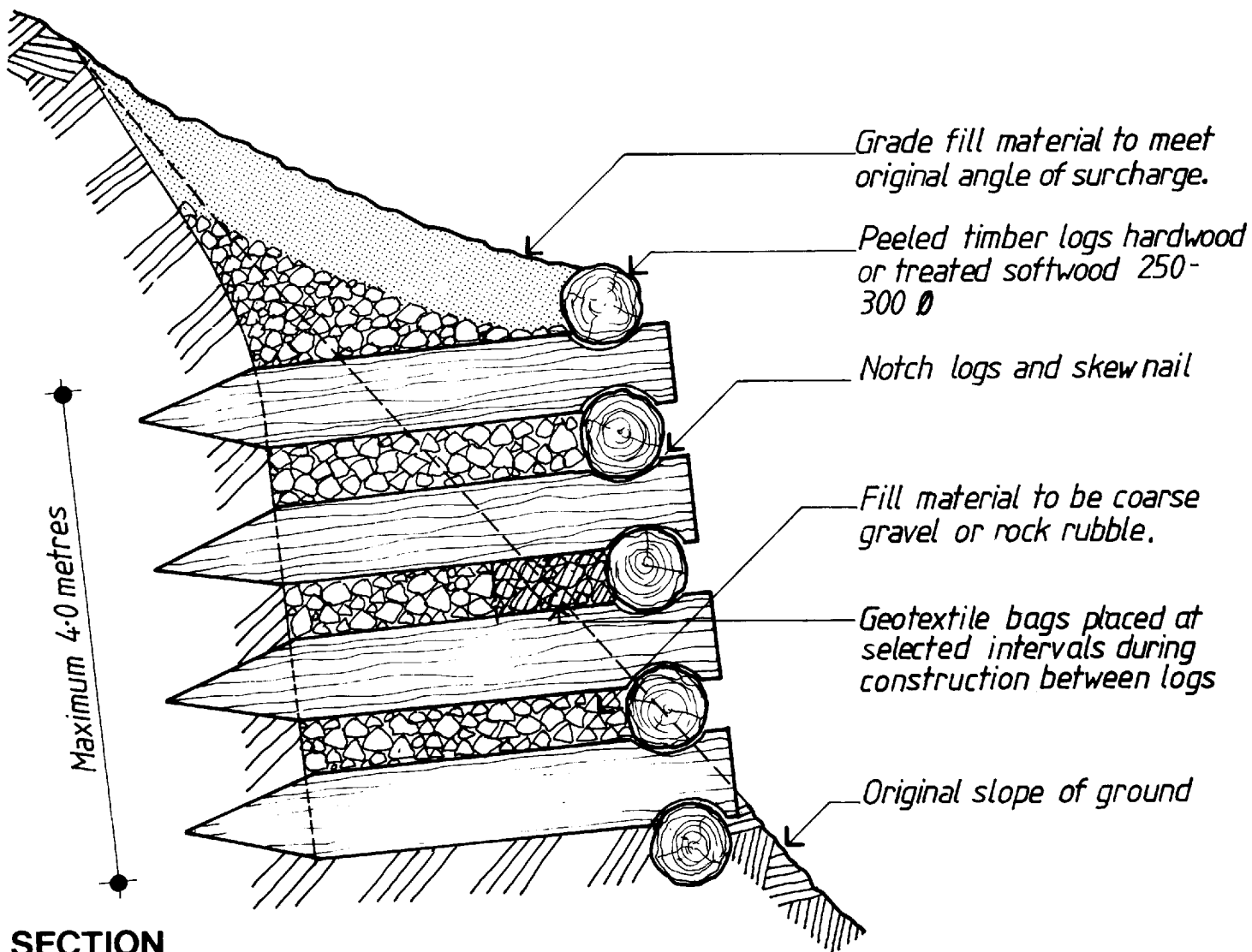
RETAINING WALL
timber stepped





Scale 1:20

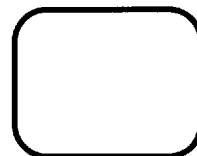
RETAINING WALL
timber crib (square)

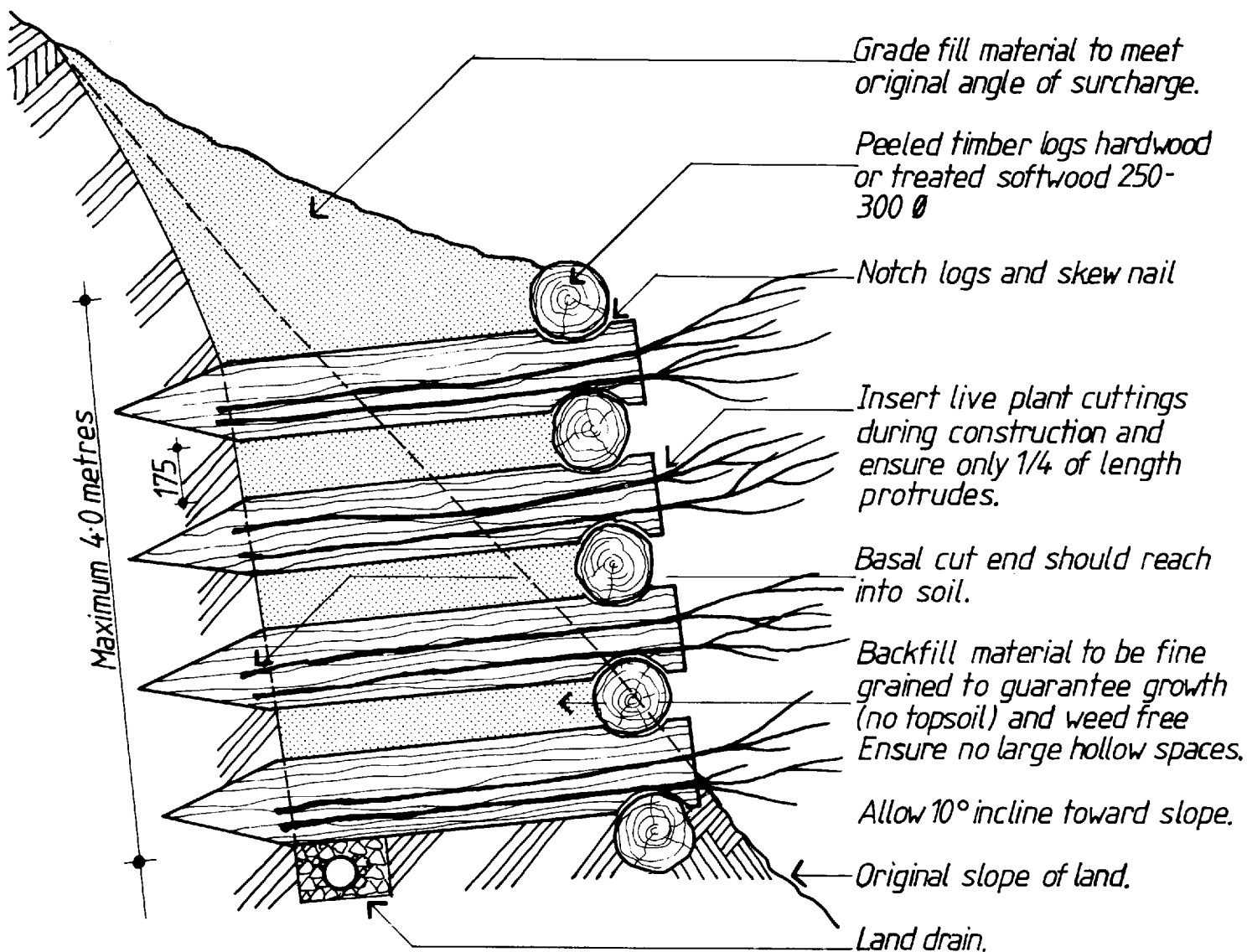


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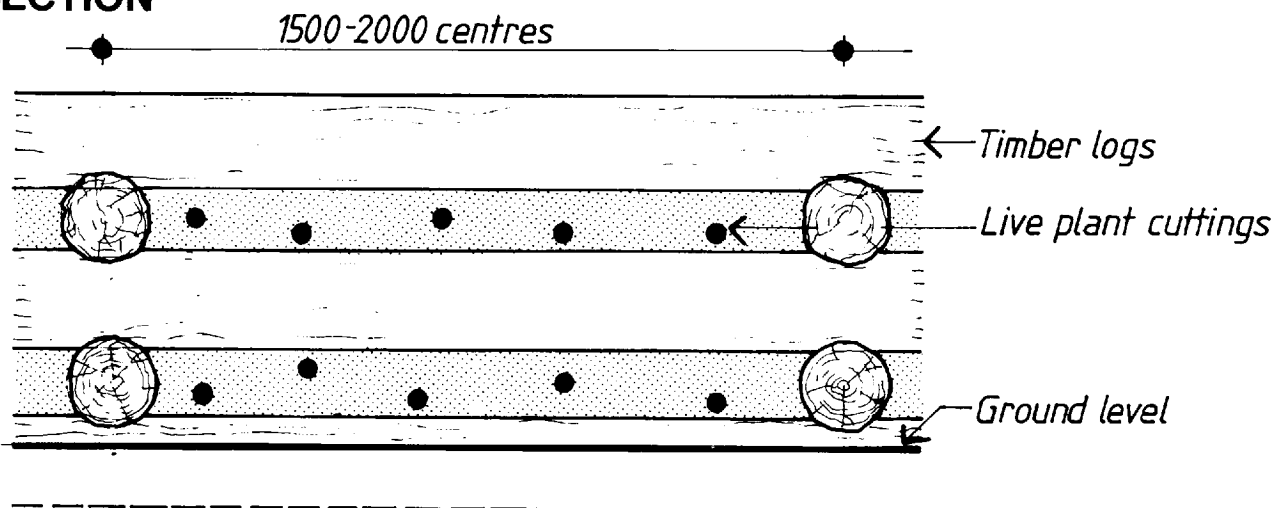


RETAINING WALL
Timber crib (round)



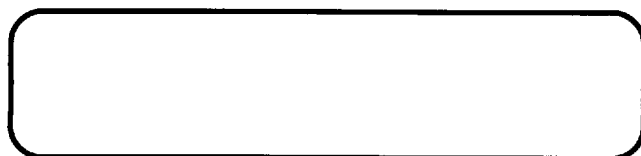


SECTION

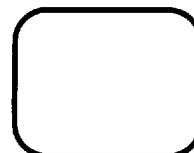


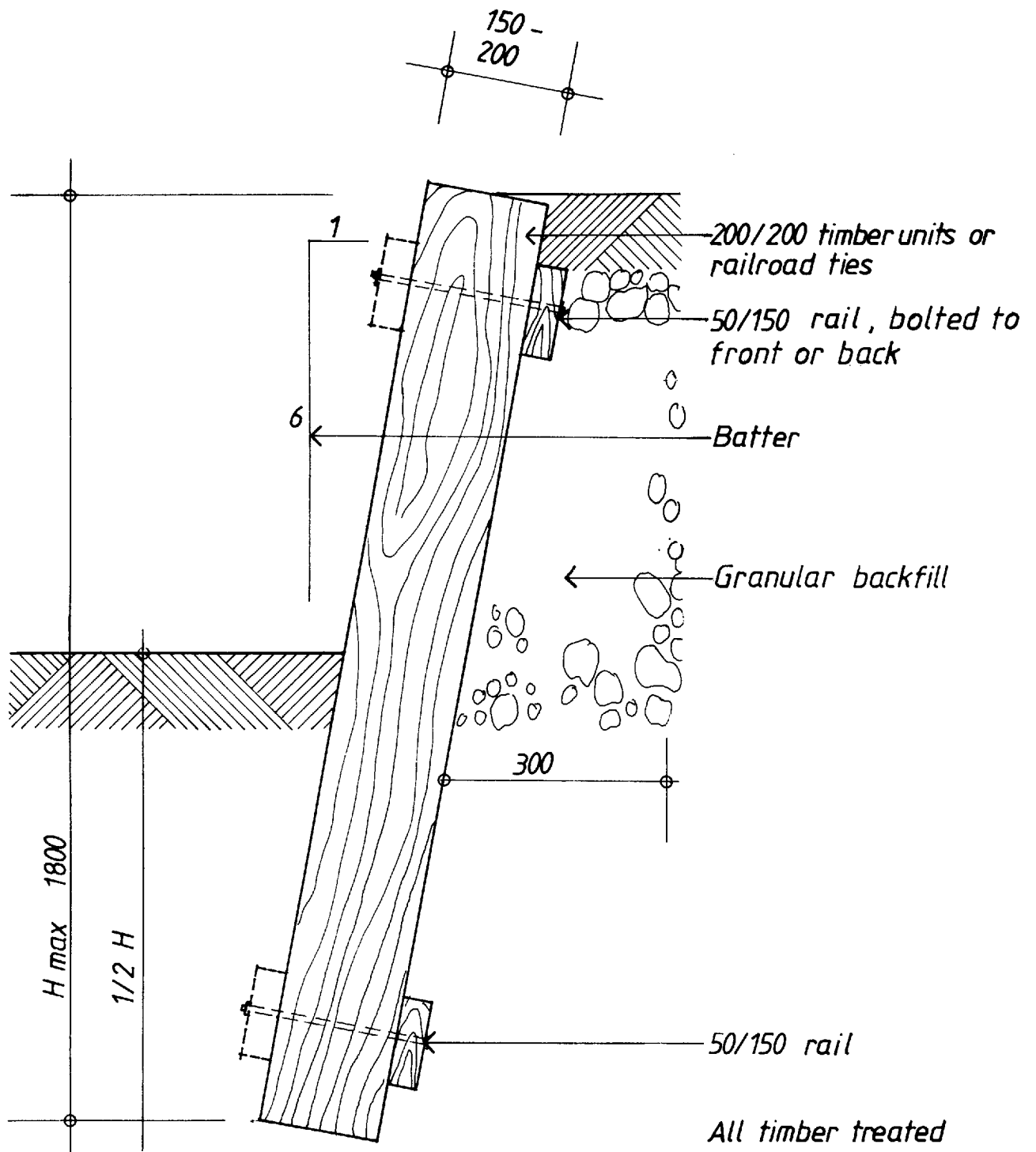
ELEVATION

Scale 1:20



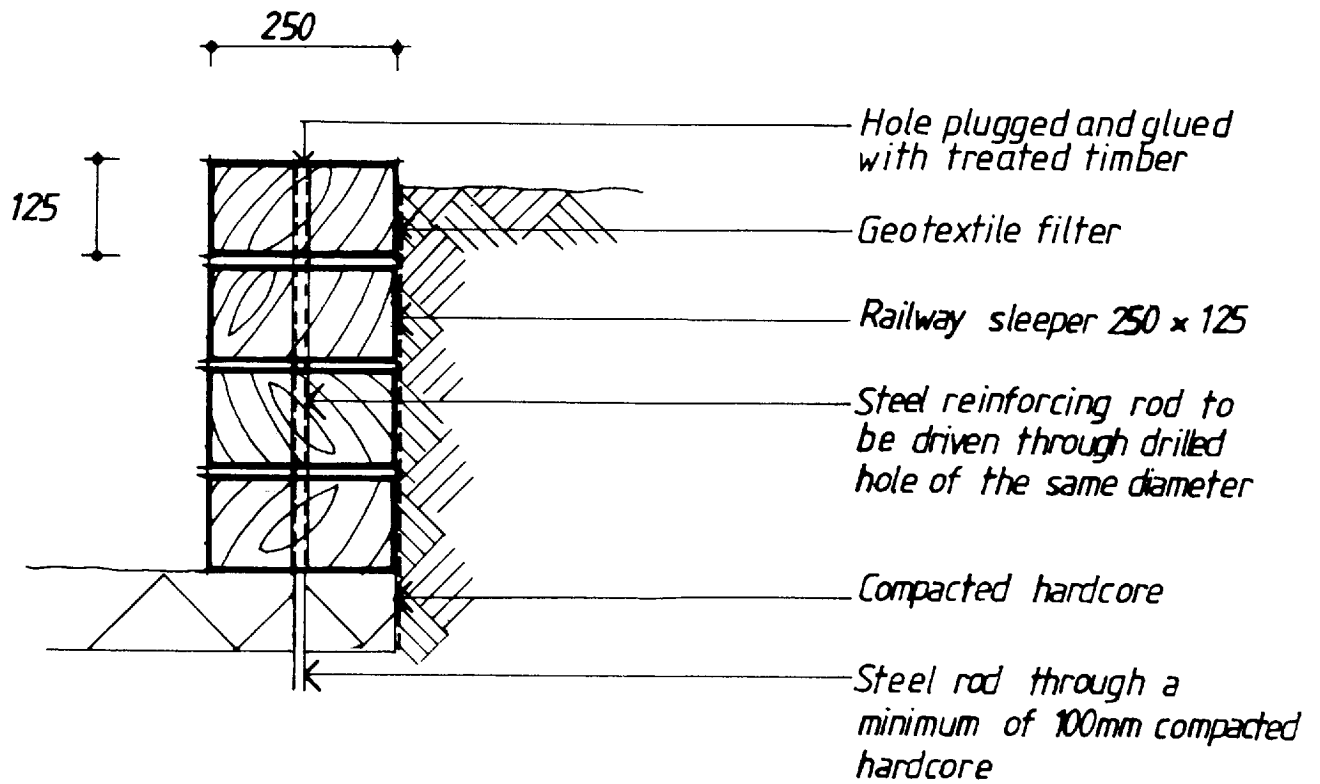
RETAINING WALL
Timber crib (live)



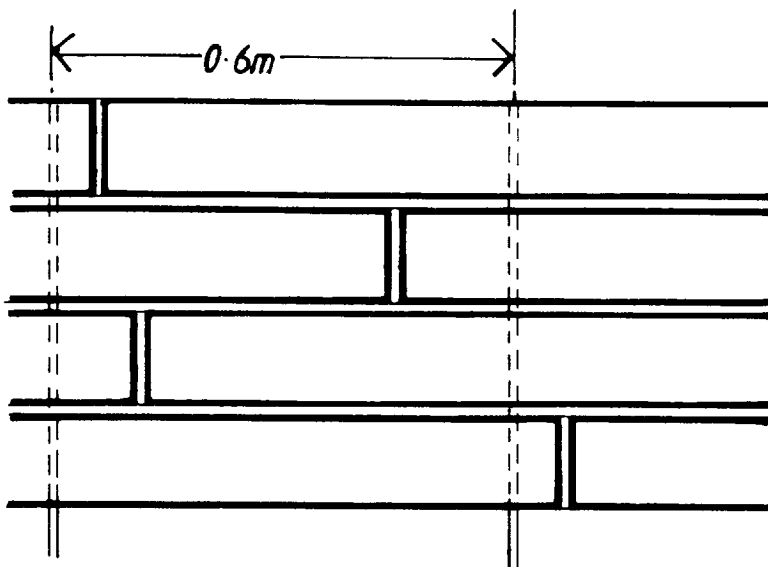


Scale 1:10

RETAINING WALL
Timber angled – vertical



SECTION



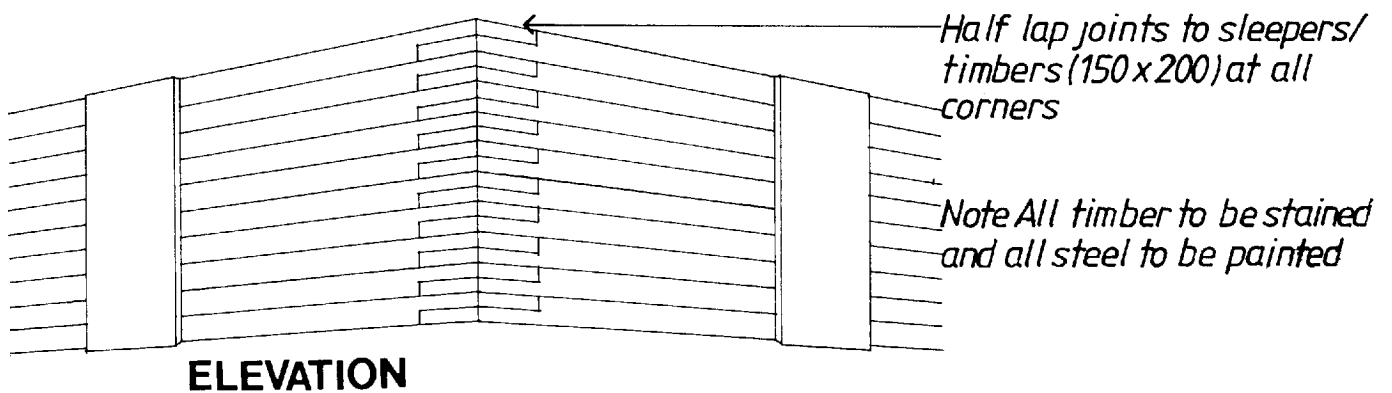
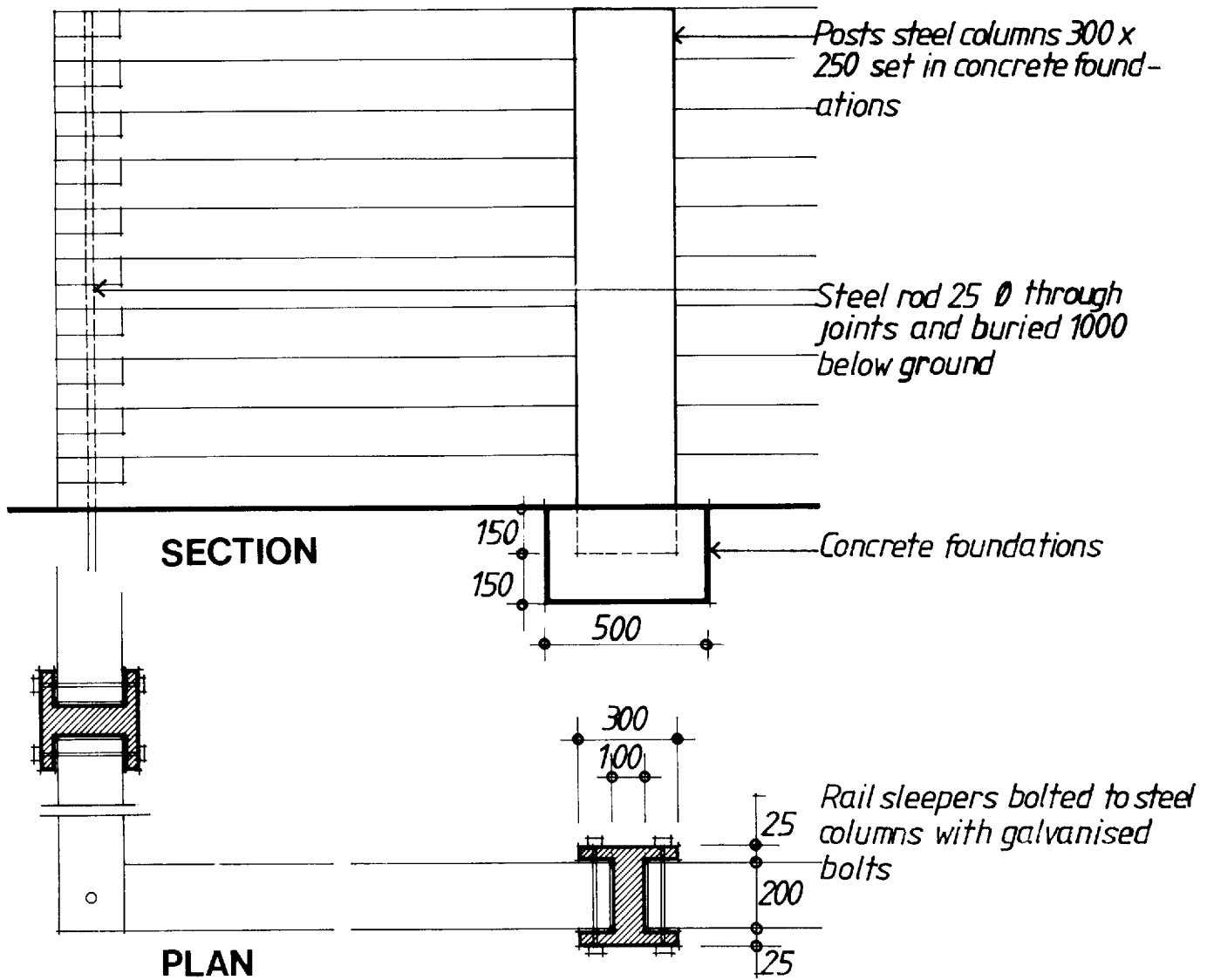
ELEVATION

Scale 1:10



RETAINING WALL
railway sleeper



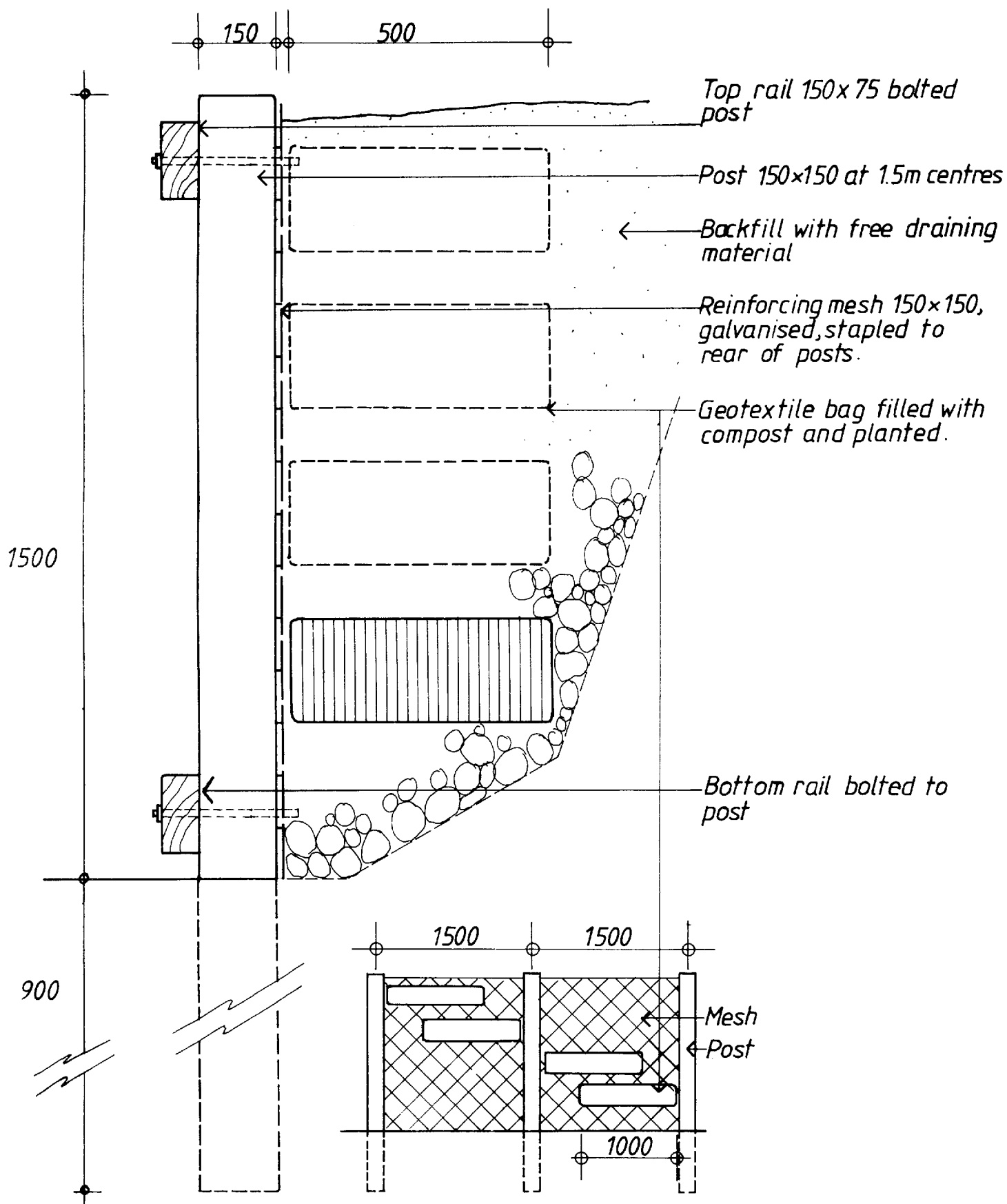


Scale 1:20



RETAINING WALL
timber & steel





ELEVATION 1:50

Scale 1:10

RETAINING WALL
timber & mesh

FENCES

GUIDANCE NOTES

Timber

Standards

BS 1722 covers many kinds of fencing. Each part of the Standard specifies in detail not only the materials required but also the method of erection. Fences are also classified according to variable characteristics, e.g. height, number of wires or rails, post material.

These are listed in table 1 of each part of the Standard, together with examples of typical uses of each type.

- Part 4 Cleft chestnut pale
- Part 5 Close boarded and oak pale
- Part 6 Wooden palisade
- Part 7 Wooden post and rail
- Part 11 Woven wood

There are, of course, many other types of fencing not yet covered by the British Standard, such as wattle, etc.

The type and height of a fence should be determined by its situation and location – urban or rural.

Fence types

Cleft chestnut spile

The timber from chestnut coppice woodland is cleft by hand with a billhook usually on site after the coppice has been cut, and consists of cleft spiles driven not less than 300 mm into the ground usually at six per metre spacing. It is connected to each other by one or two lines of galvanised wire twisted round each spile and then between end and intermediate posts.

Chestnut pale

This is made by a machine to BS 1722: Part 4 and comes in rolls for easy erection and dismantling. The pales are connected together by two or three lines of galvanised wire being spaced from 50 to 100 mm apart according to quality. Round posts are used for support. This type of fencing is recommended only for temporary use, the driven spile being far more satisfactory for permanent fences.

Close boarded and oak pale

Oak is the best timber but is expensive, especially if hand split. Machine sawn is less weatherproof and subject to warping. Softwood is cheaper but requires to be treated. Construction consists of squared posts and triangular arris or rectangular rails which generally have tongued ends set into mortises formed in the longest faces of the post. Pales or feathered edge boards are fixed vertically and lapped 13–19 mm, with galvanised rails/bolts. Additional refinements are capping and gravel boards which provide protection. BS 1722: Part 5 covers this type.

Horizontal palisade

The boards have feathered edge and lapped or plain section and jointed or spaced apart like louvres. They are fixed horizontally directly to the posts whose maximum spacing is 2.0 m, depending upon the width and thickness of the board and its spacing.

Vertical palisade

Boards are either spaced apart or butt jointed and are usually fixed to the rails on one side, although boards on both sides with the equivalent size spaces between them are quite common.

Post and rail

This fence consists of square or rectangular section posts and one or more horizontal rails in which the parts can be mortised together without the aid of nails. Sawn or wrot timber versions which are subject to warping need to be bolted or mortised and pinned together or held by metal straps and brackets. Part 7 of BS 1722 covers this type.

Interwoven

This consists of panels formed from thin slats of wood woven horizontally between framed uprights. The panels are supported between wooden posts. Quality and durability depend upon timber used – oak and cedar lasting well compared to treated softwood, which becomes brittle and may disintegrate much more quickly. Part 11 of the BS covers this type.

Wattle

Hazel handles are made from hazel coppice using single rods often woven continuously *in-situ* and following the configuration of the ground. Wattle handles and panels are made from the osier-willow rods woven in groups of four or five, all in the round and are usually 2 m long by 0.6–1.8 m high in increments of 150 mm. Wattle fencing has a limited life of approximately six years, especially in exposed positions. It should be secured to supporting posts between battens.

Preservation

The most common fencing timbers are oak, sweet chestnut and cedar. These need no preservation when used above ground. Where these timbers are used as posts, the ends should be impregnated with creosote. All softwood used in fencing should be preservative-impregnated by a vacuum process.

Post spacing

Posts should be spaced at 2,700 mm maximum centres for mortised fencing and 1,800 mm maximum centres for nailed fencing.

Depth of insertion

Under normal exposure conditions a post insertion depth of 500 mm is sufficient for a 1.0 m high fence, and 750 mm for a 1.8 m high fence. If a concrete surround is used, these dimensions can be reduced to 400 mm and 650 mm, respectively, because of the greater face area given by the concrete in certain locations.

Fixings

All fixings for fencing should be galvanised. In public areas, drive screws or wood screws should be used. Where bolts are used the nuts should be recessed into the post or the projecting threads should be burred over.

Finish

Suitable timbers, such as oak, cedar, sweet chestnut or larch, can be left unfinished. Otherwise use a two-coat application of wood dye or a coloured preservative (e.g. Cuprinol). Some brush-applied preservatives are not compatible with vacuum preservation processes which include water repellents. Opaque oil paint systems can be satisfactorily used on vacuum-impregnated timber provided that the preservative does not contain a water repellent. Where a water repellent has been used, loss of adhesion of the paint film may result unless a special primer is used. The paint manufacturer should be consulted.

Changes in level

These are accommodated by stepping each panel when using fences made from prefabricated panels. Posts at the steps will have to be increased in length. Where post and rail fencing is used, changes in level are accommodated by inclining the rails in the mortises, while keeping the posts vertical.

Metal

Metal fencing can be conveniently divided into two groups. The first comprises those types made up from stiff metal sections of flats and round or square bars, forming rigid barriers. The second group comprises wire mesh of flexible material supported by posts and sometimes framed in panels; woven, wire, chain link and welded mesh. BS 1722 lists the following:

- Part 1 Chain Link
- Part 2 Woven Wire
- Part 3 Strained Wire
- Part 8 Mild Steel Bar
- Part 9 Mild Steel Vertical Bar
- Part 10 Anti-Intruder
- Part 12 Steel Palisade
- Part 13 Tennis Court

The first group includes the painted iron railing that is found around many eighteenth-century parks, estates and large

gardens and in urban areas provides a substantial see-through barrier. Metal post and rail or continuous bar fencing is used for boundary enclosure, mainly in rural areas. It is much lighter than timber and less obtrusive. Vertical bar railings are primarily urban in character because of their strong verticality.

The second group comprises chain link, wire mesh and line wire, all of which are strained between concrete, metal or timber posts, with straining posts at changes of direction and at the ends. This type of fence is used more for security and is more functional than aesthetic, although if used in frames it could be made to look considerably better. Wire mesh can also be used for supporting climbing plants. Strained wire fences are most commonly used in rural areas as a barrier against farm animals and in urban areas as a temporary fence between private gardens. The main objective is to ensure that the posts (structural members) are reduced to the minimum while keeping the wires taut. The lighter they are, the cheaper the cost and the better the appearance.

Netting fencing should be used over large boundary runs where timber is too expensive, or where the site is to be defined but any views retained within it. Posts can be timber but concrete is better, since they must support stressing wires along the top and bottom of the mesh. Chain link fencing, which is wire or wire covered with coloured plastic, is the strongest mesh. Plastic netting is less tensile than chain link but is ideal for more general garden use, for infilling decorative panels and to support climbing plants. Decorative fences can be made in a combination of materials, such as translucent fibreglass sheets and timber, or bamboo, etc.

Sizes of members

The materials used almost exclusively for metal fencing are wrought iron and mild steel. BS 1622: Parts 8 and 9 can be used as a guide for sizes and spacing of members. Space between vertical members should not be less than 100 mm or more than 120 mm.

Intermediate posts

An insertion depth of 600 mm is suitable for a fence up to 1.8 m high. Spacing should be 2.7 m maximum. A wing plate parallel to the fence line should be welded to each standard. Standards should have a bearing plate welded to the base.

Standards for fences incorporating strained wire should have stays fixed at 60° to the horizontal at corners and ends. Because of the small face dimension of metal members, a concrete surround to the standard should always be used.

Preservation and finish

BS 1722: Part 8 specifies a range of ex-works finishes, the usual finish being a red-oxide primer or hot-dip galvanising after fabrication. The finish coating should be paint, applied as two undercoats and one gloss top coat.

Changes in level

Prefabricated metal fencing is accommodated to changes in level by stepping the panels or fabricating special inclined panels. With either method, the profile of the ground must be known in advance of fabrication.

Rigid fencing types

Continuous bar fencing

Rails can be flat section or round bars and the overall height 1,070, 1,220 or 1,370 mm. Standards can be flat, L or T section. This type of fence is also obtainable in hurdle form.

Vertical bar unclimbable fencing

This fence is also simple, consisting of either round or square vertical bars and flat top and bottom rails. Both square and round versions can be obtained in either rigid or self-adjusting forms for sloping ground lines. The tops of the fence type vary in design and can be pointed, blunt, bowed, interlaced and spear-head. Portable hurdles are obtainable in 2 m lengths.

Vertical bar wall fencing

A swarf version of vertical bar fencing is made for fixing on top of low walls, in heights from the top of the wall of 600, 750 and 900 mm.

Steel angle unclimbable fencing

Substitution of all the components in the vertical bar fence with steel angles for standards, pales and rails results in a similar type of fence with additional rigidity, as the angle pales are more resistant to bending than vertical bars. Heights range from 1,220 mm to 2,440 mm.

Corrugated steel pale fencing

This consists of corrugated steel pales fixed to angle section rails. The tops of the pales are single pointed or splayed to give triple points. The corrugated pales have much greater strength than other types and it is made in heights from 1,525 mm to 3,050 mm.

Post and rail fencing

A number of variations occur in this fence type, which is in the form of horizontal rails of stout tubular section supported by steel standards or cast iron pillars. This type of fence is useful where a strong, robust quality is required.

Post and chain barrier

This is useful for marking the boundary and forming a mild deterrent to possible trespassers. Chains can be plain link or ornamented with spikes and links. Chains are attached to each standard by eyes or hooks or they can pass through intermediate standards and be fixed intermittently. Thicknesses of chain are usually 9.5 mm diameter, but chains are also available in 6.4 mm and 7.9 mm diameter metal. As alternatives to metal, posts can be in concrete or timber.

Flexible fencing types

Although these notes are primarily concerned with metal fencing, the materials described below can be used with posts of timber or concrete in place of steel.

Chain link fence

Anti-intruder chain link fences (with cranked tips) are also available. The BS specifies various gauges of steel wire that may be used for stated purposes and allows for the alternative use of mild steel, aluminium or plastic-coated steel.

Woven wire rectangular and square mesh fence

This fence type is widely used in agriculture and is often called pig wire. It is a more open type of fence than chain link and hexagonal mesh.

Welded mesh fence

A mesh formed of wires welded together to form a rectangular mesh and of stout gauge with or without PVC bonding. It is also used in framed panels, the framing being in timber or steel sections. In the heavier gauges it can form a very strong unclimbable fence.

Post and wire

The use of spring steel or high tensile steel requires supporting timber posts at only up to 18 m centres with strainers up to 1 m apart with or without droppers at 1.8 m centres, depending upon the purpose of the fence.

Expanded metal fence

Expanded metal is made by slitting sheet metal and stretching strands so formed at right angles to the plane of the sheeting, resulting in a diamond mesh. A flattened form is available. There are a large number of mesh sizes available in aluminium, brass, copper and stainless steel as well as ordinary steel. Expanded metal is usually framed in metal or timber when used for fencing.

Perforated metal sheeting

A large range of patterns is available in this material, which is highly decorative for garden screens. The patterns depend on the shape of the perforations or cut-out portions of the sheeting.

Hexagonal mesh wire fence

Hexagonal mesh is of small-gauge wire, galvanised during manufacture. It cannot

compare with chain link in permanency and strength. Hexagonal mesh is commonly used for rabbit-proof enclosures around young plantations. Standards can be of angle steel section, peeled larch or chestnut poles. It can be strengthened by the addition of a few lines of galvanised wire.

Finish and painting of metal fencing

A steel fencing requires protection from corrosion, and unless specified to be galvanised the fencing will arrive on-site with a temporary coating, usually of one coat black varnish, although some makers supply fencing coated with boiled linseed oil. The fencing must be treated more adequately without delay, and the general treatment consists of one coat corrosion-inhibiting priming coat (no red oxide), one or two undercoats and a finishing coat of oil paint. If a galvanised finish is required it should be specified to be in accordance with the requirements of BS 443: 1969, for woven wire, or BS 729: 1971, as applicable.

Plastic-coated chain link fencing has become available in a number of colours, the best for weathering being black.

Concrete fencing

Members are always reinforced with mild steel bars. There is no British Standard covering concrete fences as a whole, although concrete posts in composite fencing are dealt with in appropriate parts of BS 1722.

The two most popular concrete fences are:

- *Concrete post and panel fence:* This consists of reinforced concrete posts grooved to receive concrete slabs 38 mm or 50 mm thick and 150, 225 or 300 mm wide, which are placed one on top of the other between posts and in the grooves, to make up a panel.
- *Concrete palisade fence:* This fence is similar to its traditional timber counterpart, consisting of posts and rails and palisades. Fixings are all concealed and the finish is in plain concrete. No gates are available.

SPECIFICATION CHECK LIST

General

Refer to the relevant British Standard for the selected fence, either timber or metal. Information should be given in all the clauses to comply with SMM 7 and to provide checks for any site supervision.

Timber types

For timber fences BS 1722 covers the following parts:

- Part 4 Cleft chestnut pale
- Part 5 Close boarded and oak pale
- Part 6 Wooden palisade
- Part 7 Wooden post and rail
- Part 11 Woven wood

Cleft chestnut pale

Type: Use table 1 from the British Standard and select reference (e.g. CW 135).

Posts and struts: From tables 2 and 3 state either concrete or timber posts, struts and size.

Centres of posts: BS 1722: Part 4 specifies the following maximum centres for intermediate posts:

Height of fence (mm)	Concrete posts (m)	Timber posts (m)
900, 1,050	3.00	2.50
1,200, 1,350	2.75	2.25
1,500, 1,800	2.25	2.00

Straining posts: 70 m in straight runs and at all ends, corners, changes of direction and acute variations in level.

Setting posts: See table 1 of the British Standard. Insert the method and depth.

Other requirements: If barbed wire is required, the details can be given here (e.g. two lines of barbed wire). BS 1722: Part 4 specifies that the posts should be increased by 150 mm in height for each line of barbed wire.

Close boarded

Type: Use table 1 from the British Standard and select reference (e.g. BCM 120). The type reference indicates whether oak or other timber is to be used, material of posts, height and whether posts are mortised or recessed to receive rails.

Height: State size. Part 5 specifies heights of 1,050, 1,200, 1,500, 1,650 and 1,800 mm.

Boards/rails: See Guidance Notes. BS 1722: Part 5 specifies feather-edged boards and arris (triangular section) or rectangular rails. Insert e.g. softwood feather-edged boards on 75 × 50 mm arris rails.

Posts: Specify concrete or timber posts' length and size. Insert type of treatment and/or desired service life.

Centres of posts: State distances between posts.

Setting posts: Insert the method and the depth (e.g. set in rammed earth to a minimum depth of 600 mm). BS 1722: Part 5 specifies concrete posts to be set in concrete and timber posts to be set in rammed earth or concrete.

Other requirements: State whether a gravel board (concrete or timber) and/or a capping is required (for use with mortised posts only). State whether barbed wire is required, the number of lines and the finish for the mild steel extension arms.

Wooden palisade

Type: Use table 1 from the British Standard and select reference (e.g. WPC 120). The type reference indicates whether oak or other timber is to be used, material of posts, whether posts are mortised or recessed to receive rails and height.

Height: State size. Part 6 specifies heights of 1,050, 1,200, 1,500, 1,650 and 1,800 mm.

Palisades and rails: Give details of the timber required and the design of palisade and rails, e.g. 75 × 20 mm softwood with pointed tops and arris rails. BS 1722: Part 6 specifies palisades of 75 × 20 mm, 65 × 20 mm and triangular section cut from 50 × 50 mm. Arris (triangular section) or rectangular rails can be specified.

Posts: Specify concrete or timber posts, length and size.

Preservation treatment: Insert type of treatment and/or desired service life.

Centres of posts: State distances between posts.

Setting posts: Insert method and depth. The British Standard specifies concrete posts to be set in concrete and timber posts to be set in rammed earth or concrete.

Other requirements: State whether barbed wire is required, the number of lines and the finish of the mild steel extension arms. A single line of barbed wire can be fixed by stapling to the top of timber posts. BS 1722: Part 6 specifies barbed wire for fixing to the top of 1800 mm high fences only.

Wooden post and rail

Type: Use table 1 from the British Standard and select reference (e.g. MPR 11/3). There are two basic constructions – mortised and nailed.

Height: State size. Part 7 specifies only two heights of fence – 1,100 and 1,300 mm.

Timber: Specify type of timber.

Preservation treatment: Insert type of treatment and/or desired service life (e.g. CCA category B).

Centres of posts: BS 1722: Part 7 specifies: 2.85 m for mortised fences, 1.80 m for nailed fences.

Setting posts: BS 1722: Part 7 specifies setting in concrete or rammed earth, or posts can be driven. Mortised fences have an intermediate ('prick') post which is set by driving to a depth of 450 mm.

Wooden panel

Type: Use table 1 from the British Standard and select reference (e.g. WW 60).

Infill: Options include:

- Woven wood
- Waney edged overlapping horizontal boards
- Square sawn overlapping horizontal or vertical boards
- Feather-edged overlapping horizontal or vertical boards.

BS 1722: Part 11 specifies preservative treatment for panels other than those made from Western red cedar. State which is required.

Height: State size. Part 11 specifies heights of 600, 900, 1,200, 1,500 and 1,800 mm.

Posts: Insert details (either timber or concrete).

Preservation treatment: Part 11 specifies preservative treatment for all timber other than Western red cedar.

Setting posts: Insert method and the depth.

Metal types

For metal fences BS 1722 covers the following parts:

- Part 1 Chain Link
- Part 2 Woven Wire
- Part 3 Strained Wire
- Part 8 Mild Steel Bar
- Part 9 Mild Steel Vertical Bar
- Part 10 Anti-Intruder
- Part 12 Steel Palisade
- Part 13 Tennis Court

Chain link

BS 1722: Part 1 specifies chain link fencing for domestic and commercial use with heights ranging from 900 to 1,800 mm (2,130 mm with barbed wire extension arms).

Type: Select reference from BS 1722: Part 1, tables 1 or 2 (e.g. GLS 140C).

Height: Specify measurements. Heights given in BS 1722: Part 1 are based on commonly available widths of chain link mesh.

Mesh and wire: Insert the type of wire and finish required. Mesh size and wire gauge are specified in BS 1722: Part 1 for each height and application of the fencing and need not be given unless a different size and gauge are required.

Posts and struts: Insert details (e.g. concrete, wood or metal). Proprietary non-standard types of post (e.g. lattice pylons) can be specified as an alternative.

Treatment: Insert (e.g. galvanised). For preservative treated timber insert type of treatment required and/or desired service life (e.g. CCA category B).

Centres of posts: State distances between posts. These requirements are specified in BS 1722: Part 1.

Setting posts: See table and insert details.

Woven wire

Woven wire fencing to BS 1722: Part 2 is mainly used for agricultural and forestry use. It is in two parts:

- General pattern.
- Specialised types (with high tensile spring steel wire support) for rabbit and deer control in forestry protection.

Type: Select reference from BS 1722: Part 2, table 1 (e.g. C6/90/30W).

Height: State size and specify measurements.

Mesh: Insert requirements selected from BS 1722: Part 2, table 6 (e.g. hexagonal mesh above C6/90/30W woven wire).

Posts and struts: See table in BS 1722: Part 2 which specifies:

- Concrete.
- Steel angle, tube and rectangular hollow section.
- Square sawn or round timber.
- Round or sawn timber for posts of high tensile pattern fences.

Treatment: Insert details (e.g. galvanised for metal, creosote for timber).

Centres of posts: Specify dimensions.

Intermediate posts should be at not greater than 25 m centres for rabbit fencing and 14 m centres for stock, sheep and deer fencing.

Setting posts: See tables in BS and specify dimensions.

Strained wire

BS 1722: Part 2 specifies two basic forms of strained wire fencing – general pattern and dropper pattern. The use of droppers permits greater spacing of intermediate posts while still maintaining the correct spacing of the wires.

Type: Select from BS 1722: Part 3, table 1 (e.g. SW10.B).

Height: Specify size.

Wire: BS 1722: Part 3 specifies:

- Mild steel wire (4, 4.5 and 5 mm diameter).

- High tensile steel wire (3.15 mm diameter) and steel barbed wire.
- Plastics-coated high tensile steel wire.
- Spring steel wire (dropper pattern fences only).
- Barbed wire.

All wire is required to be zinc coated.

The choice of method of fixing the wires to the posts can usually be left to the fencing contractor. However, if a particular method is required (see BS 1722: Part 3, clause 3.3) it should be specified.

Droppers: BS 1722: Part 3 specifies cleft chestnut pale, timber battens and galvanised or zinc-coated steel. If droppers are not required, delete the item.

Posts and struts: See tables in BS 1722: Part 3 which specify:

- Concrete.
- Steel angle, tube and rectangular hollow section.
- Square sawn, round and cleft timber.

Treatment: Insert (e.g. galvanised). For preservative treated timber insert type of treatment required and/or desired service life (e.g. CCA category B).

Centres of posts: Specify dimensions. General pattern intermediate posts are spaced at 3.5 m centres, dropper pattern intermediate posts at 12 m centres.

Setting posts: See table and insert details. If posts for dropper pattern fences are set in concrete, the hole should be filled to the top.

Mild steel continuous bar

Select reference from BS 1722: Part 8, table 1 (e.g. MSCB/105/4).

Height: State size. Part 8 specifies three heights – 1,050, 1,200 and 1,350 mm.

Bars: Insert 'round top bar' and 'flat lower bars' or 'all to be round'.

Posts: Insert 'flats' or 'tee section'. Hollow section pillars would be provided at all ends and may be specified for corners. In soft or loose ground pillars should have thrust plates. Insert details.

Treatment: Insert details (e.g. galvanised).

Centres of posts: Specify dimensions from table 1 in British Standard.

Method of setting: BS 1722: Part 8 specifies the following minimum depths (mm):

Height of fence (mm)	Flat posts with pronged feet	Tee posts	Pillars
1,050	300	400	450
1,200	350	500	550
1,350	350	600	600

Mild steel vertical bar

Also known as 'unclimbable fencing': insert 'welded' or 'self-adjusting' for the type.

Height: Specify size. BS 1722: Part 9 specifies heights ranging from 1,200 to 2,100 mm in 150 mm increments.

Verticals: Dimensions are specified in BS 1722: Part 9. Verticals can be round or square with various tops (e.g. blunt, pointed, spear, bow, double bow, interlaced bow). In addition, square verticals can be fixed 'square to view' or 'angle to view'.

Centres of verticals: Select from Table 5.13.5

Posts: BS 1722: Part 9 specifies flat bar, butt tees, angles and H sections can also be used. If a sufficiently strong section is used for the posts, the stays can be omitted to give a less cluttered appearance.

Treatment: Insert details (e.g. galvanised)

Centres of posts: State dimensions.

Setting posts/stays/legs: There are many proprietary designs of railing, and methods of setting in the ground vary accordingly. BS 1722: Part 9 specifies posts to be driven, stays and legs to be set in rammed earth. If the ground is not suitable, posts, stays and legs can be set in concrete. Posts should be set at the following depths:

Height of fence (mm)	Minimum depth (mm)
1,200, 1,350	530
1,500, 1,650, 1,800	610
1,950, 2,100	760

BS 1722: Part 9 does not specify complete requirements for posts and stays set in concrete. A suggested specification is: 'Concrete in 300 × 300 mm holes filled to not less than half the depth with post holes not less than (e.g.) 600 mm deep.' Vertical bar railings are often mounted on low walls and can be set between brick pillars instead of posts. In such cases the ends of posts, stays, legs and rails, as appropriate, should be suitable for grouting or running in lead. See BS 5390, Code of Practice for stone masonry, which considers the problems of fixing steel and iron into masonry. Insert appropriate requirements.

Anti-intruder chain link

Type: BS 1722: Part 10 specifies chain link fencing topped with barbed wire, 2.4 m high to the top of the chain link and 2.9 m high to the top line of barbed wire.

Mesh and wire: Insert 'galvanised' or 'plastics coated'. BS 1722: Part 10 specifies plain galvanised wire and Grade A plastics-coated wire (plastics coating on galvanised wire).

Posts and struts: Insert details. State whether posts are to have cranked tops, extension arms or both, to support the barbed wire.

Centres of posts: State dimensions.

Setting posts: From BS 1722: Part 10 specify requirements. Also note:

- Holes for concrete gate posts to be filled up to ground level.
- Holes for steel gate posts to be filled and the foundation brought up to a height of 50 mm above ground level and finished with a weathered top.

Bottom of fencing: Insert details. BS 1722: Part 10 specifies that galvanised mesh, which is to be buried, should be coated with black paint or bitumen solution. The bottom 450 mm is dipped and the bottom 300 mm is then buried vertically. PVC-coated mesh does not need to be dipped.

Other requirements: State any.

Steel palisade

In situations of extreme exposure or

abnormal loading it will be necessary to agree special requirements with the manufacturer (e.g. the provision of stays).

Type: Select type from BS 1722: Part 12, table 1 (general-purpose fencing) or table 2 (security fencing) (e.g. SP 30).

Height: State size. BS 1722: Part 12 specifies the following heights:

- General-purpose fencing 1,800 mm and 2,100 mm.
- Security fencing 2,400, 3,000 and 3,600 mm.

Top of pales: Various patterns are available: plain pointed, pointed, split and fanged, rounded, etc. BS 1722: Part 12 specifies tops of pales for security fencing to be pointed, split and fanged.

Treatment. Fencing components and accessories: Provide details. *Fastenings:* Fastenings should be galvanised steel or stainless steel. Sheradised or plated fastenings will normally be adequate if fence is to be painted. BS 1722: Part 12 requires bolts to be burred over to prevent removal and, if galvanised, to be touched up with zinc-rich primer.

Centres of posts: Specify dimensions. BS 1722: Part 12 specifies 2.75 m centres generally and 1.85 m centres for 3,600 mm high fencing.

Post holes: Provide details. Specify the depth as 'equal to one third of the height of the fence above ground level'.

Other requirements: For example, continuous concrete sill to security fencing (see BS 1722: Part 12, clause 4.9, Removable panels).

Tennis court chain link

Height: State size. BS 1722 specifies a minimum of 2,750 mm.

Mesh and wire: Mesh size and wire gauge are specified in the British Standard and need not be given unless a different size and gauge are required. However, it is necessary to specify the type of wire required and finish (e.g. 'green plastics-coated galvanised mesh and line wire').

Posts and struts: BS specifies steel angle and rectangular hollow section (including the sizes).

Centres of posts: State dimensions.
Setting posts: Insert the method and depth.
 For posts set in concrete insert also the plan size of the post hole and the requirement to fill the hole not less than half full. BS 1722: Part 13 specifies sizes of holes for straining, struts and intermediate posts to be set in concrete. BS 1722: Part 13 permits alternative methods of setting posts when ground conditions vary. All posts can be set in rammed earth and in poor ground intermediate posts can be set in concrete.

Tables 5 to 8

Tables 5 to 8 summarise the detailed requirements of BS 1722 for setting posts in the ground. Specify the method and depth required in the relevant clause. For posts set in concrete, insert also the plan size of the post hole and the requirement to fill the hole to not less than half full.

Acknowledgements to National Building Specification Ltd.

TABLE 5. POSTS FOR CHAIN LINK FENCING

	Straining posts	Struts	Intermediate posts
Method of setting concrete posts	Concrete	Concrete	Concrete
Method of setting steel posts	Concrete or rammed earth	Concrete or rammed earth	1,800 high – concrete Others – concrete, rammed earth or driven
Method of setting timber posts	Concrete or rammed earth	Concrete or rammed earth	Concrete, rammed earth or driven
Plan size of post holes for setting in concrete (mm)	450 × 450 or 300 diameter	300 × 450	To give 75 mm cover all round or 300 diameter
Minimum depth of post holes for any method of setting:			
Height of fence (mm)	900, 1,200, 1,400 over 1,400	All heights	900, 1,200, 1,400 over 1,400
Depth of hole (mm)	600	750	450 600 750

TABLE 6. POSTS FOR WOVEN WIRE FENCING

	Straining posts	Struts	Intermediate posts
Method of setting concrete posts*	Concrete Concrete	Concrete (loose base plate in rammed earth)	Rammed earth (concrete)
Method of setting steel and timber posts*	Concrete (rammed earth)	Concrete (rammed earth)	Driven
Plan size of post holes for setting in concrete (mm)	450 × 450	300 × 450	To give 75 mm cover all round
Minimum depth of post holes for any method of setting	750	450	600

*BS 1722: Part 2 specifies one method for each type of post. It also suggests alternatives which can be specified if warranted by the ground conditions – these are shown in parentheses.

TABLE 7. POSTS FOR STRAINED WIRE FENCING

	Straining posts	Struts	Intermediate posts
Method of setting concrete posts	Concrete	Concrete	Concrete
Method of setting steel and timber posts	Concrete or rammed earth	Concrete or rammed earth	Concrete, rammed earth or driven
Plan size of post holes for setting in concrete (mm)	450 × 450 or 300 diameter	300 × 450	To give 75 mm cover all round or 300 diameter
Minimum depth of post holes for any method of setting:			
Height of fence (mm)	900 1,050–1,350 1,800, 2,100	All heights	Less than 1,800 1,800, 2,100
Depth of hole (mm)			
– for concrete	600 750 900	450	600 750
– for rammed earth	900 1,050 1,200	450	600 750

TABLE 8. POSTS FOR CLEFT CHESTNUT PALE FENCING

	Straining posts	Struts	Intermediate posts
Method of setting concrete posts	Concrete	Concrete	Concrete
Method of setting timber posts	Concrete or rammed earth	Concrete or rammed earth	Driven
Plan size of post holes for setting in concrete (mm)	300 × 300 or 300 diameter	300 × 450	To give 75 mm cover all round or 300 diameter
Minimum depth of post holes for any method of setting:			
Height of fence (mm)	Less than 1,500 1,500, 1,800	All heights	Less than 1,500 1,500, 1,800
Depth of hole (mm)	600	750	450 600 750

DETAIL SHEETS

General

Cappings (2)
 Timber post foundations (2)
 Osier hurdles
 Woven wood
 Waney edge board
 Close board – medium
 Close board – high
 Close board – double-sided
 Close board – thick and thin board
 Close board – with brick
 Timber palisade – medium
 Timber palisade – high
 Timber palisade – double-sided
 Timber palisade – diagonal board
 Timber palisade – with PVC
 Palisade – angled
 Horizontal double slatted
 Horizontal open slatted
 Cleft chestnut
 Cleft chestnut and wire
 Post and rail – staggered joints
 Post and rail – butt joints
 Lincolnshire post and rail
 Sussex two rail
 Post and wire
 Plastic mesh

Low mesh (2)

Timber and mesh

Timber poles

Metal

Vertical bar – medium (2)

Vertical bar – tall

Bow topped – low

Bow topped – medium

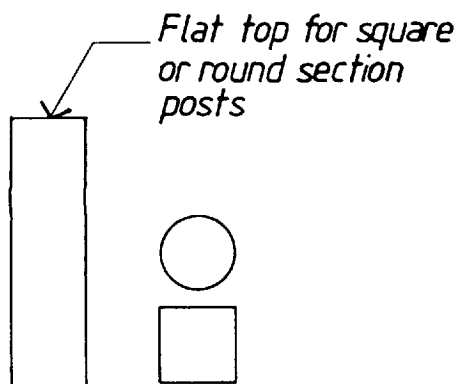
Metal – flat section (2)

Chain link 1800

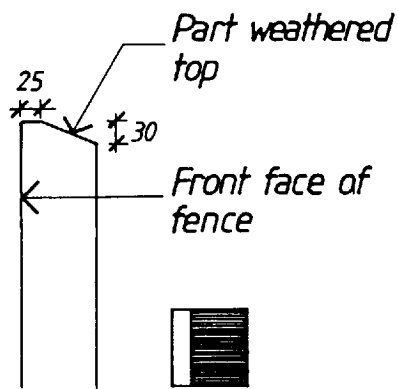
Chain link 3000

General

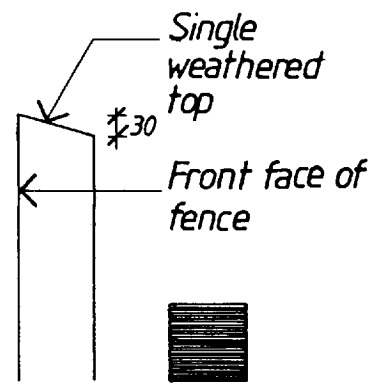
Concrete palisade



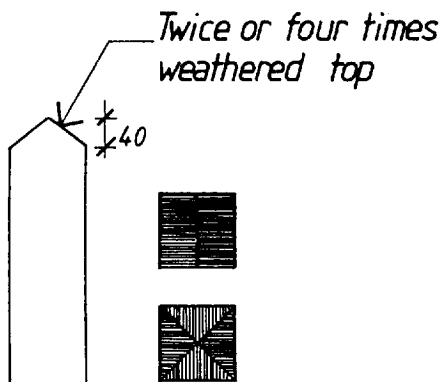
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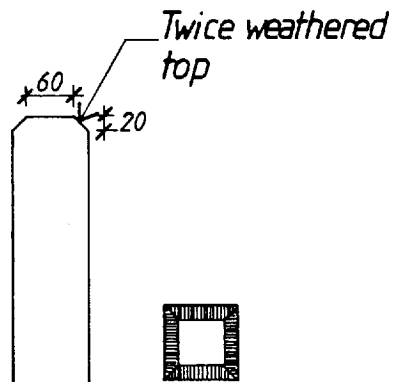
B.



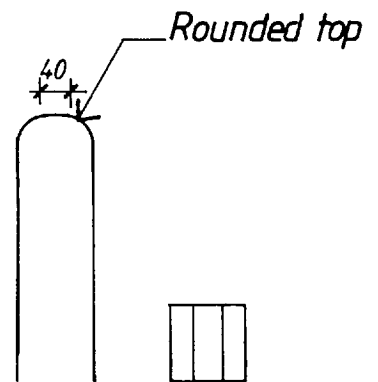
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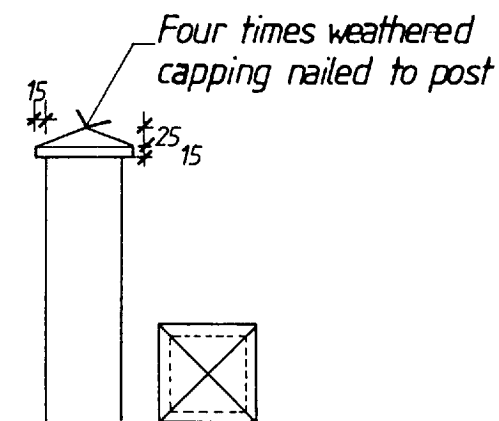
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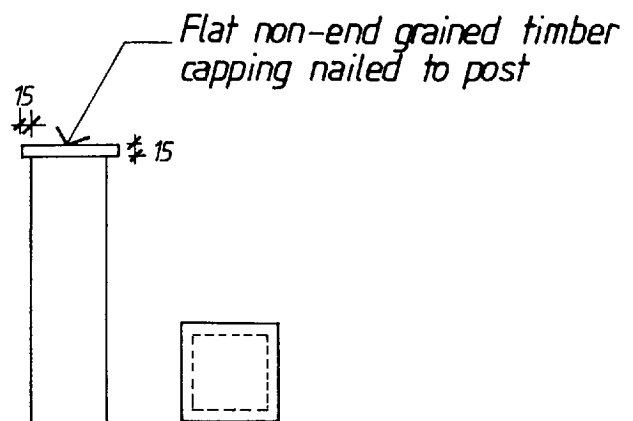
E.



F.



G.



H.

PLAN

PLAN

SECTION

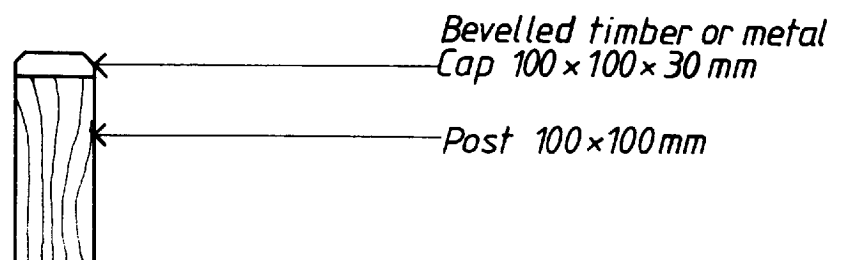
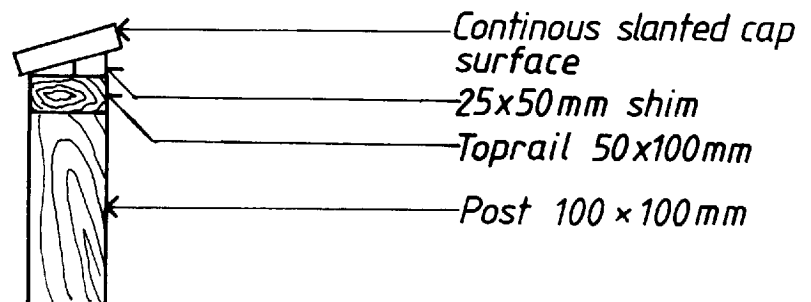
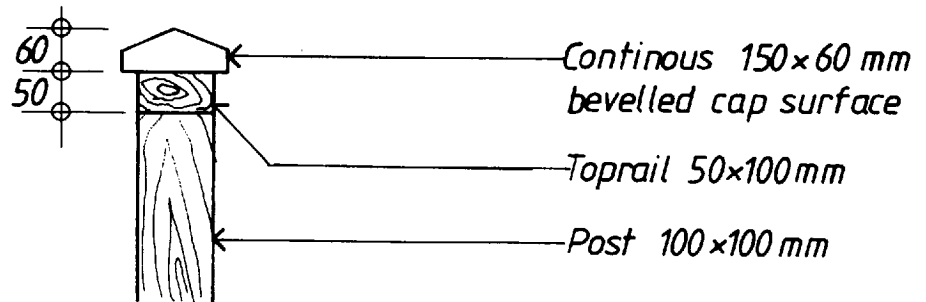
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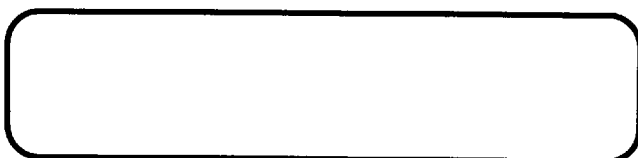


FENCE timber posts
cappings





Scale 1:10



FENCE
caps



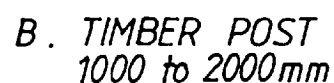


Diagram illustrating the installation of a timber post into the ground, showing cross-section details:

- Timber post selected as required
- Backfill around post to depth of 50mm with soil
- 500 (Dimension indicating the length of the augered hole)
- Augered hole, backfilled with well-rammed coarse aggregate, to within 50mm of surface
- Coarse aggregate beneath post to facilitate drainage

Timber post selected as required

Backfill around post to depth of 50mm with soil

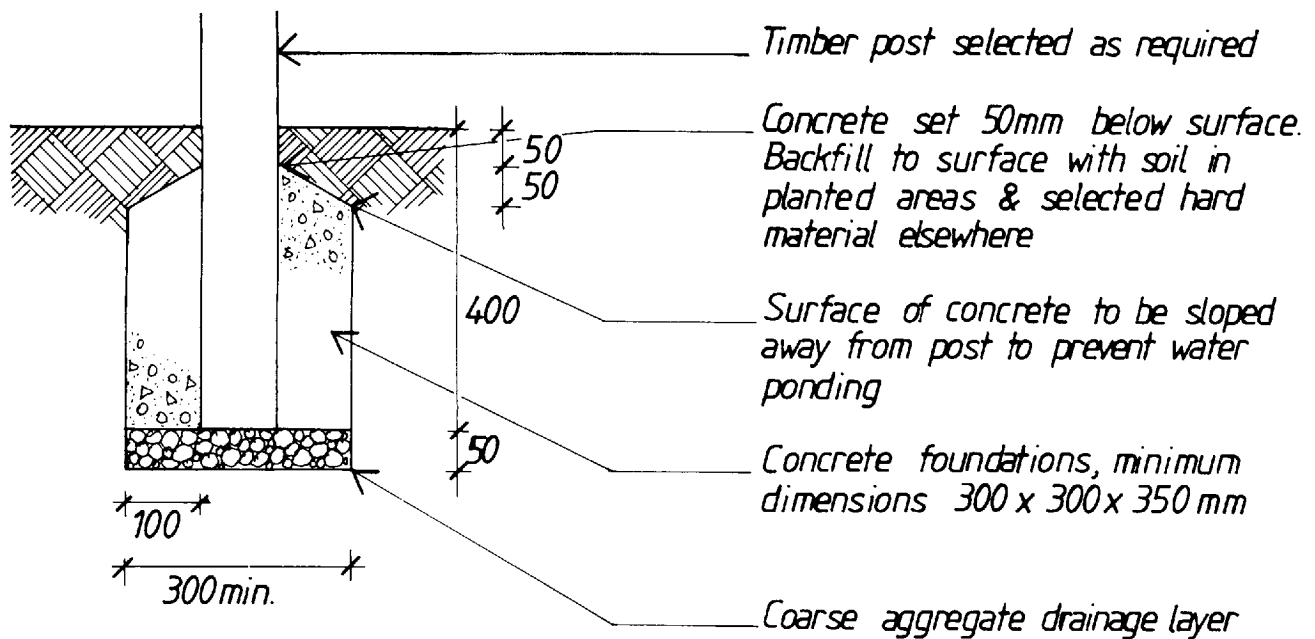
750

Augered hole, backfilled with well-rammed coarse aggregate, to within 50mm of surface

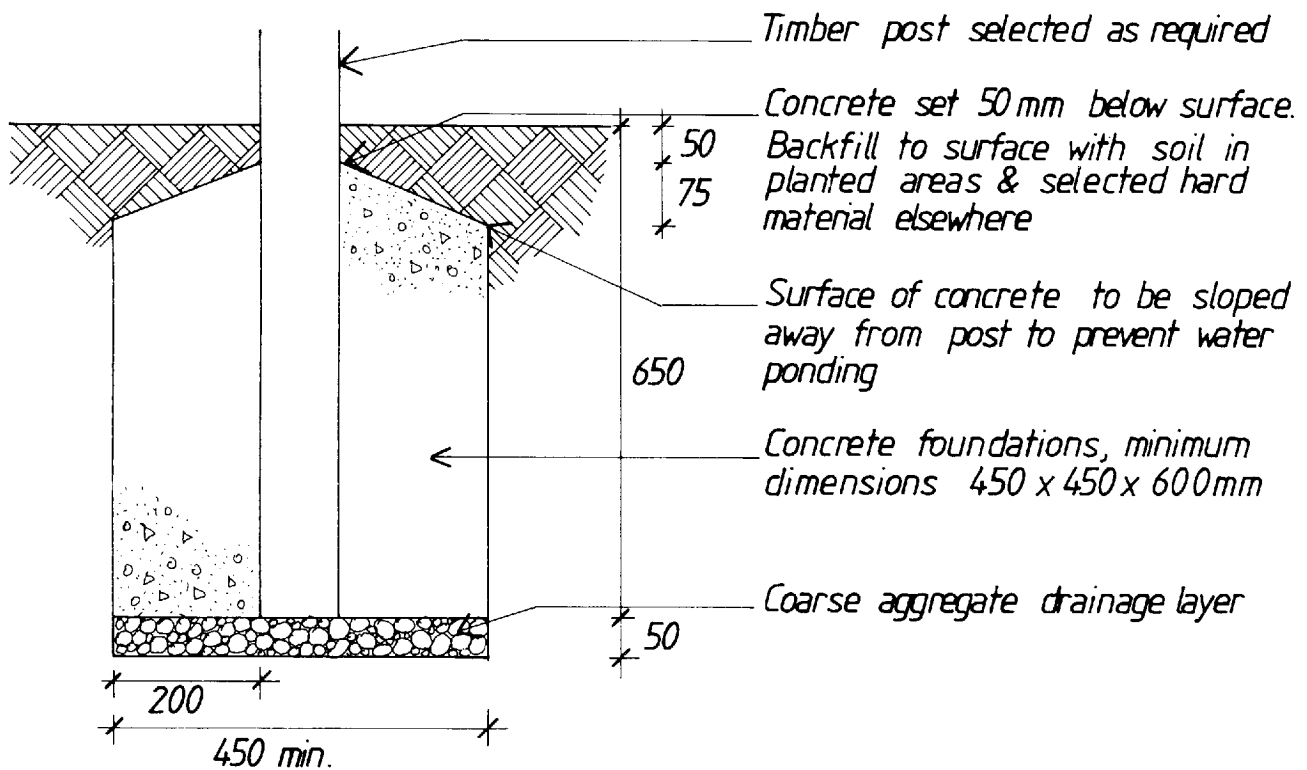
Coarse aggregate beneath post to facilitate drainage

2. RAMMED

FENCE timber posts
foundations



A. TIMBER POST upto 1000mm



B. TIMBER POST 1000 to 2000mm

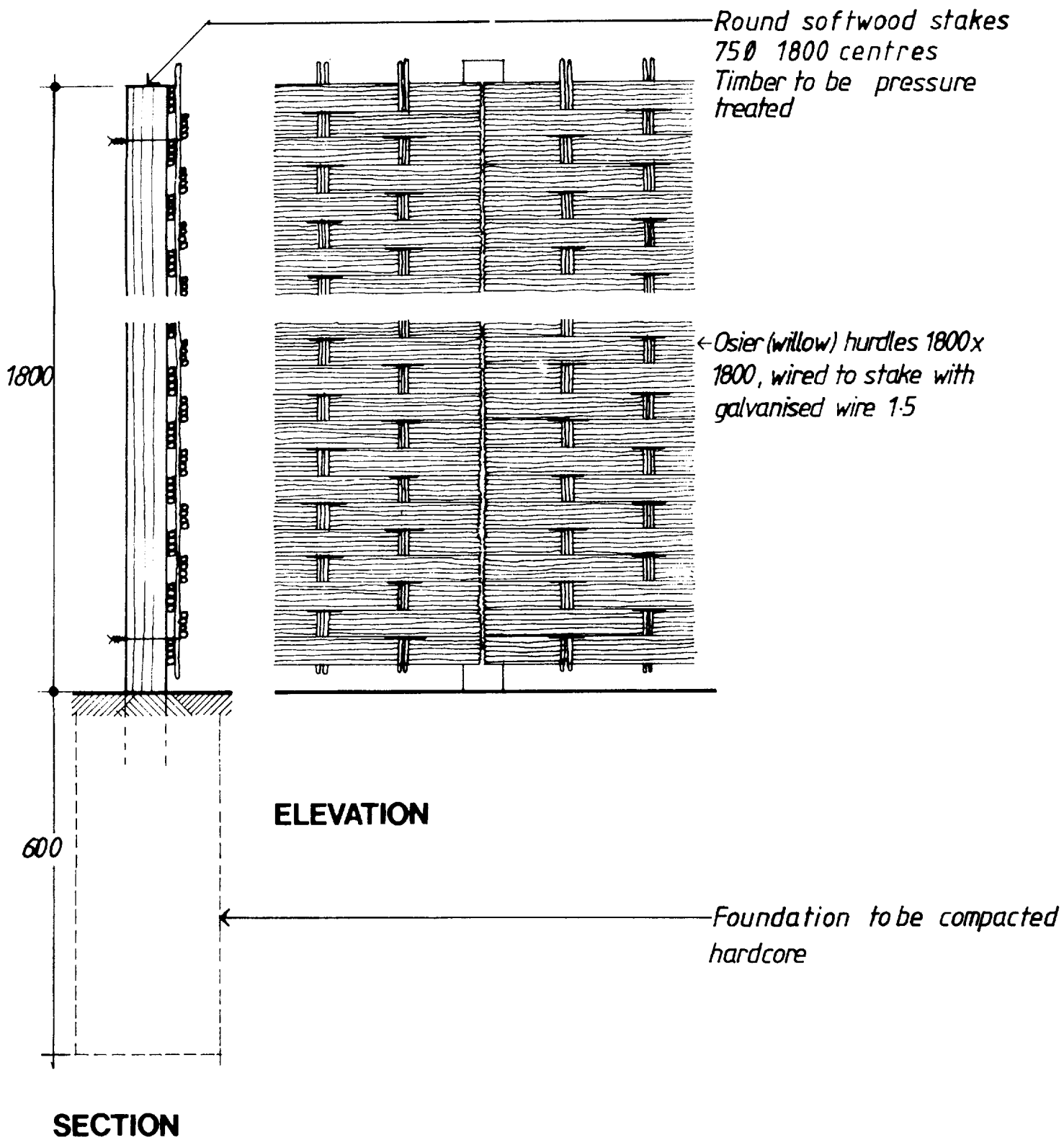
3. CONCRETE

Scale 1:10



FENCE timber posts
foundations

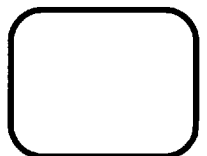


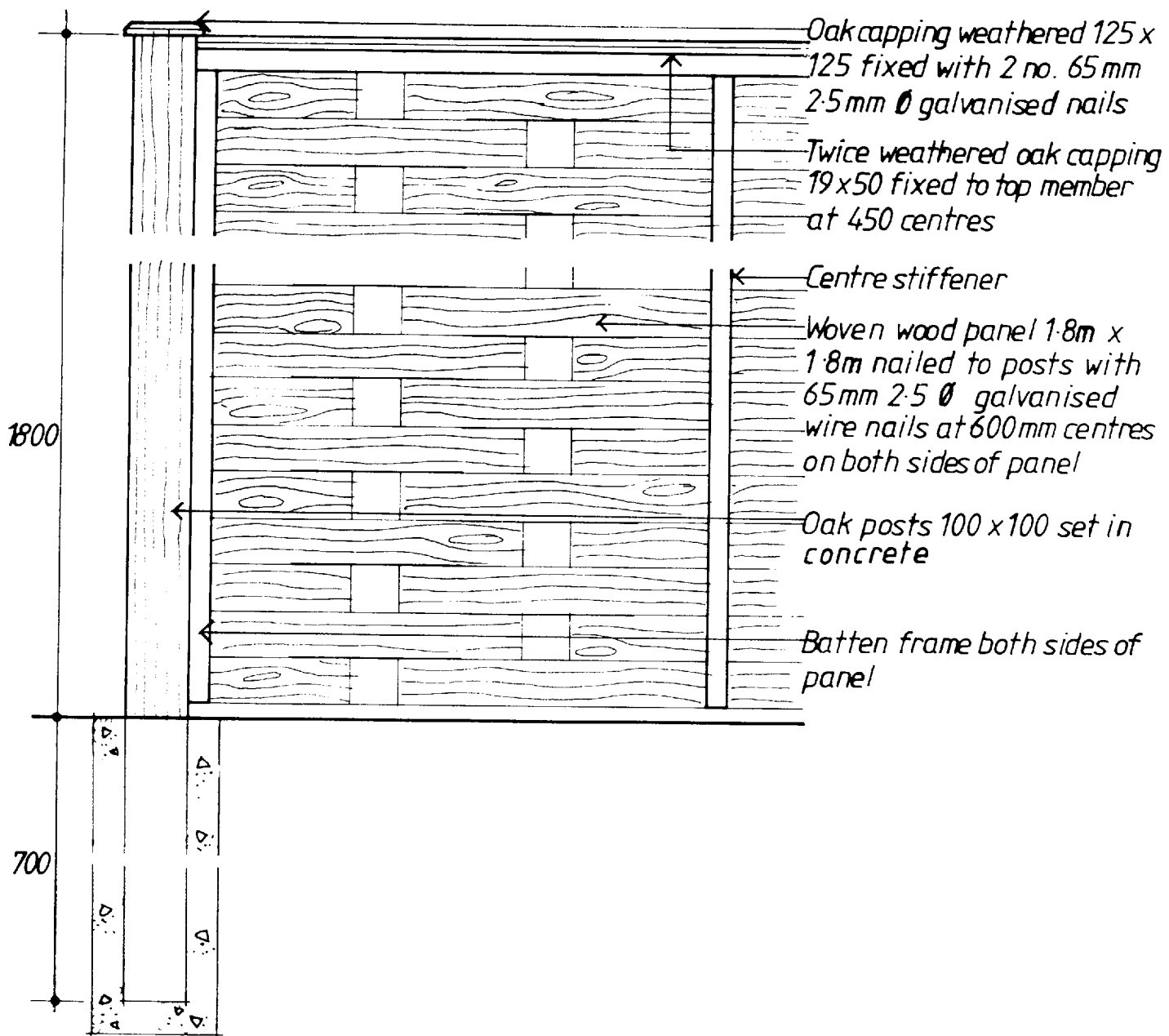


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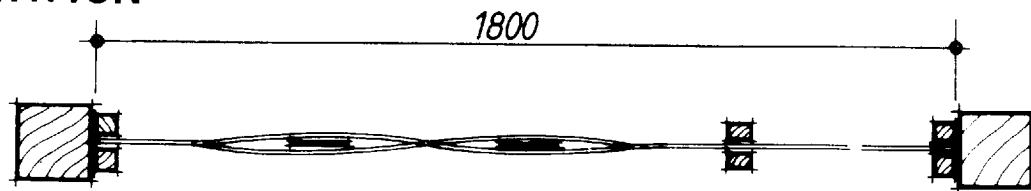


FENCE
osier hurdles



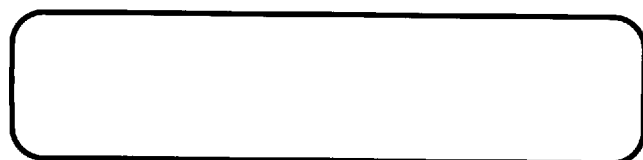


ELEVATION

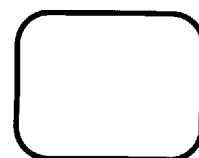


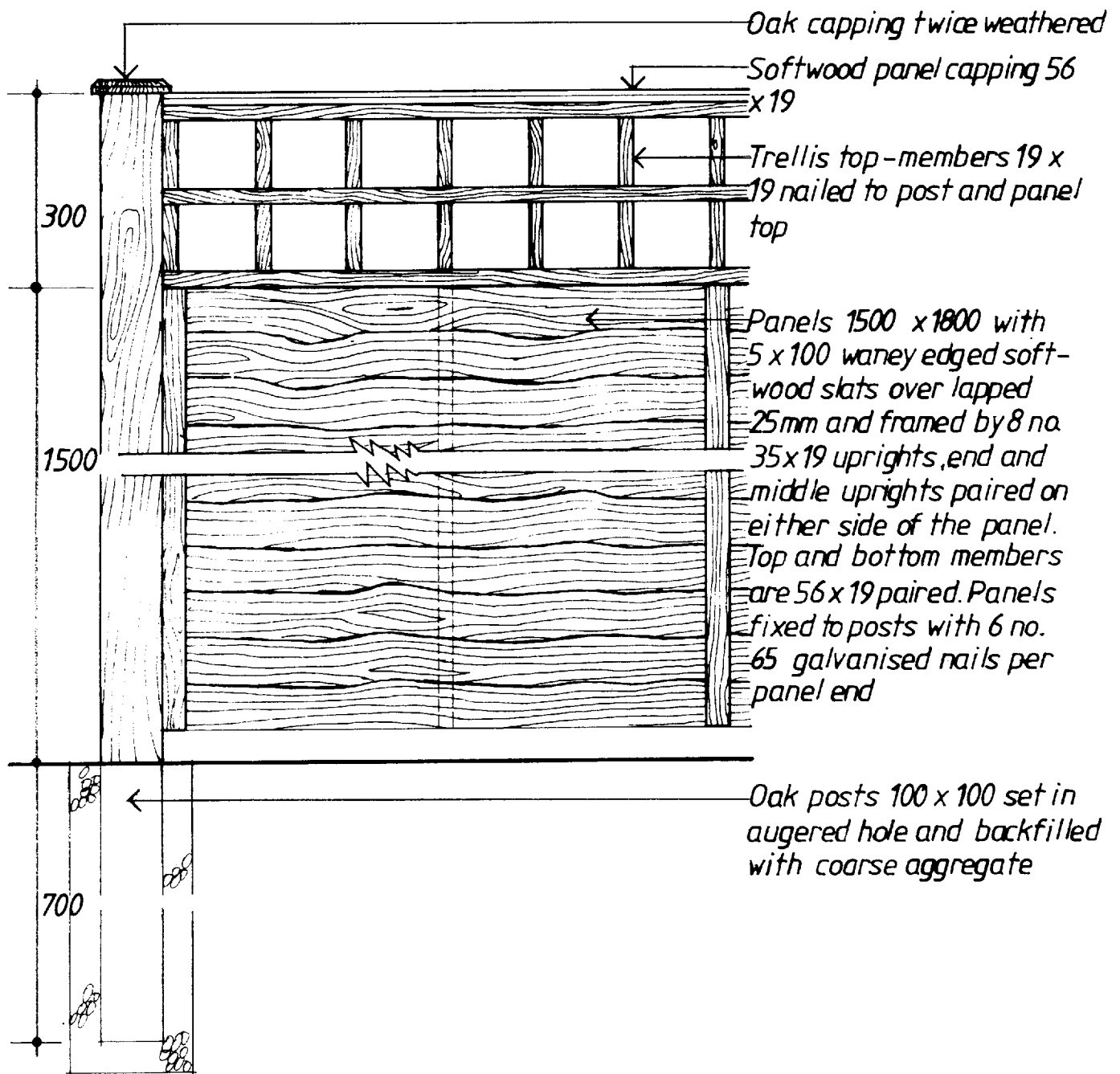
PLAN

Scale 1:10



FENCE
woven wood

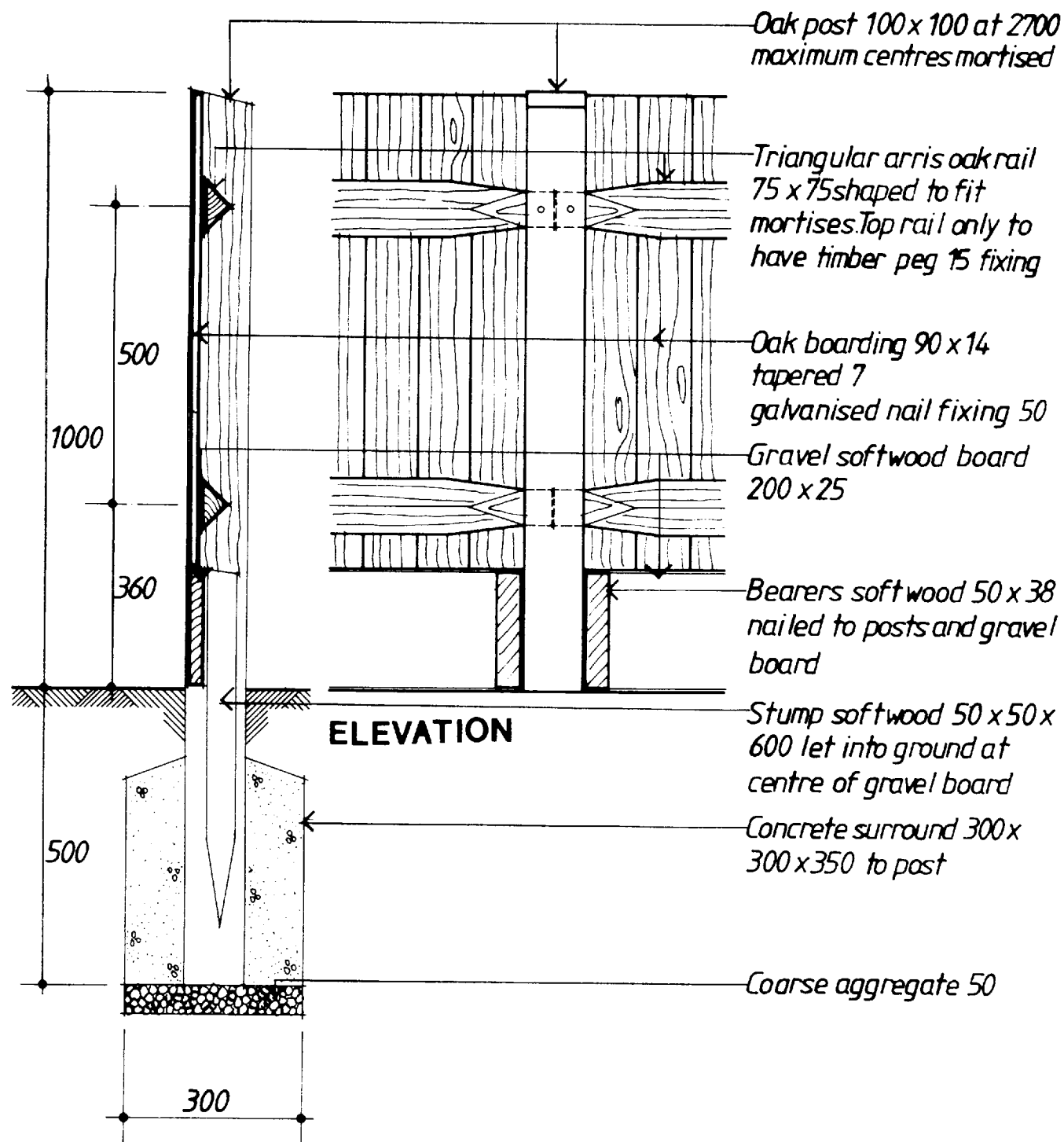




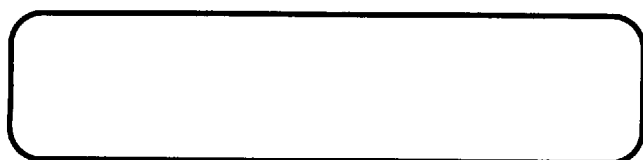
ELEVATION

Scale 1:10

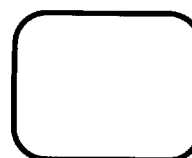
FENCE
waney edge board

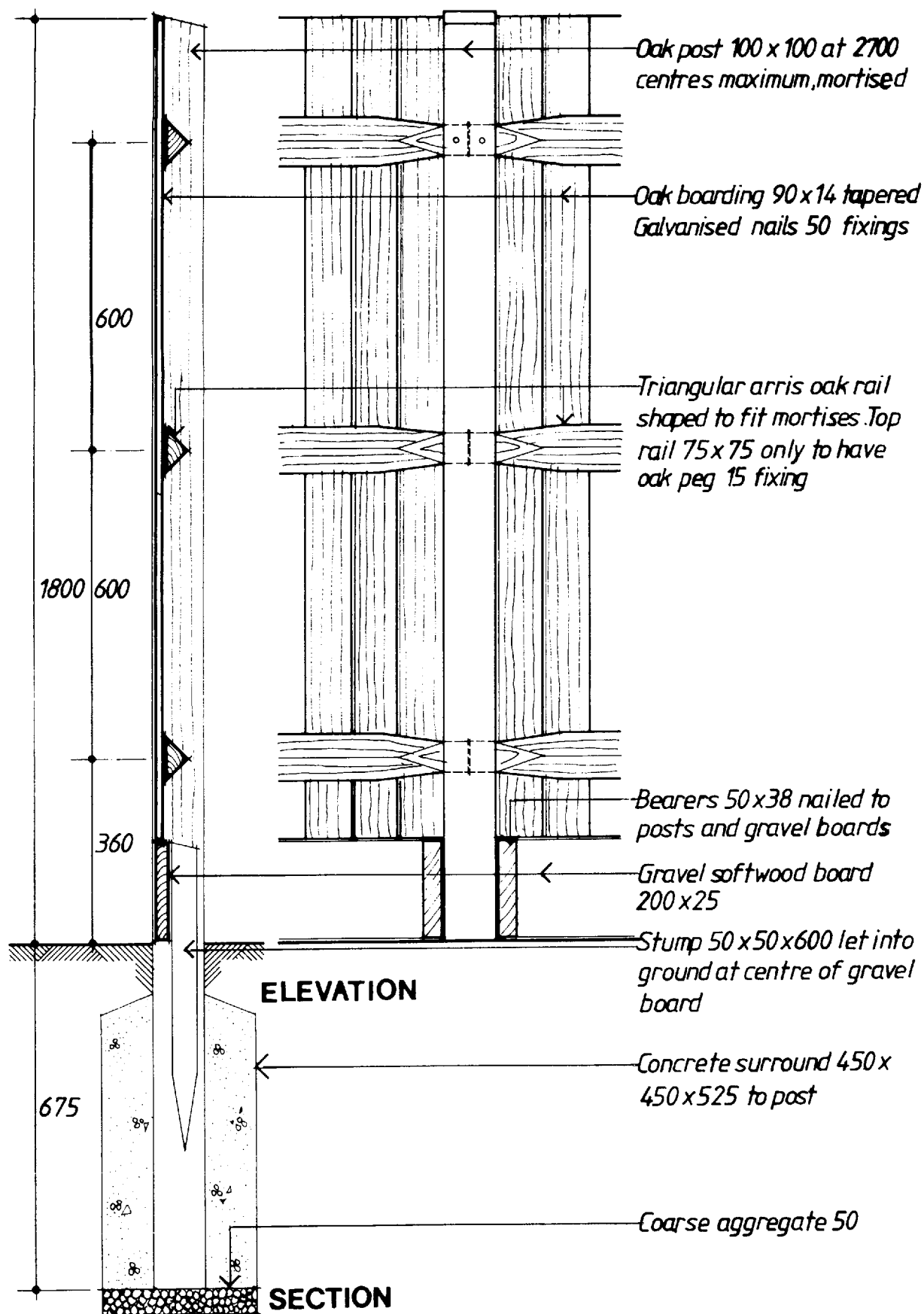


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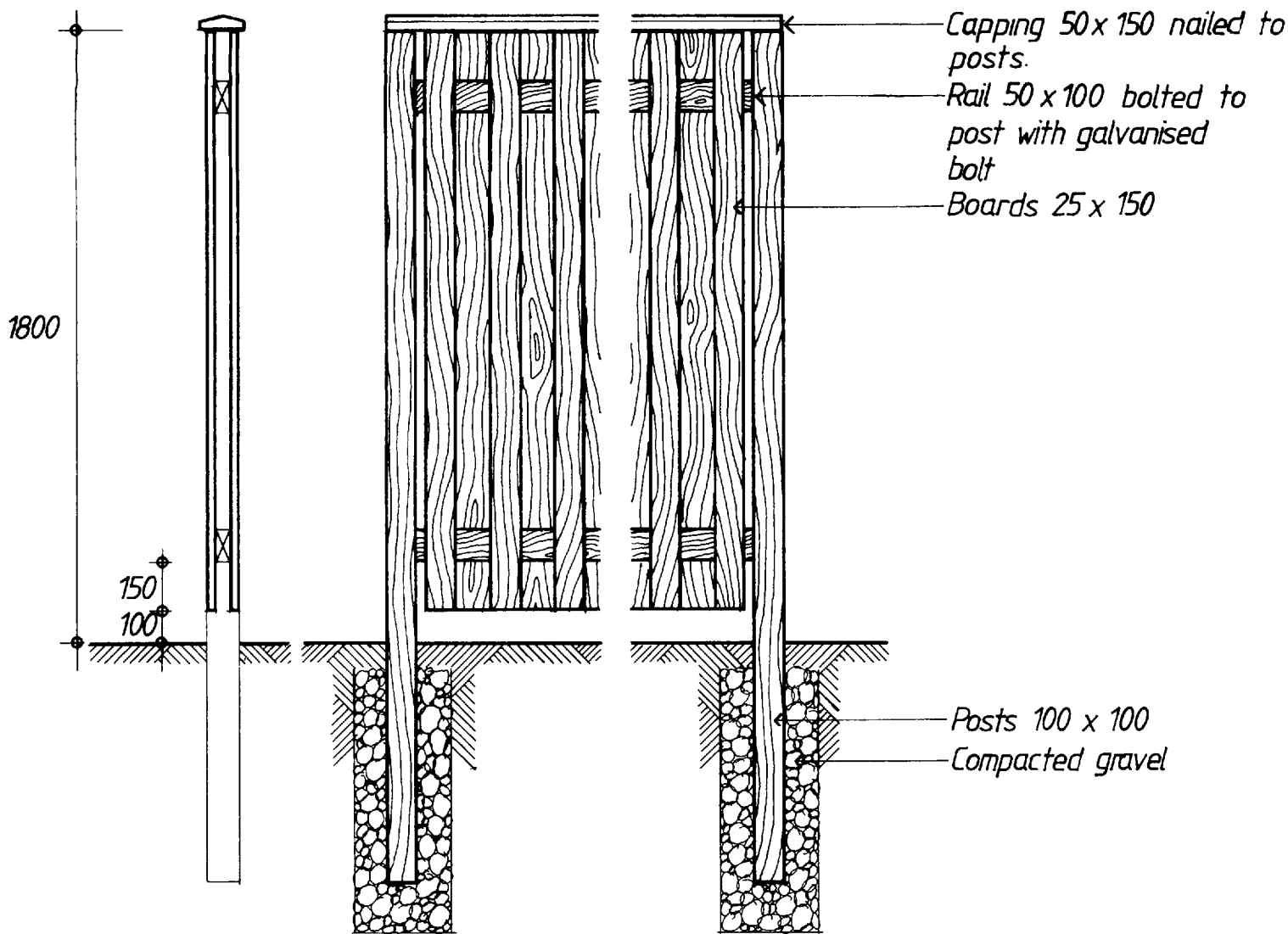
FENCE
close board – medium





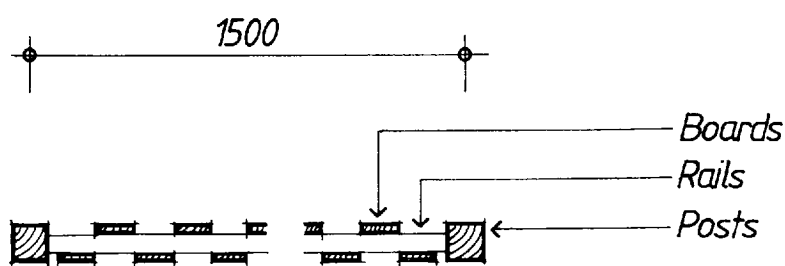
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FENCE
close board – high



SECTION

ELEVATION



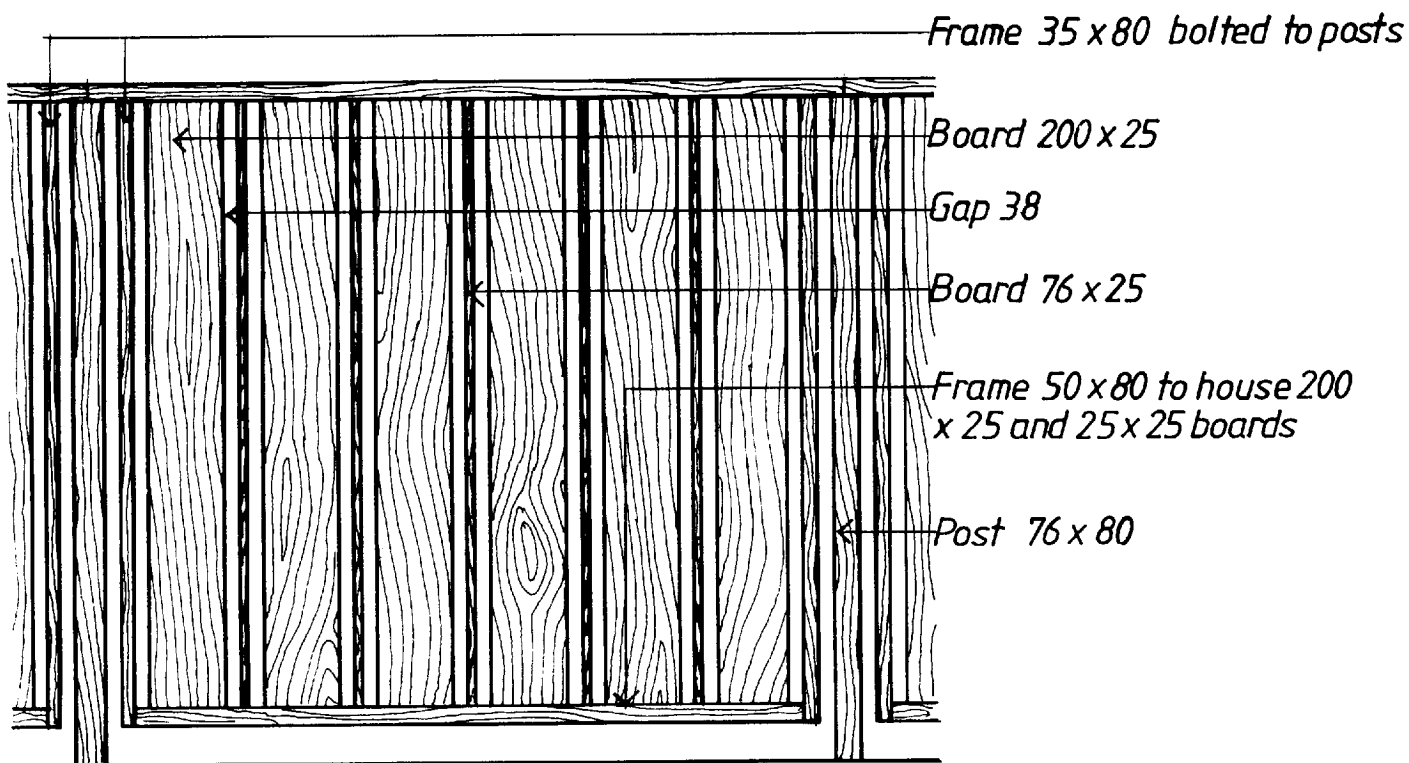
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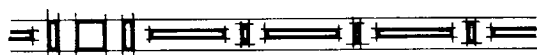


FENCE
close board –
double-sided

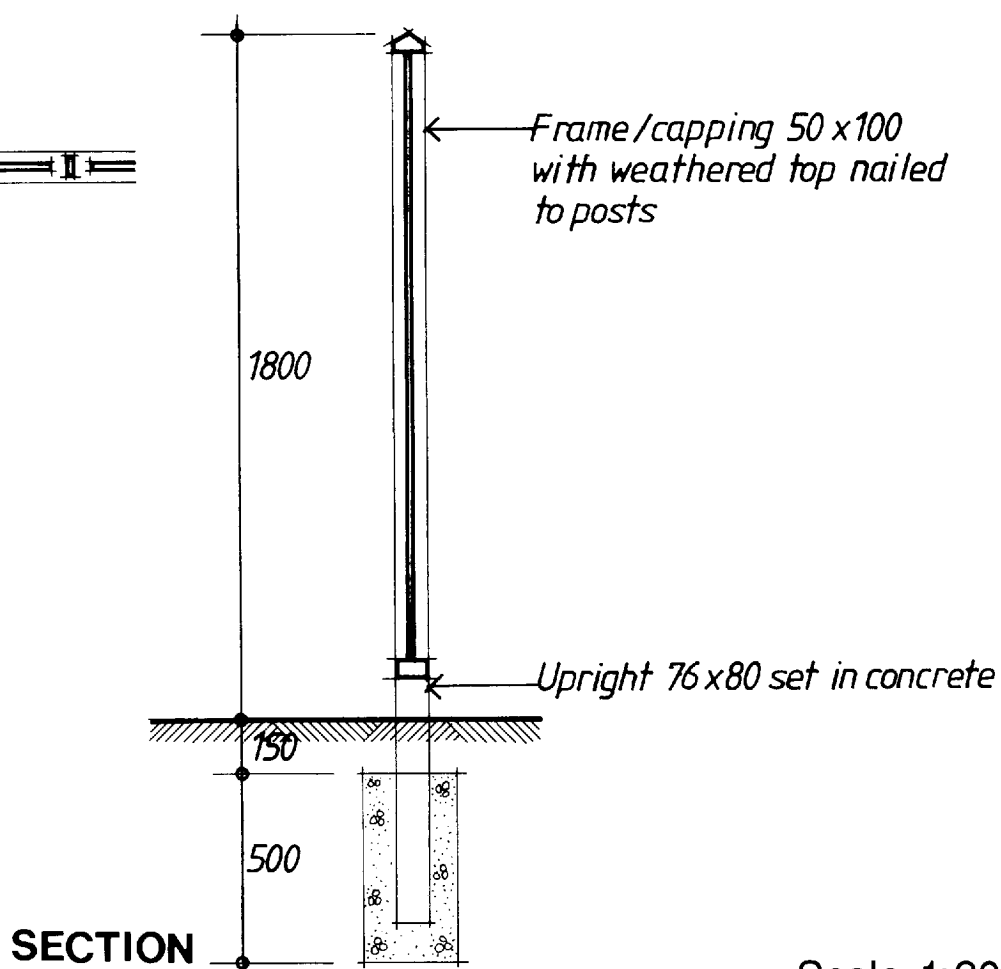




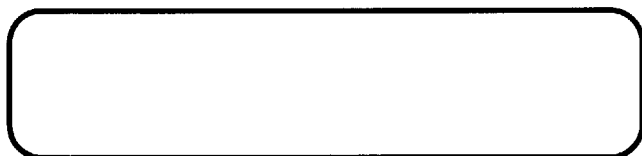
ELEVATION



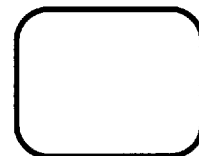
PLAN

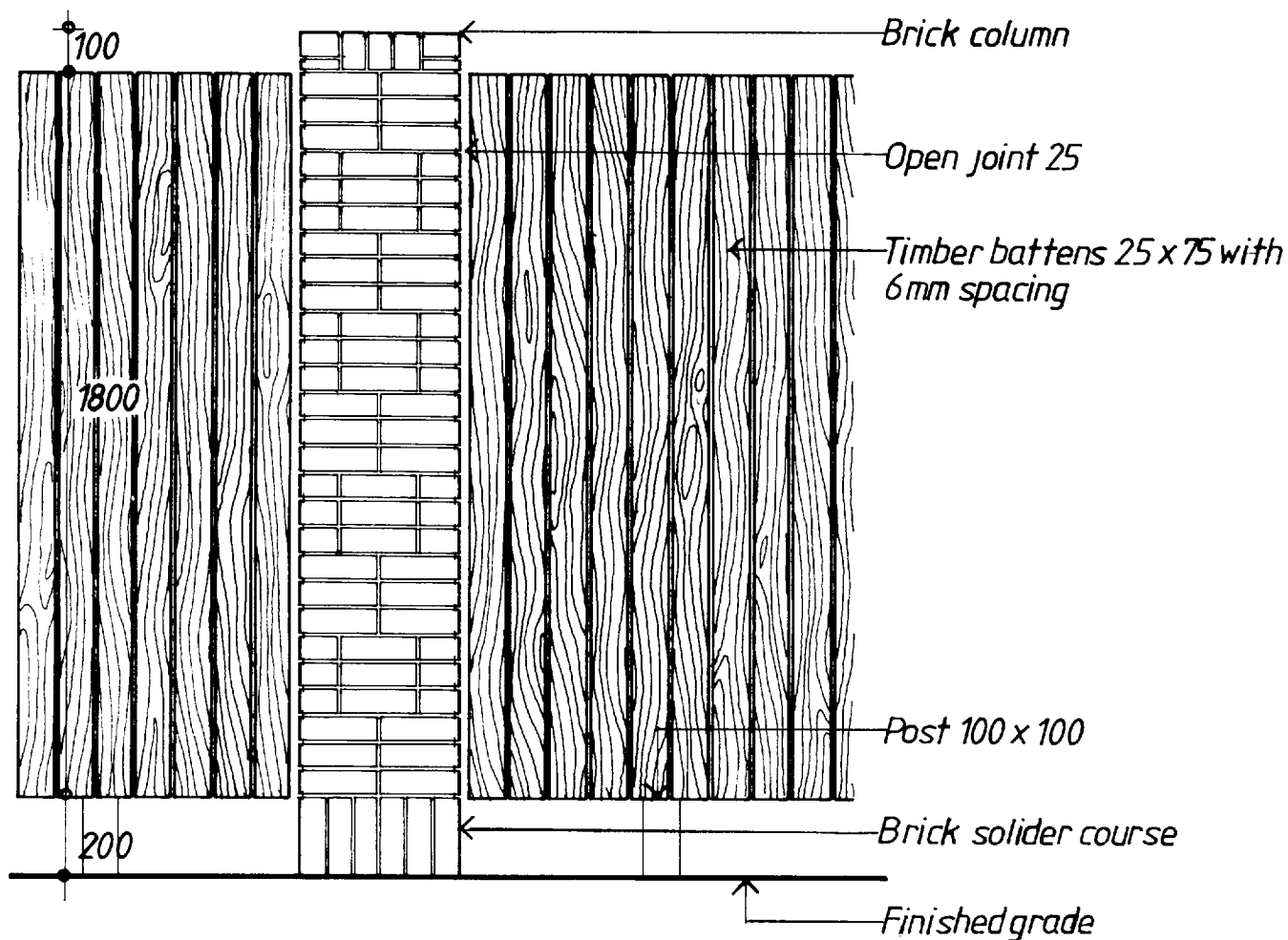


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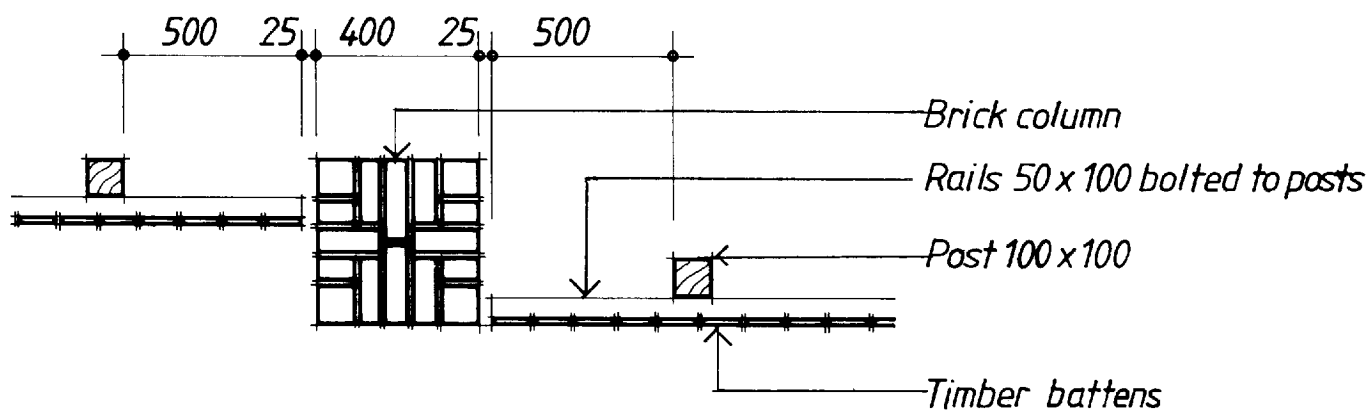


FENCE
thick & thin board



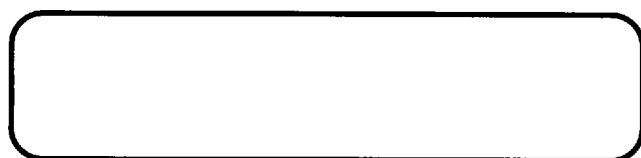


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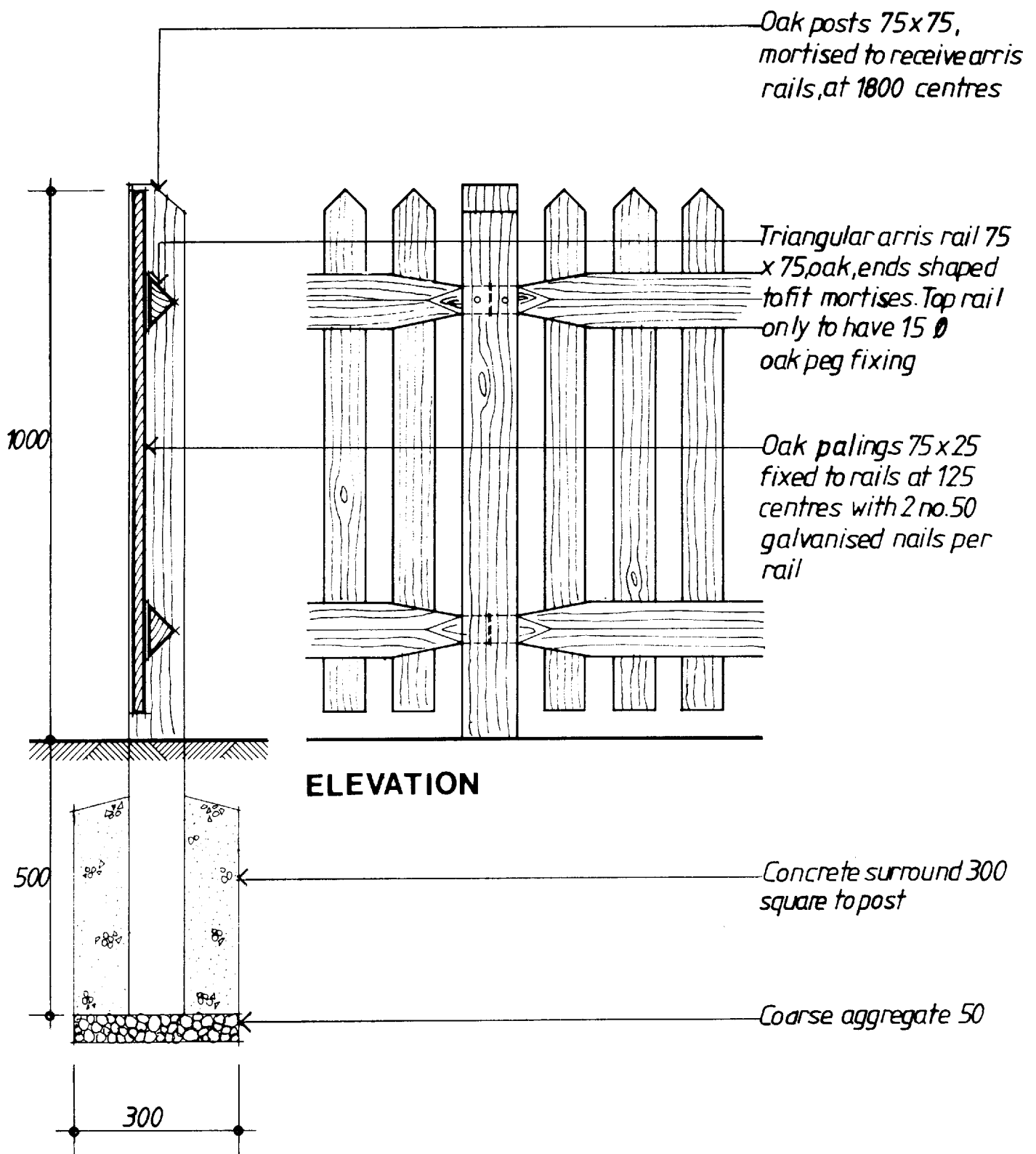
PLAN

Scale 1:20



FENCE
brick & wood



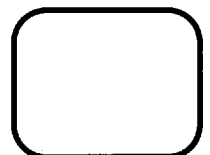


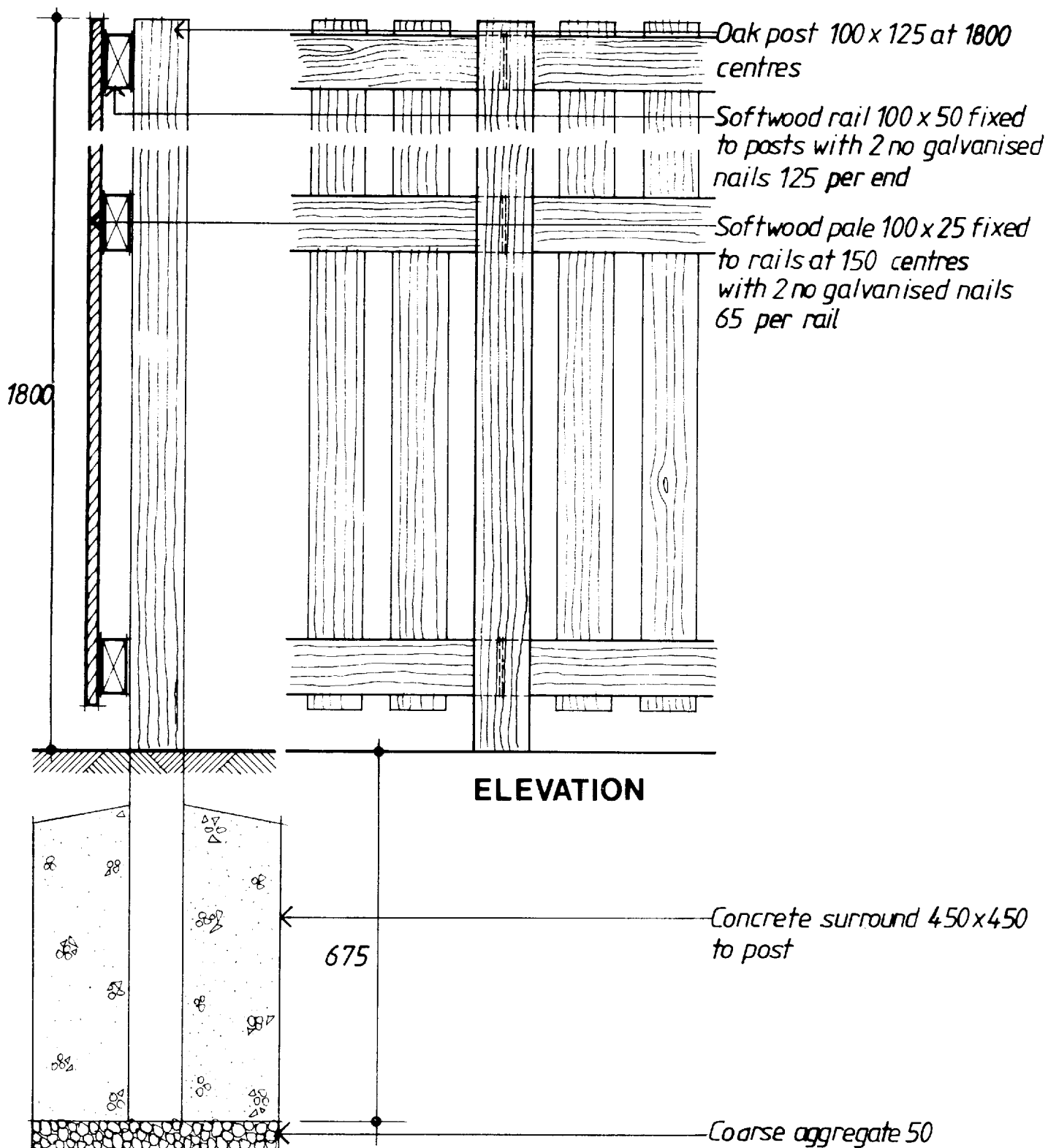
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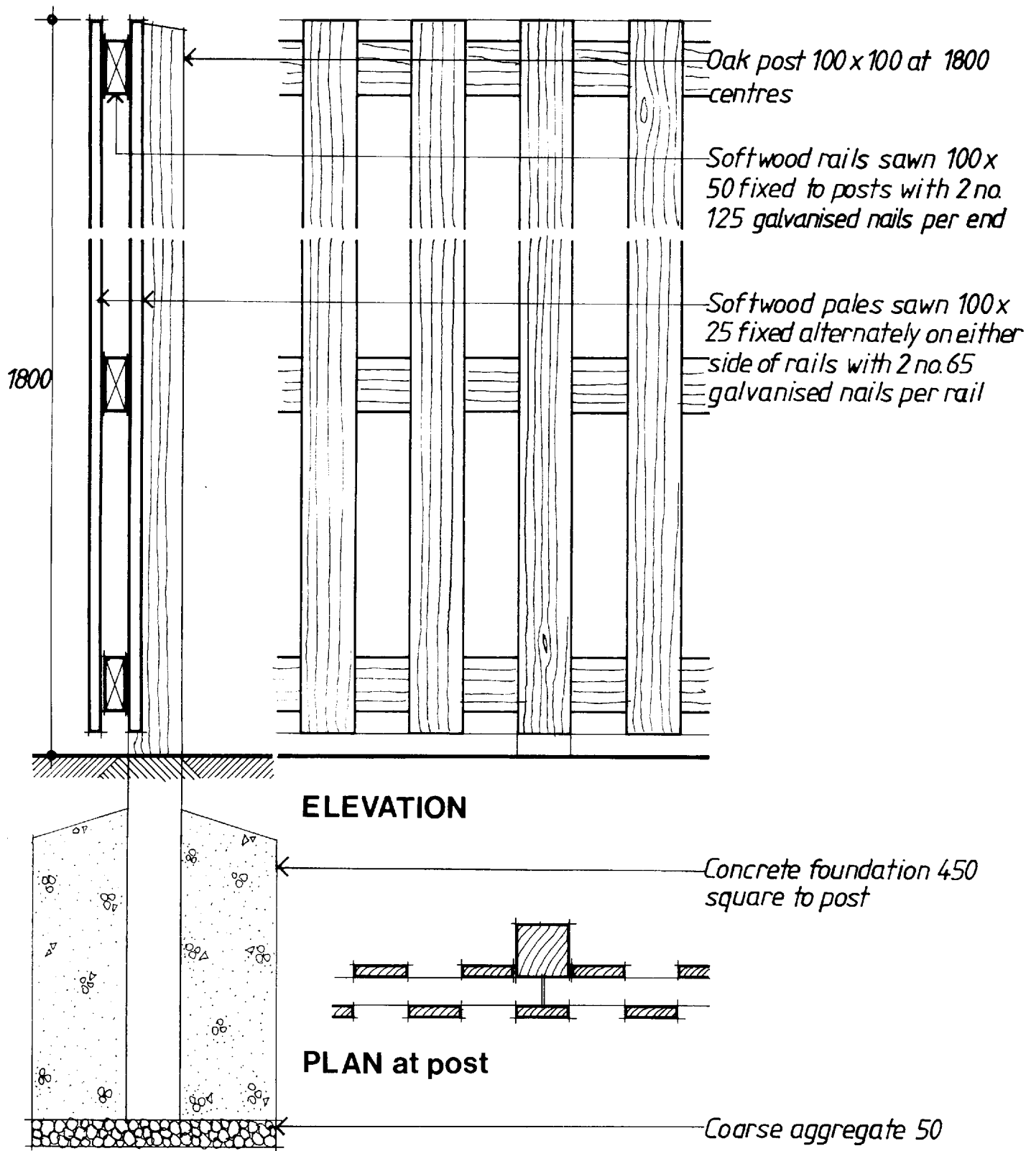
FENCE
timber palisade –
medium





Scale 1:10

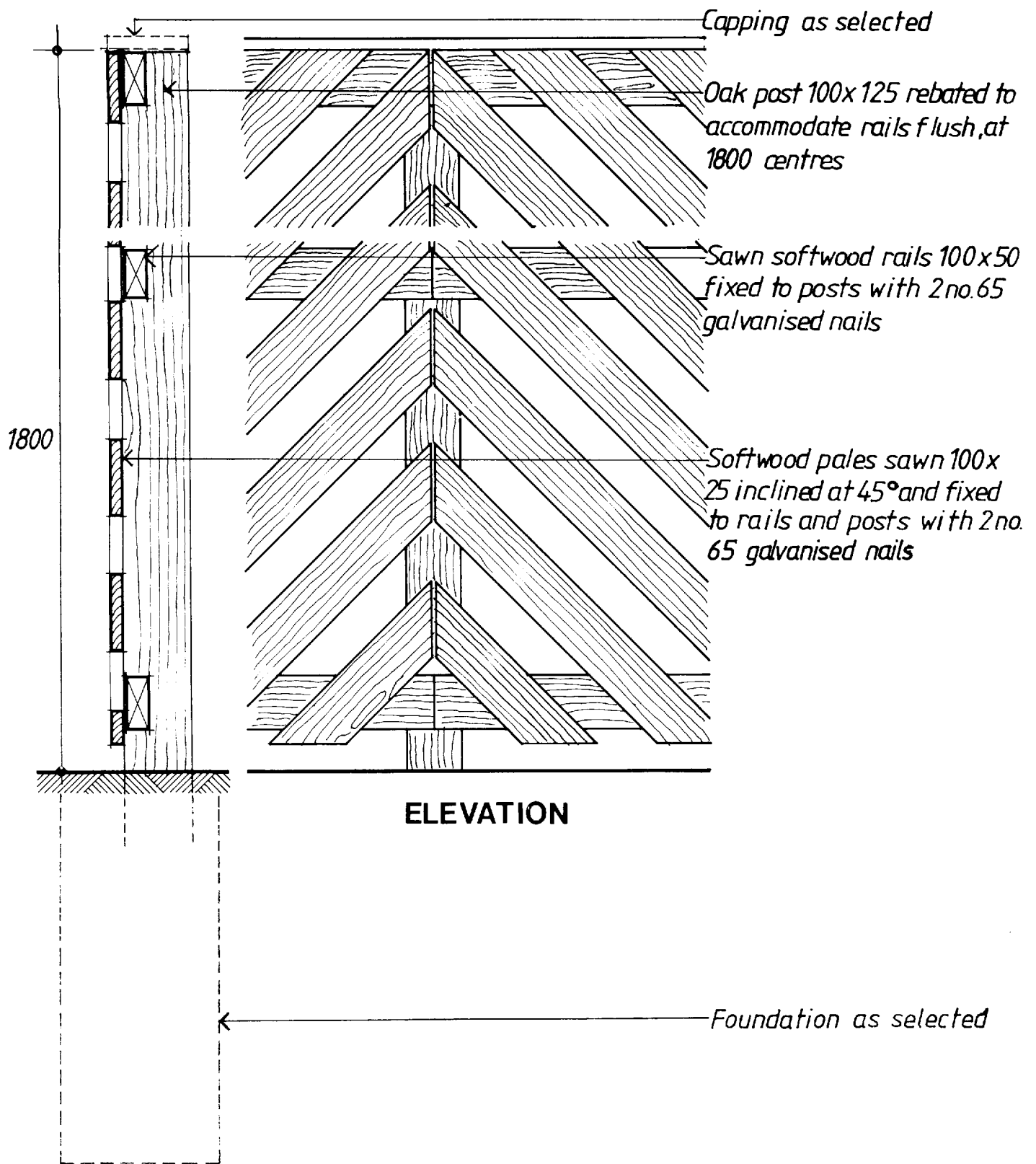
FENCE
timber palisade –
high



SECTION

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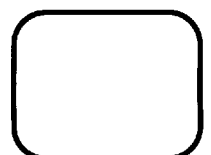
FENCE
double sided palisade

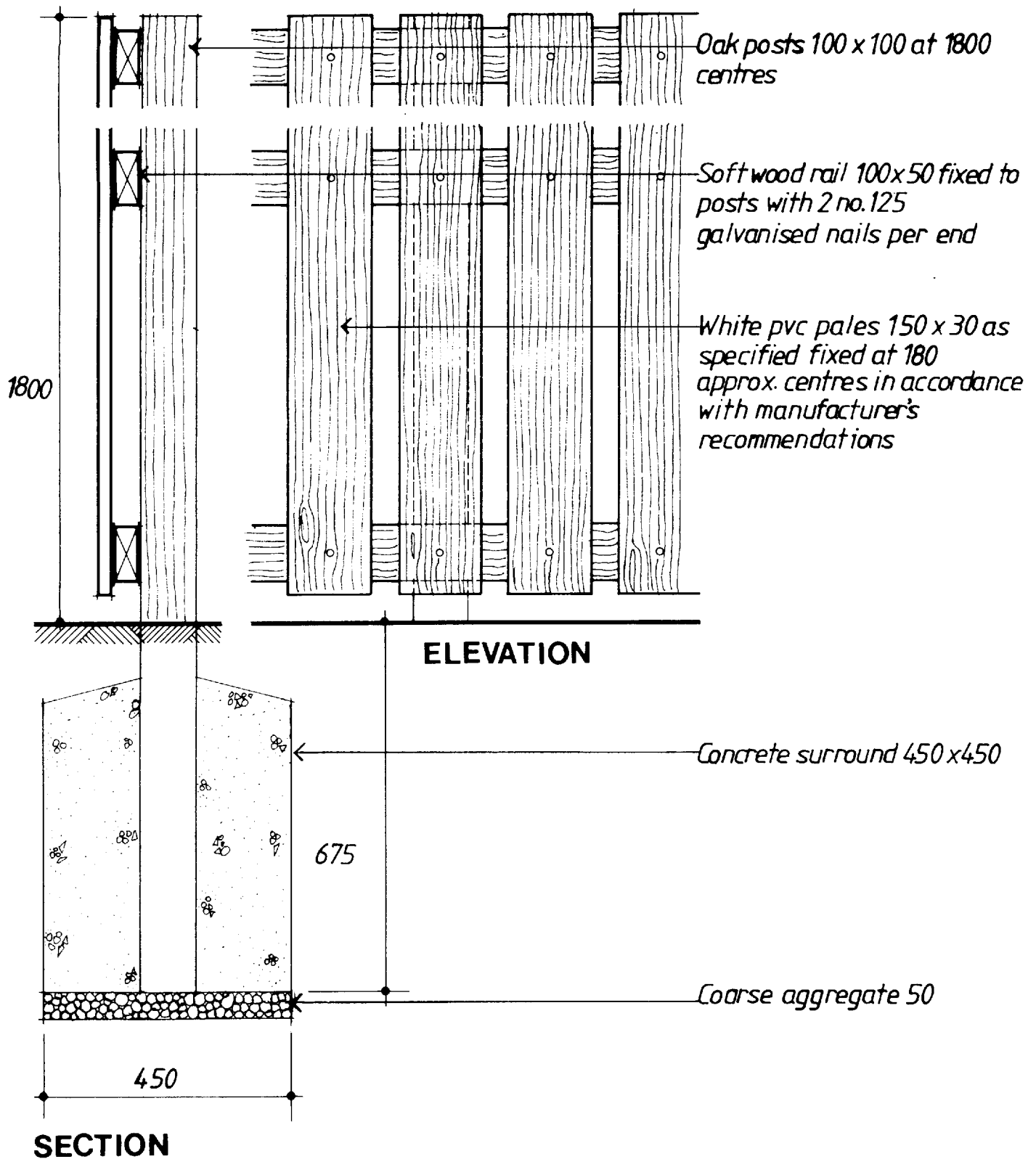


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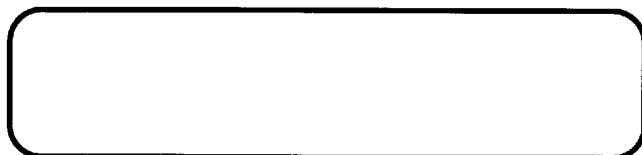


FENCE
diagonal board

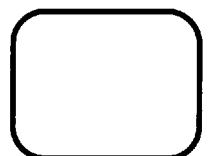


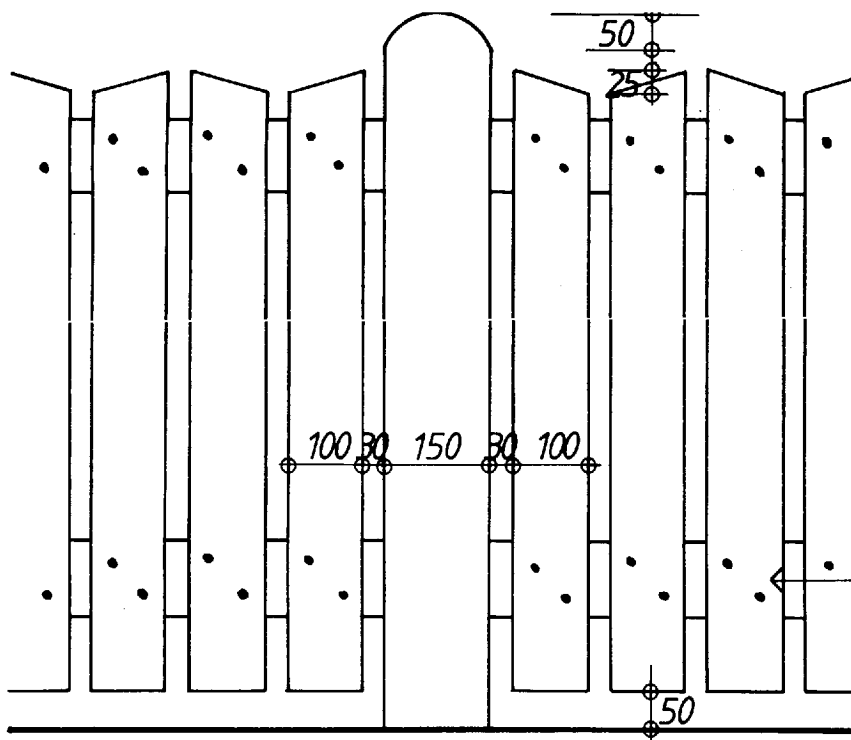


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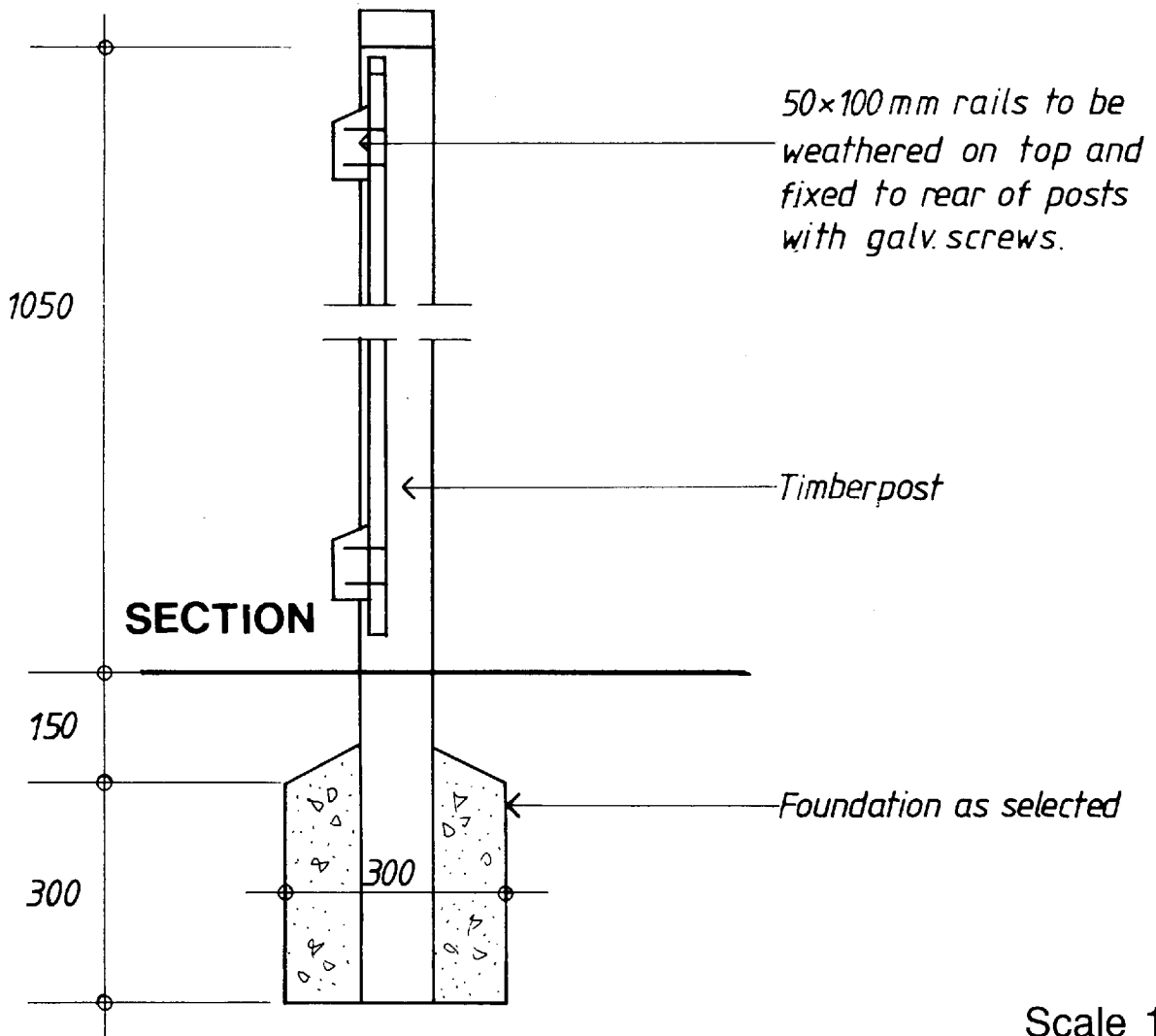
FENCE
timber & pvc palisade





Slats to be fixed to rails with threaded nails

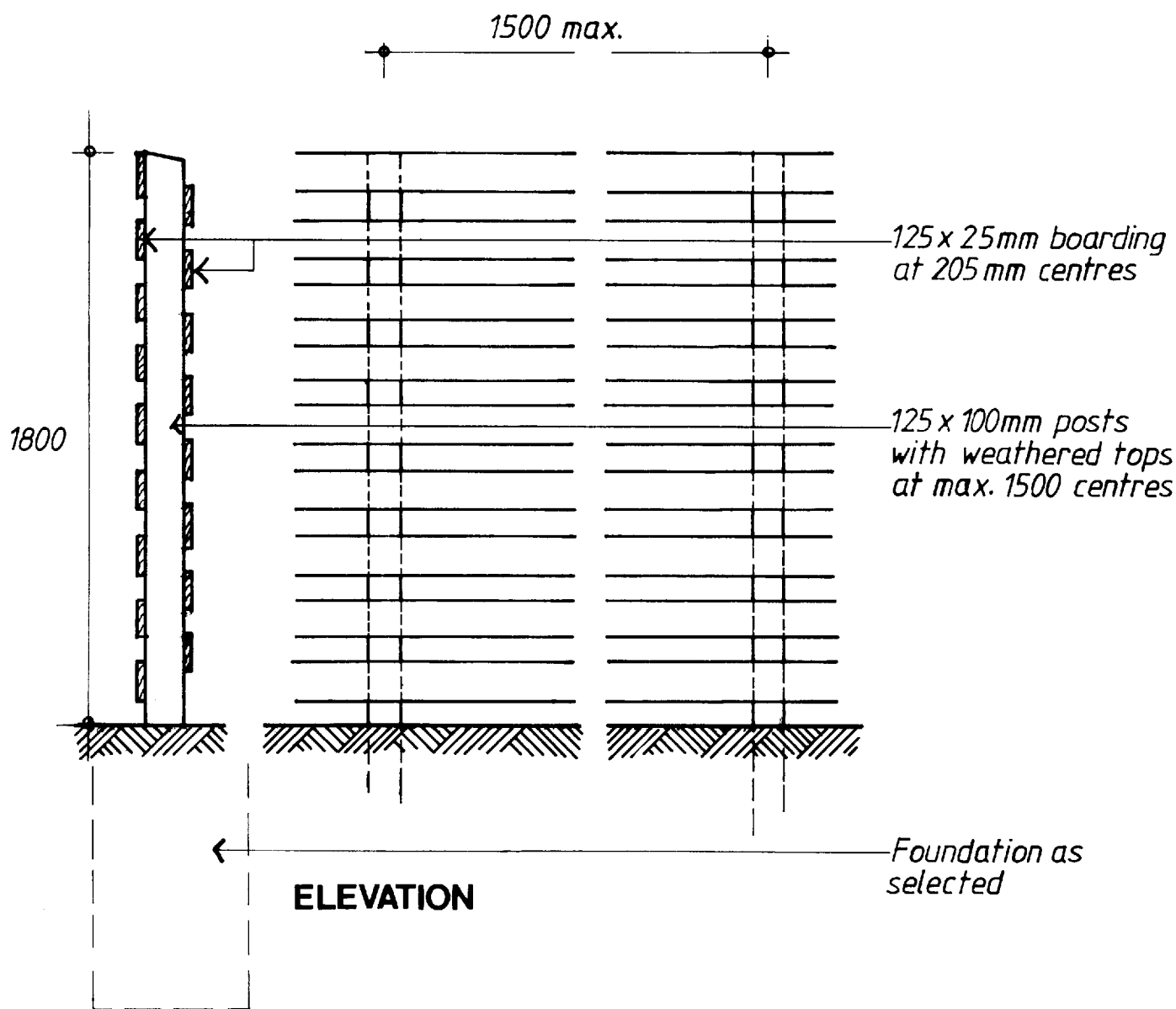
ELEVATION



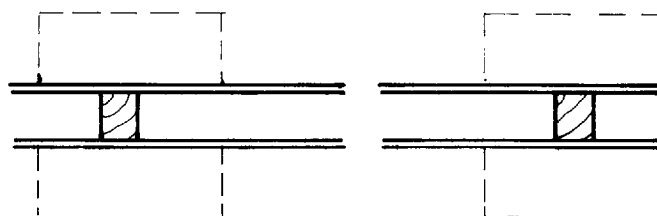
SECTION

Scale 1:10

FENCE
palisade – angled

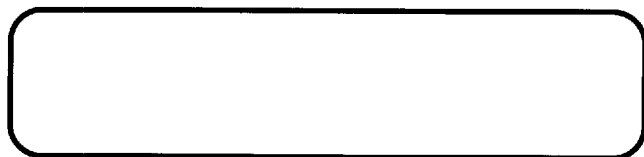


SECTION



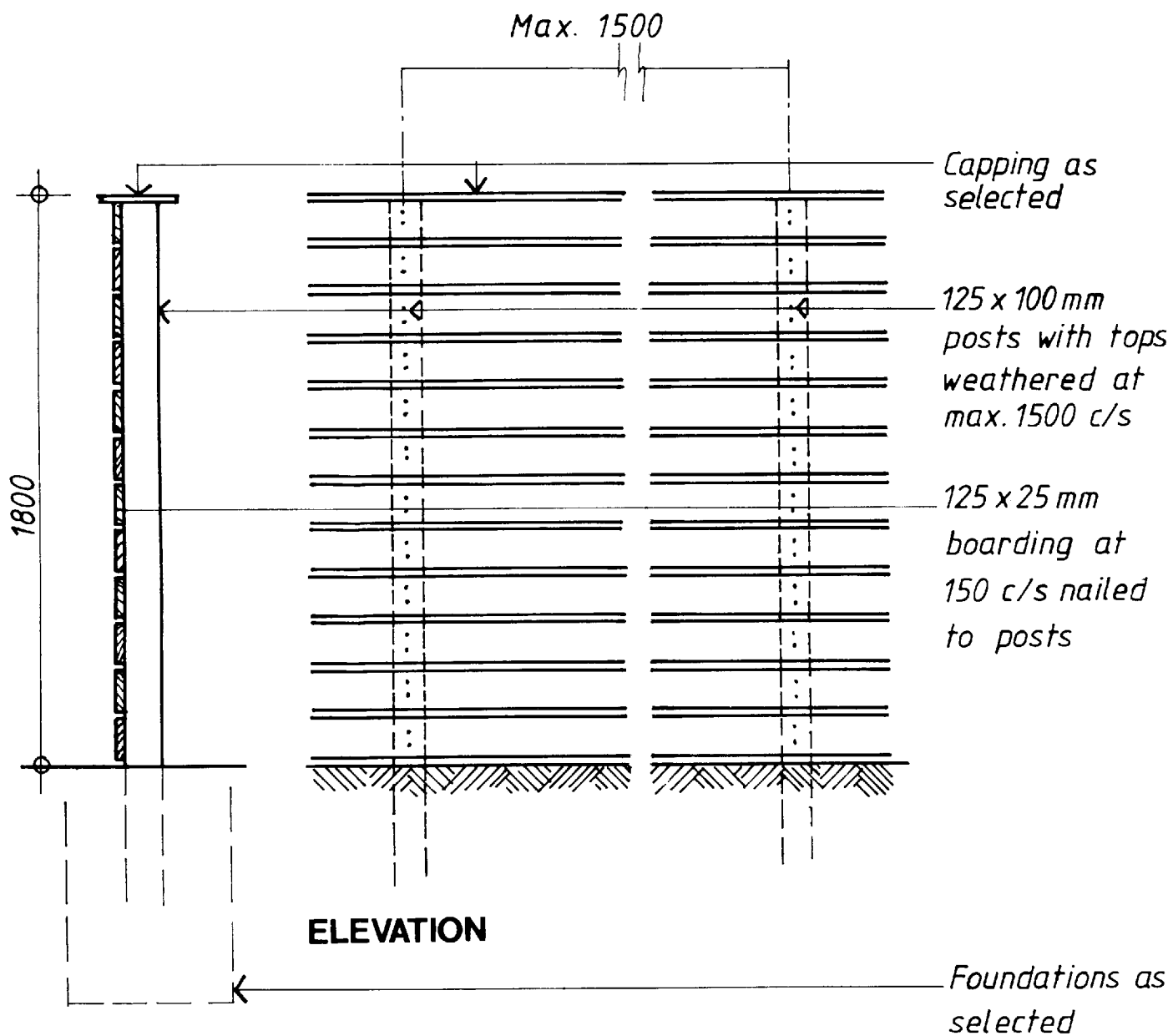
PLAN

Scale 1:20



FENCE
horizontal double
slatted



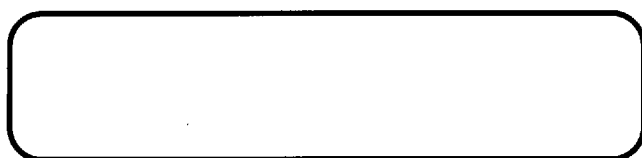


SECTION

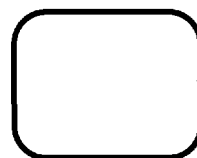


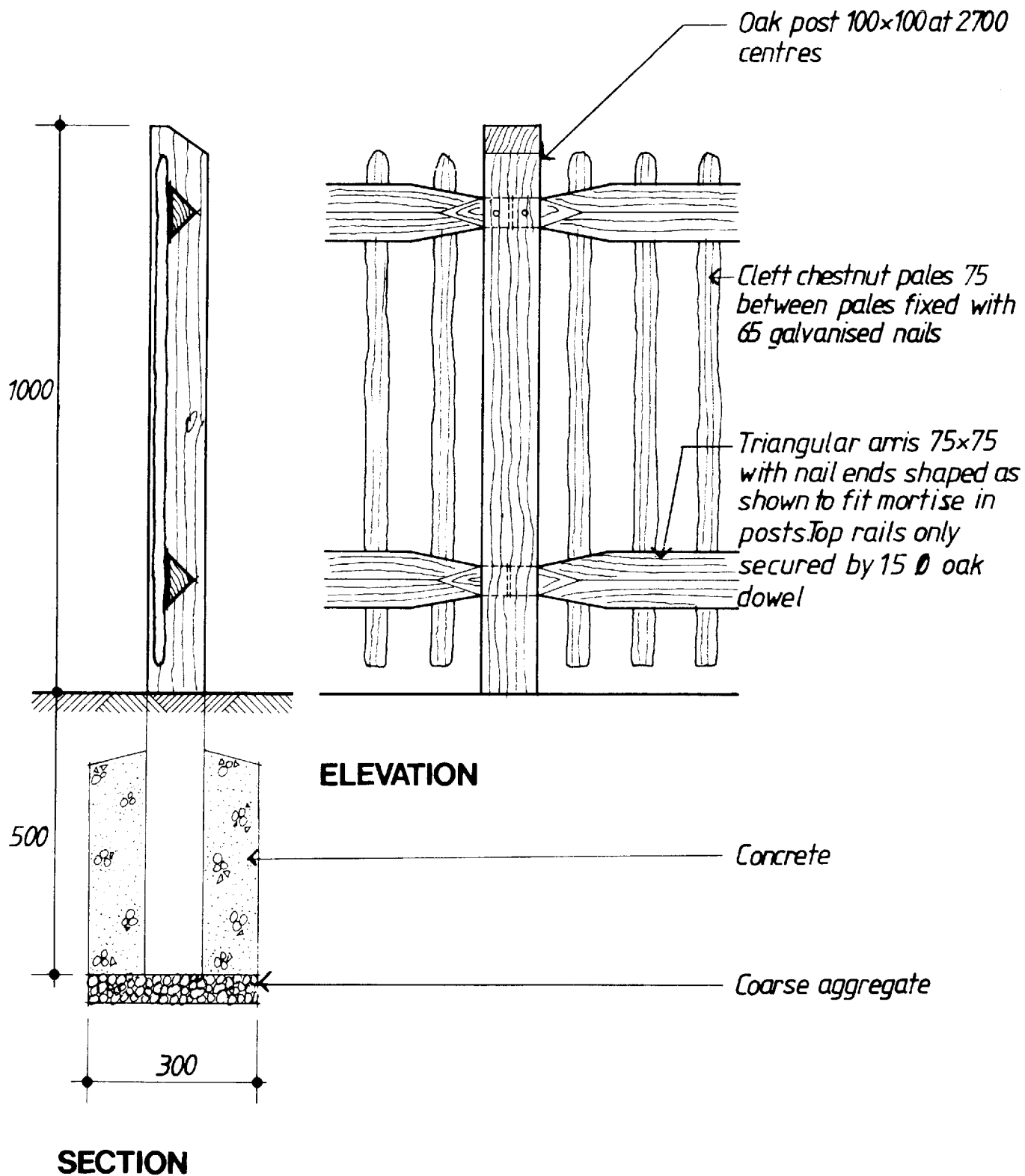
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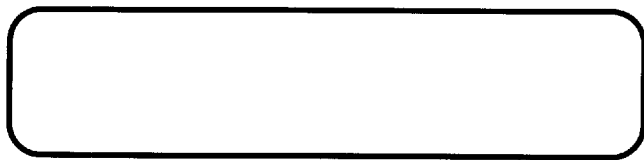


FENCE
horizontal open
slatted

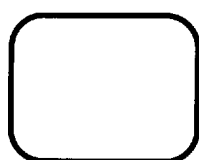


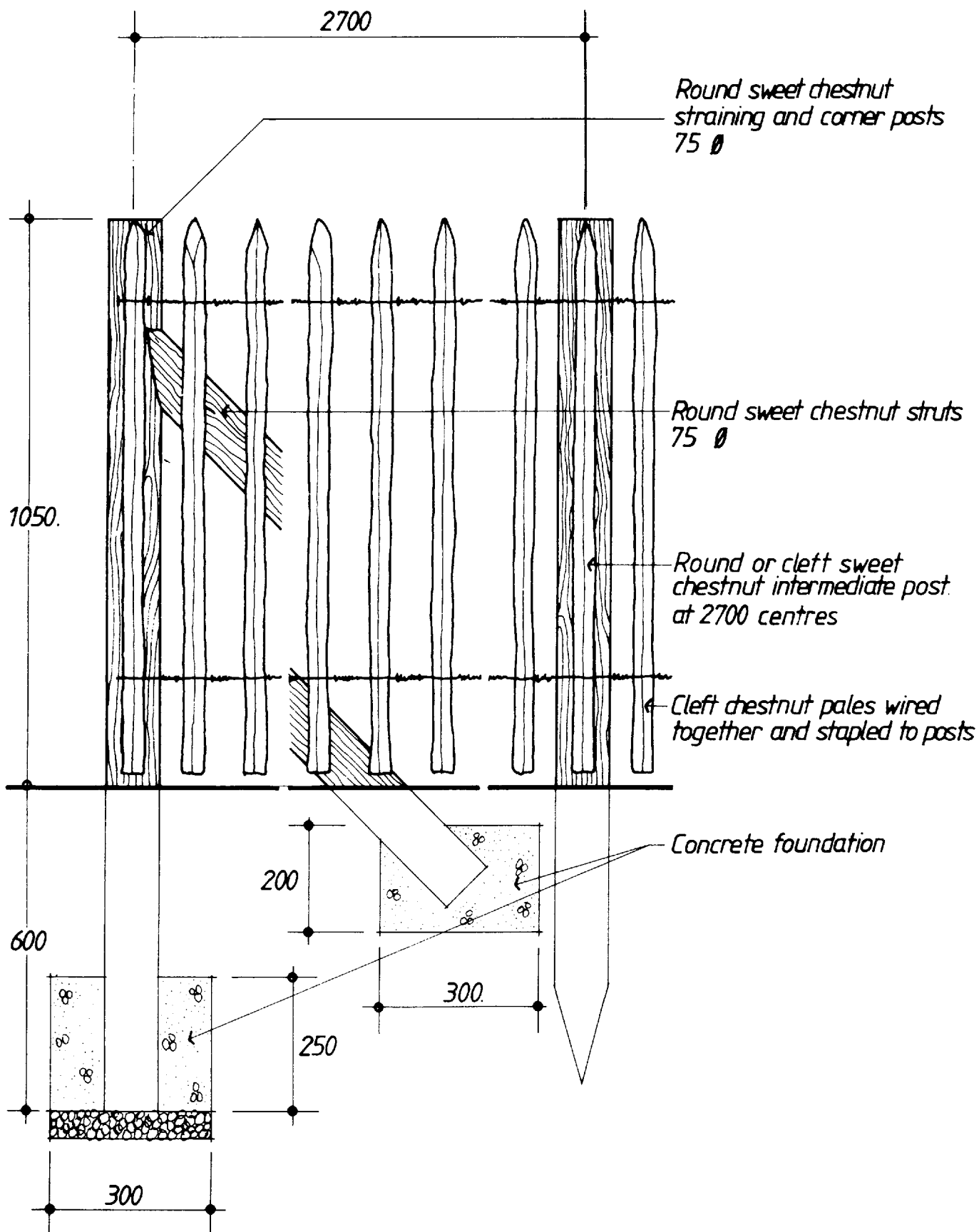


Scale 1:10



FENCE
cleft chestnut

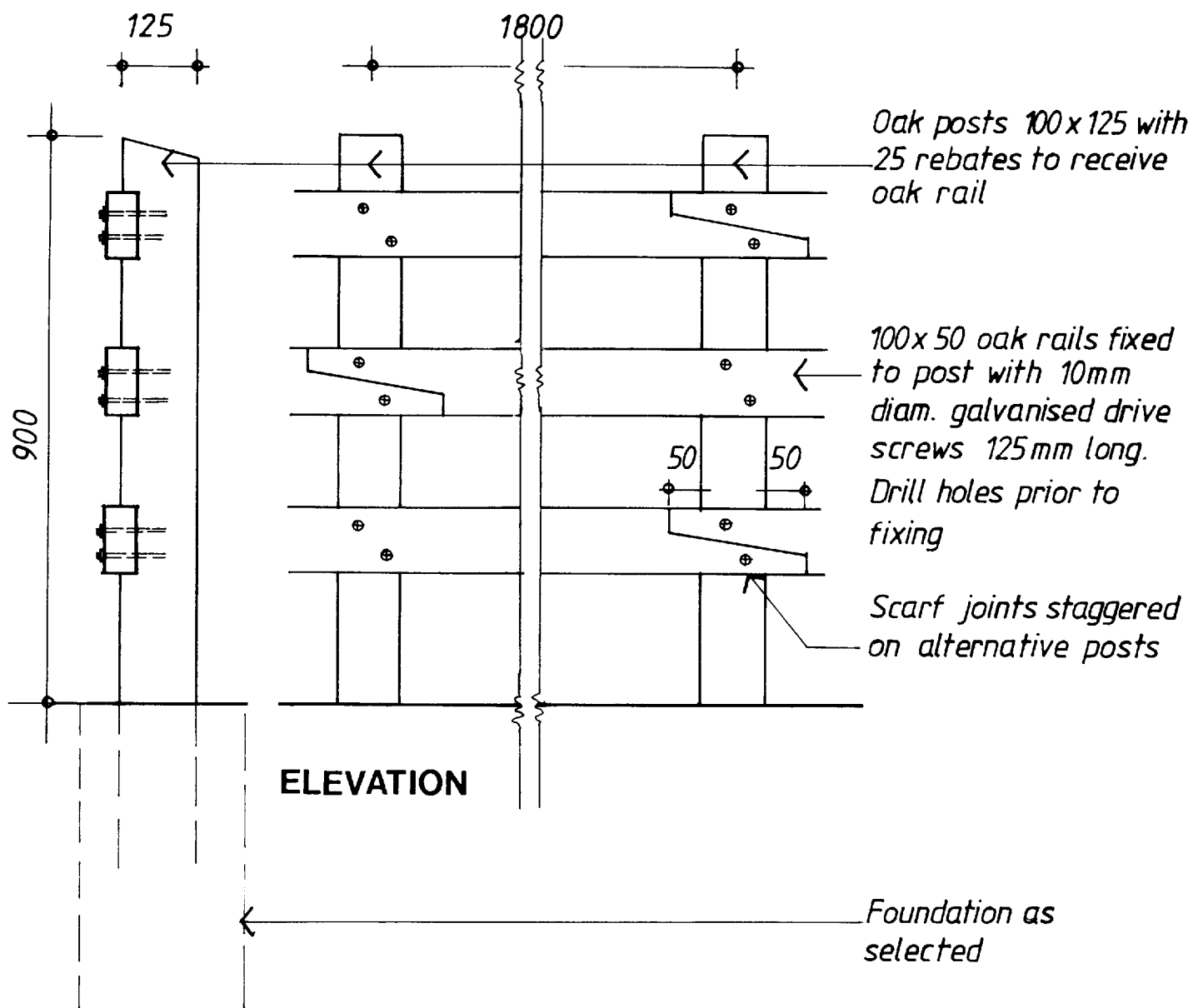




ELEVATION

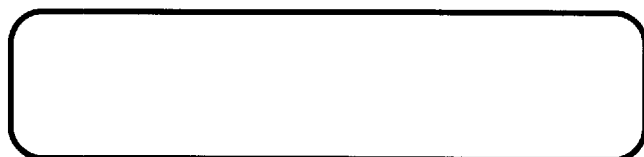
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FENCE
cleft chestnut & wire

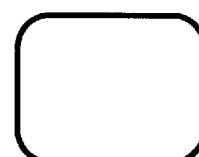


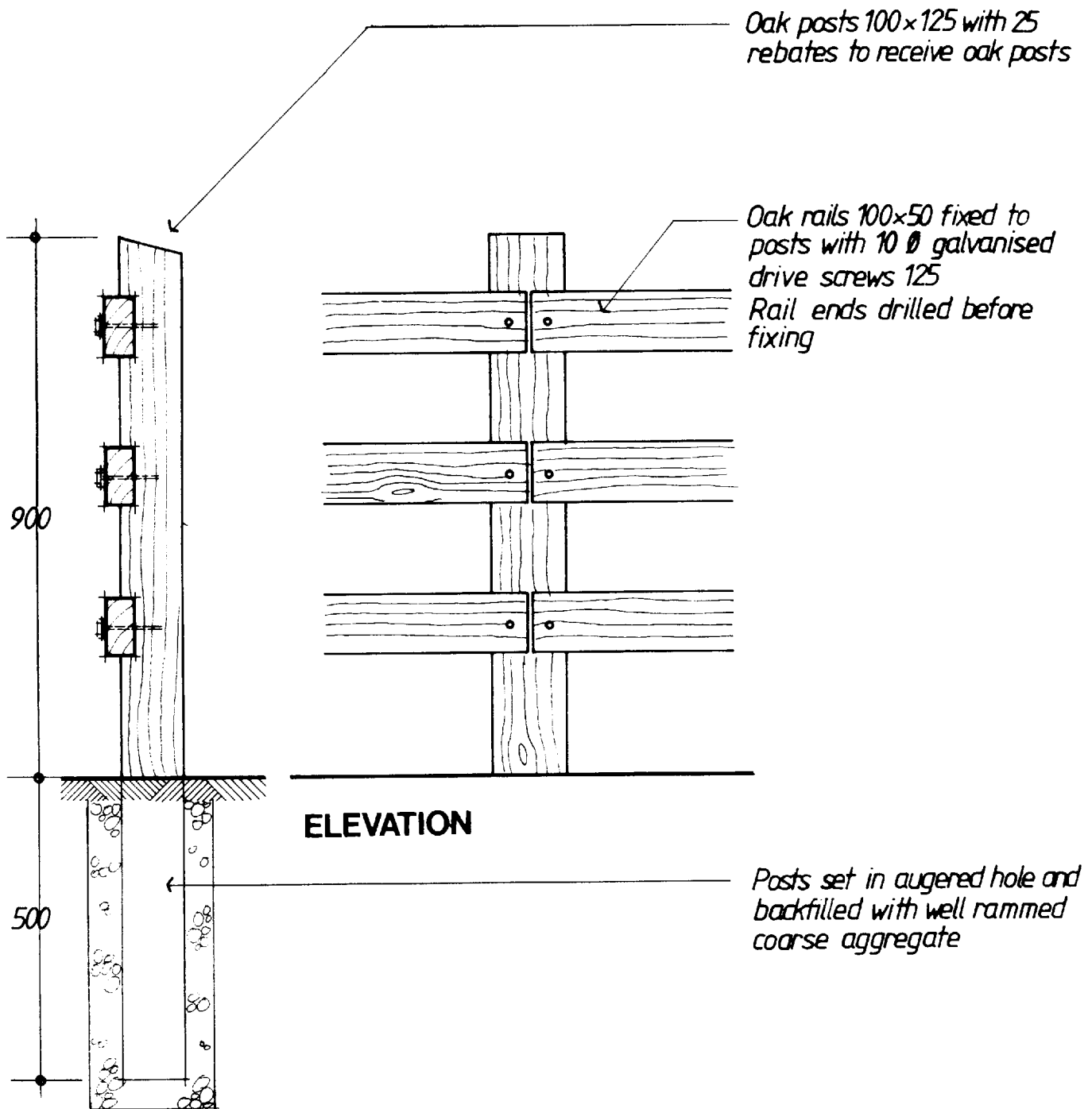
SECTION

Scale 1:10



FENCE
post & rail –
staggered joints



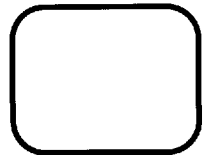


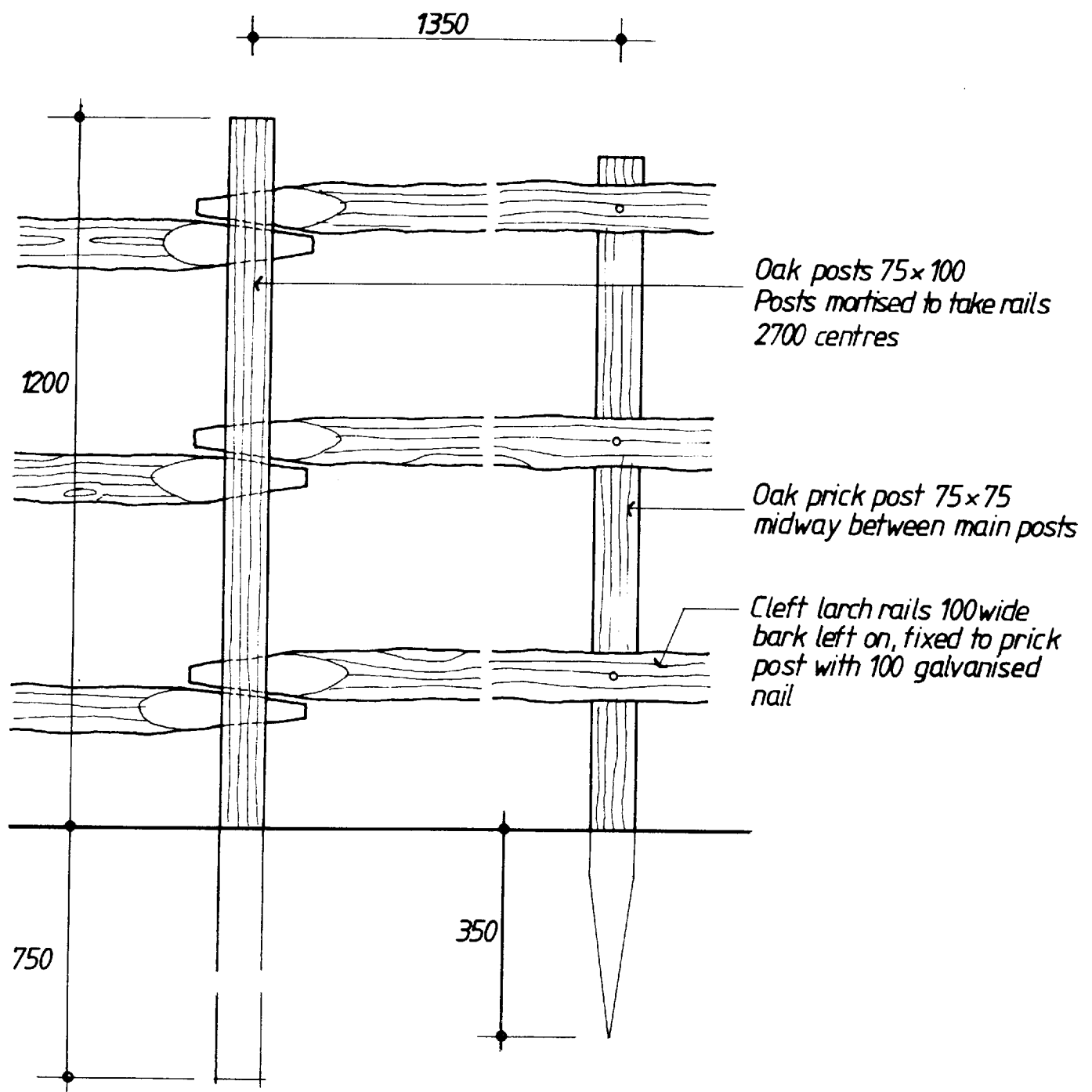
SECTION

Scale 1:10

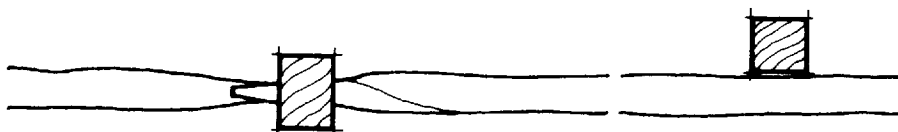


FENCE
post & rail –
butt joints



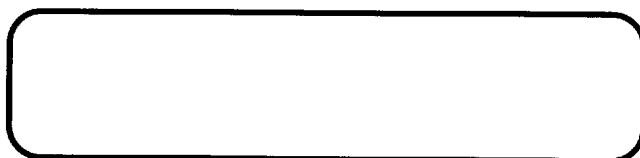


ELEVATION



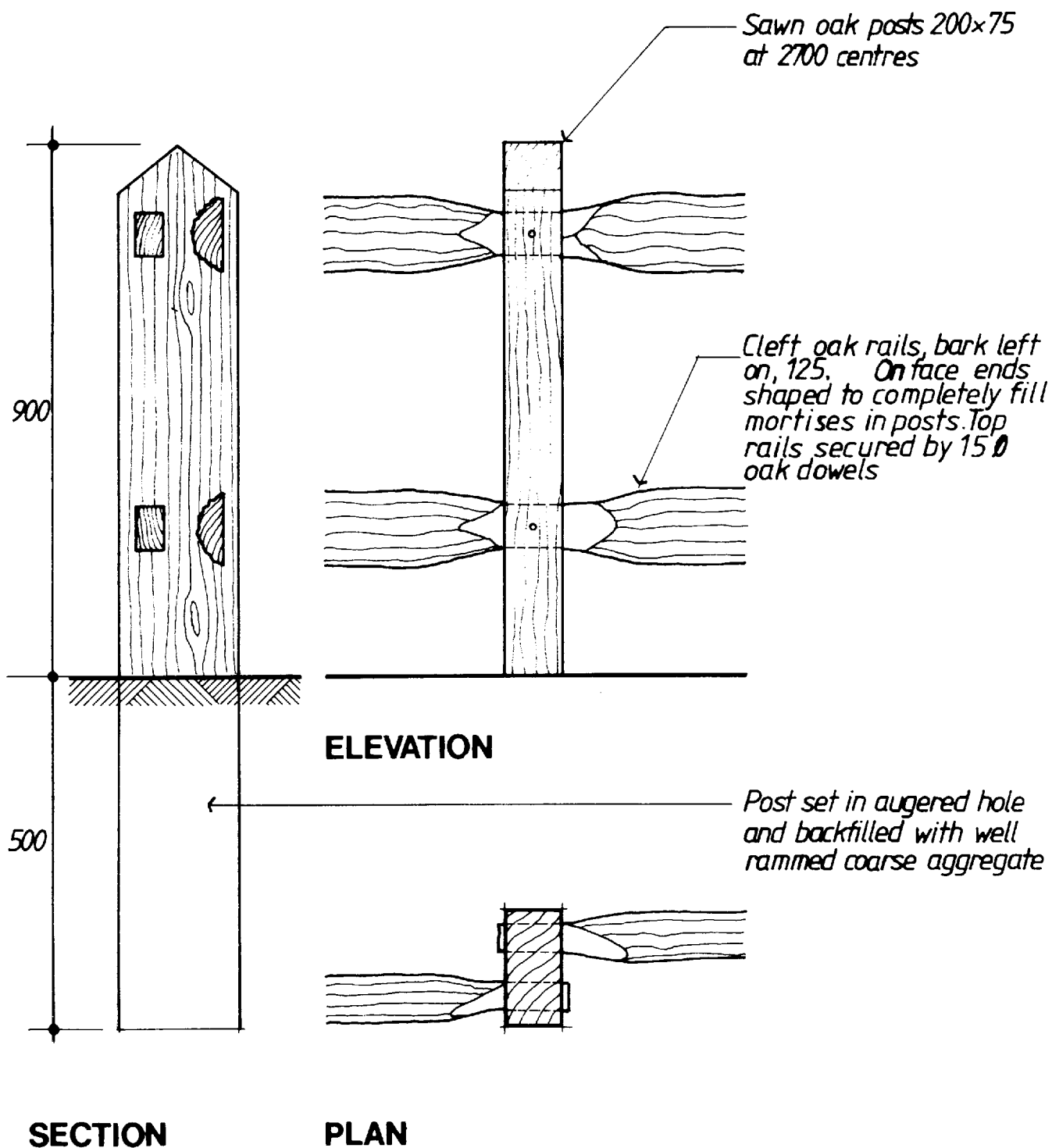
PLAN

Scale 1:10

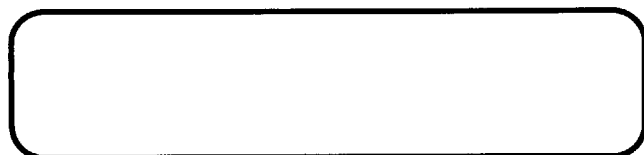


FENCE
Lincolnshire post & rail

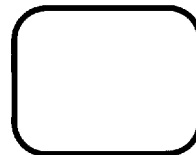


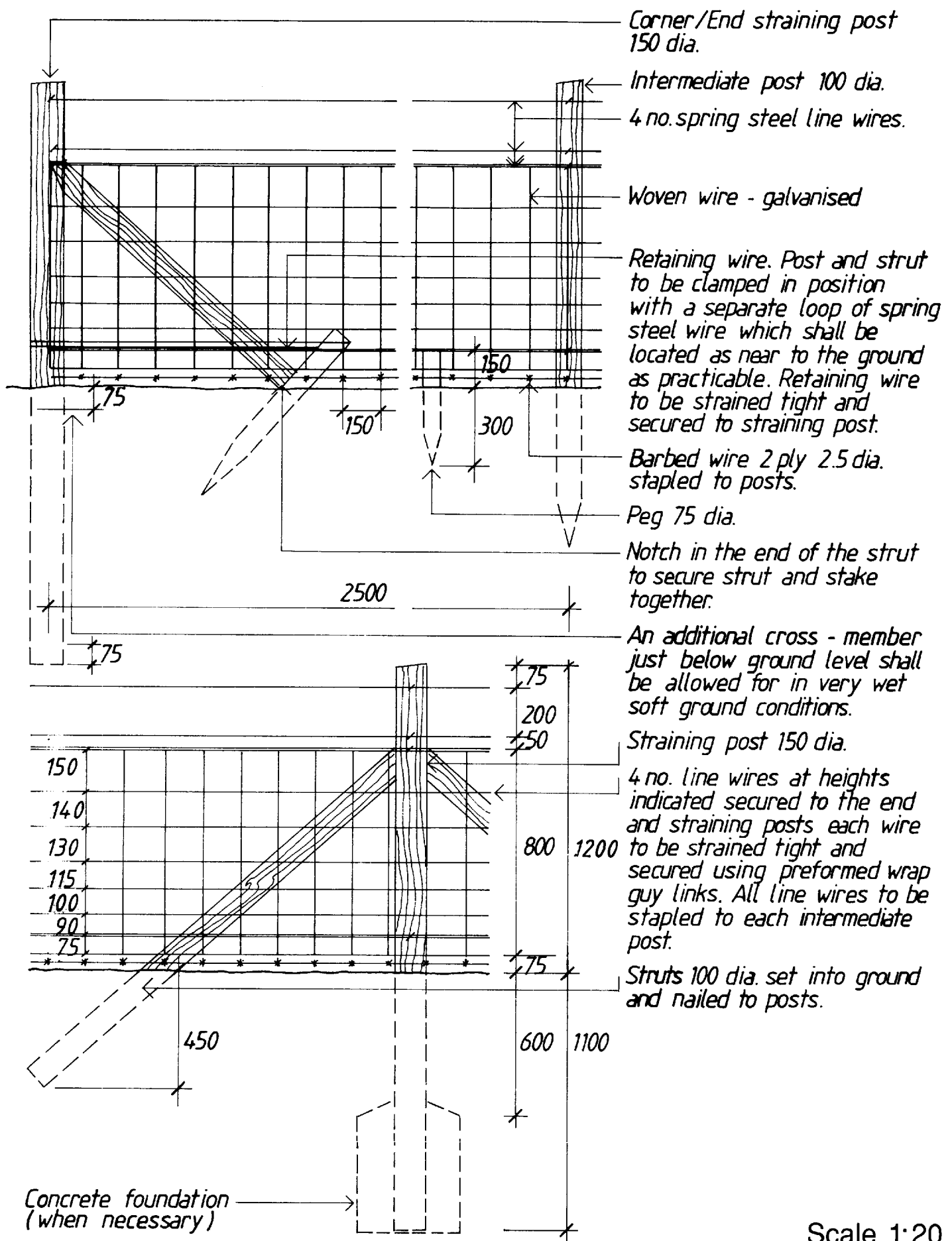


Scale 1:10

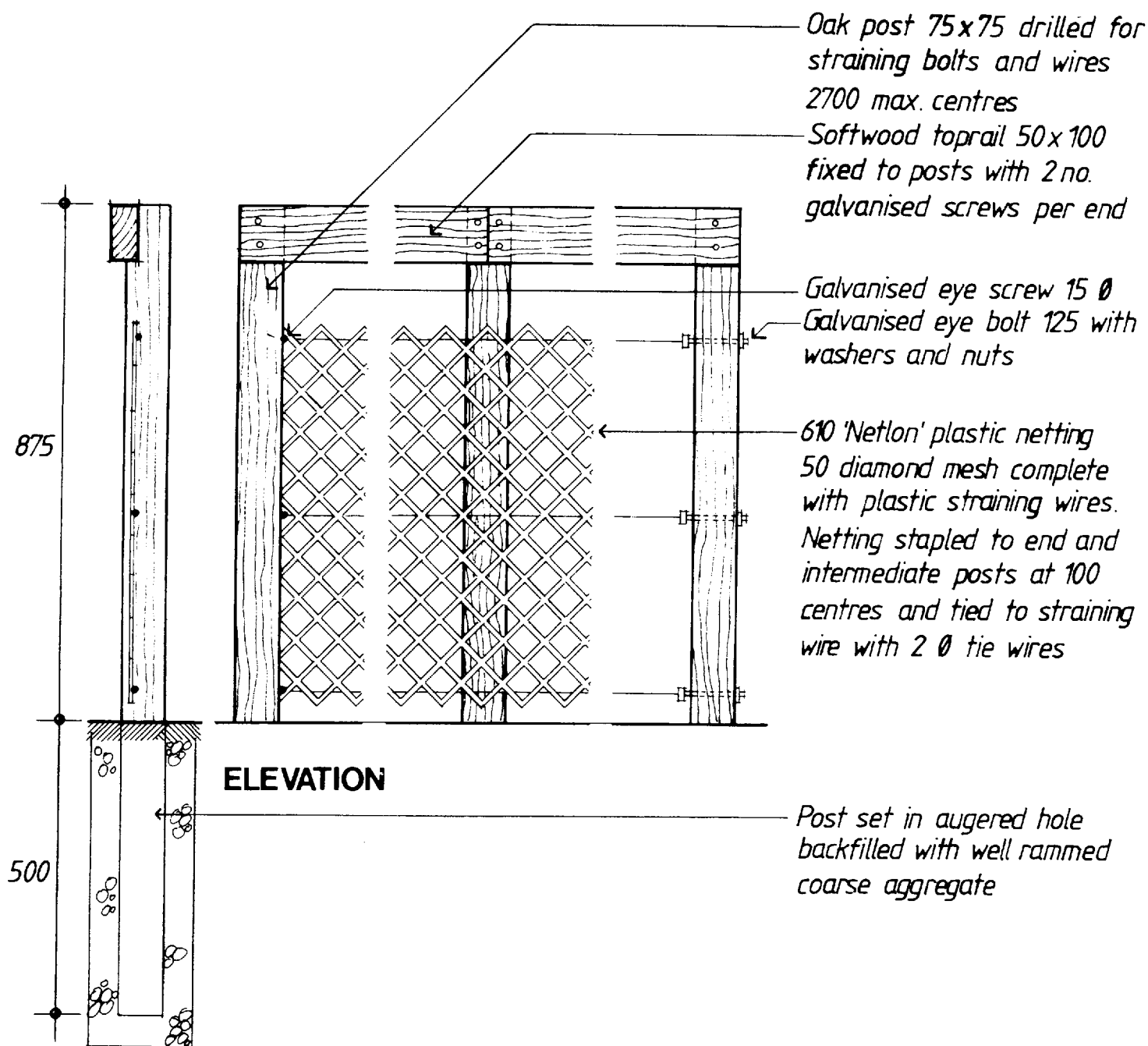


FENCE
two rail Sussex



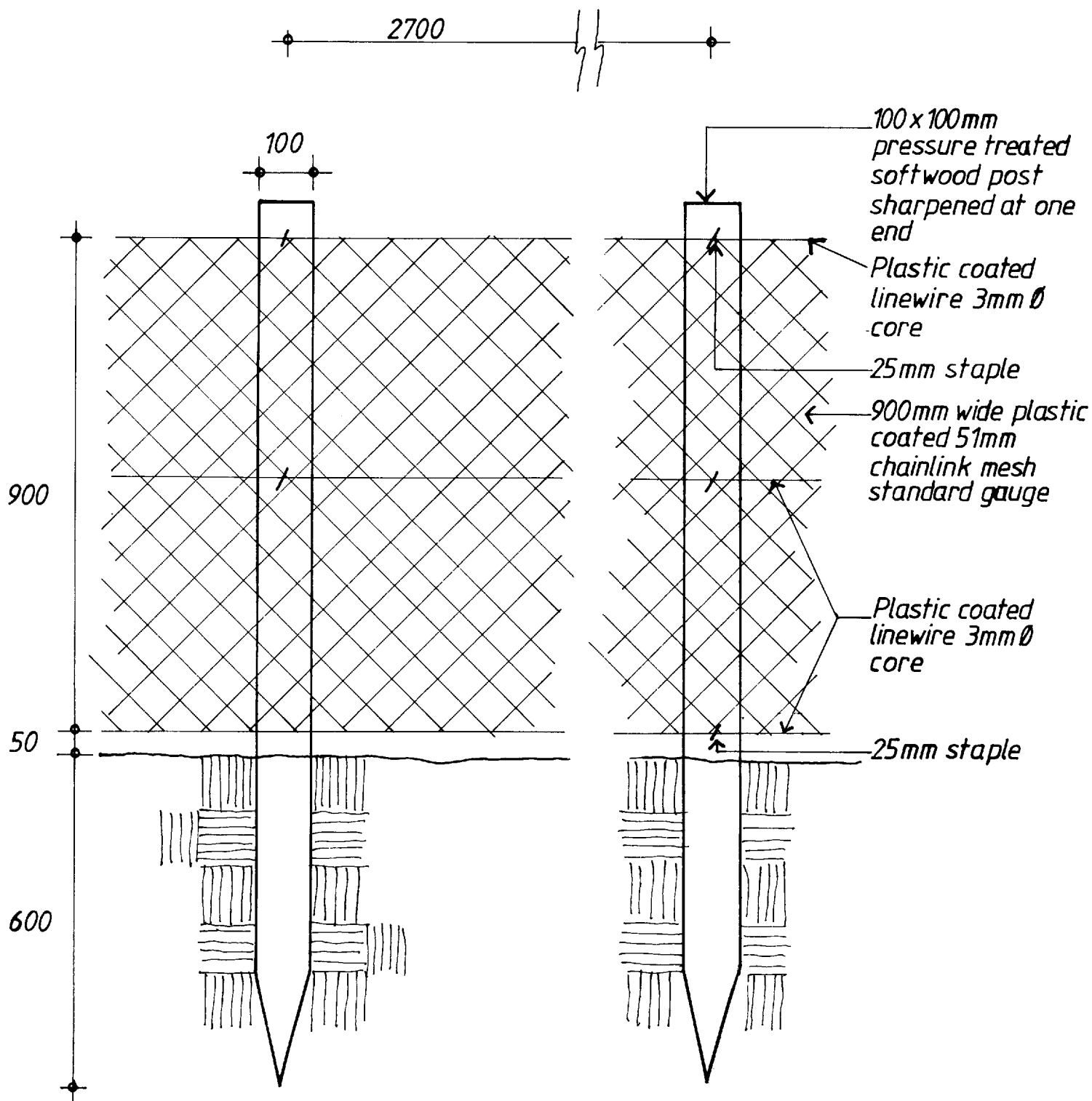


FENCE
post and wire



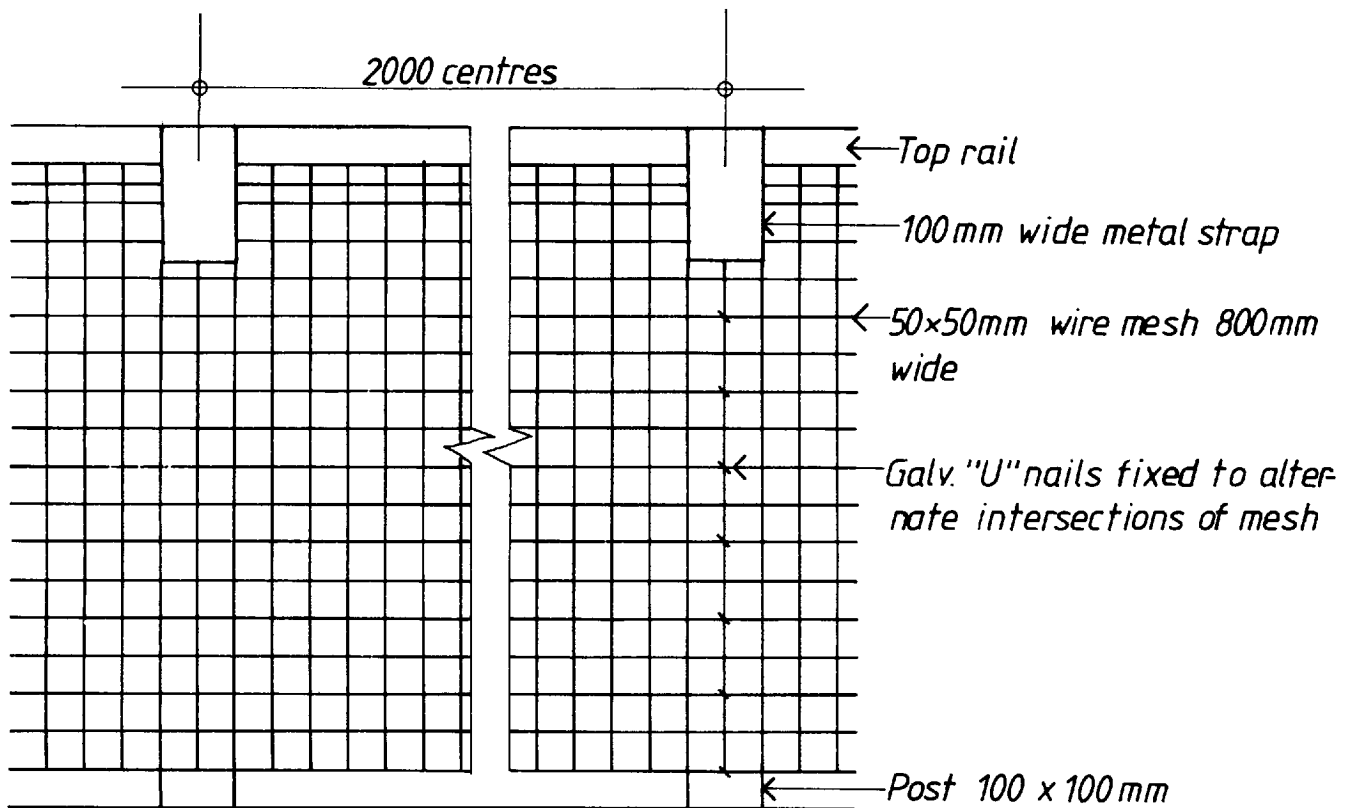
Scale 1:10

FENCE
timber and plastic mesh

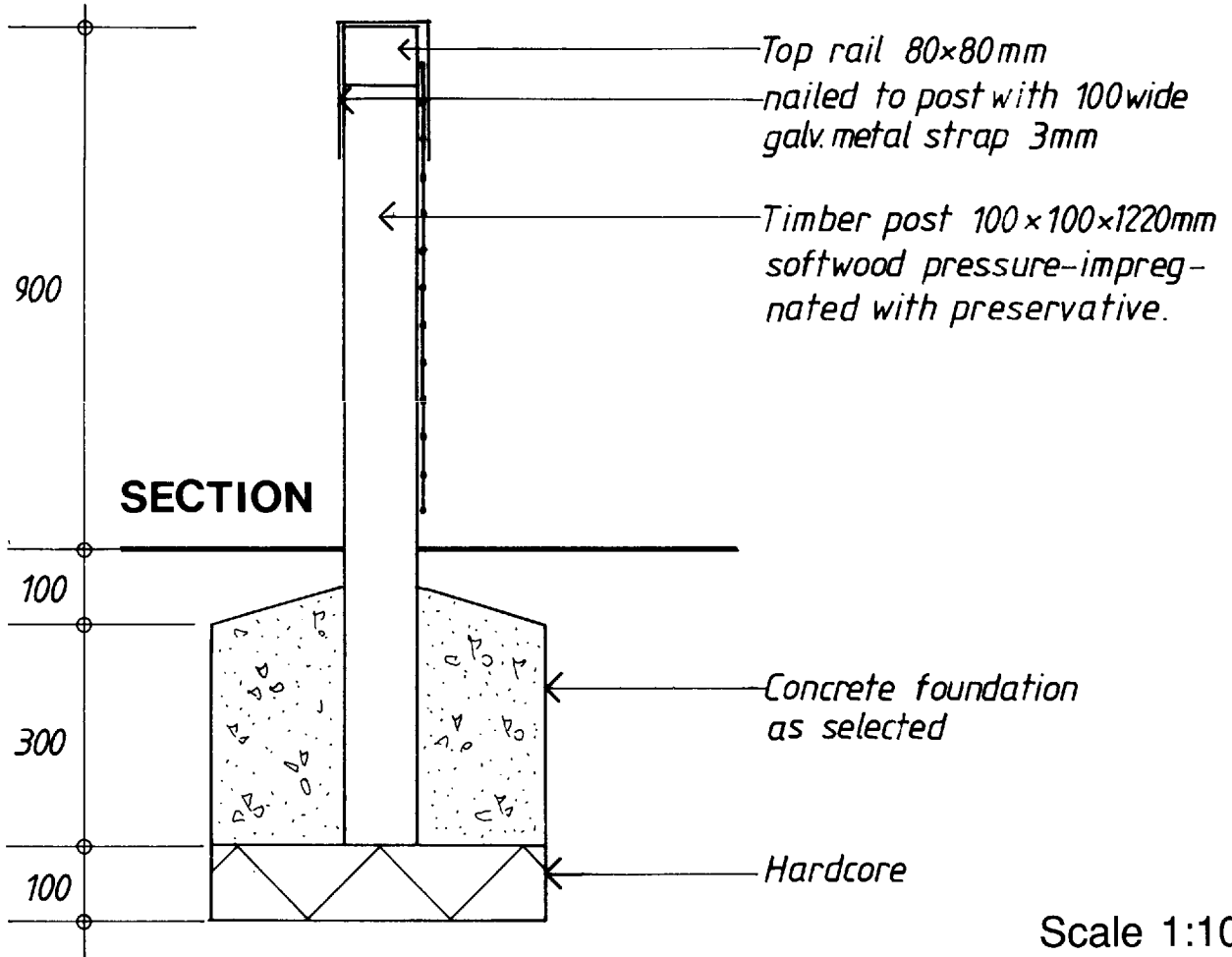


Scale 1:10

FENCE
low mesh



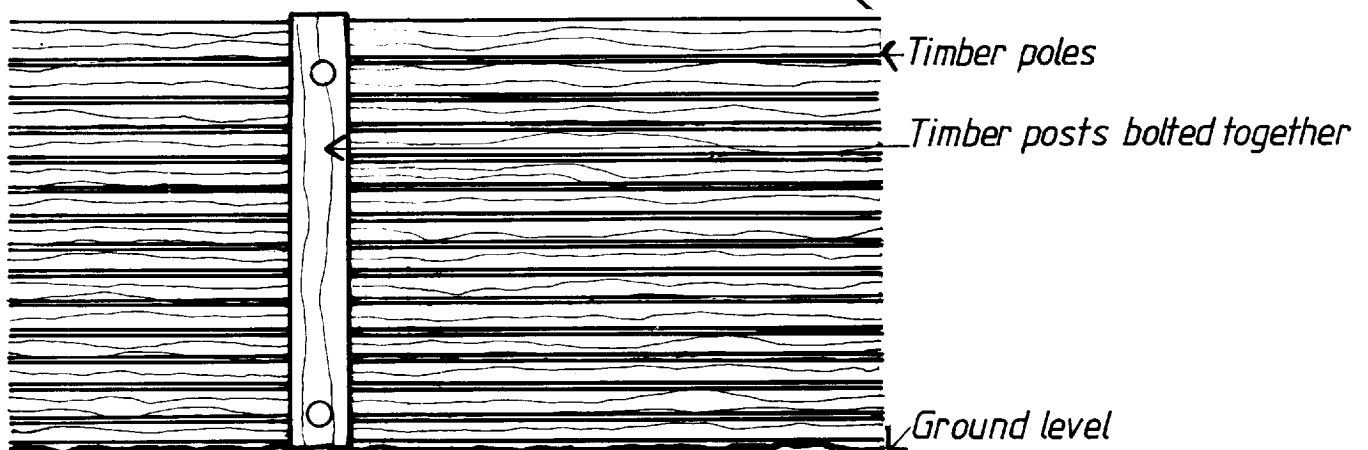
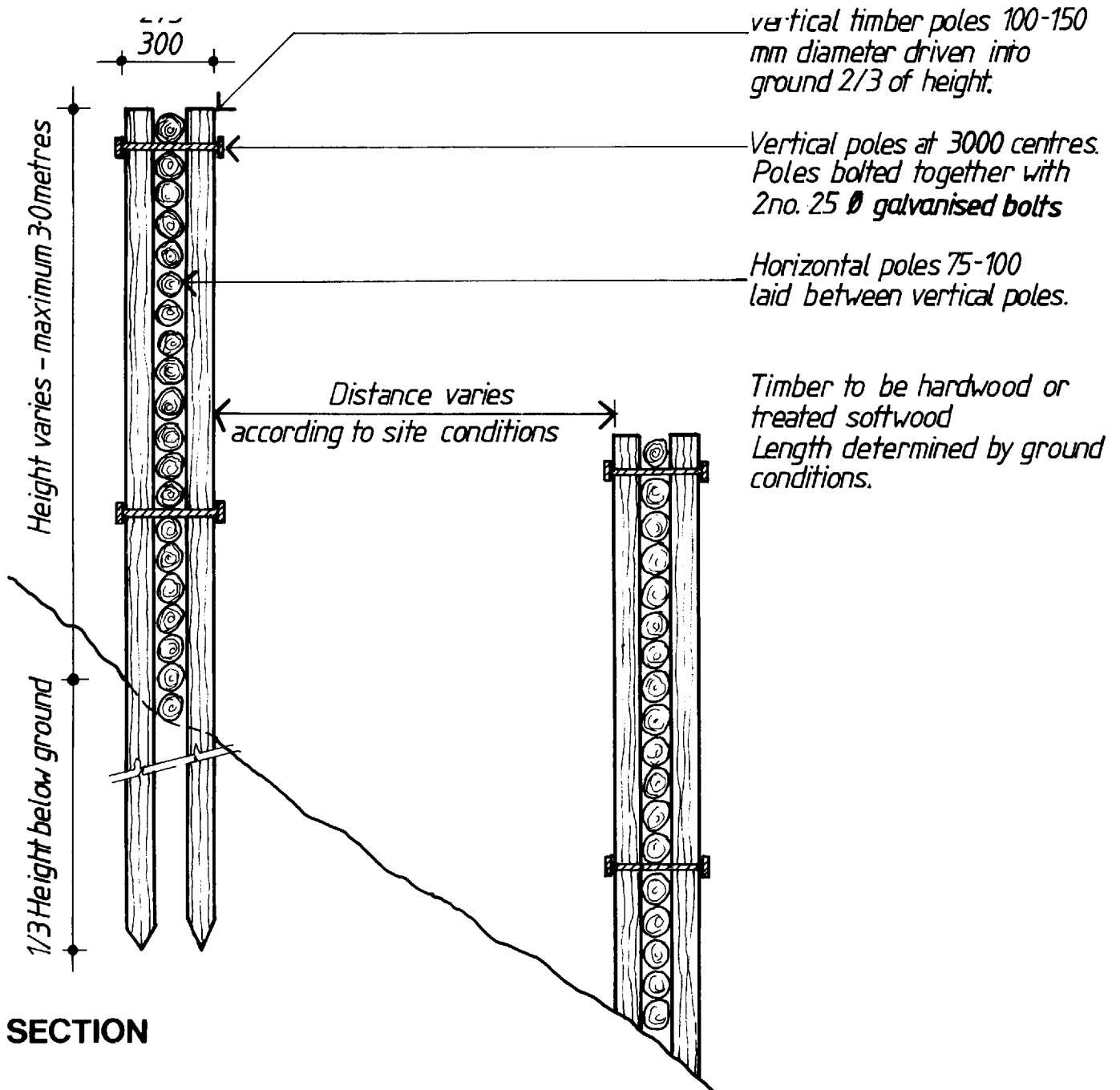
ELEVATION



SECTION

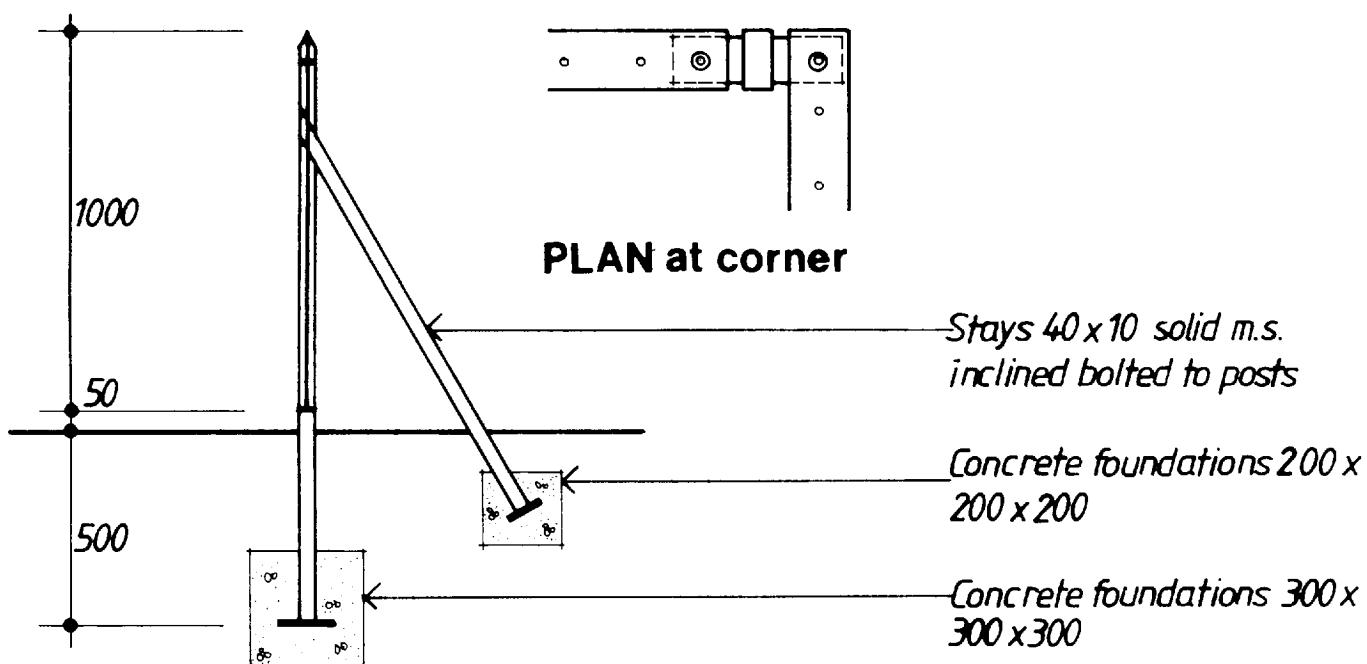
Scale 1:10

FENCE
low mesh

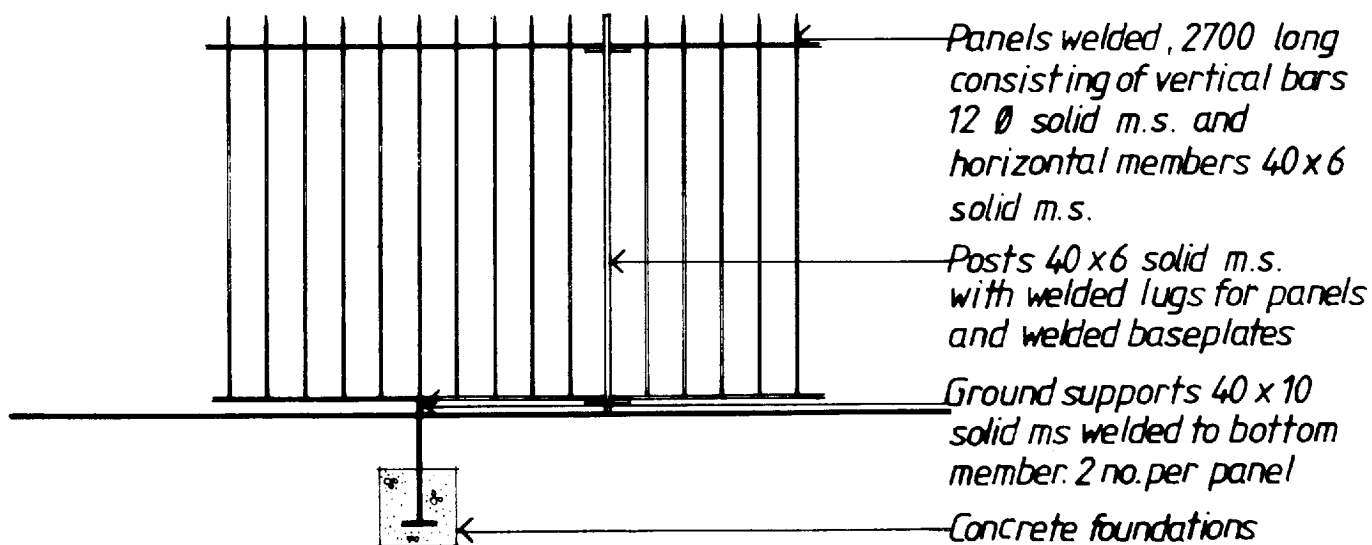


Scale 1:20

ROCK RETENTION
Fence – timber poles



SECTION

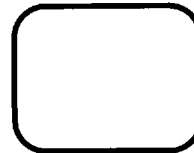


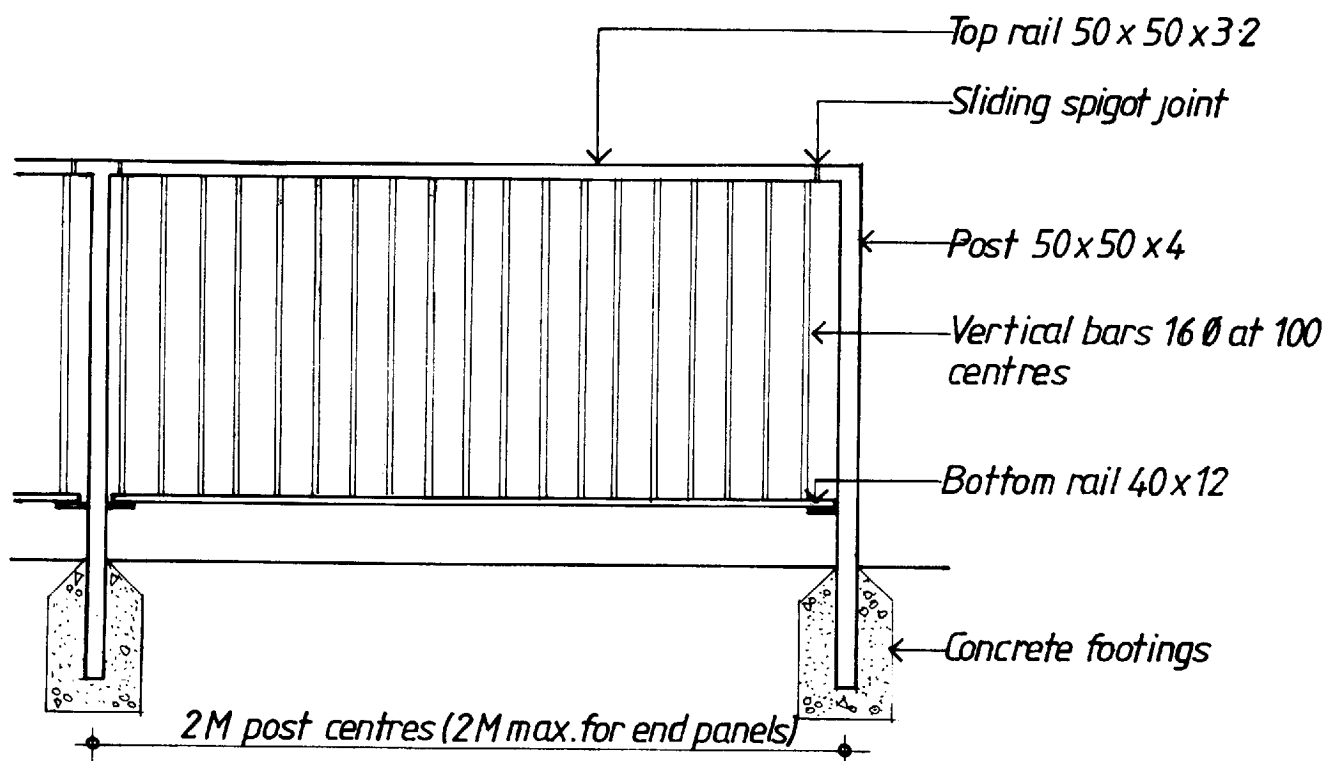
ELEVATION

Scale 1:20

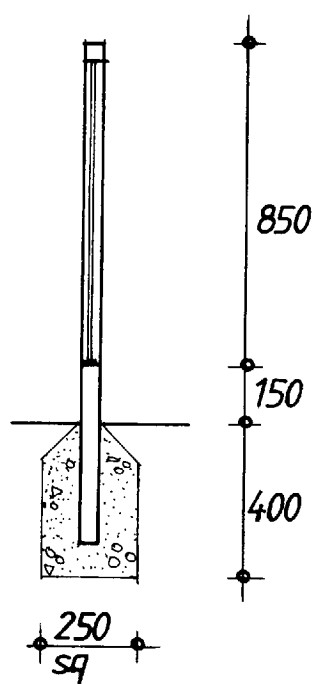


FENCE
vertical bar – medium



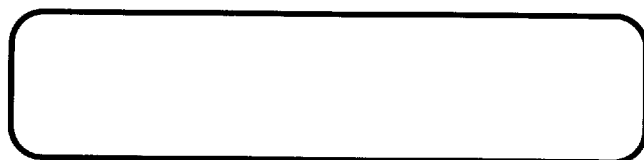


ELEVATION

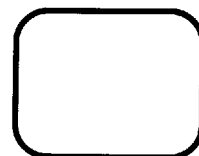


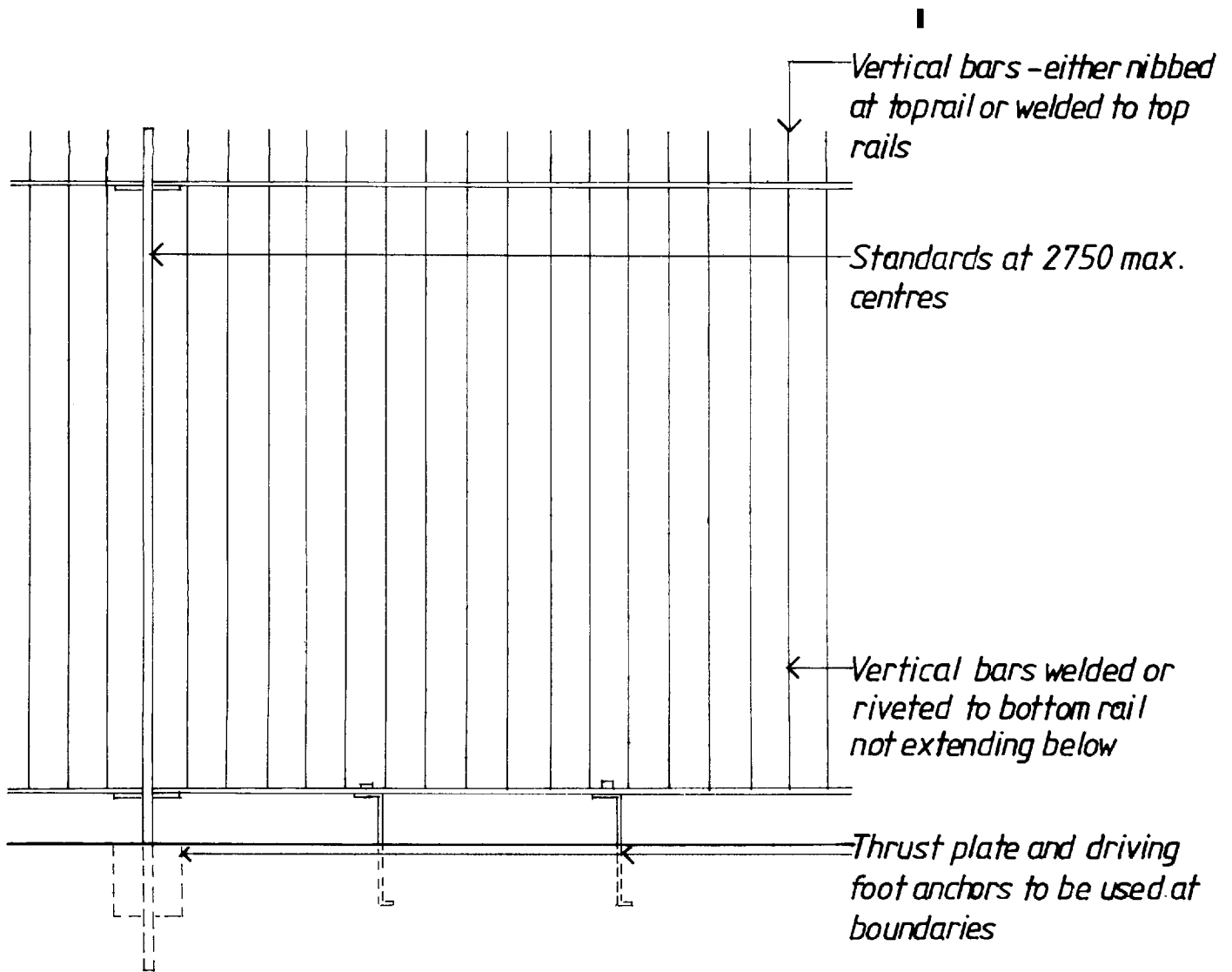
SECTION

Scale 1:20



FENCE
vertical bar – medium





ELEVATION

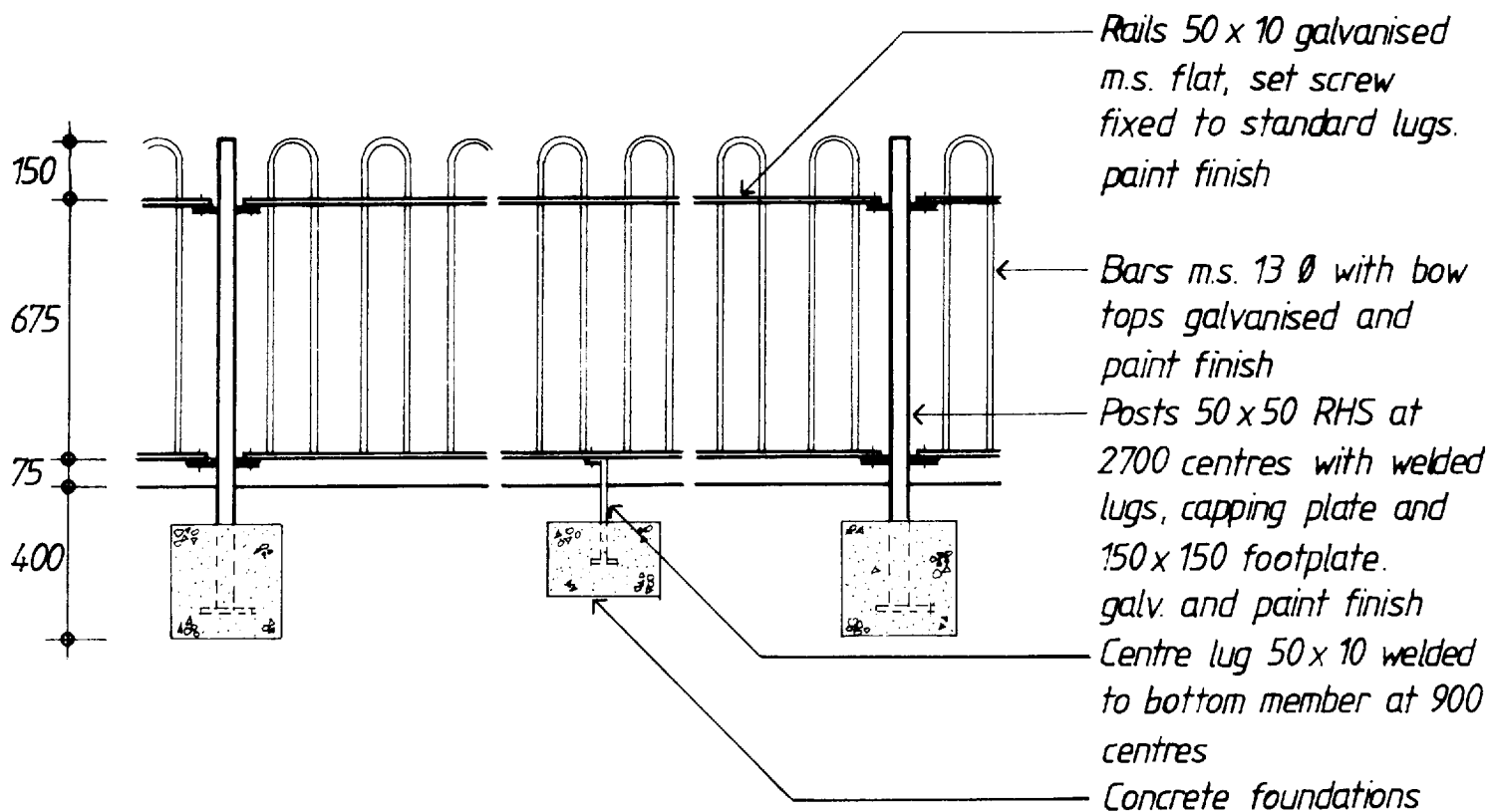
Height of railings vary from approx. 1200 - 2200

Scale 1:20

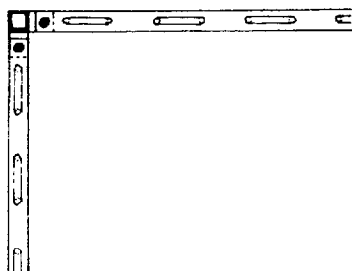


FENCE
vertical bar – tall



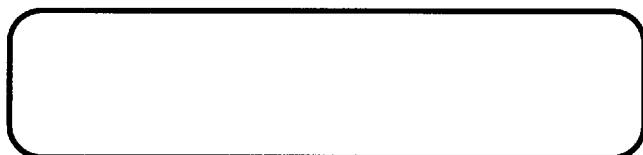


ELEVATION



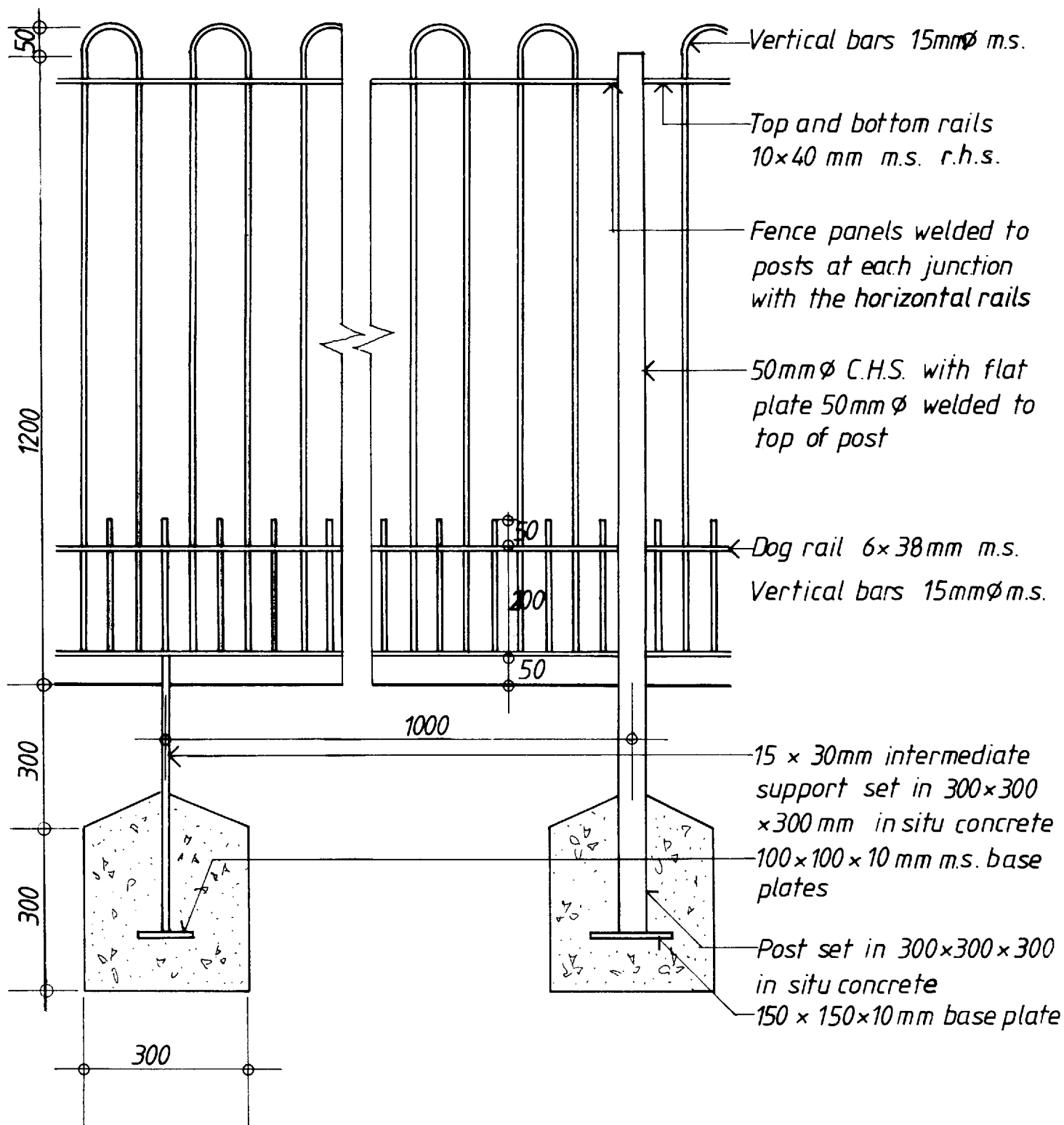
PLAN at corner

Scale 1:20



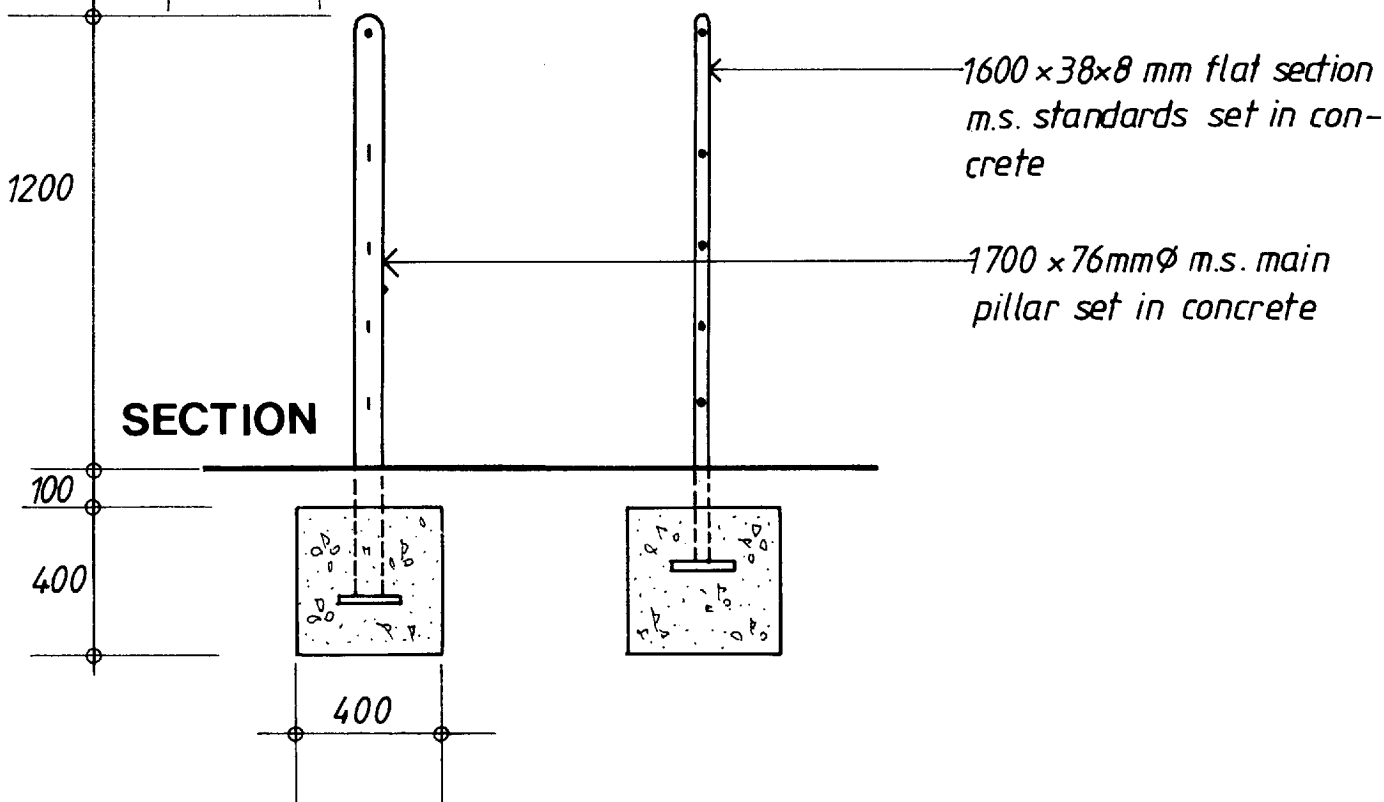
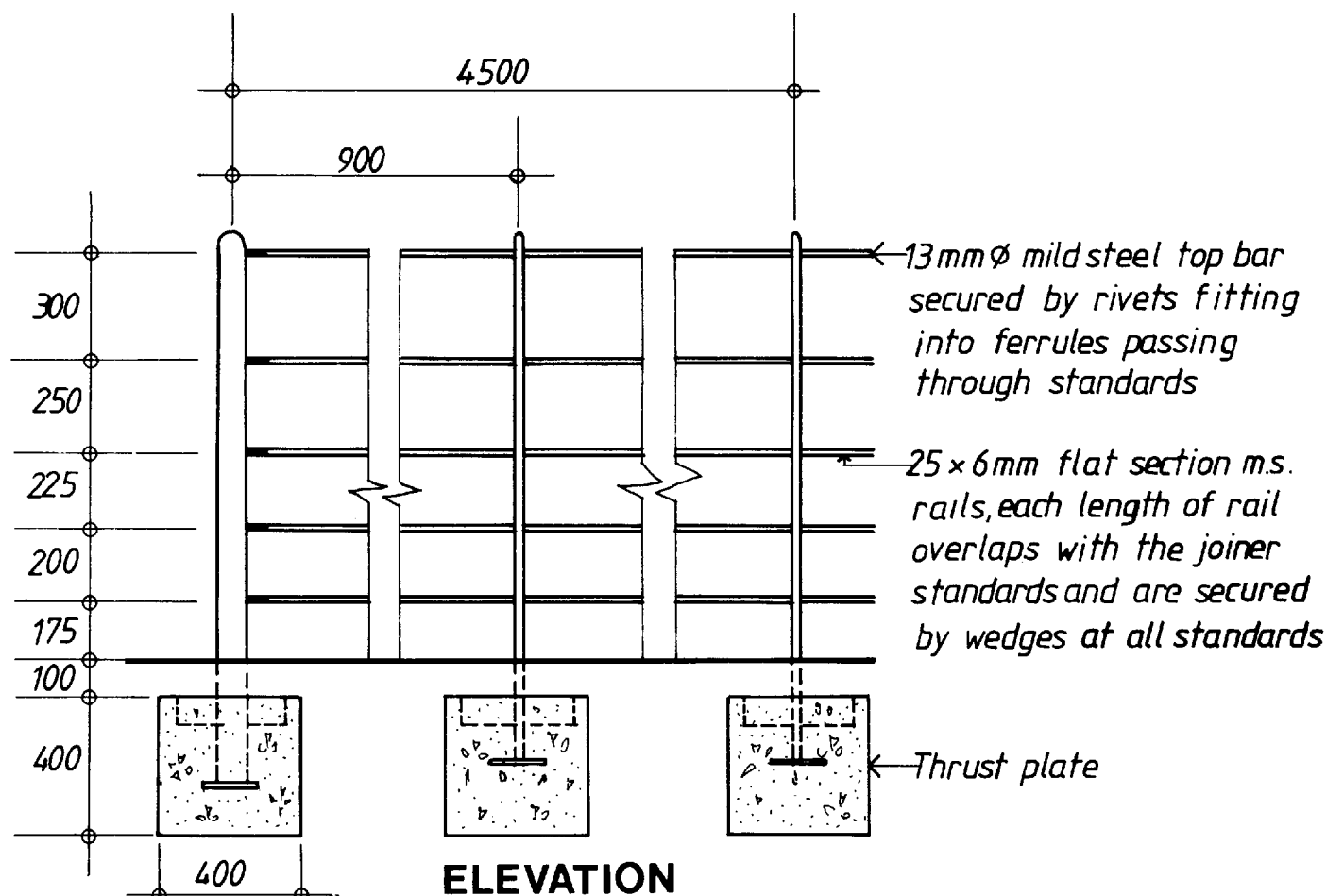
FENCE
vertical bow topped –
low





Scale 1:10

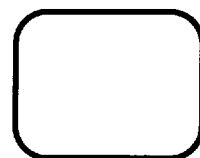
FENCE
vertical bow topped –
medium

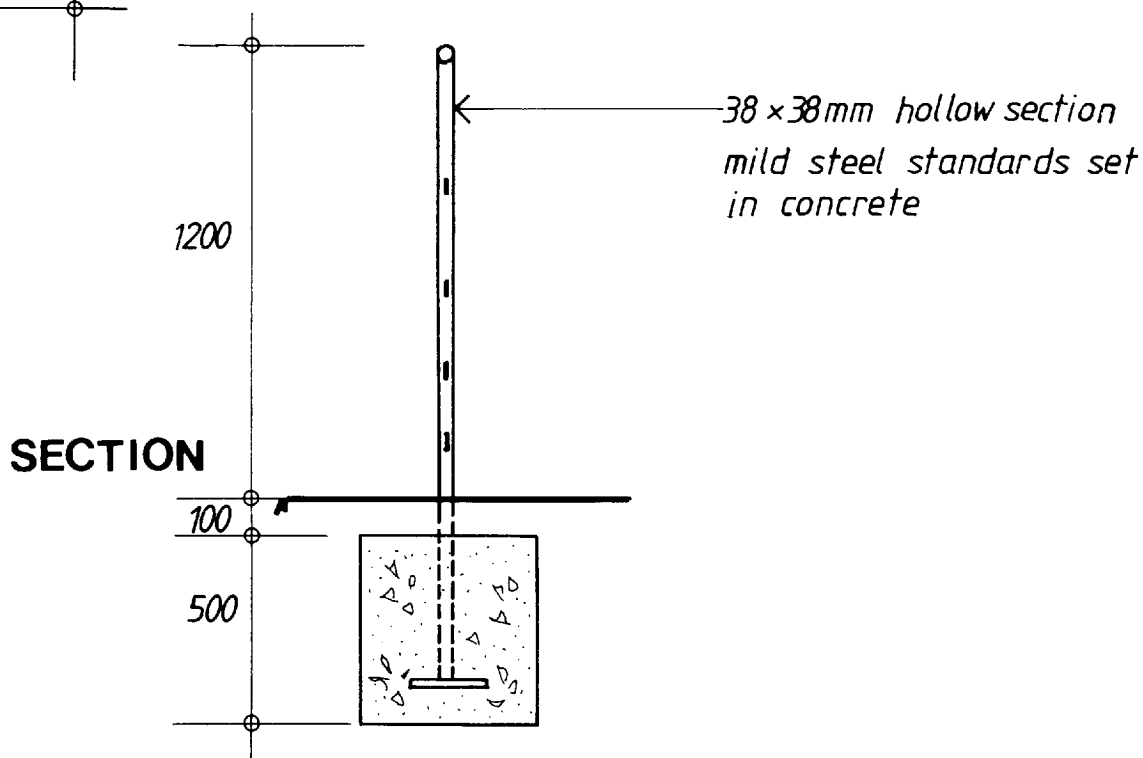
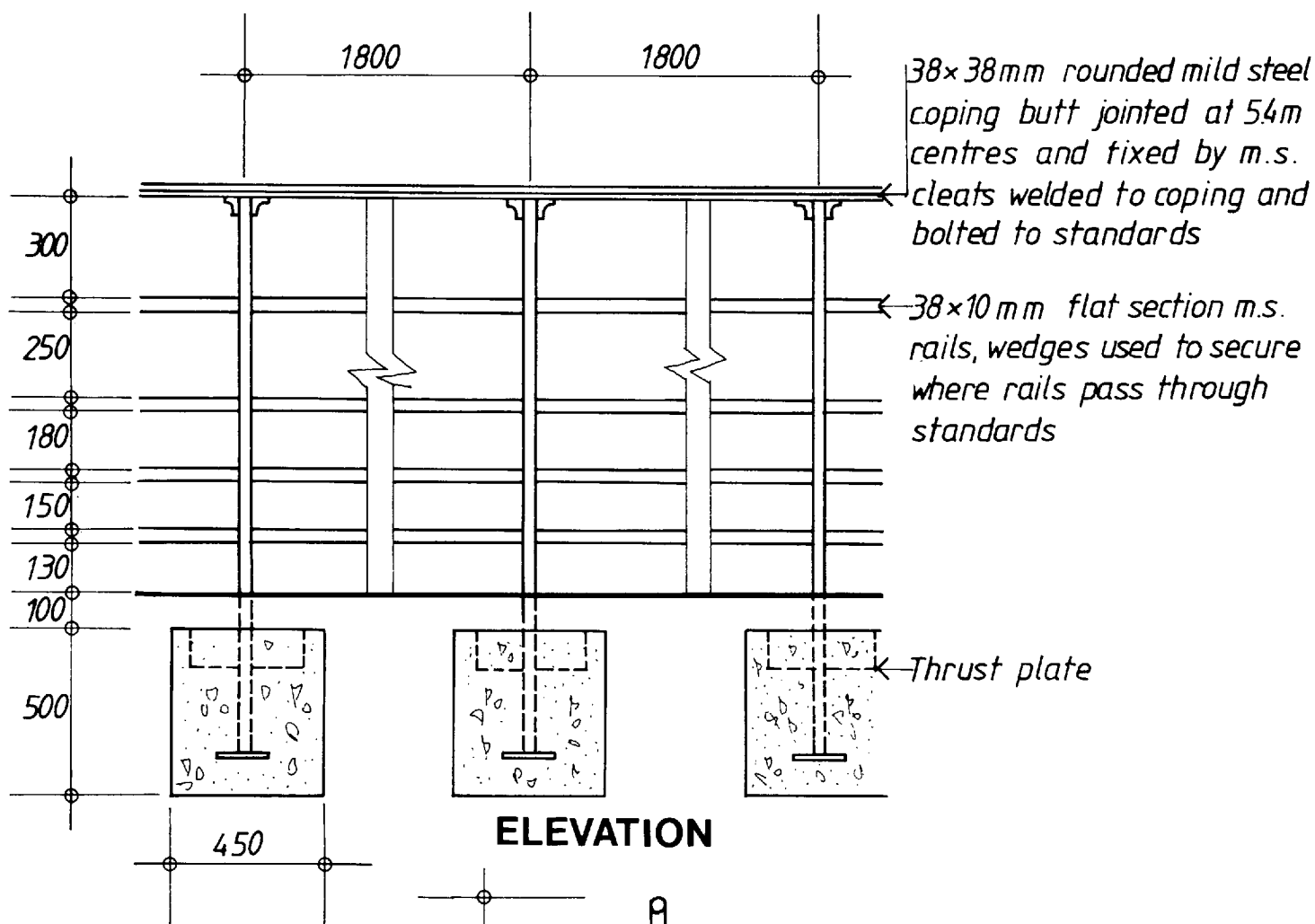


Scale 1:20



FENCE
metal – flat section

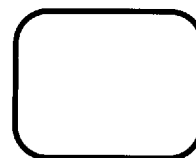


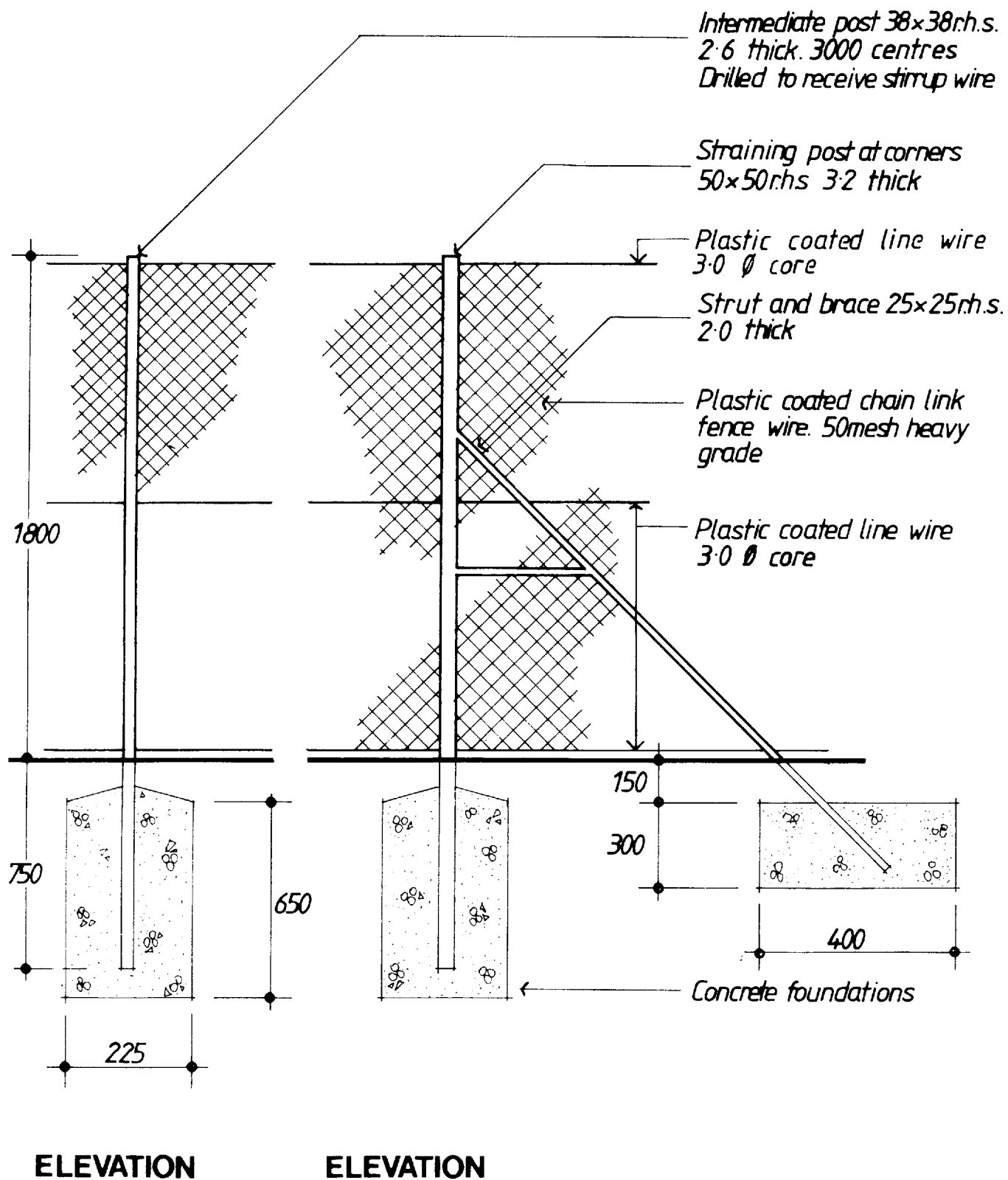


Scale 1:20



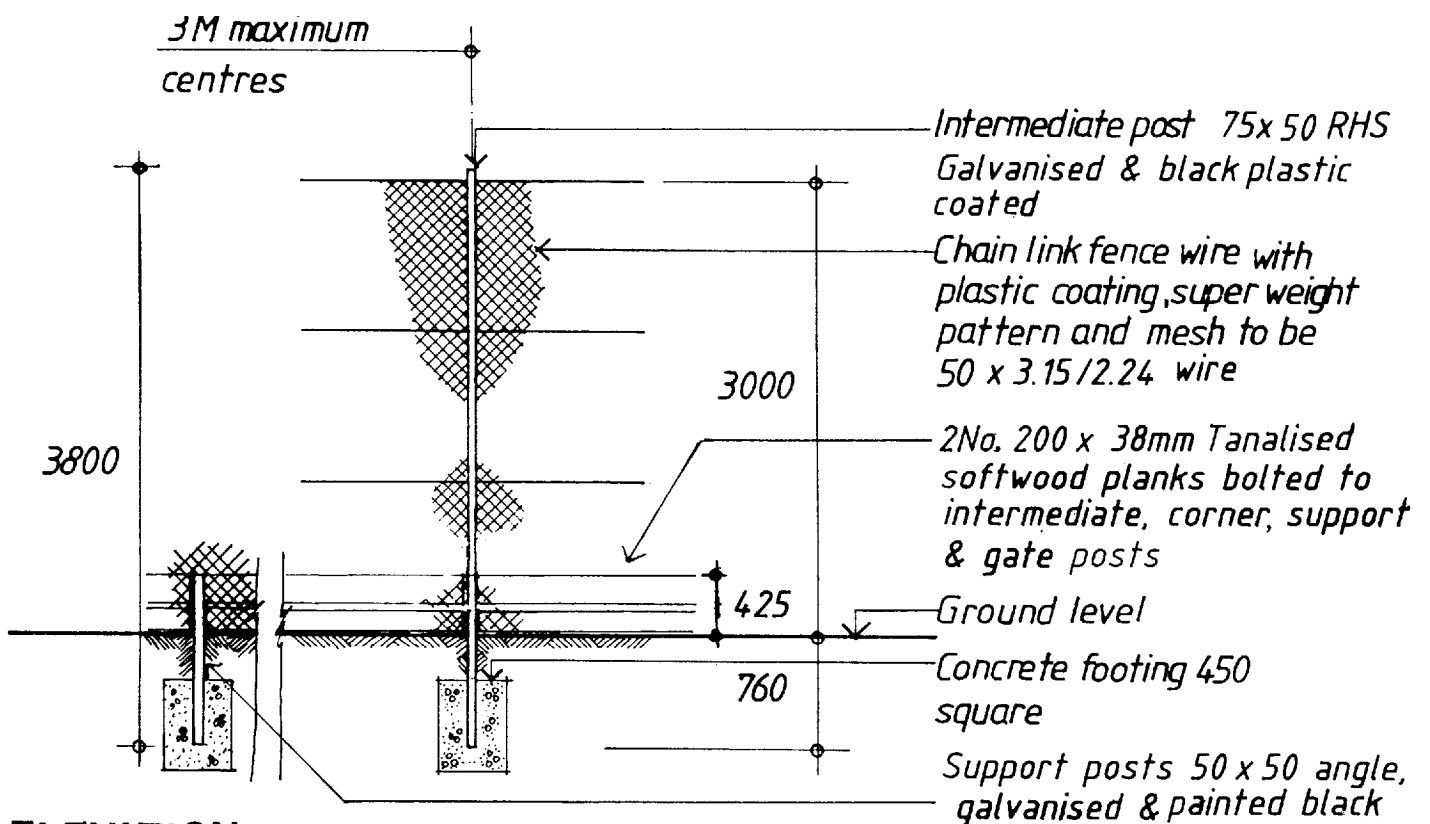
FENCE
metal – flat section



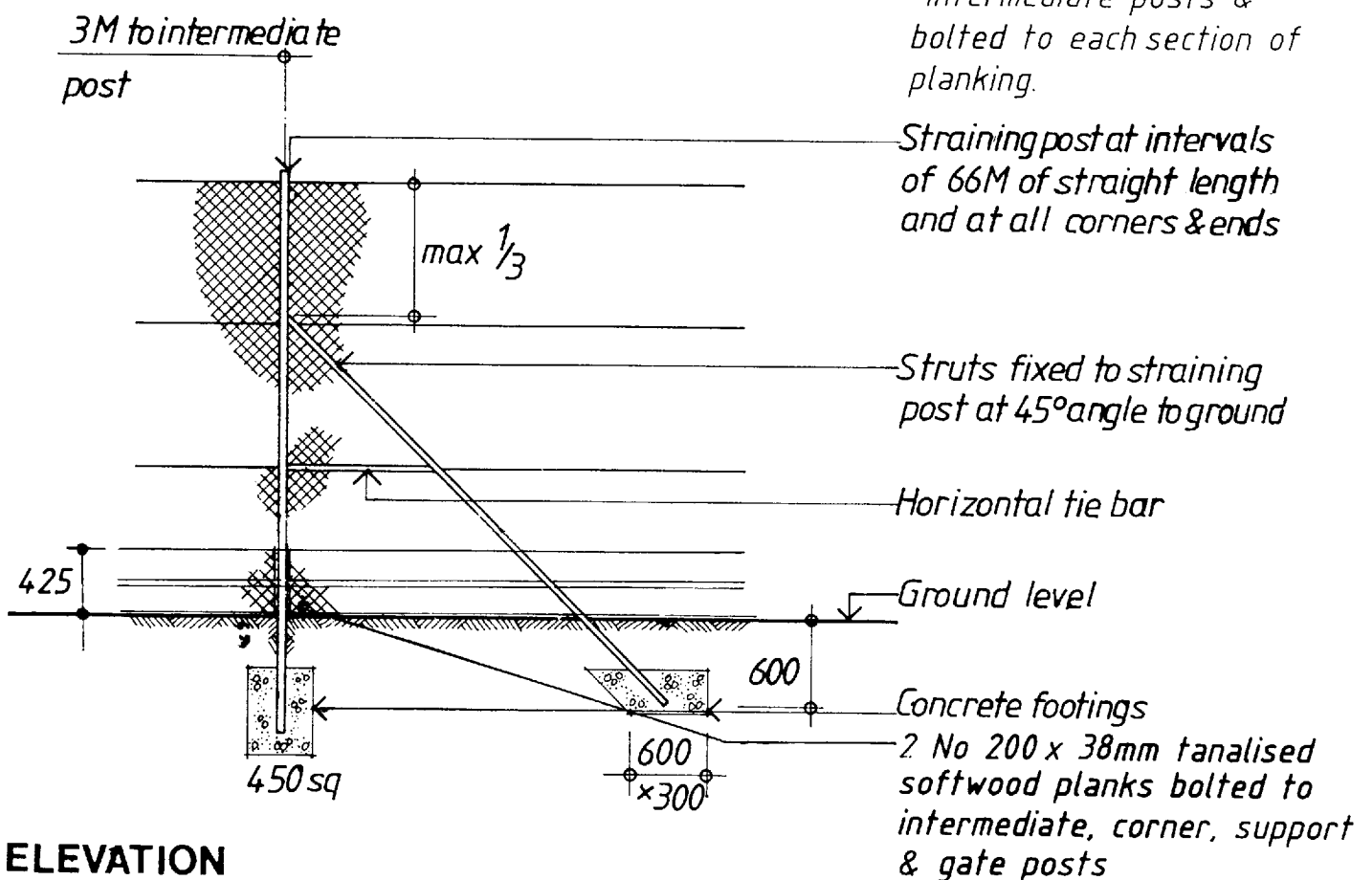


Scale 1:20

FENCE
chain link 1800



ELEVATION



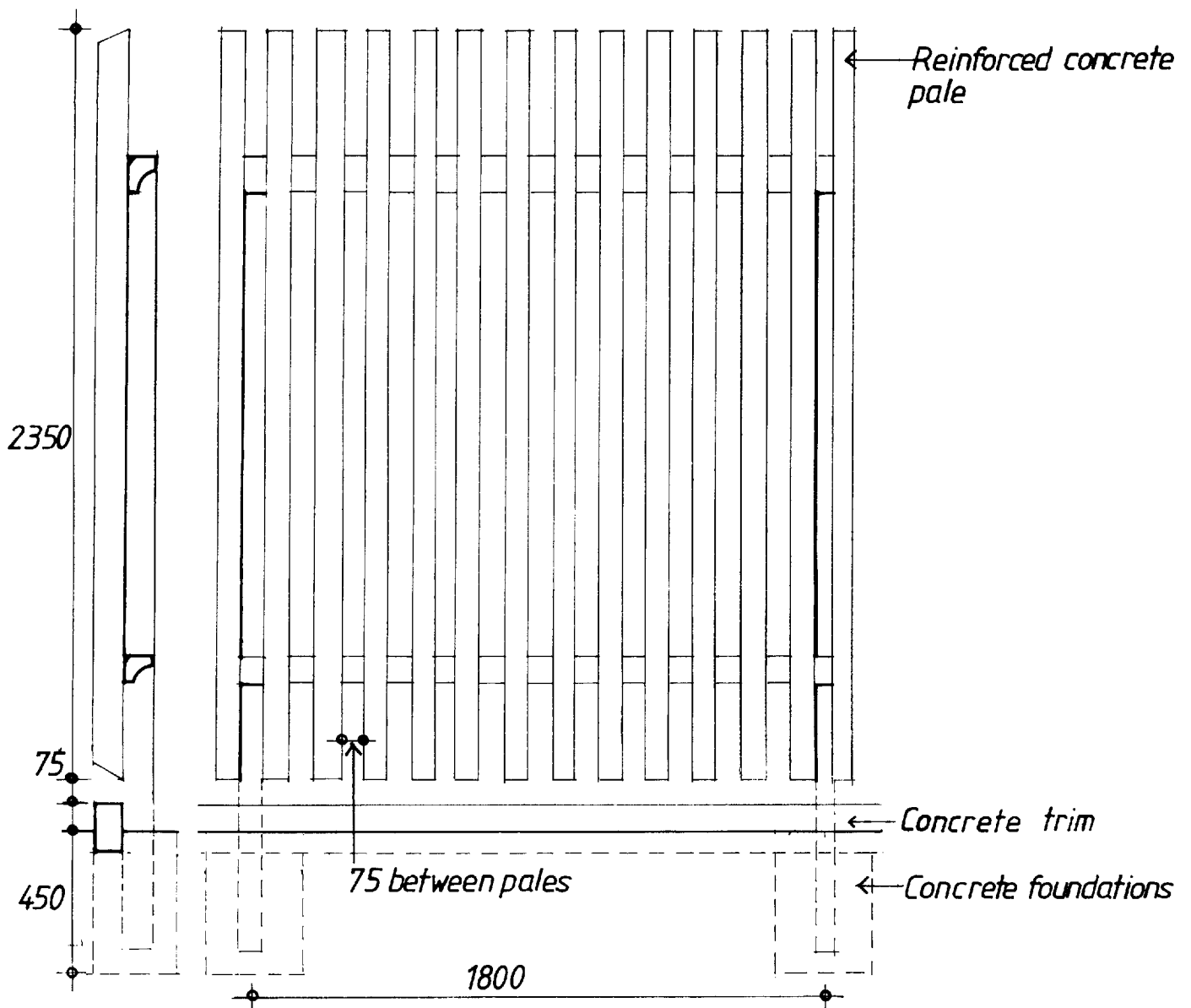
ELEVATION

Scale 1:50



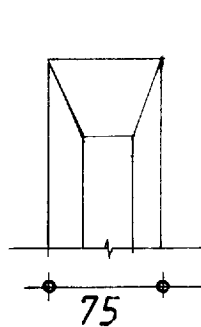
FENCE
chain link 3000



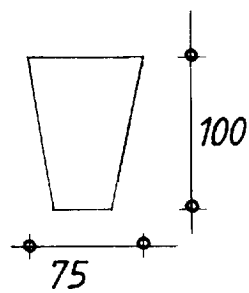


SECTION

ELEVATION

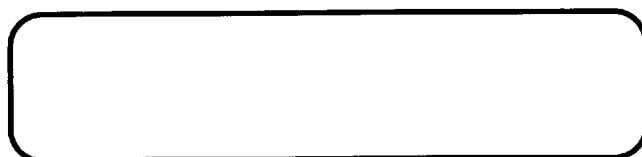


PALE ELEVATION

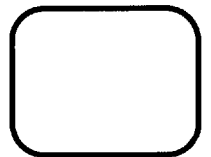


PALE PLAN

Scale 1:20



RAILINGS
concrete palisade



GATES AND STILES

GUIDANCE NOTES

Design

The character of a gate also reflects or offsets the character of the enclosure it pierces. The relationship between the proportions of fence or wall to gate is most important. For example, a heavy gate will need a strong post, pier or column to balance it. The material used for a wall will also play a key part in deciding the type of gate and its material.

If a gate looks stronger than the wall it will contradict its function. A gate must appear as a way through a fence or a wall and part of its function is to state this visually. No gate should be higher than the wall or fence which flanks it; there are exceptions to this rule which do work, but usually in the more open landscape.

Standards

BS 3470 covers the requirements for timber and steel field gates, with timber, steel and concrete gate posts. The Standard does not specify any particular design, but does set minimum units on sizes of various members. For gates in chain link and steel palisade fencing, see the relevant BS fence type.

BS 4092: Part 1 covers domestic entrance gates in tubular metal, mild or wrought steel. It specifies dimensions and certain functional requirements, but not detailed design. Part 2 of the same Standard describes domestic wooden gates. General design requirements of stiles, bridle gates and kissing gates are given in BS 5709: 1979.

Construction

The main point about gate construction is to ensure that adequate cross bracing be provided to prevent the gate from

dropping. The usual method of achieving this with timber construction is to provide a diagonal brace from the bottom corner on the hinge side so that the brace is in compression. Tall gates are provided with two separate braces. Metal gates of bolted construction are braced in the same manner. Welded metal gates are not usually cross braced.

Gate stops should be provided. The latch should not be relied on to stop the gate. Double gates should be provided with rebated meeting stiles.

Width

For pedestrian use the structural opening width should not be less than 900 mm. Except for special designs (e.g. gallows gates) the maximum leaf size should be 1,200 mm. For openings larger than this double gates should be used. For large mower access to protected grass areas, a clear opening width of 2,400 mm will be required. Vehicular control of estate roads or parking areas can be affected by the use of gallows gates with opening widths of up to 5,200 mm.

Height

Gate height should be equal to or slightly lower than the adjacent enclosure.

Hinges, locks and latches

Hinges

The usual type of gate hinge is the steel bend hinge. There are two basic types:

1. Hook and band
2. Reversible

Hook and band are available as light or heavy duty, straight or cranked. Reversible hinges are available as light or heavy duty.

Both types are available hot-dip galvanised or sheradised. Tee hinges are also used. These are usually available in a japanned finish, but can be specially ordered with a galvanised finish. All hinge types are described in BS 1227: Part 1A: 1967. They should be strong enough to prevent the gate from sagging.

Locks

Locks for gates usually take the form of barrel bolts. Where these are secured with a padlock they are called padbolts. Double gates should be fitted with bolts to engage in sockets set in concrete or tarmac. A centre stop should be used in conjunction with the bolts. These can be of the fall-down type if the passage of lawn mowers is anticipated. Provision should be made for securing gates in the catches.

Latches

The type of gate latch fitted will depend on the gate height. Low gates can have an automatic latch fitted to the inside of the gate only. High gates can be fitted with a Suffolk-type thumb latch or a ring handle gate latch, both of which are operable from either side of the gate. BS 1331 describes latches suitable for gates.

Either slip-bolts or latches, sometimes called fall-catches, are provided for fastening metal gates; the slip-bolts having the advantage of being capable of receiving a padlock. With double gates a drop-bolt should be fitted to one leaf with a stop let into the ground. Expensive stops have counterbalanced lids which shut as the gate is opened. Catches can be fitted for holding back large gates.

Gate posts

Gate posts of timber or metal can be set in earth or concrete, and BS 3470 specifies the minimum length of post to be used in each case.

They must be on both sides of the opening and be strong enough to support the gates. Dimensions will depend upon the height and weight of the gate. The tops of timber

posts should be weathered or rounded. Posts can also be of concrete, brick or masonry.

Brick or stone piers are frequently inserted in fencing to support gates. Large and heavy gates can impose a considerable strain on such piers, which have been known to break off at their base. They should be built with a substantial foundation, taken well into the ground, and with a reinforced concrete core, the reinforcing bars of which should carry down into the foundations.

SPECIFICATION CHECK LIST

BS 3470 specifies requirements for timber and steel field gates with timber, steel and concrete gate posts. The Standard does not specify any particular design but does set minimum limits on sizes of various members. BS 5709 covers Bridle and Kissing Gates. BS 1186 classifies timber.

Drawing reference number: This item can be deleted if a proprietary design of gate is being specified.

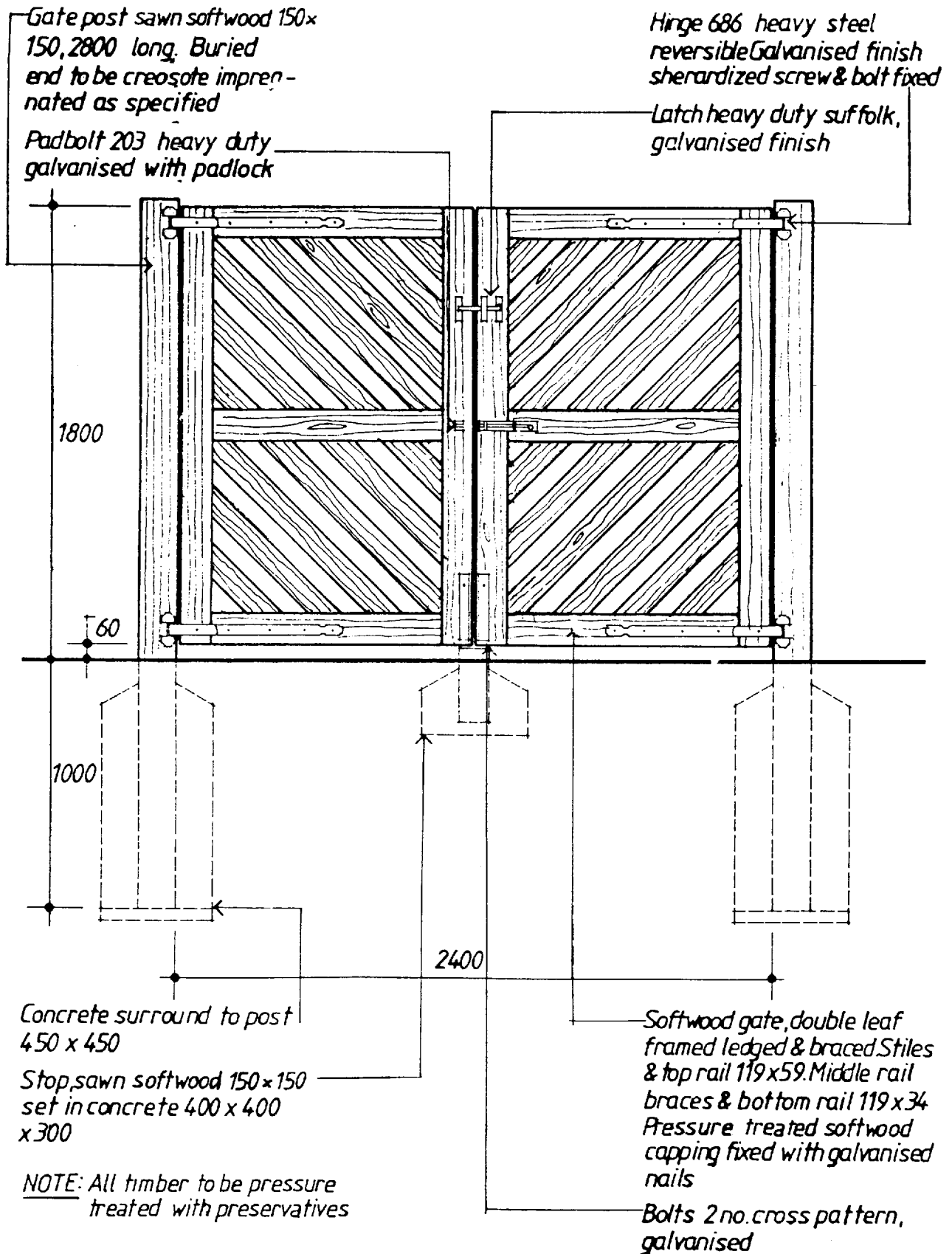
Manufacturer and reference: This item can be deleted if a purpose-designed field gate is being specified.

Size(s): The BS for Field Gates specifies a height of 1.1 m and widths of 2.4, 2.7, 3.0, 3.3, 3.6 and 4.2 m. For openings over 4.2 m gates should be made in two leaves. For kissing gates see BS 5709 for minimum height and width.

Materials: BS 3470 specifies timber and steel for gates and timber, steel and concrete for gate posts. State type of material(s). For all gates insert sizes of all component gate parts and posts.

Treatment: If timber gates are to be treated with preservative insert details. Steel gates can be supplied red oxide primed, sprayed with zinc or aluminium, hot-dip galvanised or coated with black bitumen paint.

Fittings: These include strap hinges and hooks, catches and bolts. Fittings can be supplied with various finishes but hot-dip galvanising is strongly recommended.



ELEVATION

Scale 1:20

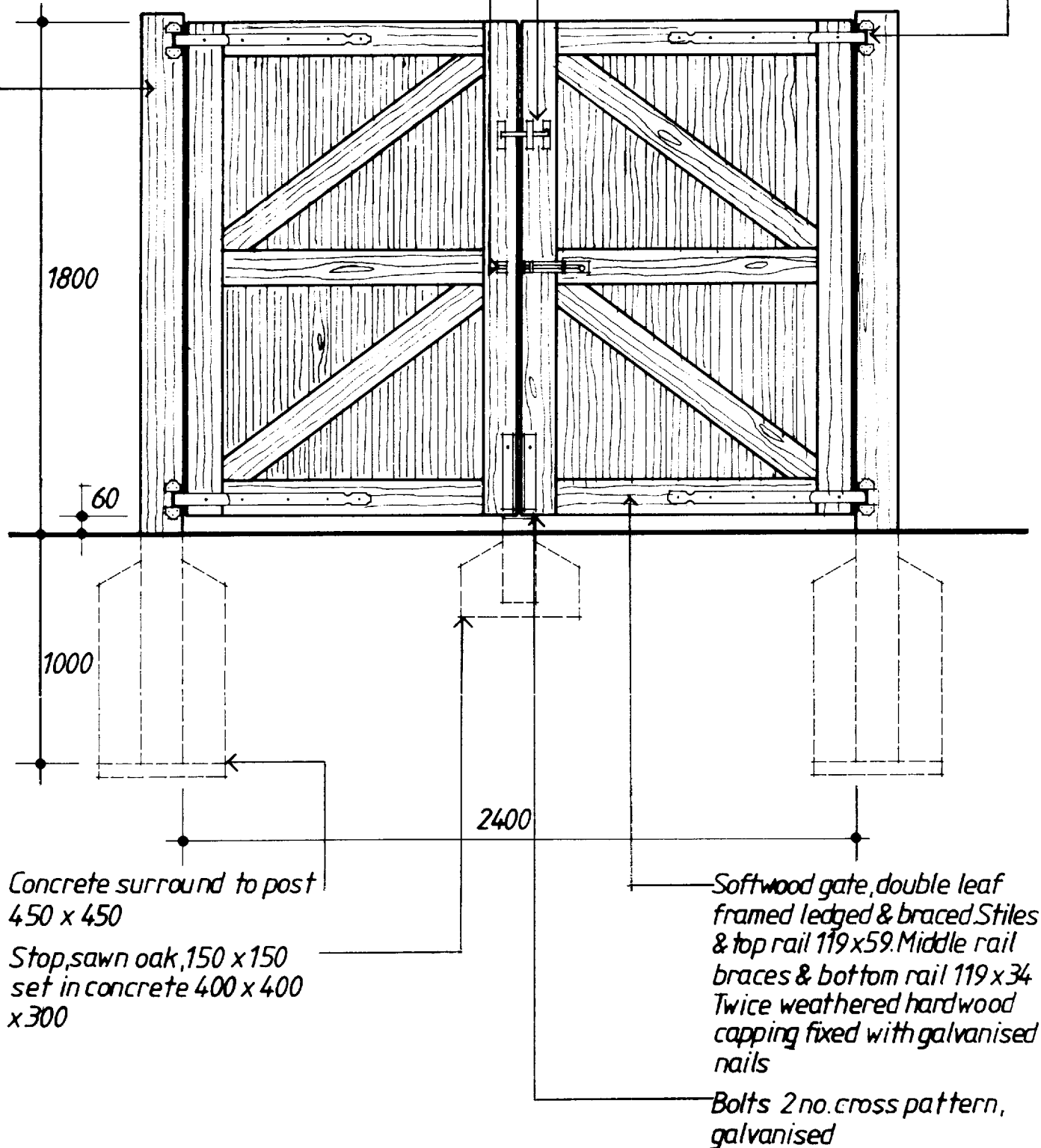
DOUBLE GATE
timber diagonal

Gate post, oak sawn, 150 x 150, 2800 long. Buried end to be creosote impregnated as specified

Padbolt 203 heavy duty galvanised with padlock

Hinge 686 heavy steel reversible Galvanised finish sherardized screw & bolt fixed

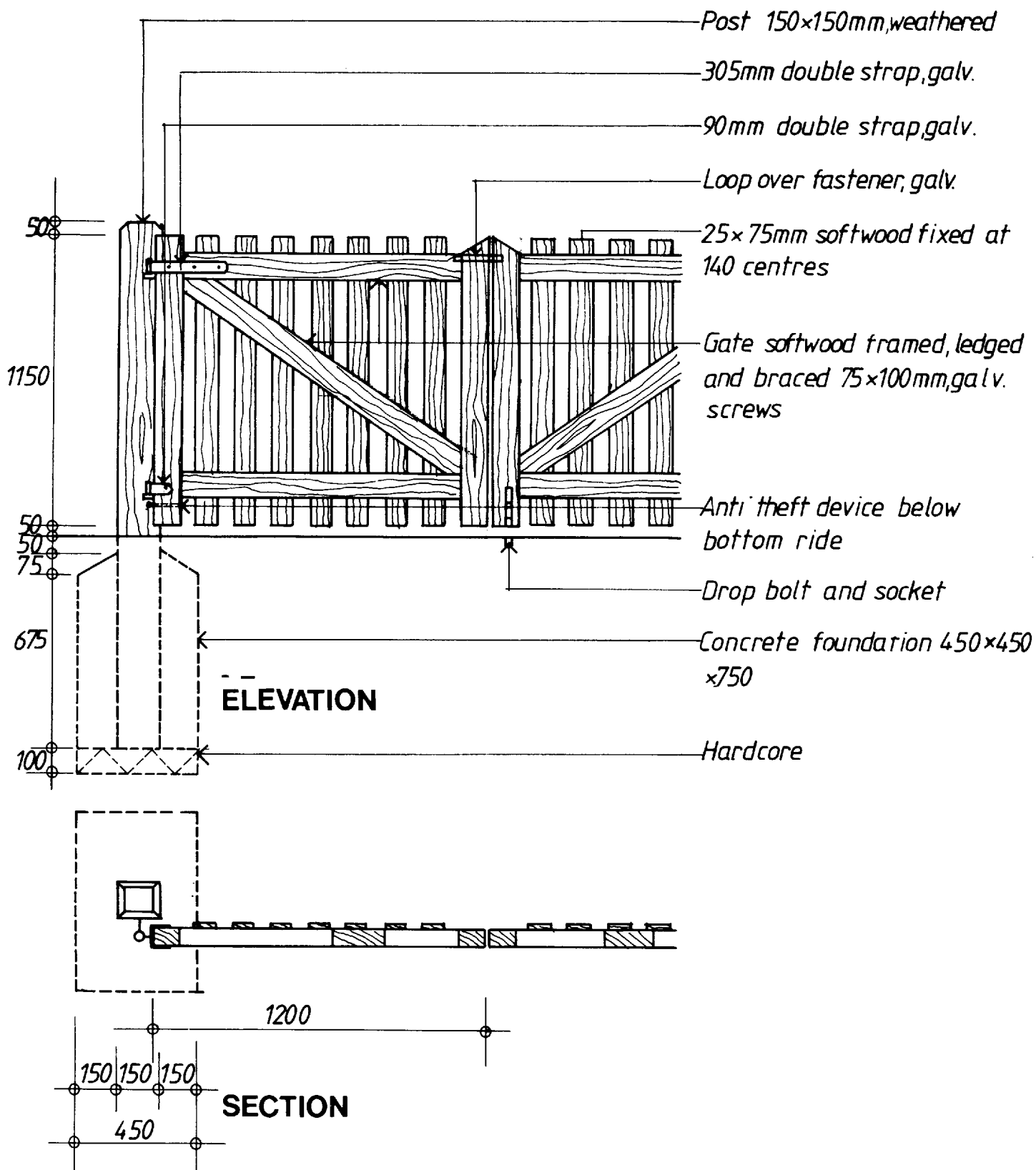
Latch heavy duty suffolk, galvanised finish



ELEVATION

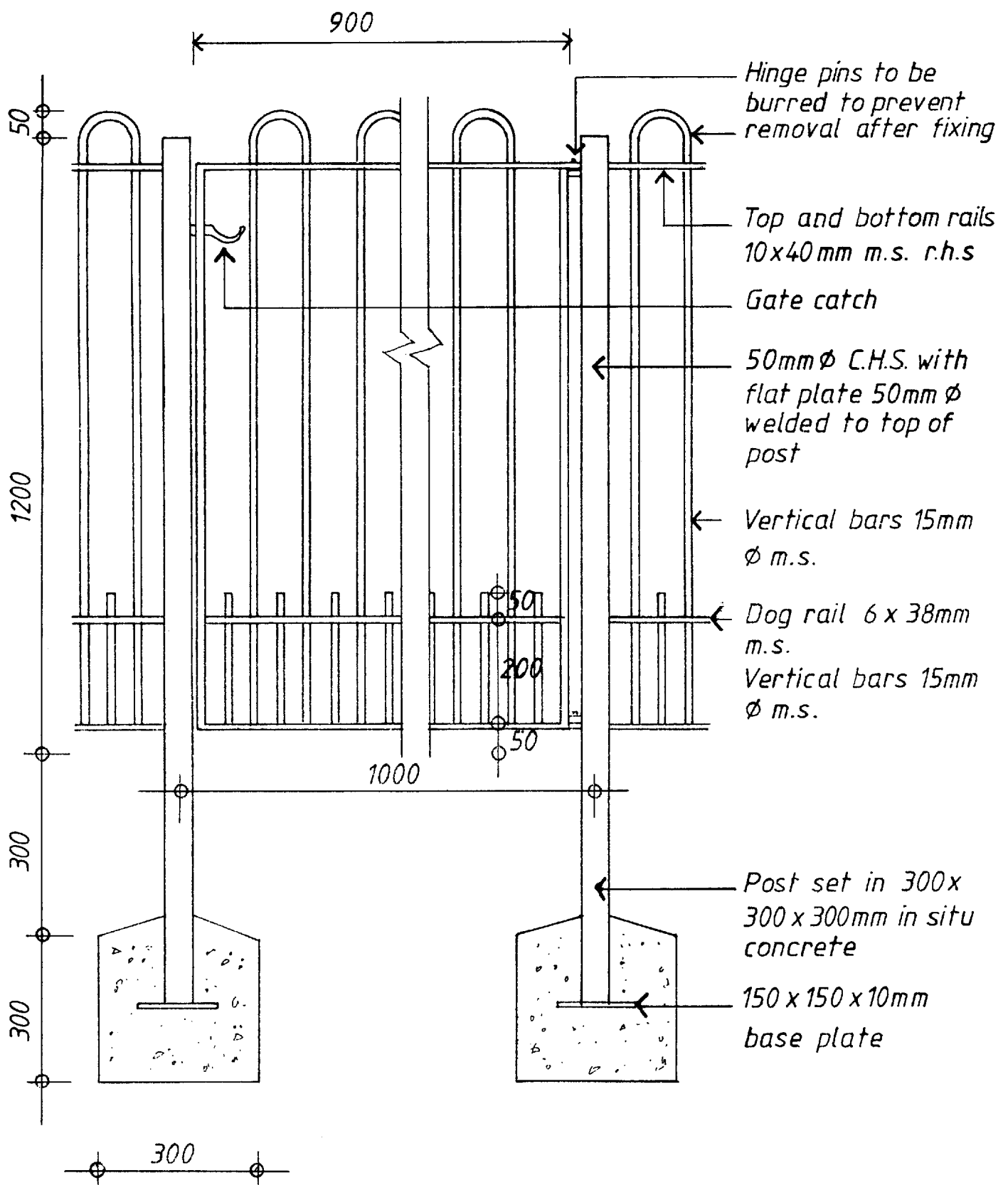
Scale 1:20

DOUBLE GATE
timber matchboard



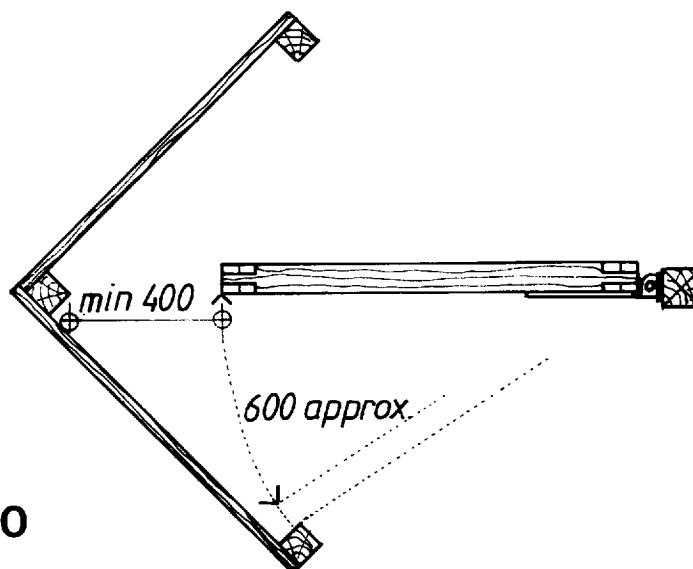
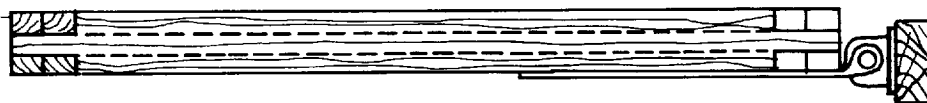
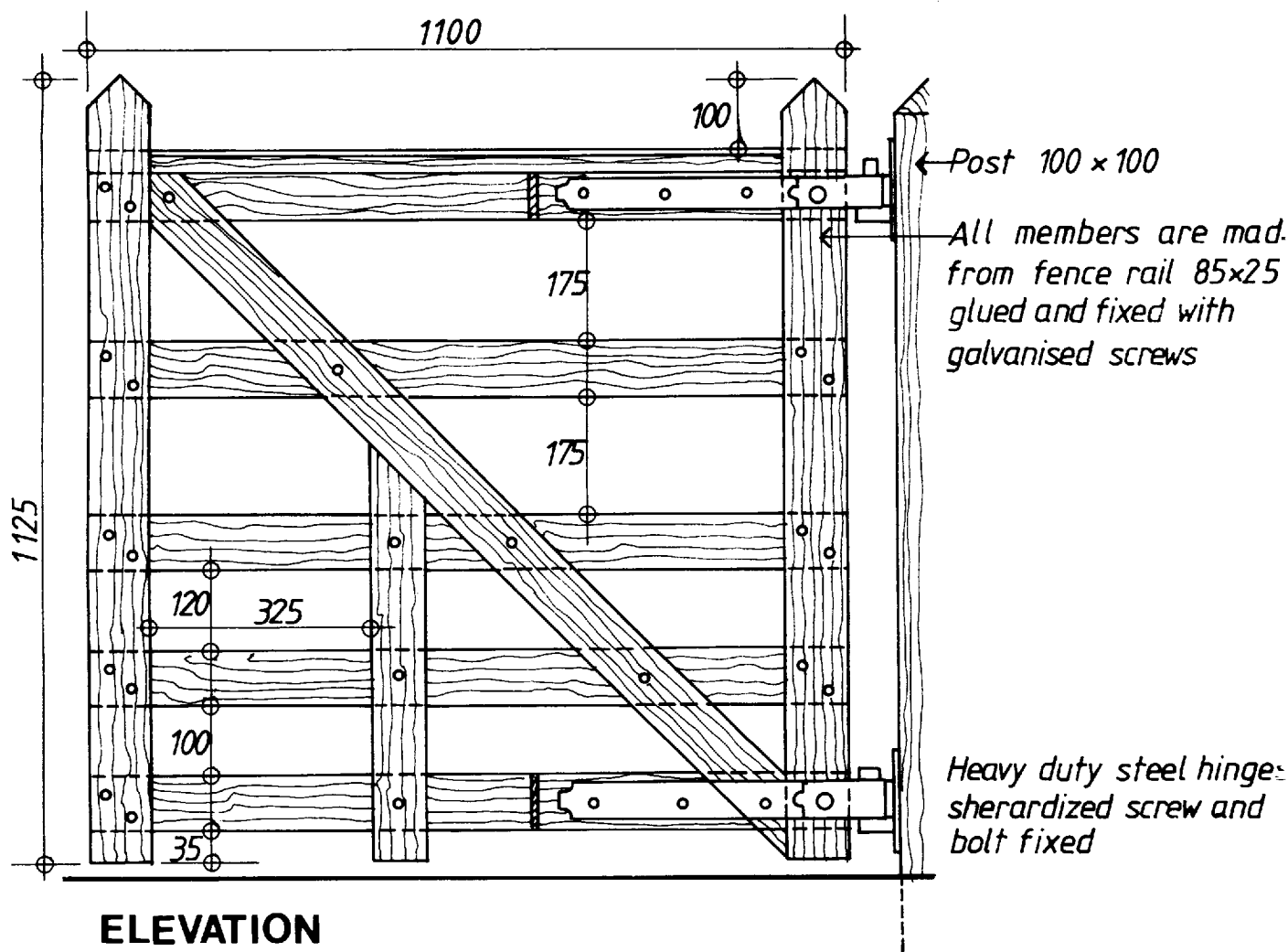
Scale 1:20

DOUBLE GATE
timber palisade



Scale 1:10

GATE
metal bow topped



Scale 1:10

GATE
timber kissing gate

Timber field gate. All joints morticed and tenoned, all rails and cross braces bolted.

Hanging stile 125 x 75

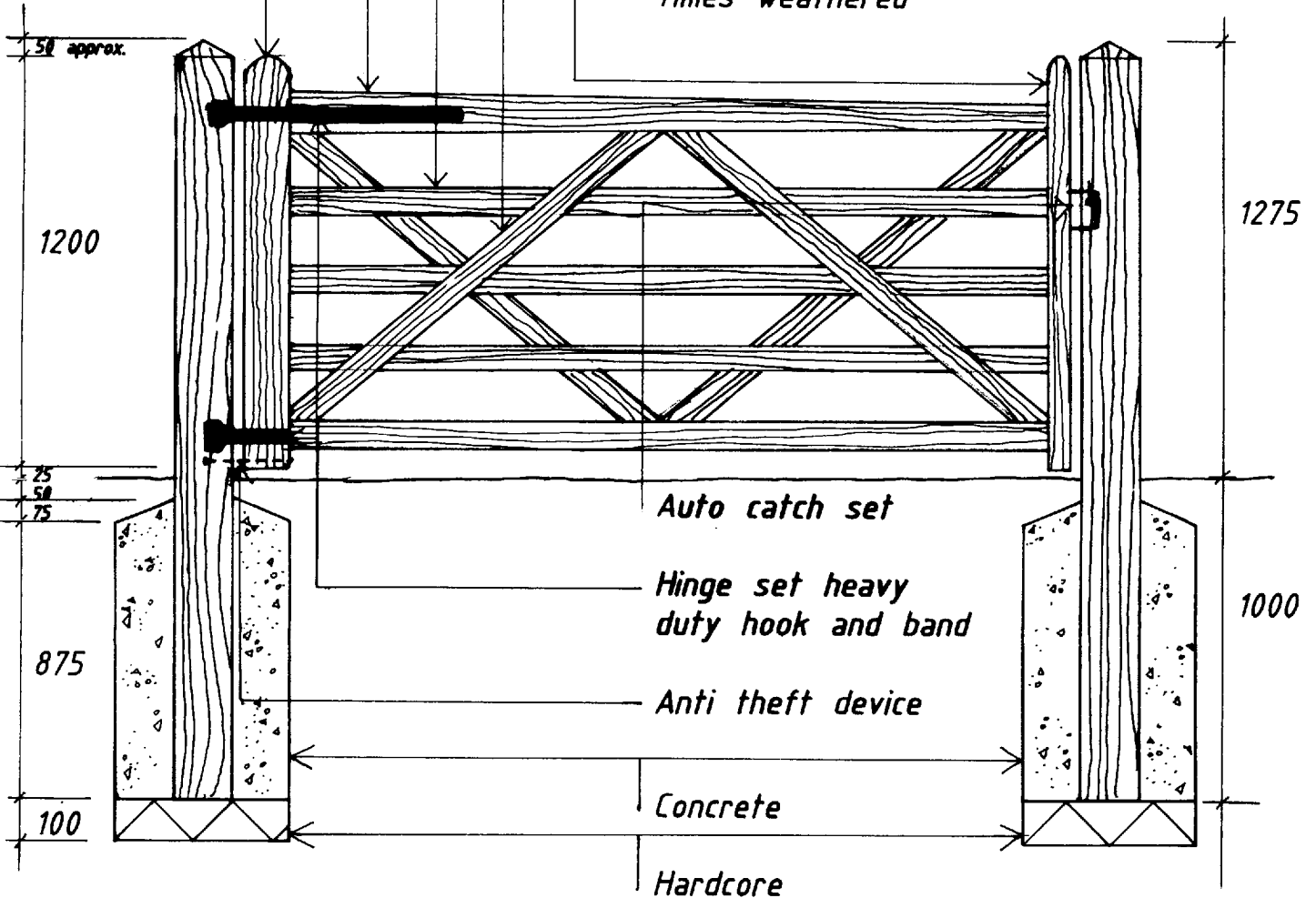
Top rail 125 x 75 tapered to 75 x 75

Under rails 75 x 75

Cross braces 75 x 25

Slam stile 75 x 75

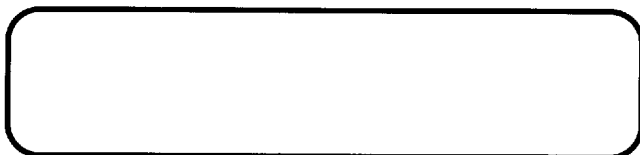
Gate posts 175 x 175 tops to be four times weathered



GENERAL

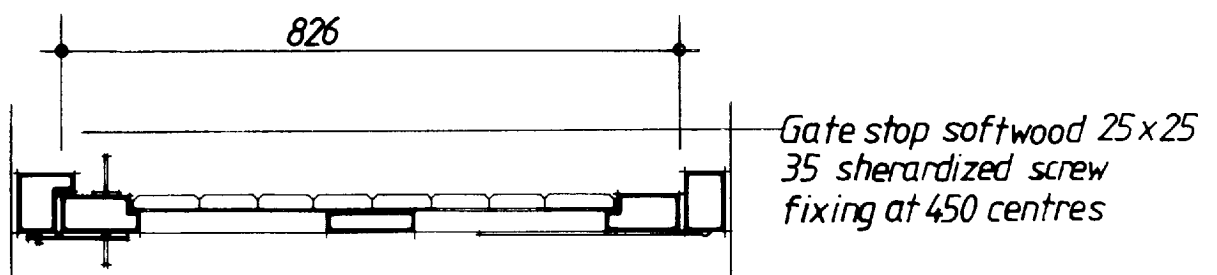
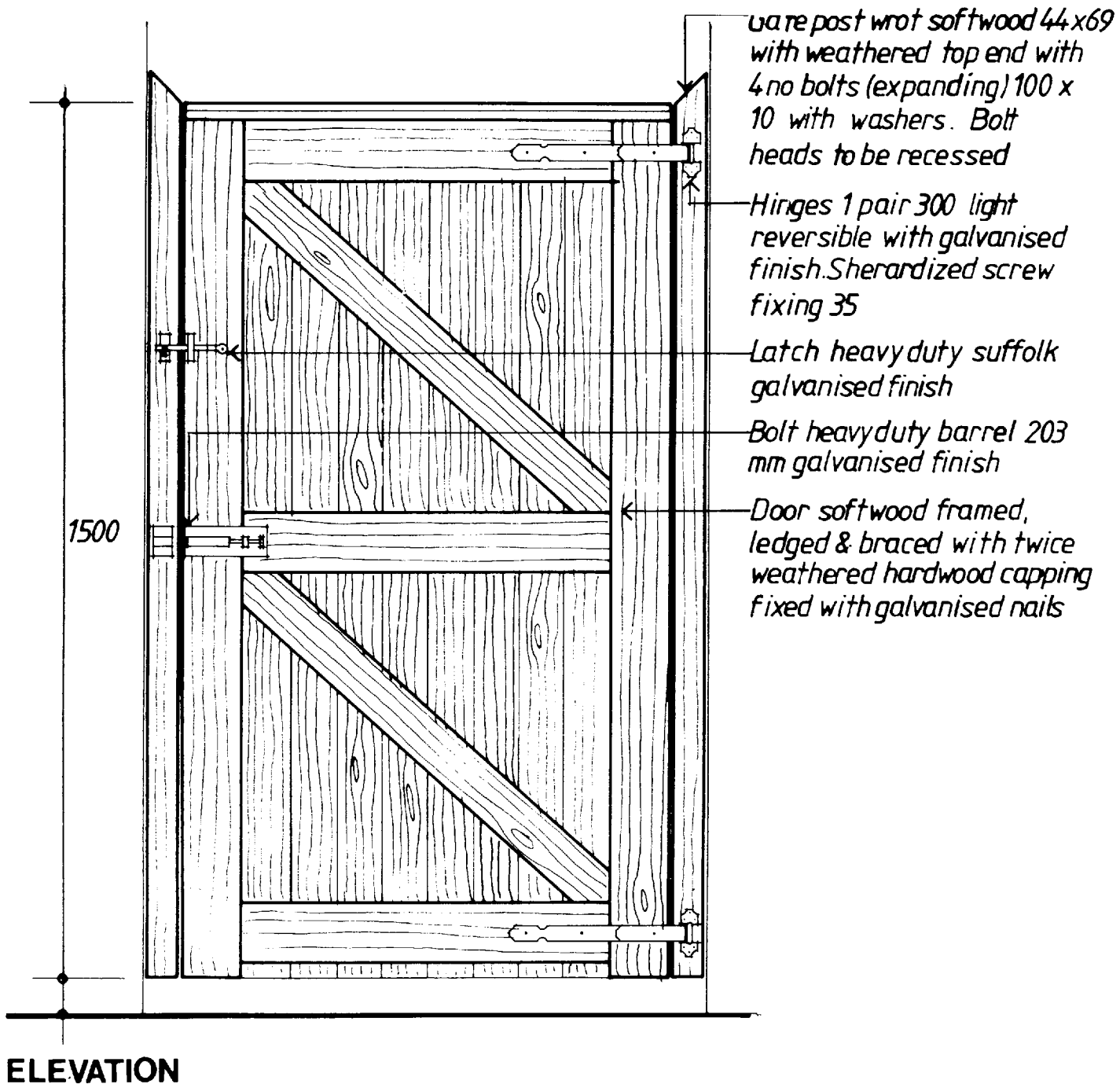
All timber to be Keruing hardwood planed and chamfered.

Scale 1:20



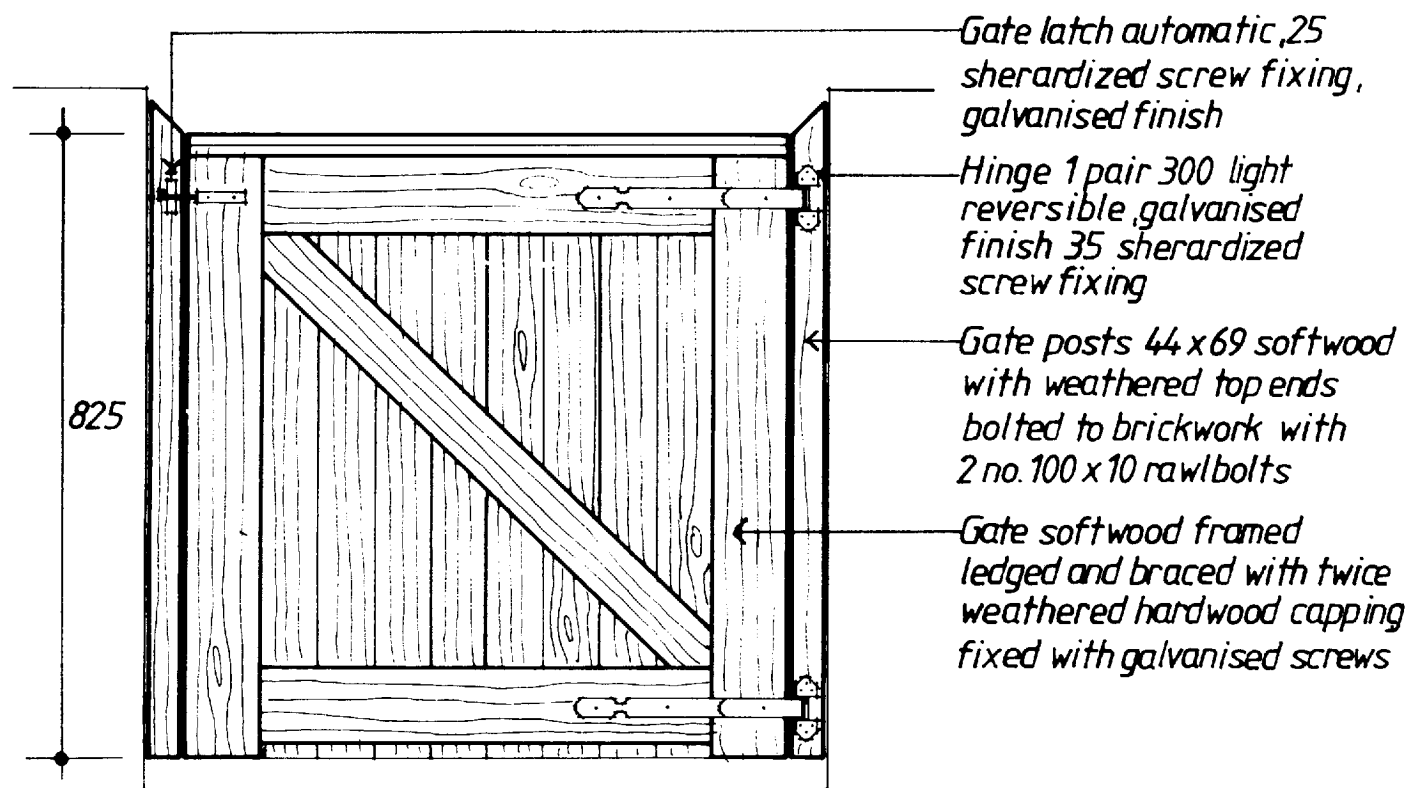
GATE
timber field



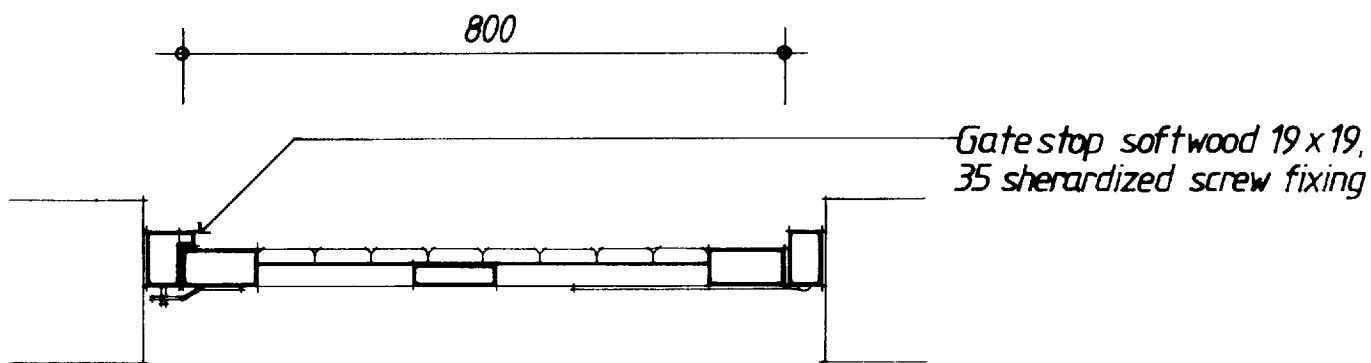


PLAN

Scale 1:10



ELEVATION

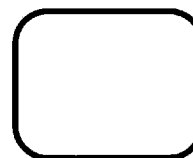


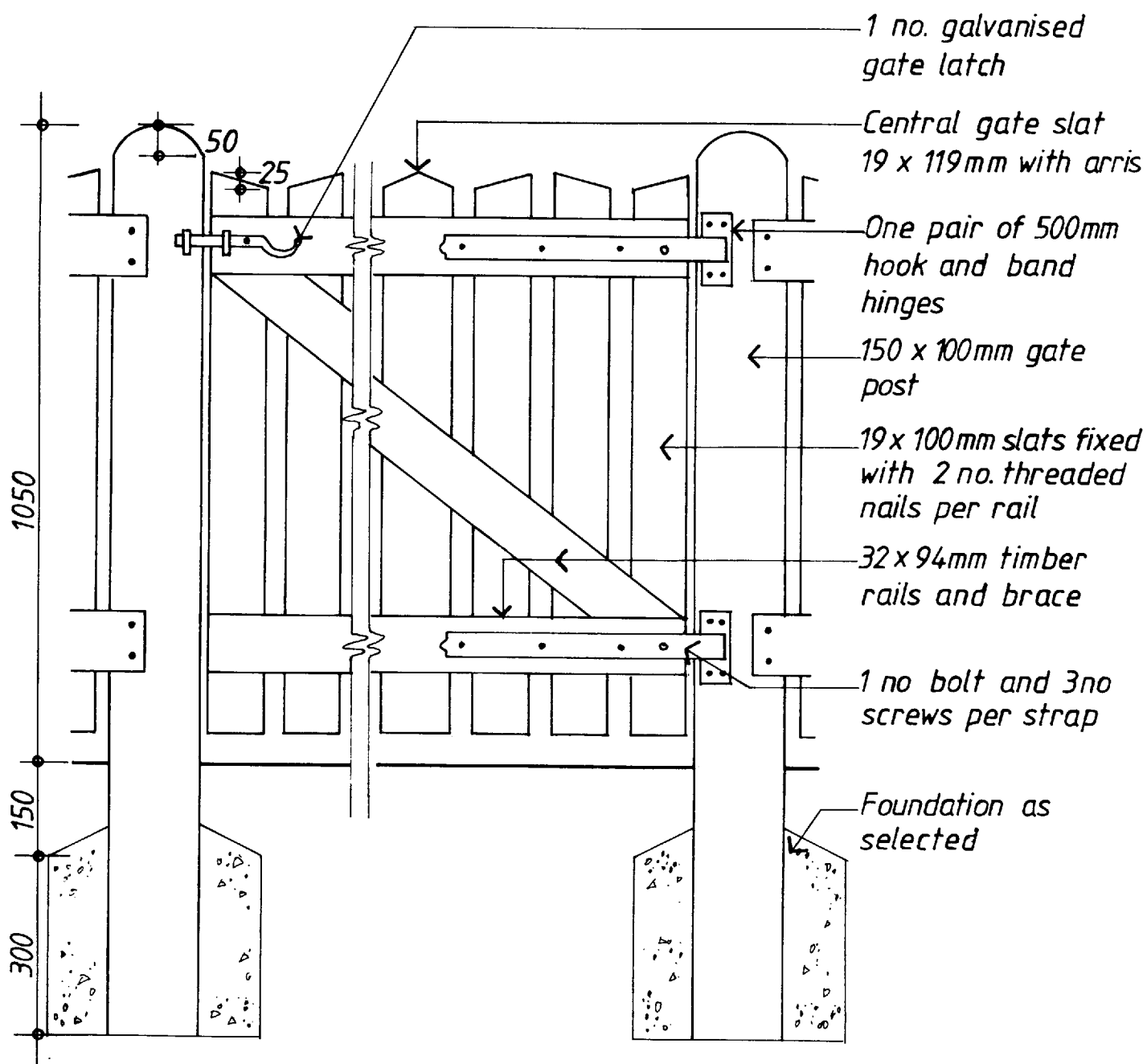
PLAN

Scale 1:10



SINGLE GATE
timber matchboard
(low)

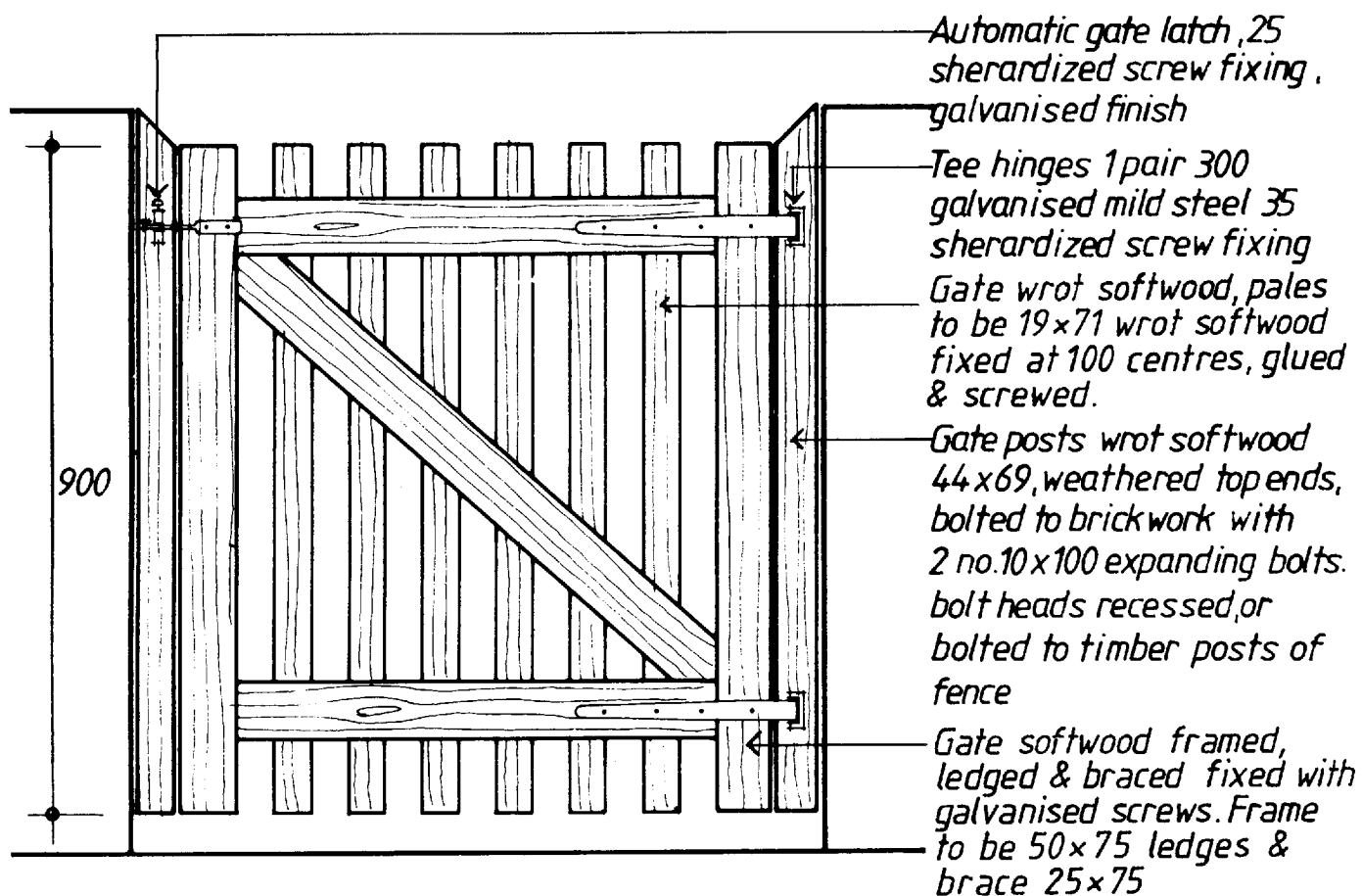




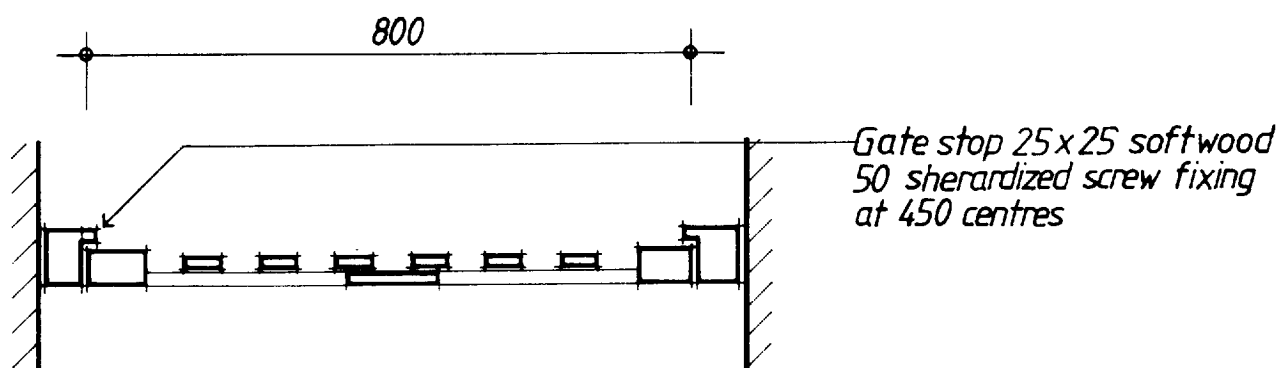
ELEVATION

Scale 1:10

GATE
timber palisade (low)



ELEVATION

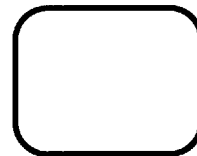


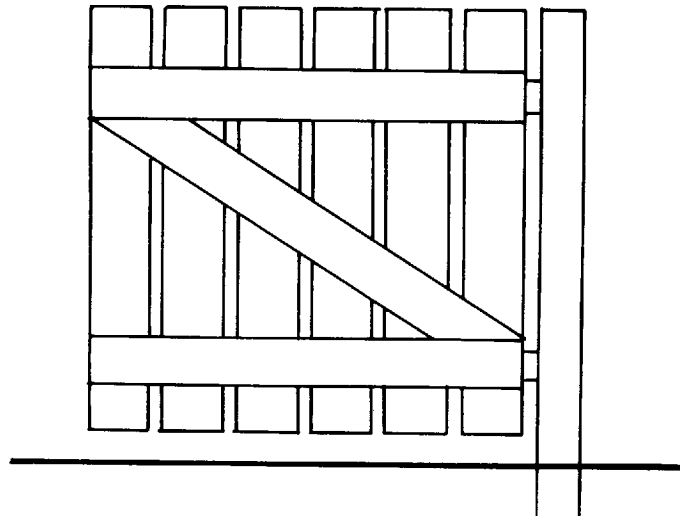
PLAN

Scale 1:10

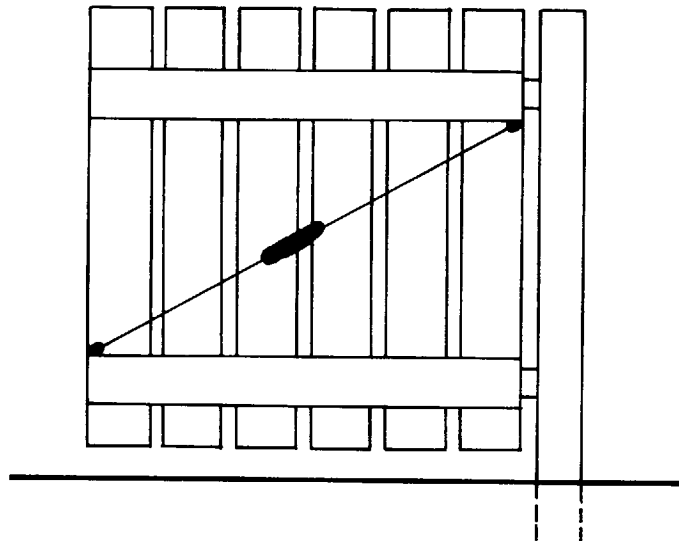


SINGLE GATE
timber palisade (low)

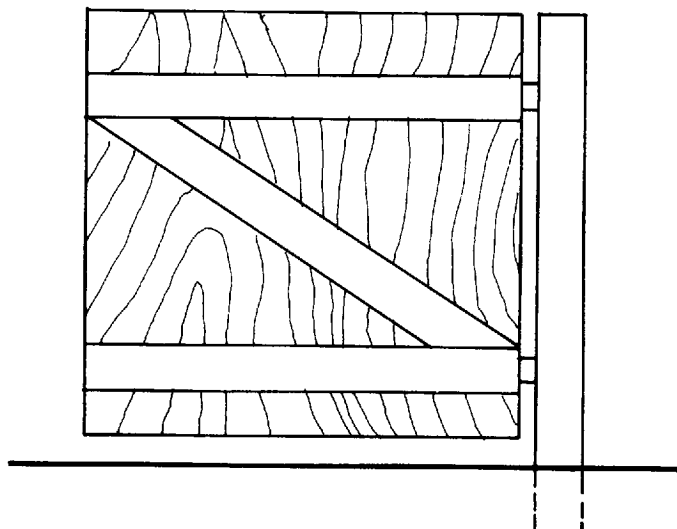




Wood brace

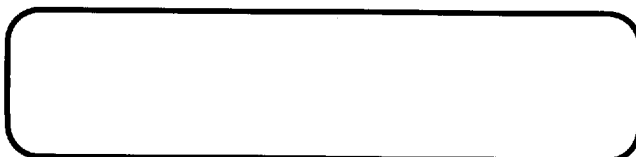


*Steel rod or wire with
turnbuckle*



Plywood panel

Scale 1:10



SINGLE GATE
brace detail



Setting posts: Gate posts can be set in earth or concrete. Specify the minimum length of post to be used in each case. The size of concrete foundation should be shown on a drawing and/or given in the specification.

Finish: Specify type of finish – plain, stained or painted, colour and number of coats.

DETAIL SHEETS

Single gates

Brace detail

Timber palisade (low) (2)

Timber matchboard (low)

Timber matchboard (high)

Timber field

Timber kissing gate

Metal bow topped

Double gates

Timber palisade

Timber matchboard

Timber diagonal

Mild steel bow topped

Metal

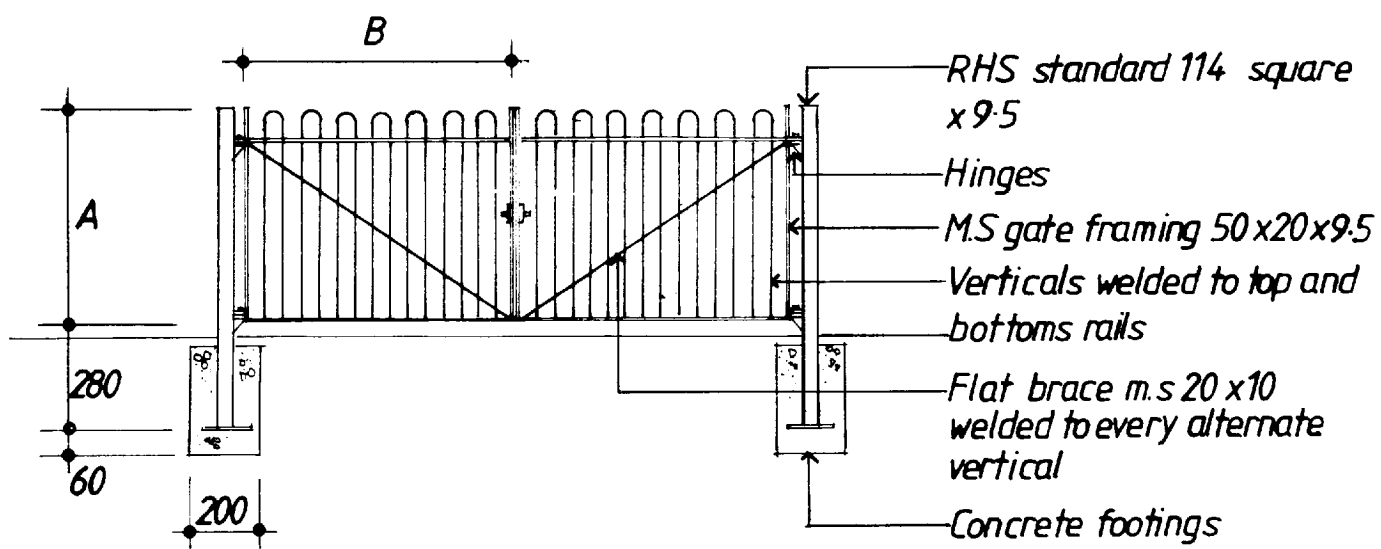
Gallows

Stiles

Timber V-shape (2)

Timber stepped (2)

Stone

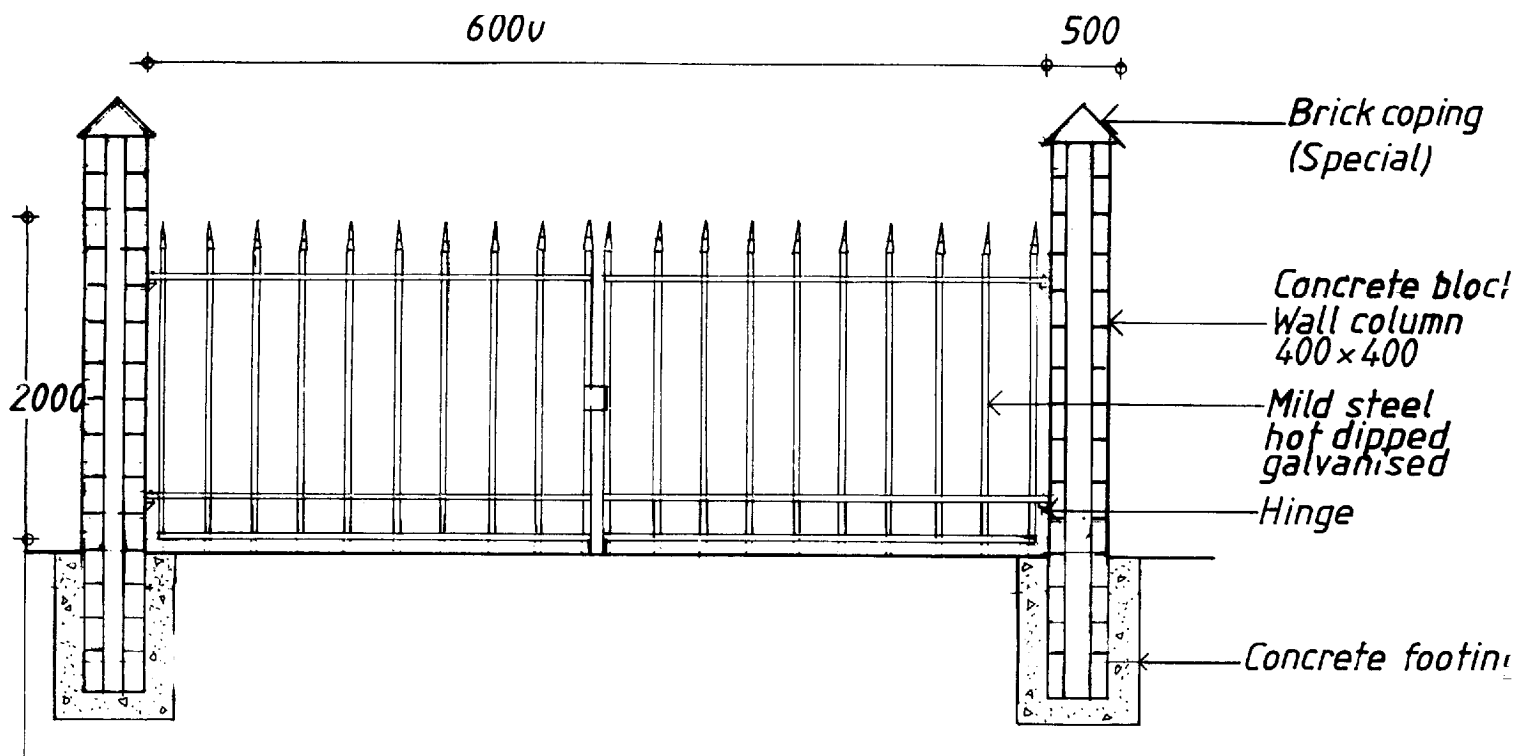


ELEVATION

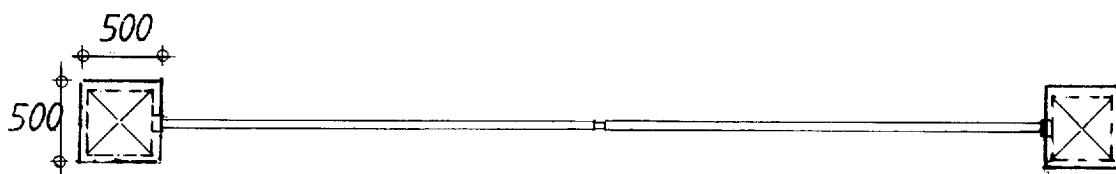
Nominal Size	A	B	No of verticals/ gate & size(dia)
3800 x 1200	1125	1818	14 x 130
3800 x 1400	1375	1818	14 x 130
3800 x 1800	1725	1818	14 x 190
4400 x 1200	1125	2118	16 x 130
4400 x 1400	1325	2118	16 x 130
4400 x 1800	1725	2118	16 x 190

Scale 1:20

DOUBLE GATE
mild steel bow topped

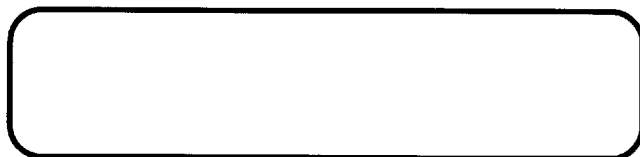


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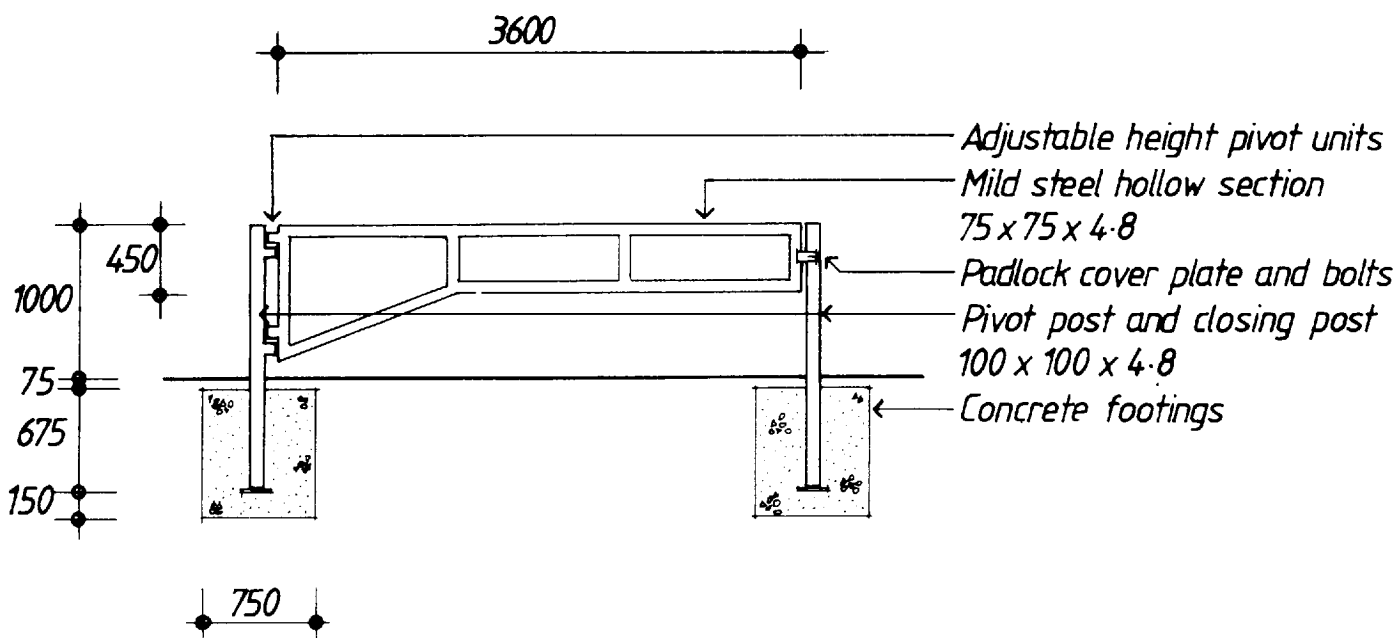
PLAN

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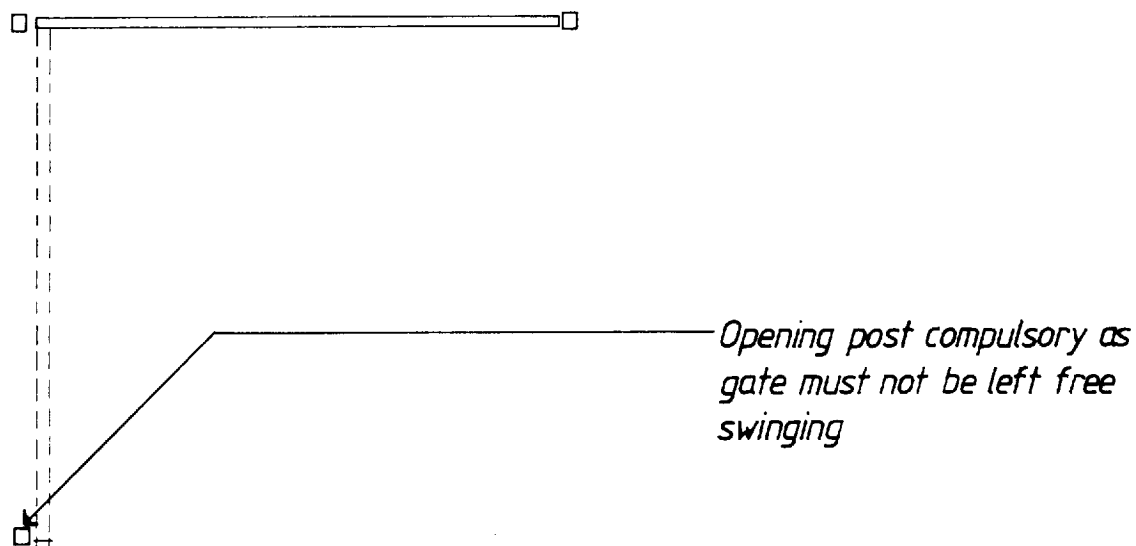


GATES
metal



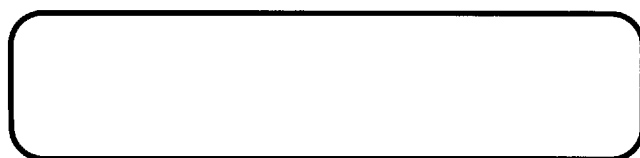


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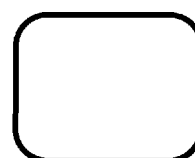


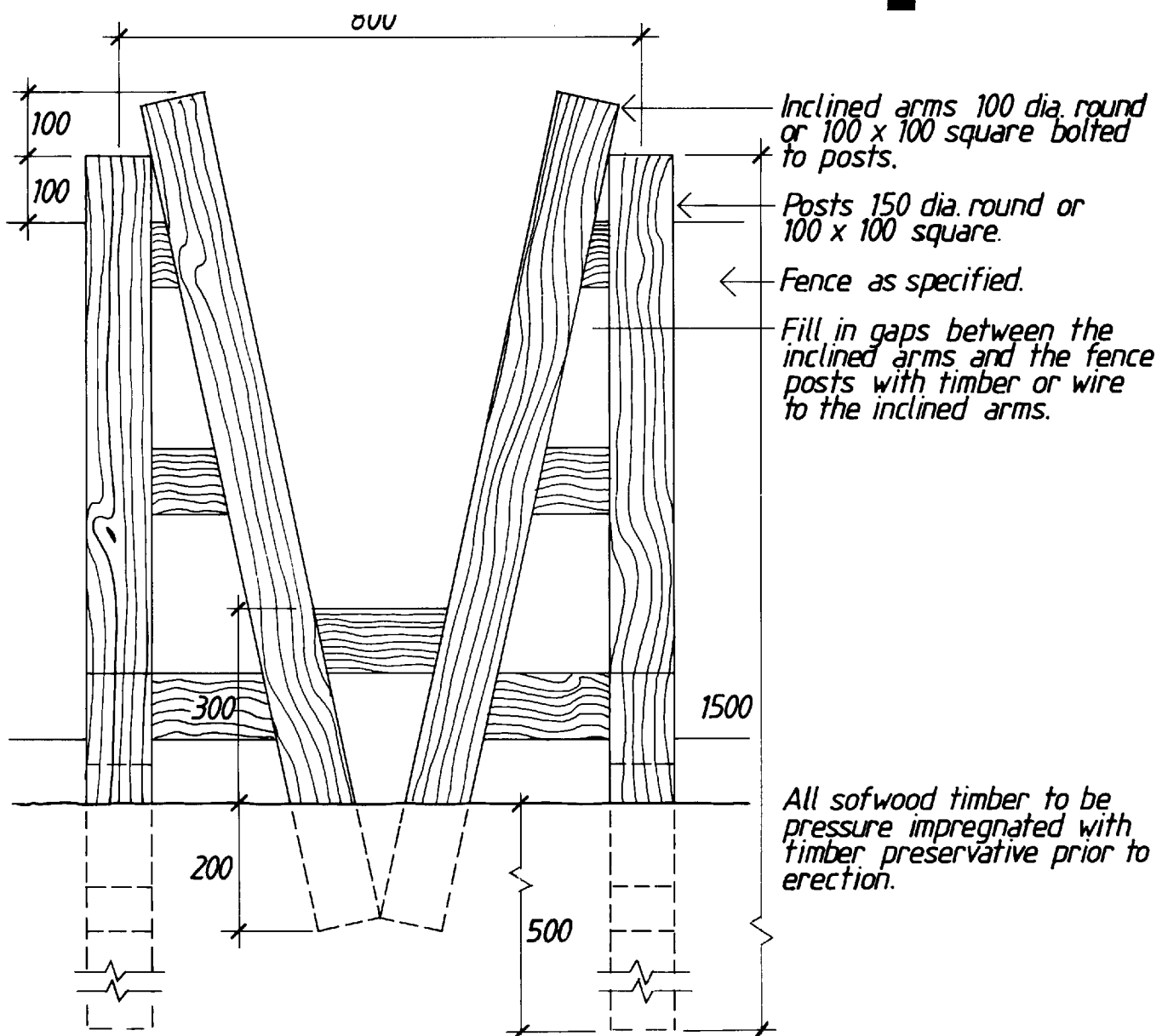
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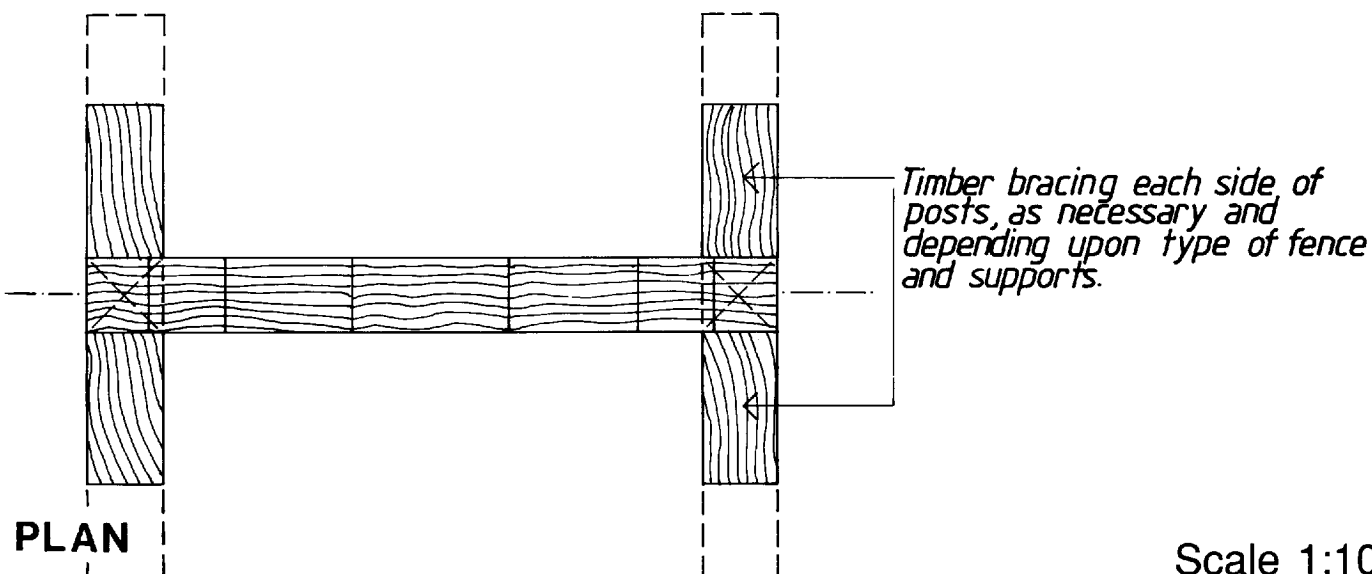


GALLOWS GATE
metal



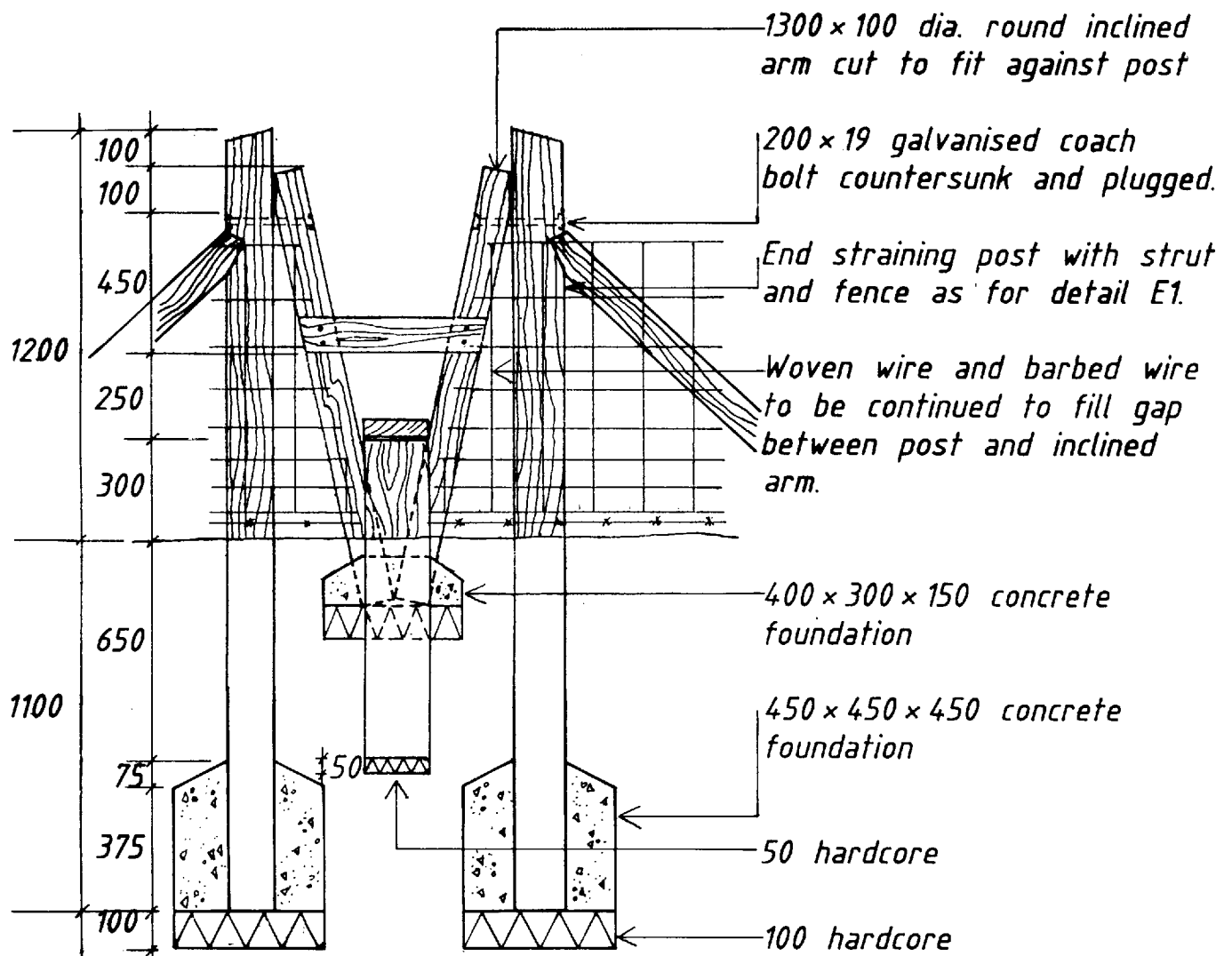


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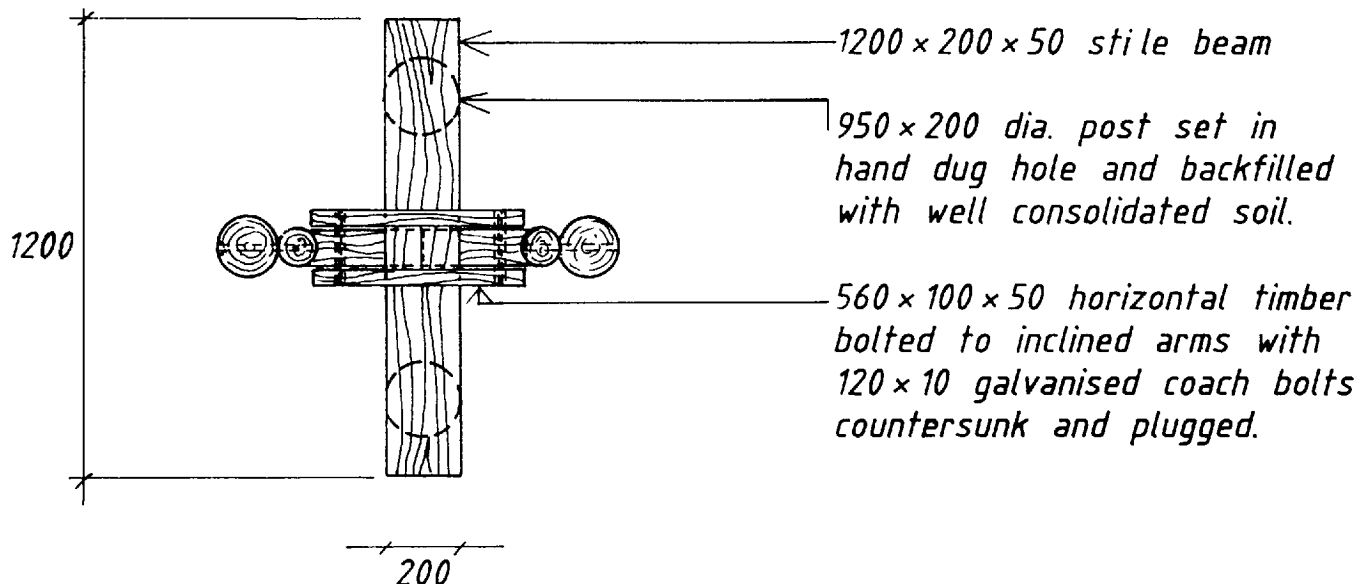


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STILE
timber V-shape



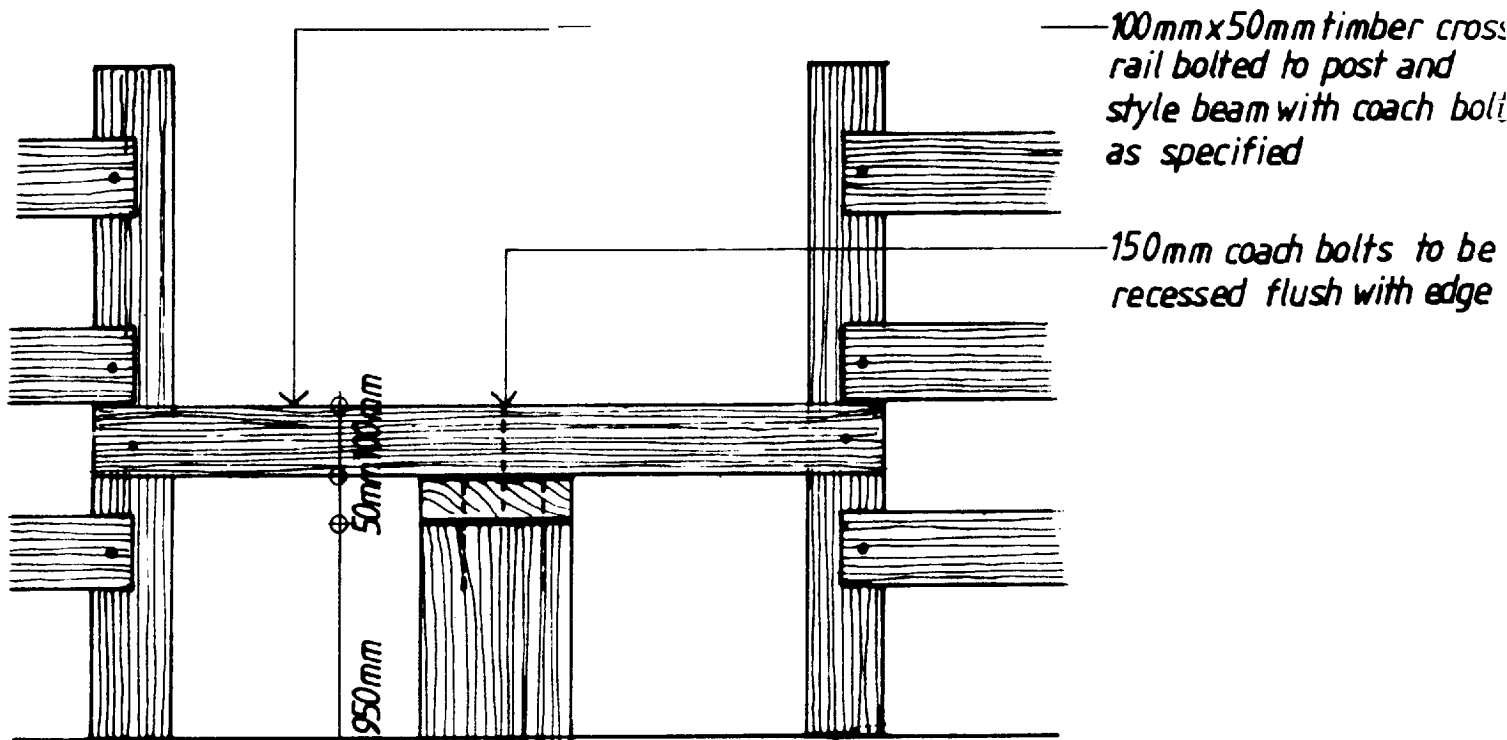
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PLAN

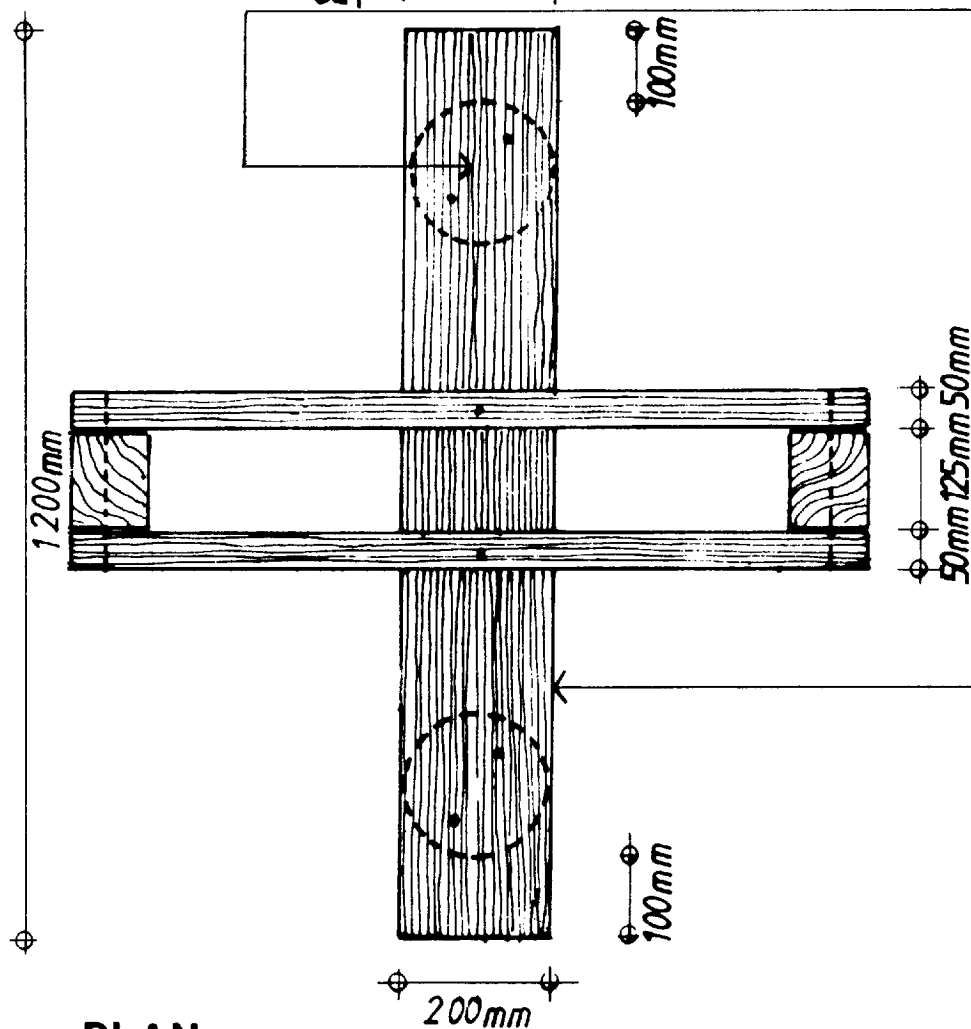
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STILE
timber V-shape



SECTION

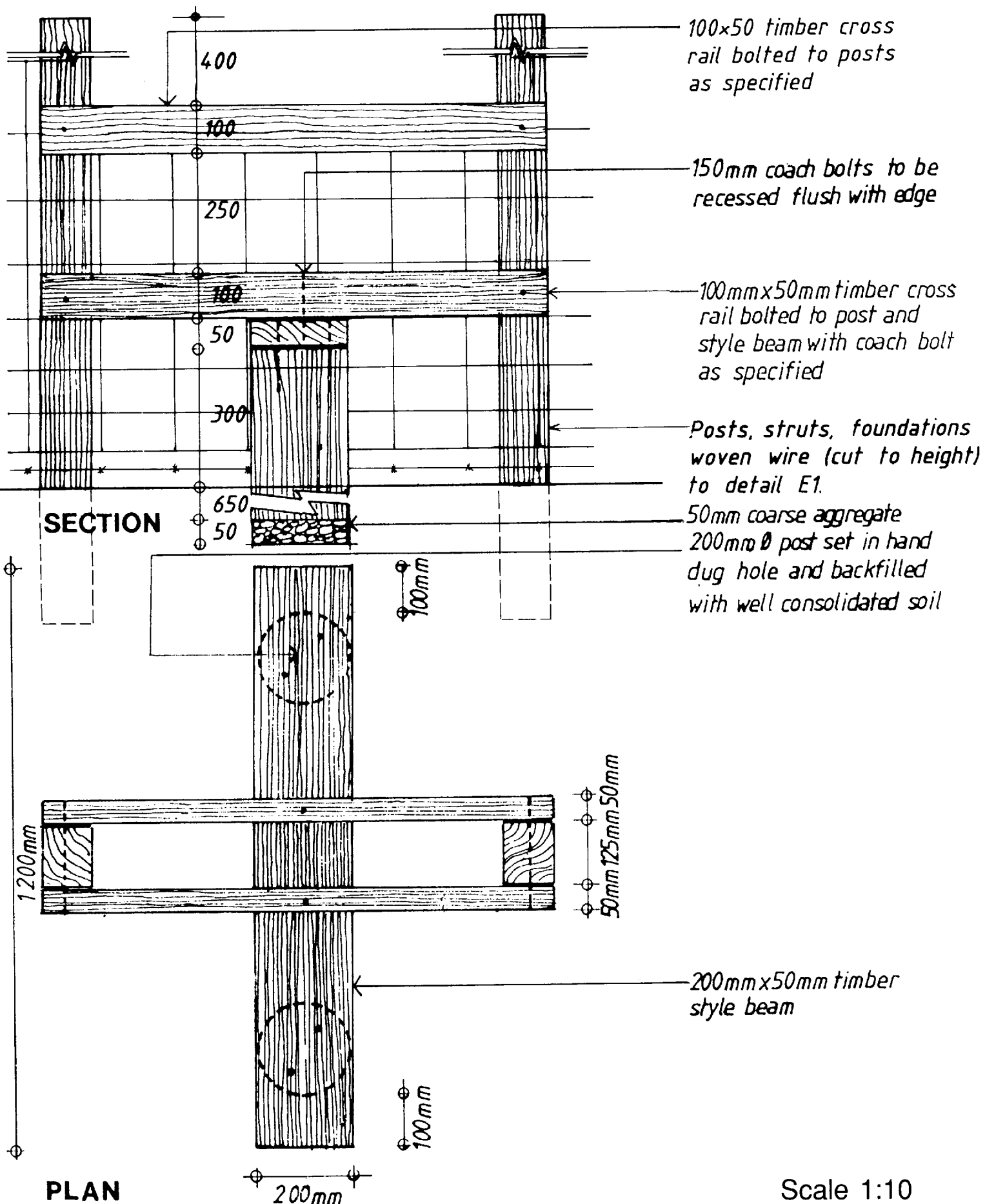
50mm coarse aggregate
200mm $\varnothing$ post set in hand dug hole and backfilled with well consolidated soil



PLAN

Scale 1:10

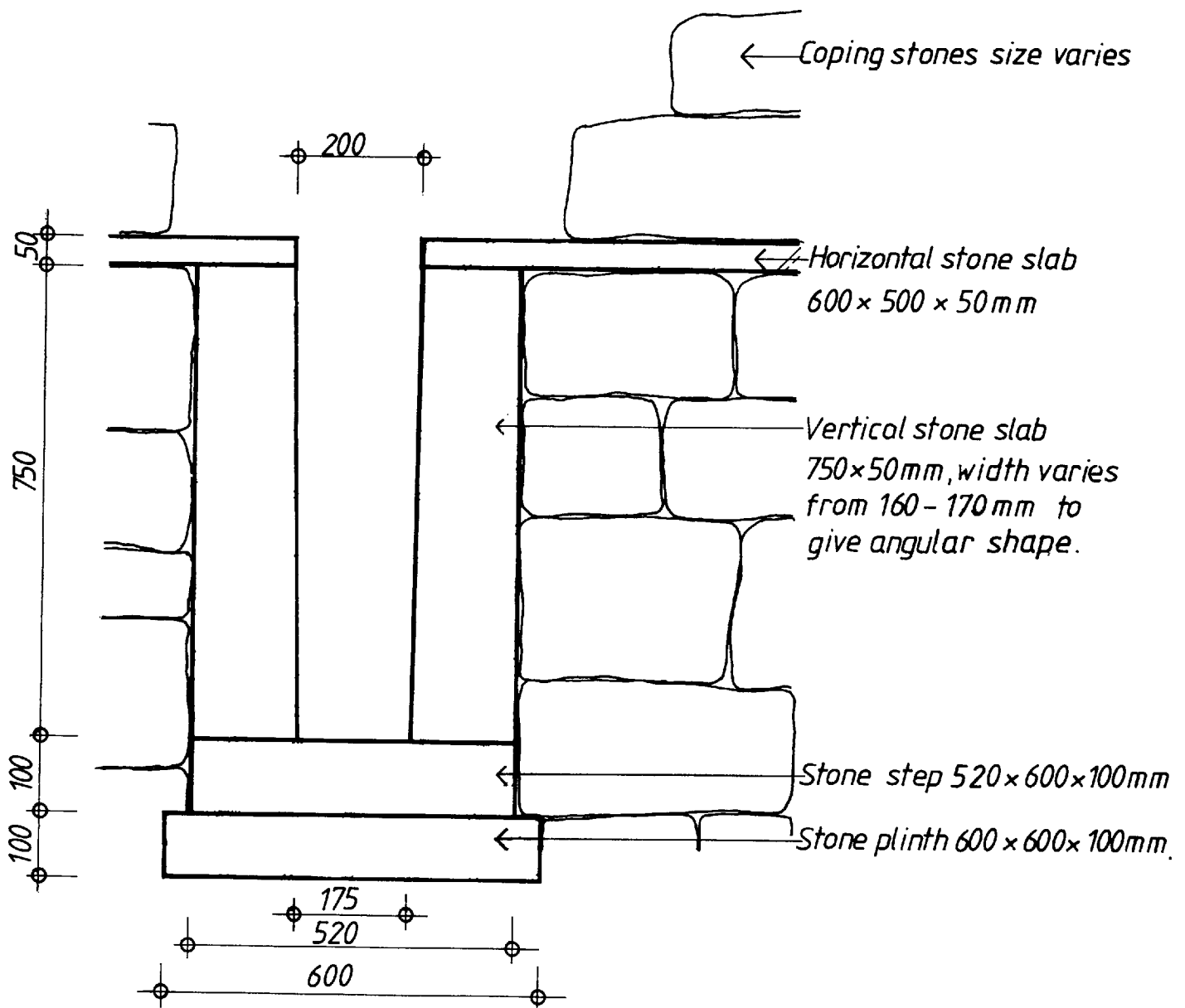
STILE
timber stepped



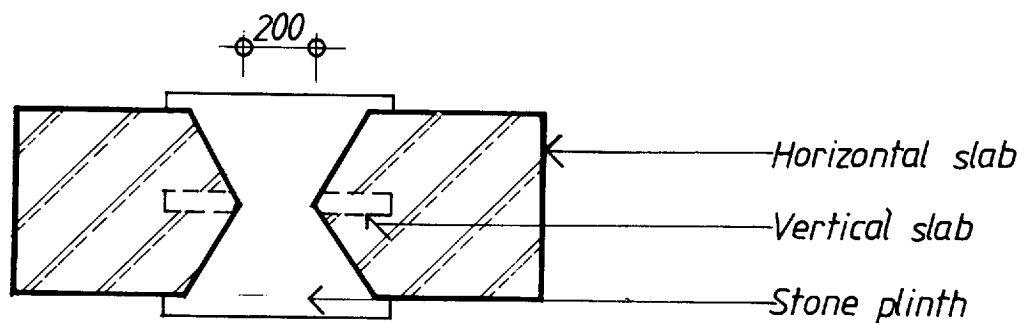
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STILE
timber stepped

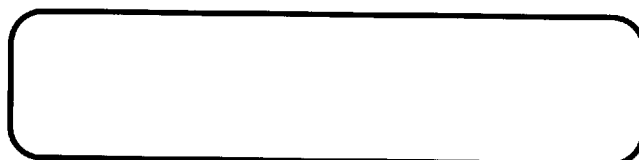


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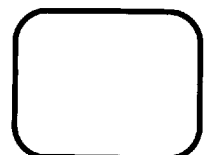


SECTION 1:20

Scale 1:10



STILE
stone



BARRIERS

GUIDANCE NOTES

Low rails

Low rails are used as barriers to protect grass or planted areas from pedestrians, except when some are used as traffic barriers in certain circumstances. As they are not more than 600 mm high they are a visual rather than a physical deterrent and whenever possible are preferable to standard fencing, because they are far less conspicuous.

When the posts of low rails are set in grass they should be set in a concrete mowing strip to avoid hand trimming. Low rails, particularly those in metal, will take curves more easily and cheaply than other materials.

Appearance

The advantage of having a low rail is that it enables a barrier to be used which does not visually intrude to any great extent, depending upon its design. Some rails have to be built to withstand the impact of vehicles and therefore they must have far greater strength than a light rail used to keep pedestrians away from an area. The height of a rail also plays an important part.

Low timber rails

Size

Because of their height (up to 400 mm), the designer of low rails should bear in mind their potential for being sat or balanced upon. The design should either cater for, or discourage, such use.

Post spacing, depth of insertion

In view of the factors mentioned above, post centres should be closer (usually 1,000–1,500 mm) than in fencing intended solely as a barrier.

For pedestrian use, a post insertion depth of 300 mm is sufficient if a concrete surround is provided. For controlling vehicular traffic, an insertion depth of 750 mm is required.

When used as a barrier for vehicles or for wheel stops in a car park, timber posts and rails are preferable. The rail must be strong, i.e. 150 × 50 mm with half-lapped joints over stout posts (100 × 75 mm or 100 × 100 mm), chamfered on top. A far stronger barrier is required to prevent vehicles running off a main road.

Joints in top rail

The solution to the problem will vary with the design. The following general points should be observed:

1. Lapped or scarfed joints are preferable.
2. Bolts or screws should not be inserted within 25 mm of board ends.
3. Where the rail consists of two or more members; the joints in each should be staggered.

Corner detail and changes in direction

Where corners and changes in direction are infrequent, the best solution is to duplicate the post support. Where a top rail is used, adequate bearing must be arranged for a mitred joint.

Low metal rails

Size

The remarks made in regard to timber low rails apply equally to metal low rails.

Spacing of standards

The standards for low metal rails are generally spaced at 1,200 mm centres. A surround of concrete must be used because of the small surface area of metal sections.

Metal protection and finish

BS 1722: Part 8 specifies a range of ex-works finishes, the usual finish being a red-oxide primer or hot-dip galvanising after fabrication.

The finished element should be shop welded, construction made with a loose end

to the rail, which is jointed on-site using a screwed sleeve or plate.
With metal rails, changes in direction are usually made in the rail; the space between standards should be closed in order to increase support at this point.

Bollards

These are vertical barriers and are one of the most unobtrusive ways of preventing access by vehicles encroaching on to pedestrian areas. Bollards can sub-divide paved areas into those for people to move quickly and those where they can congregate in a more leisurely manner. Bollards also mark boundaries and protect property, such as corners of buildings. There are a variety of bollards in different materials: metal, wood, concrete, cast iron and natural stone.

As a barrier to vehicles, bollards should be located at 1,500 mm minimum centres. Bollards should be set 300–450 mm in the ground on a concrete or consolidated hardcore base and surrounded by concrete. If bollards are set in precast paving, careful cutting of the units will be required, unless cobbles are used around them.

Precast concrete

Precast concrete bollards are used more frequently in urban areas because they are relatively cheap and also very strong. There is a variety of pleasant proprietary designs on the market obtainable in different finishes and colours.

Timber

Timber is used more in rural areas and should not be less than 100 mm diameter or square. Hardwood should always be used and the top capped or weathered to shed water.

Metal

Steel and cast iron bollards are becoming more popular in urban areas, especially in historic towns and cities. Removable bollards are usually made of steel.

SPECIFICATION CHECK LISTS

Low rails

General

State purpose – pedestrian or vehicular.

Height

Specify height of rail above ground level.

Material

Timber: Specify type, e.g. hardwood (oak) or softwood (pine). Insert method of treatment and/or desired life (e.g. CCA category 13). State size of timbers to be used for posts and rails.

Metal: Specify type, i.e. RHS – rectangular/round hollow section, RSA – rectangular section angled, etc. Insert if metal is to be hot-dipped galvanised or primed with red-oxide primer. State size of posts and rails.

Workmanship

Centres of posts: State distances between posts.

Setting of posts: Insert method and depth depending upon use, e.g. pedestrian low rail timber posts could be set in rammed earth but metal posts should be set in concrete as should all vehicular rails/barriers timber and metal.

Construction: Describe method of construction of rails to posts, especially joining lengths of rails.

Finish: Specify type of finish, e.g. timber to be stained, metal to be painted.

Bollards

General

State purpose of bollard, especially if it is for a vehicular barrier. Fixed or removable.

Height

Specify height above ground level.

Material

Timber: Specify type, e.g. hardwood (oak) or softwood (pine). Insert method of treatment and/or desired life (e.g. CCA category 13). State size of timbers to be used. Describe any special type of shaping or grooving.

Metal: Specify type and shape.

Workmanship

Centres of bollards: State distances between bollards.

Setting of bollards: Insert method and depth depending upon use, i.e. pedestrian or vehicular.

Finish: Specify type of finish, e.g. timber to be stained or painted, metal to be primed and painted.

Removable: Describe method of making bollard removable.

DETAIL SHEETS

Barriers

Timber foundations – 300 high

– 350 high

– 400 high (4)

– 500 high

– 550 high

Metal (300–500 high) (3)

Metal and timber (450 high)

Bollards

Fixed timber – 750 high

– 950 high

– 1,000 high

– 1,200 high

Concrete – 600 high

– 600 high

– 750 high

– 1200 high

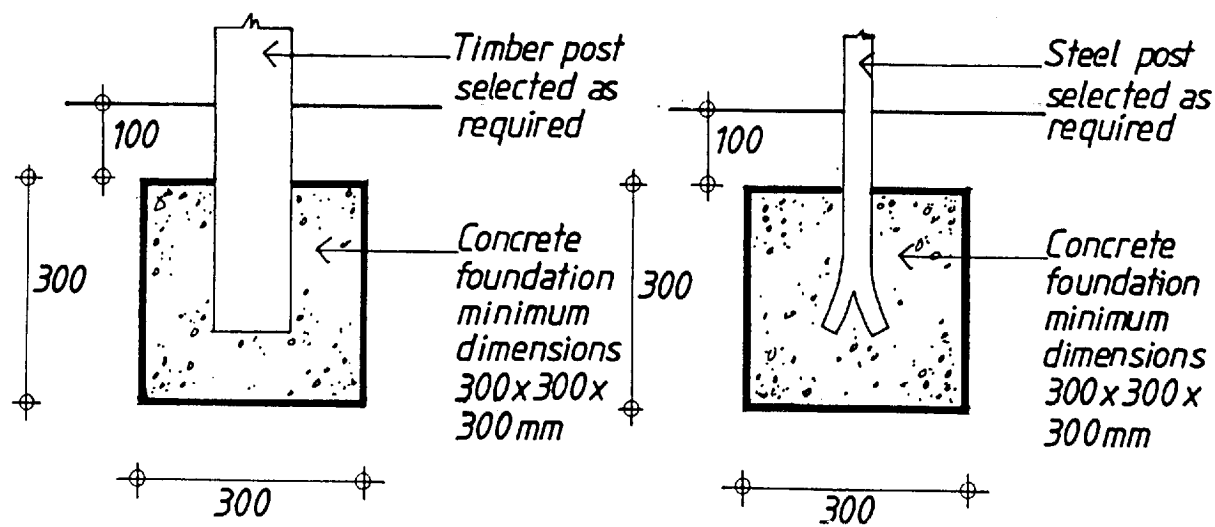
Metal – 750 high

Removable timber – 400 high

– 900 high

– 1,000 high

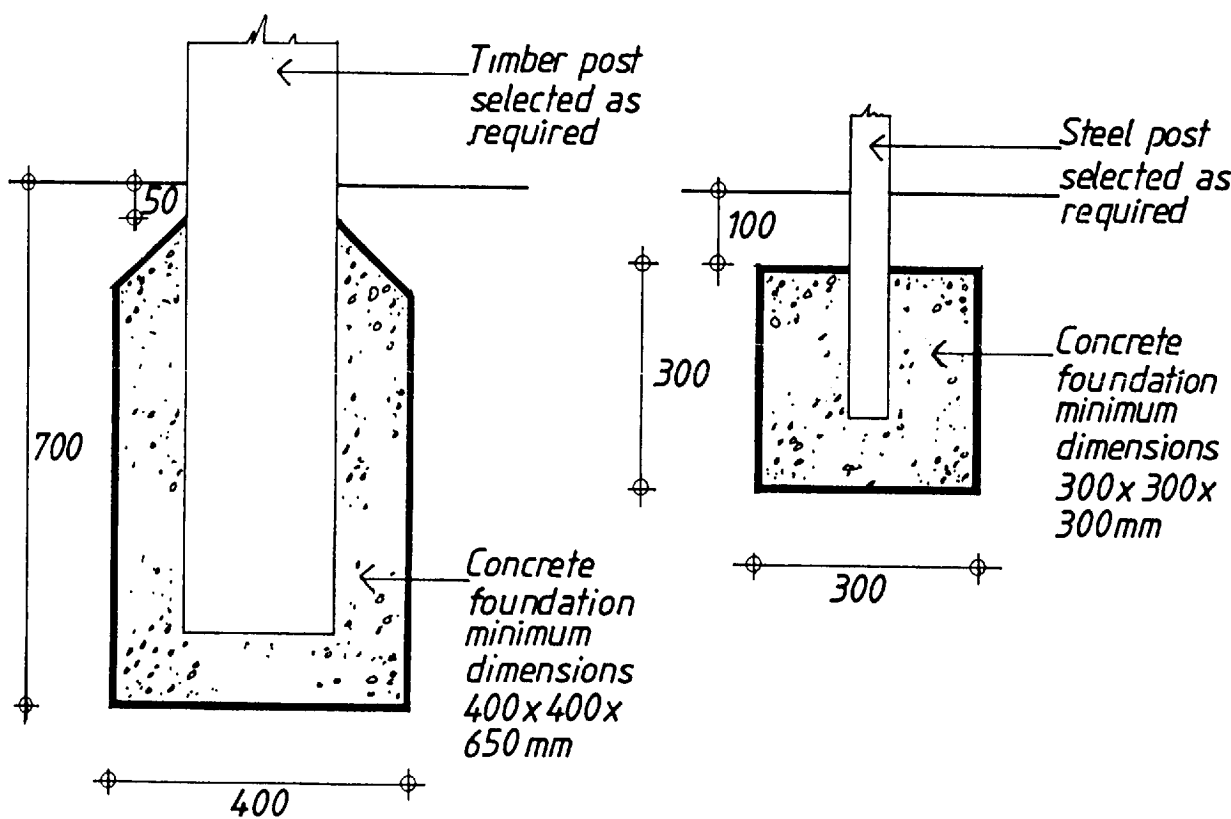
PEDESTRIAN



TIMBER POST

STEEL POST

VEHICULAR

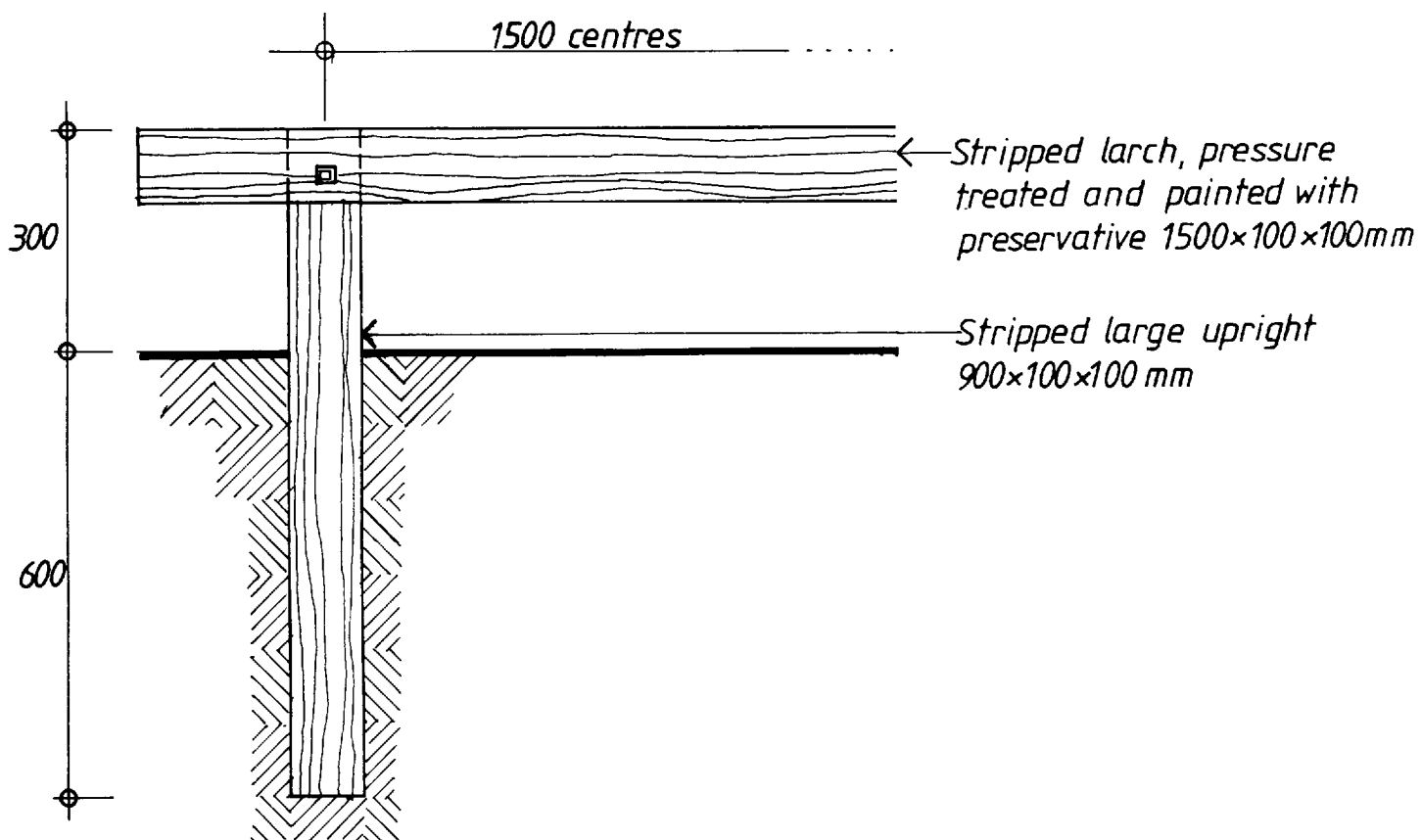


TIMBER POST

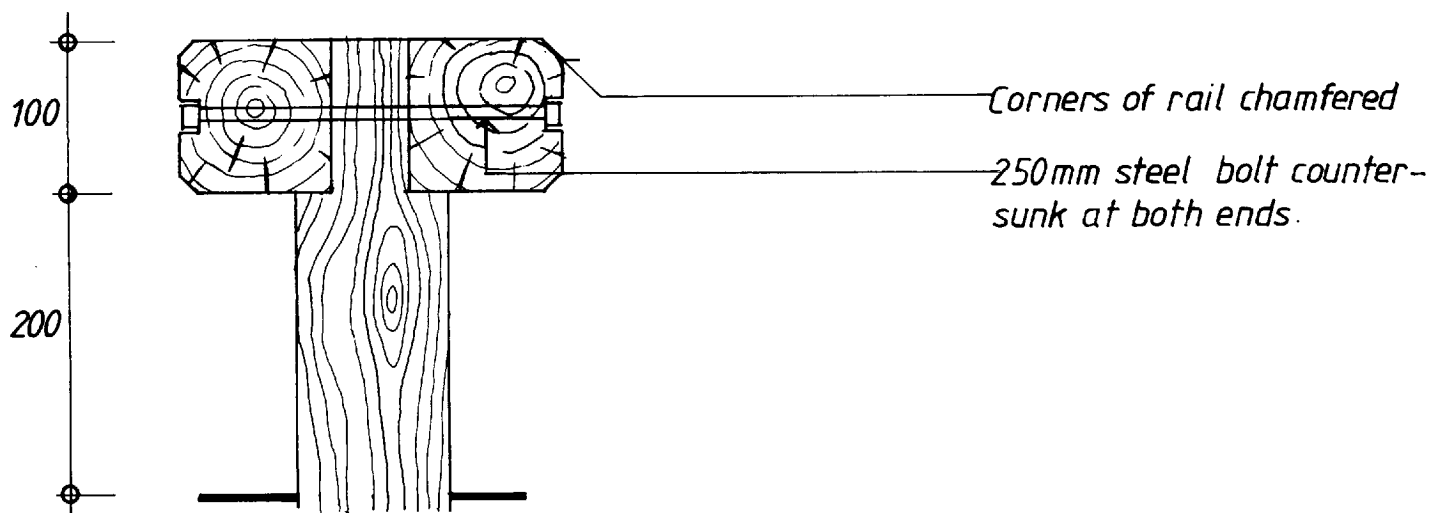
STEEL POST

Scale 1:10

LOW RAILS/BARRIERS foundations



ELEVATION



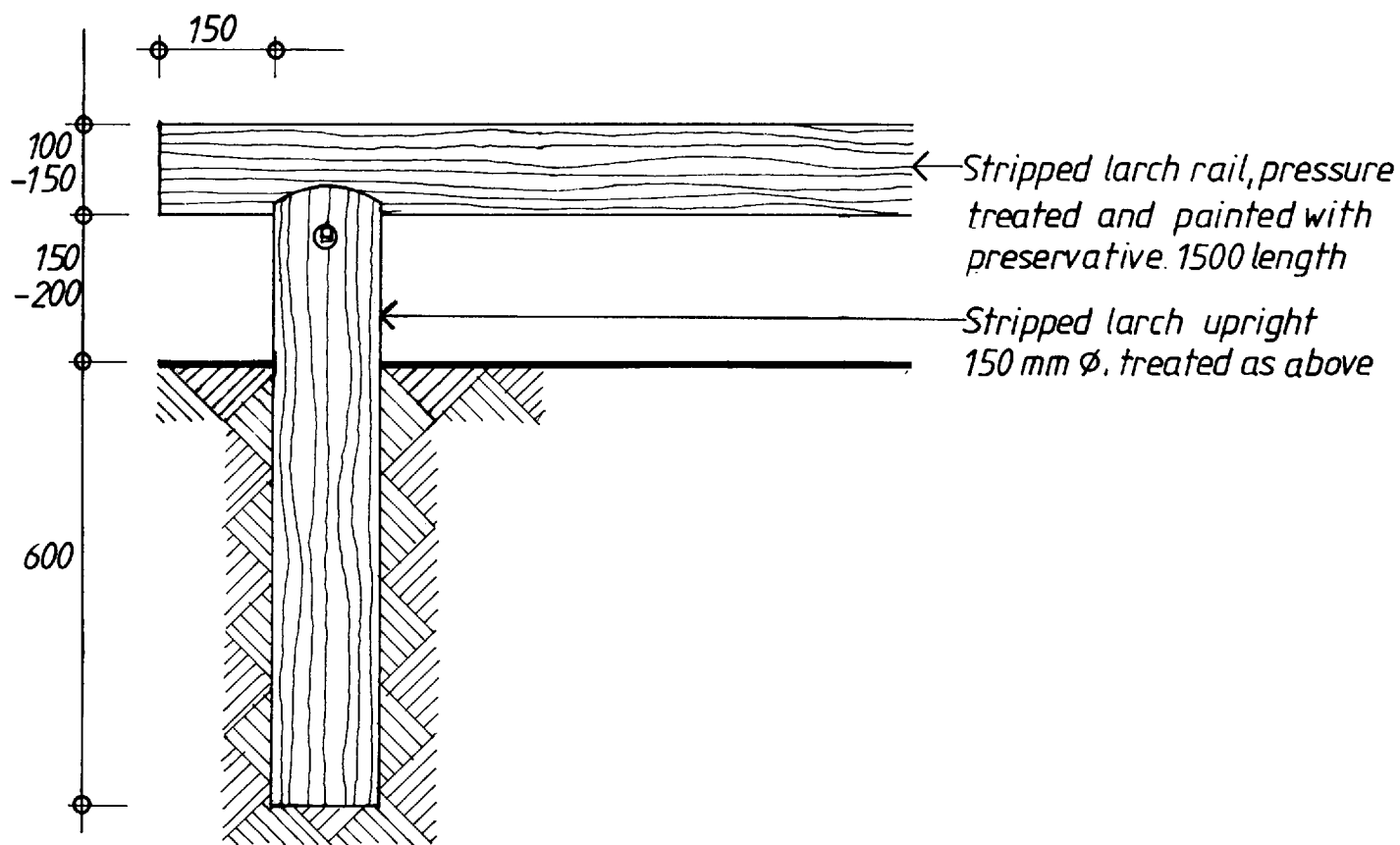
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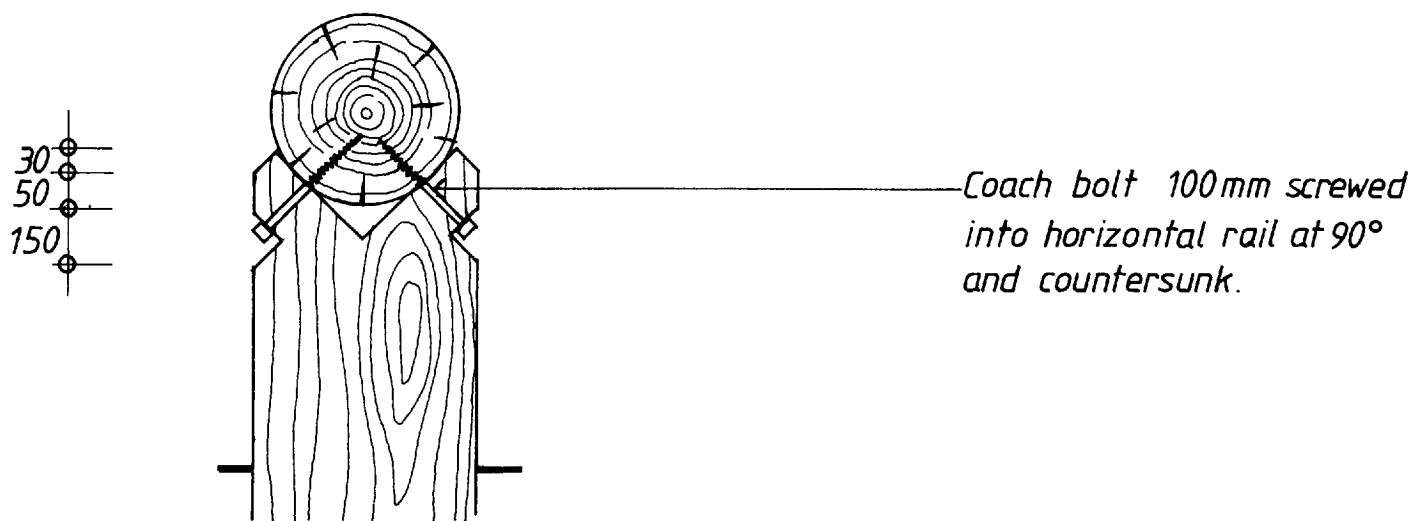


LOW RAIL
timber





ELEVATION

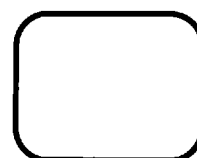


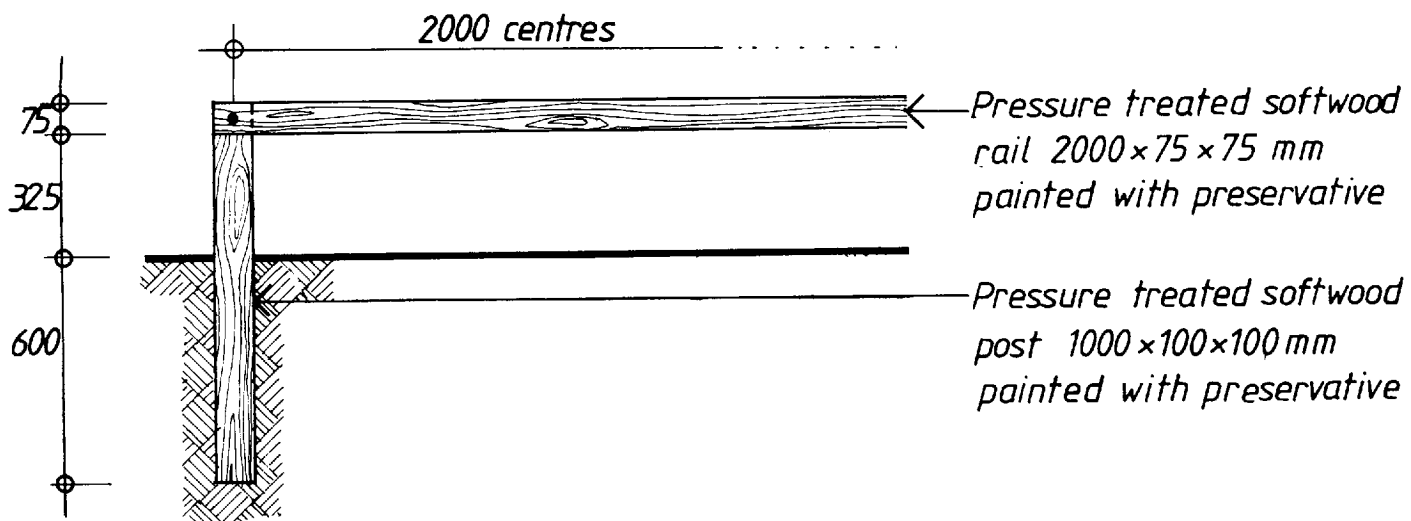
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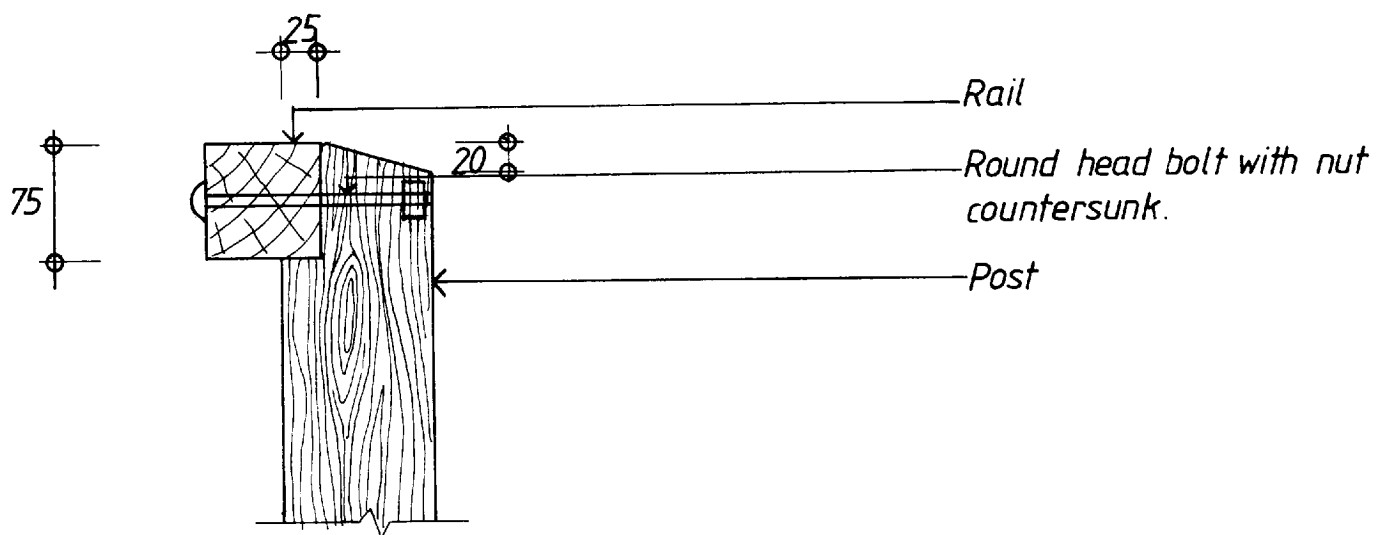


LOW RAIL
timber





ELEVATION

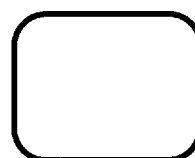


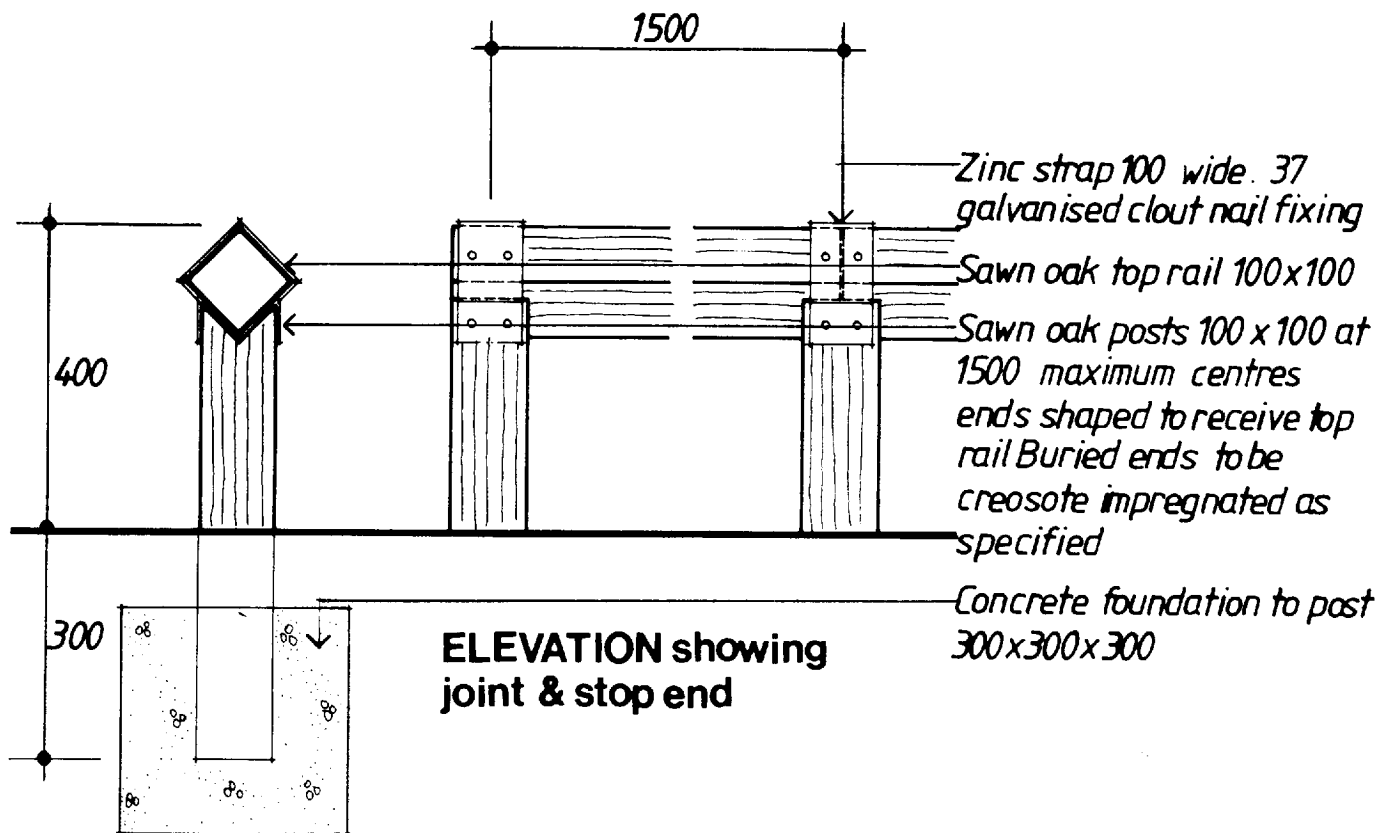
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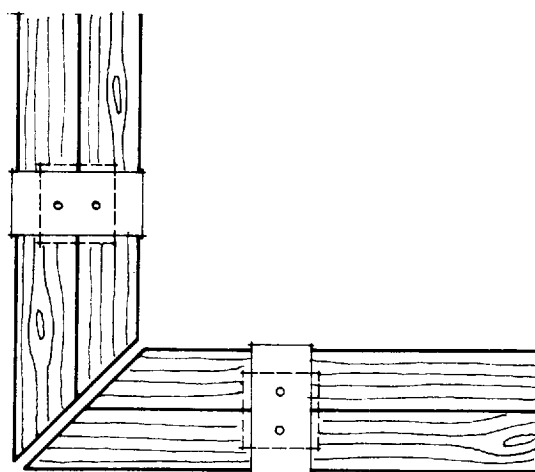


LOW RAIL
timber





SECTION



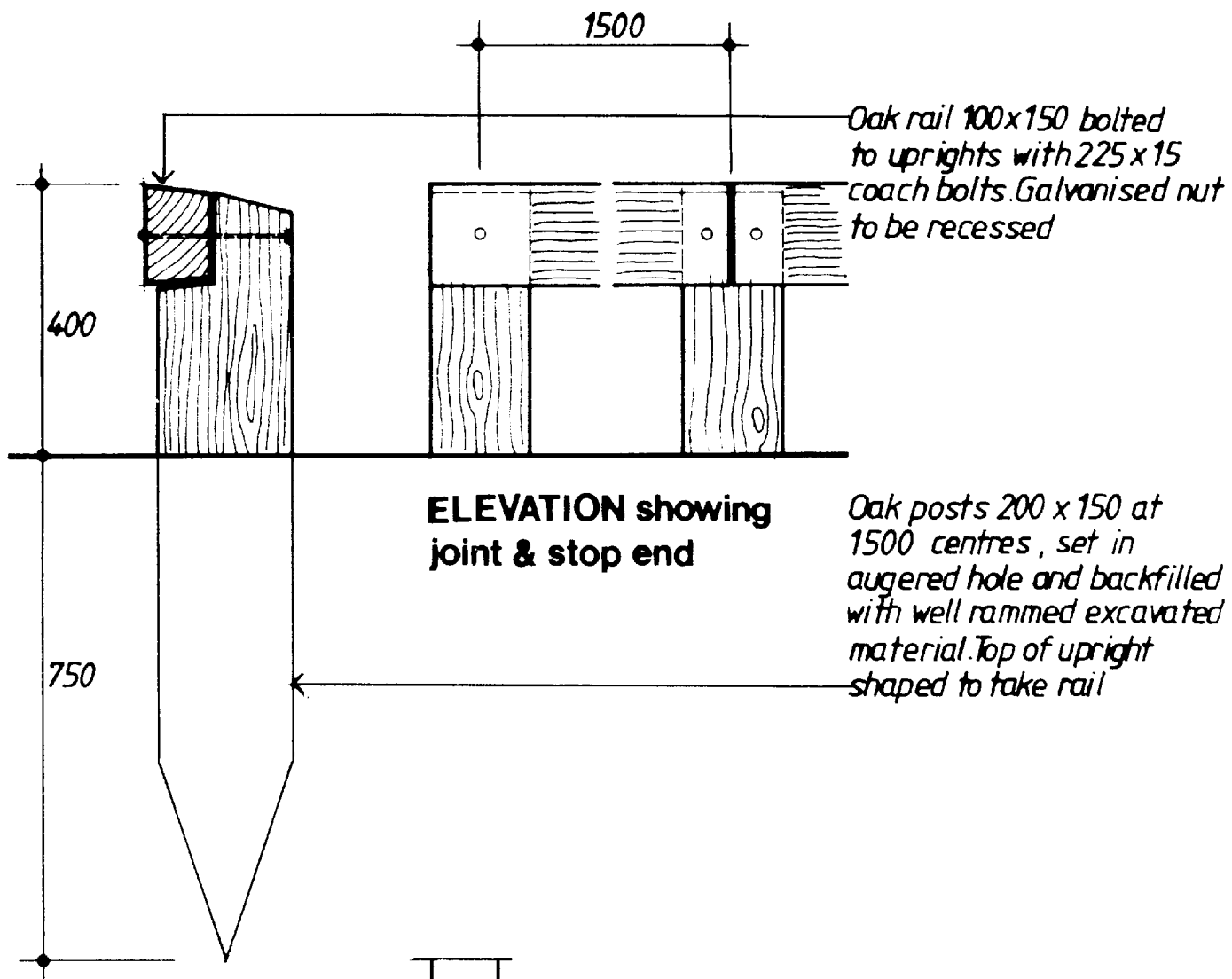
PLAN at corner

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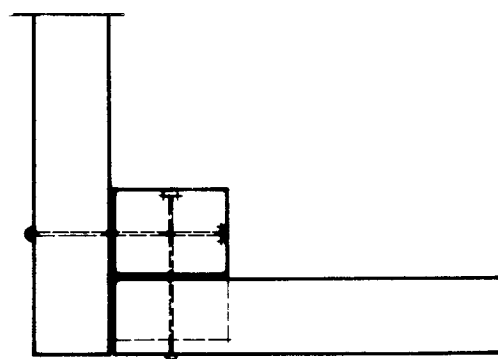


LOW RAIL
timber



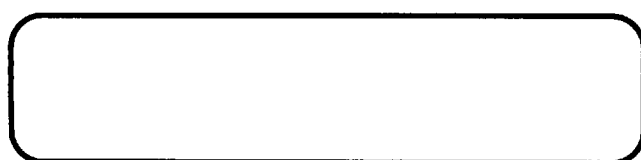


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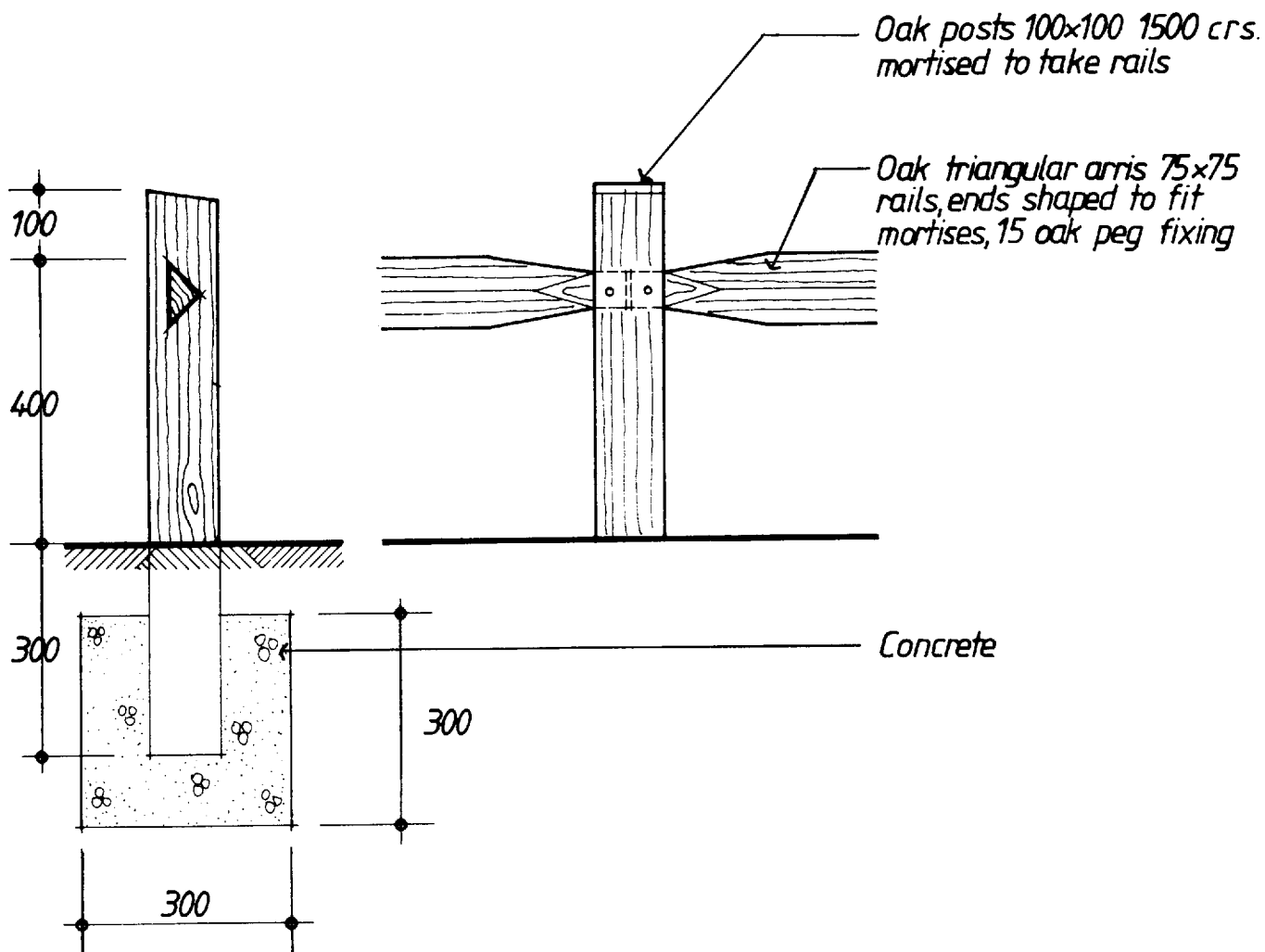
PLAN at corner

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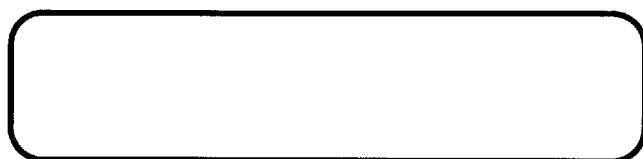


LOW RAIL
timber

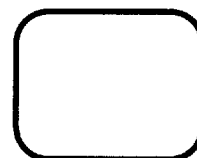


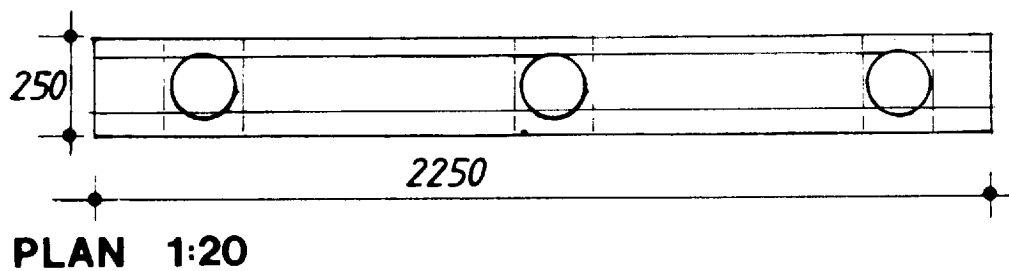
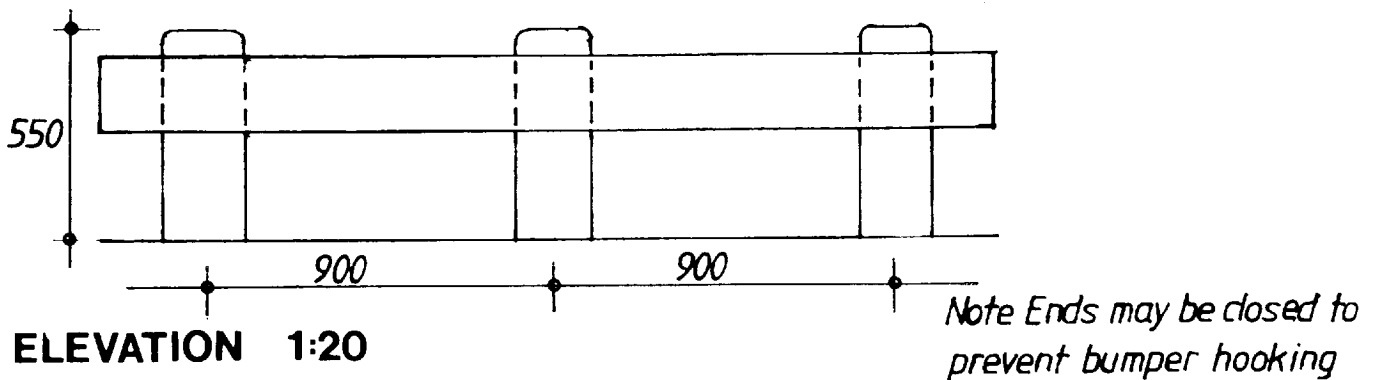
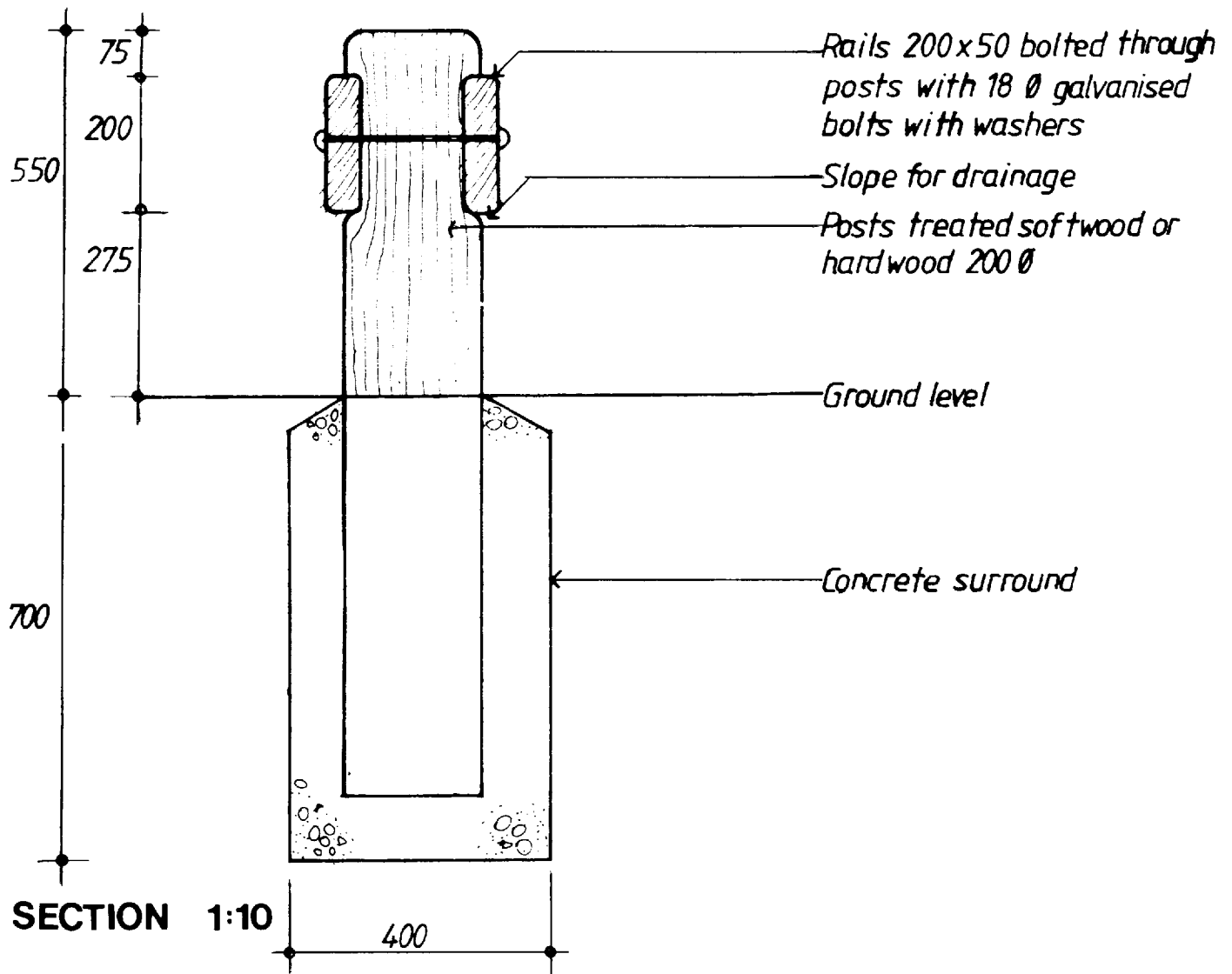


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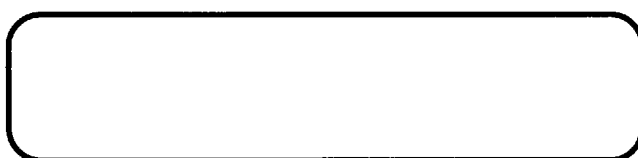


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timber

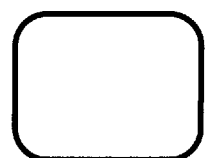


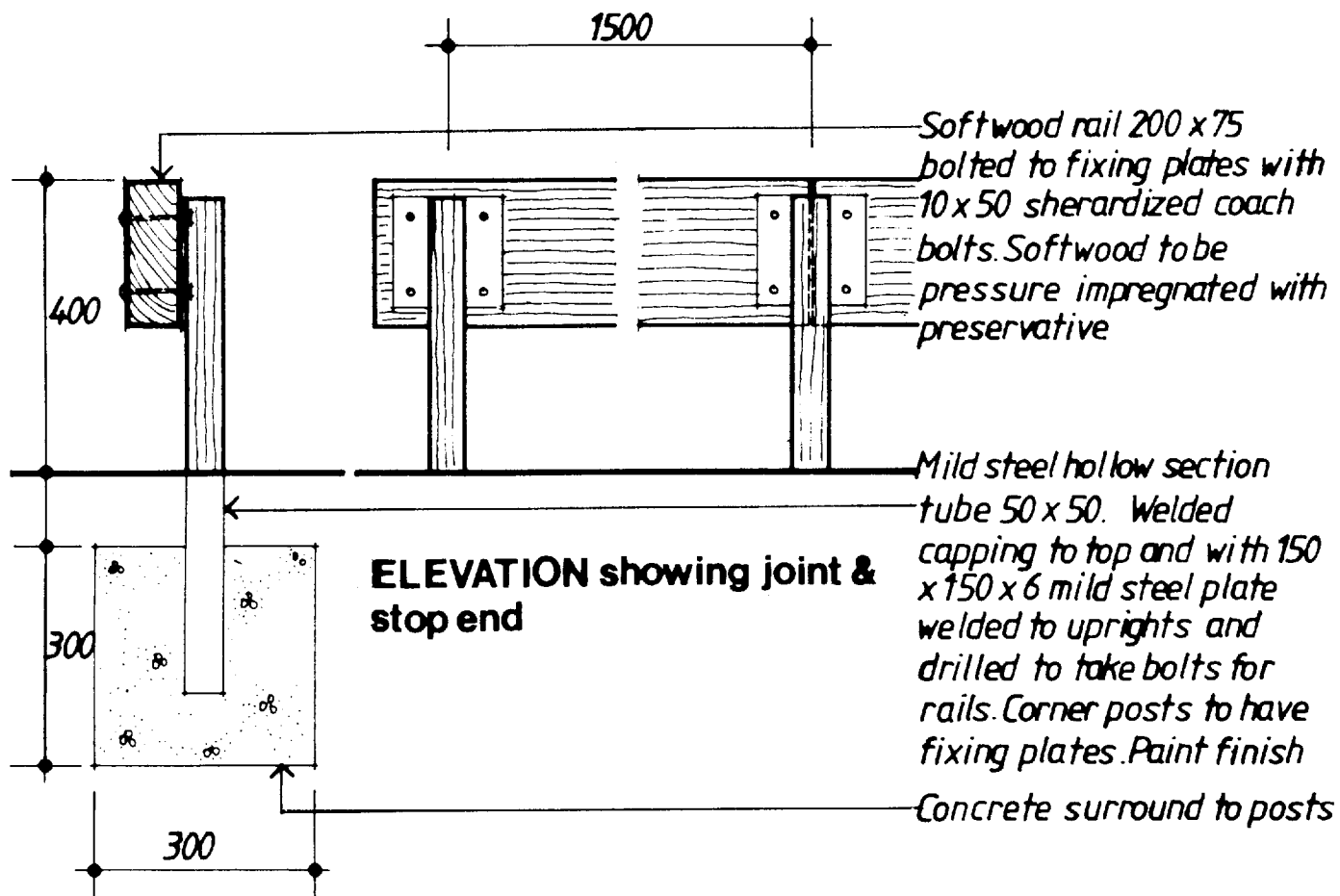


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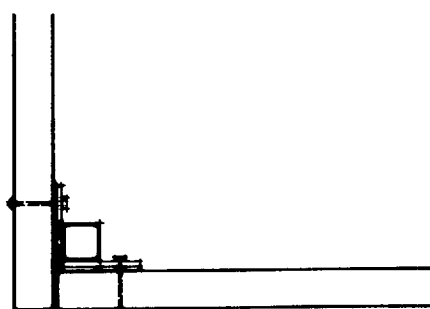


LOW RAIL
timber



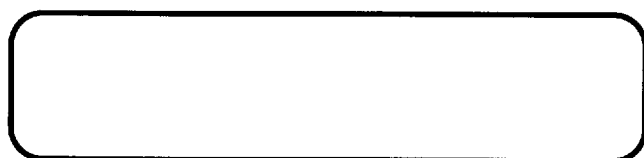


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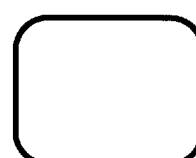


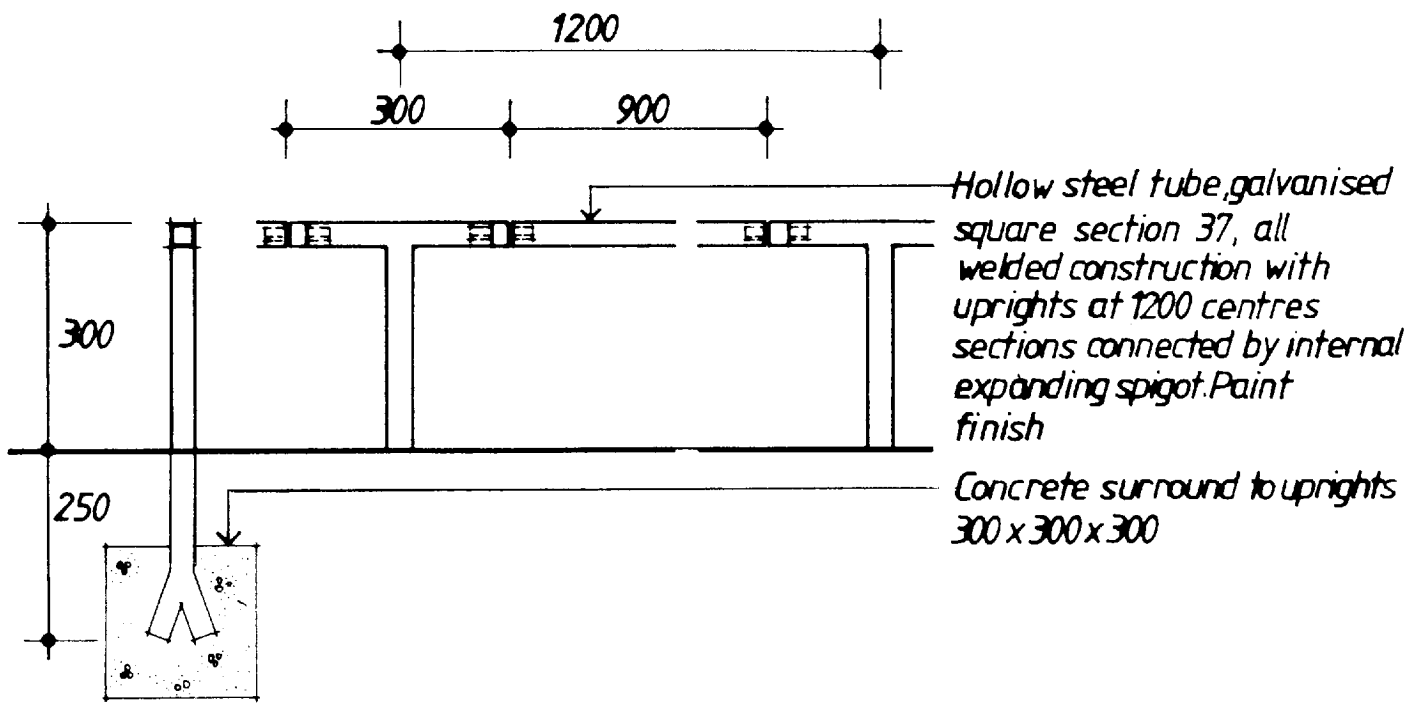
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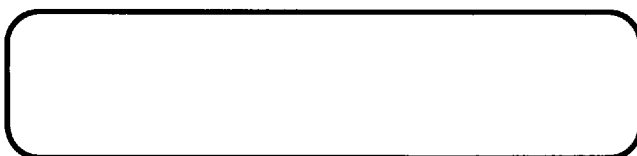
LOW RAIL
timber & steel





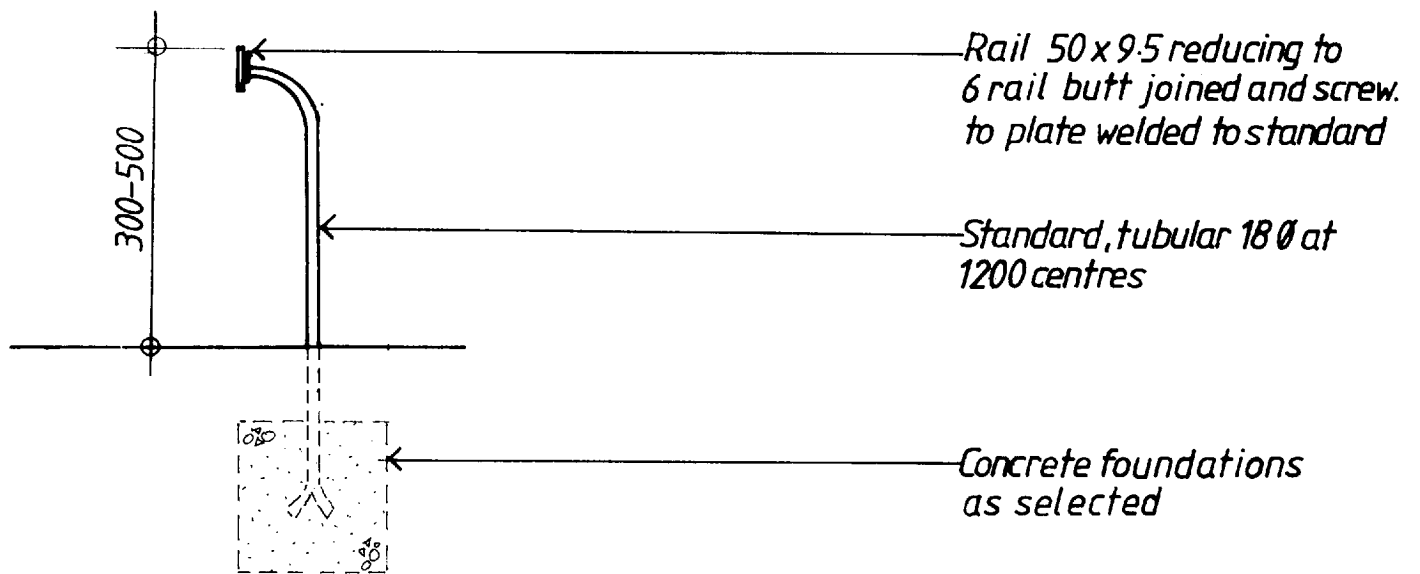
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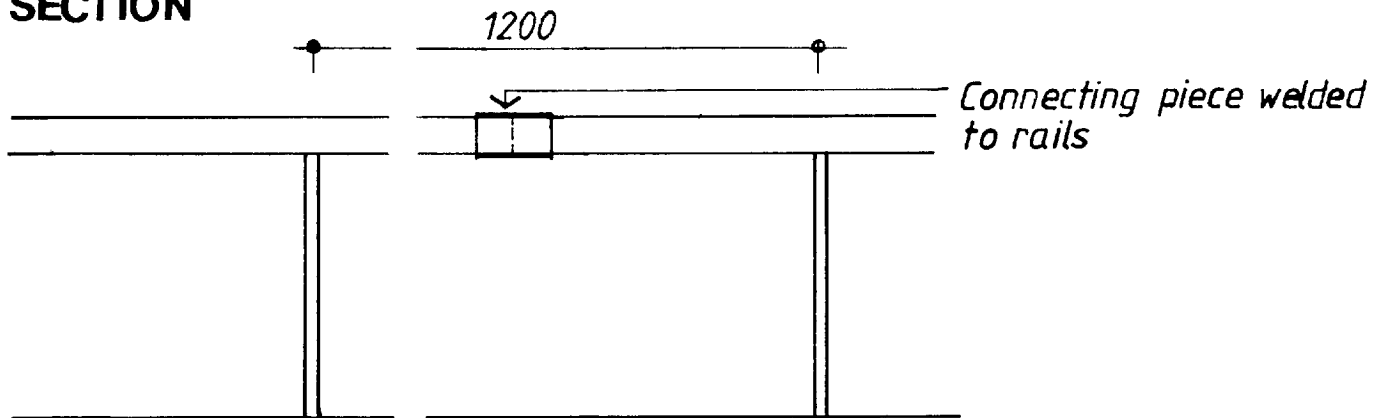


LOW RAIL
steel tube



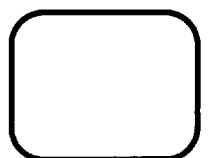
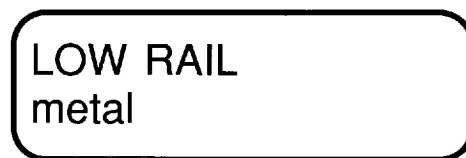


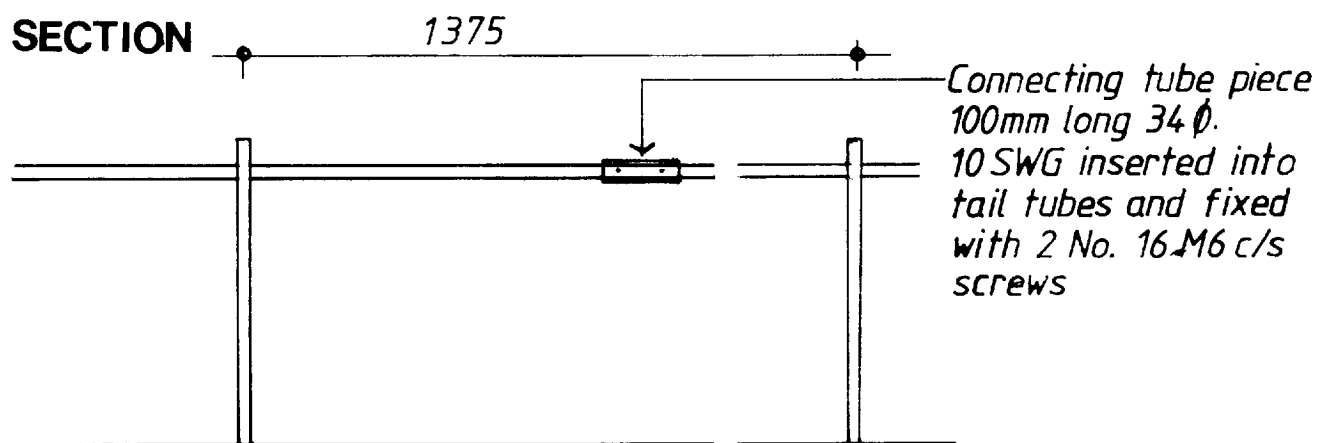
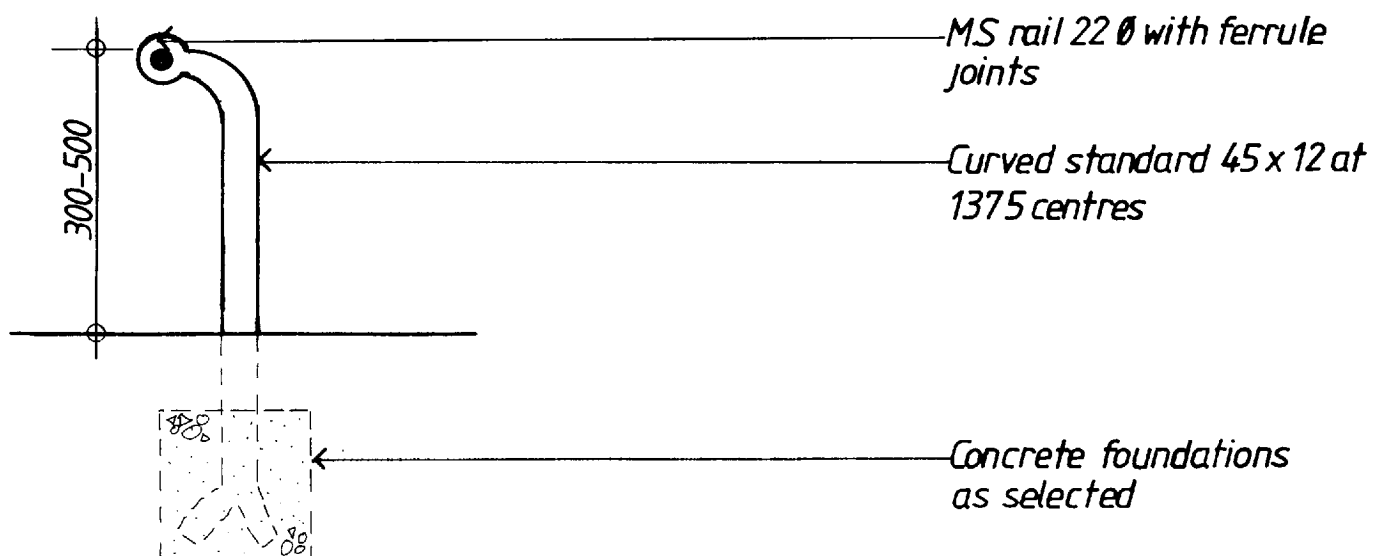
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ELEVATION

Scale 1:10



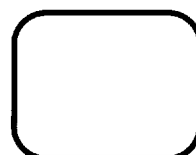


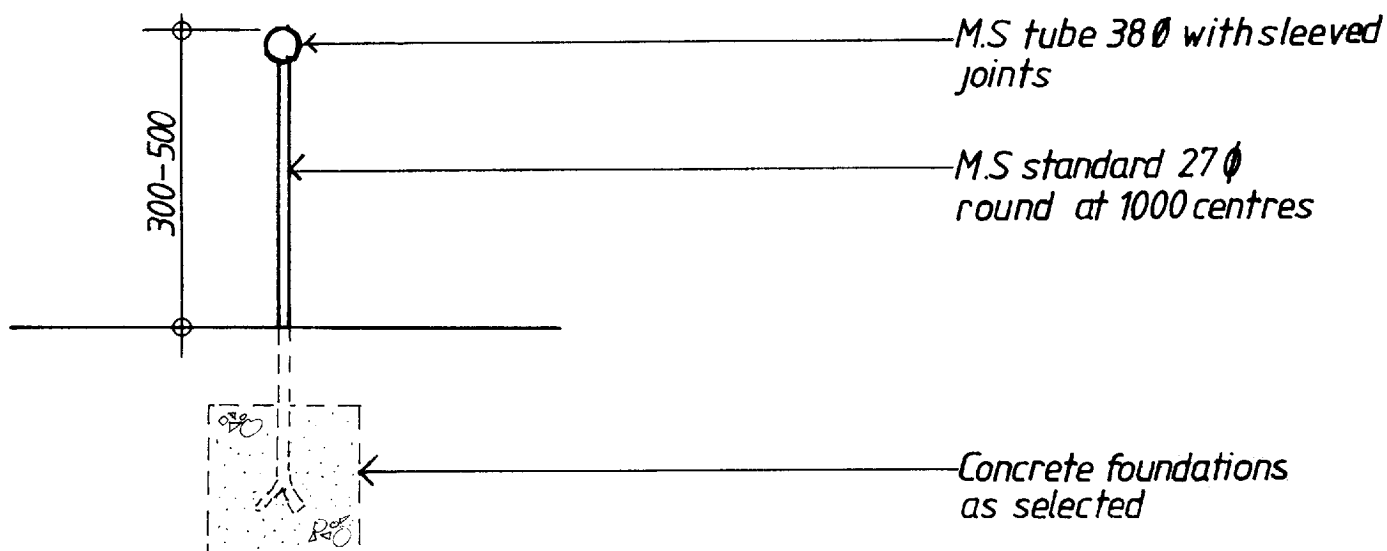
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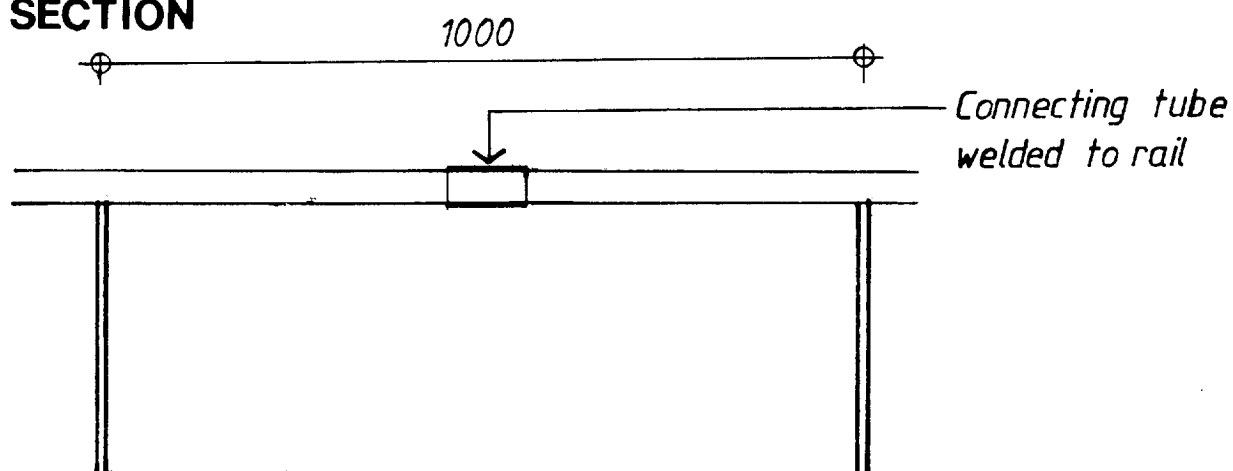


LOW RAIL
metal



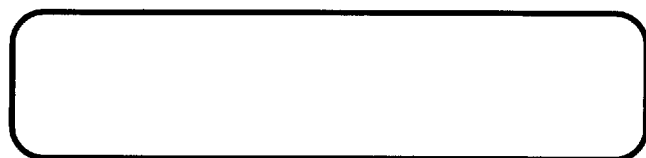


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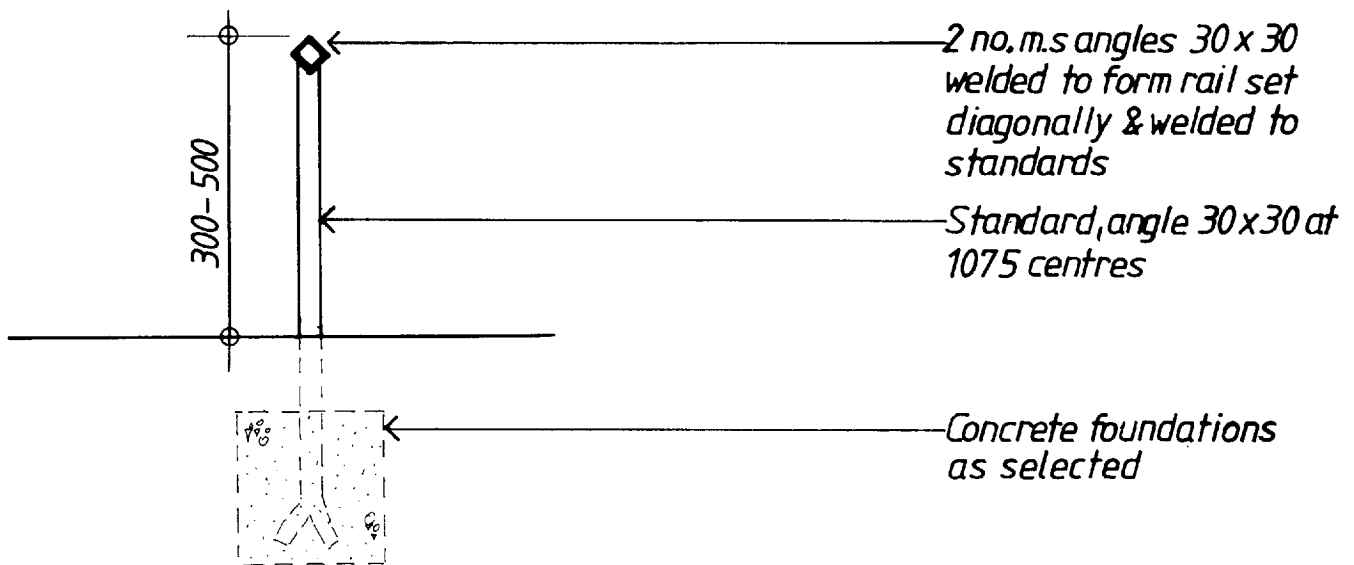
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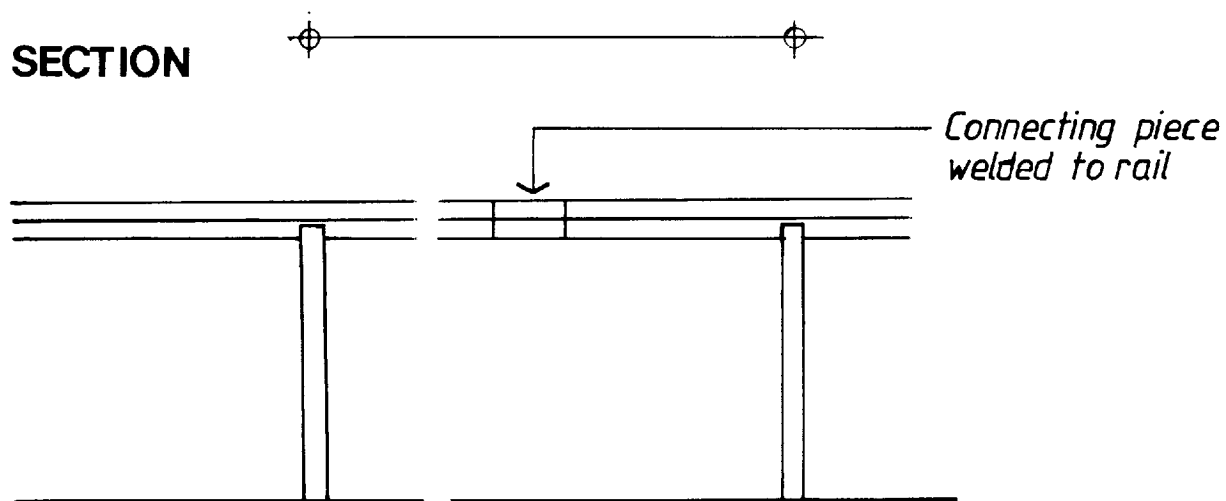


LOW RAIL
metal round





SECTION



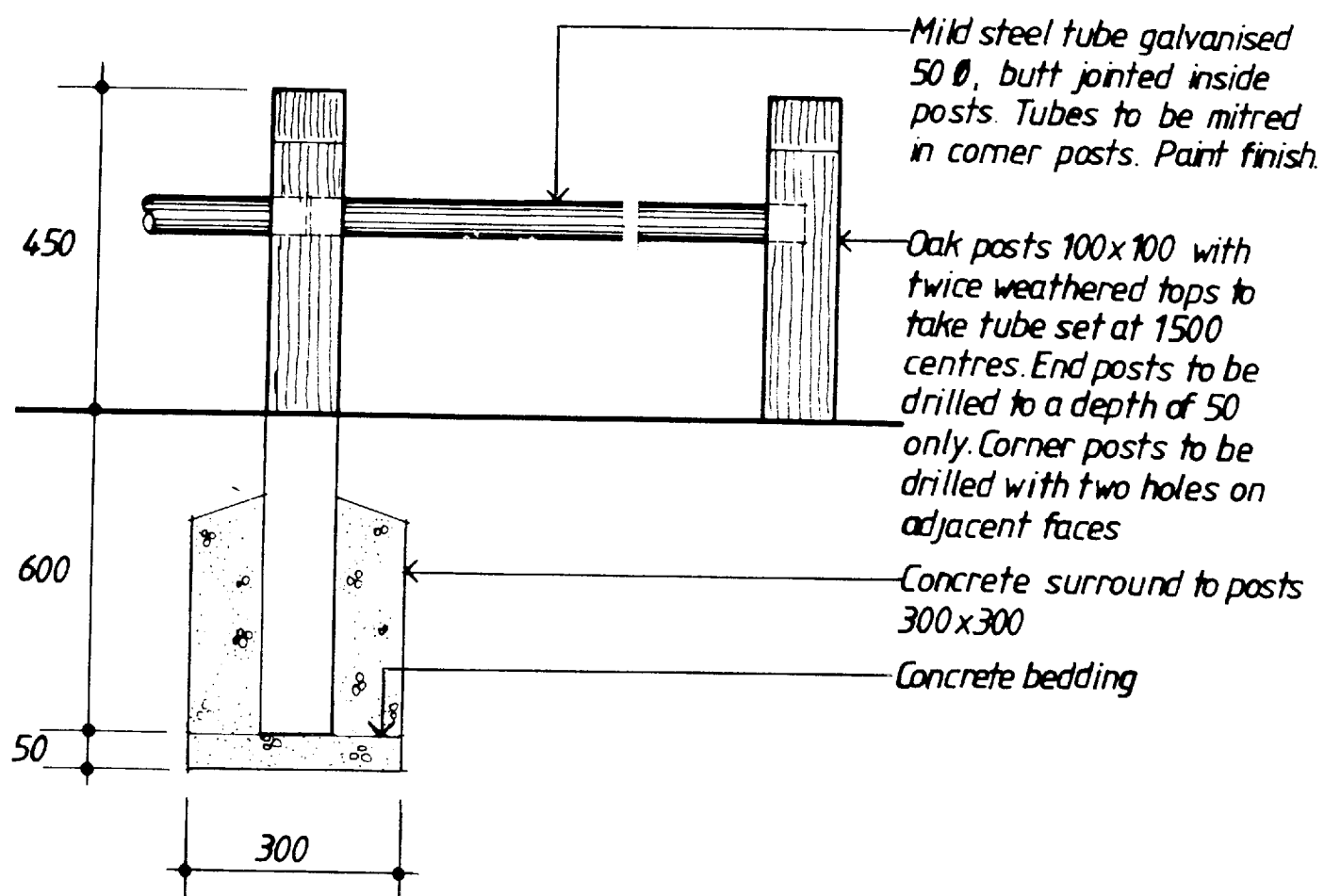
ELEVATION

Scale 1:10



LOW RAIL
metal angular

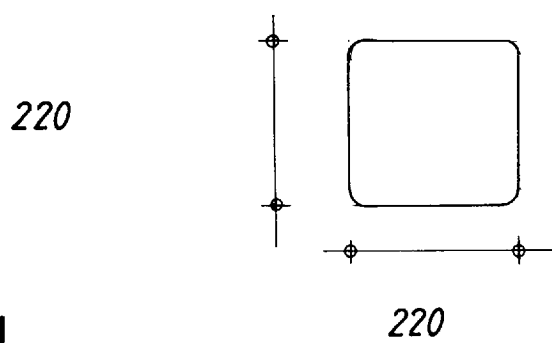
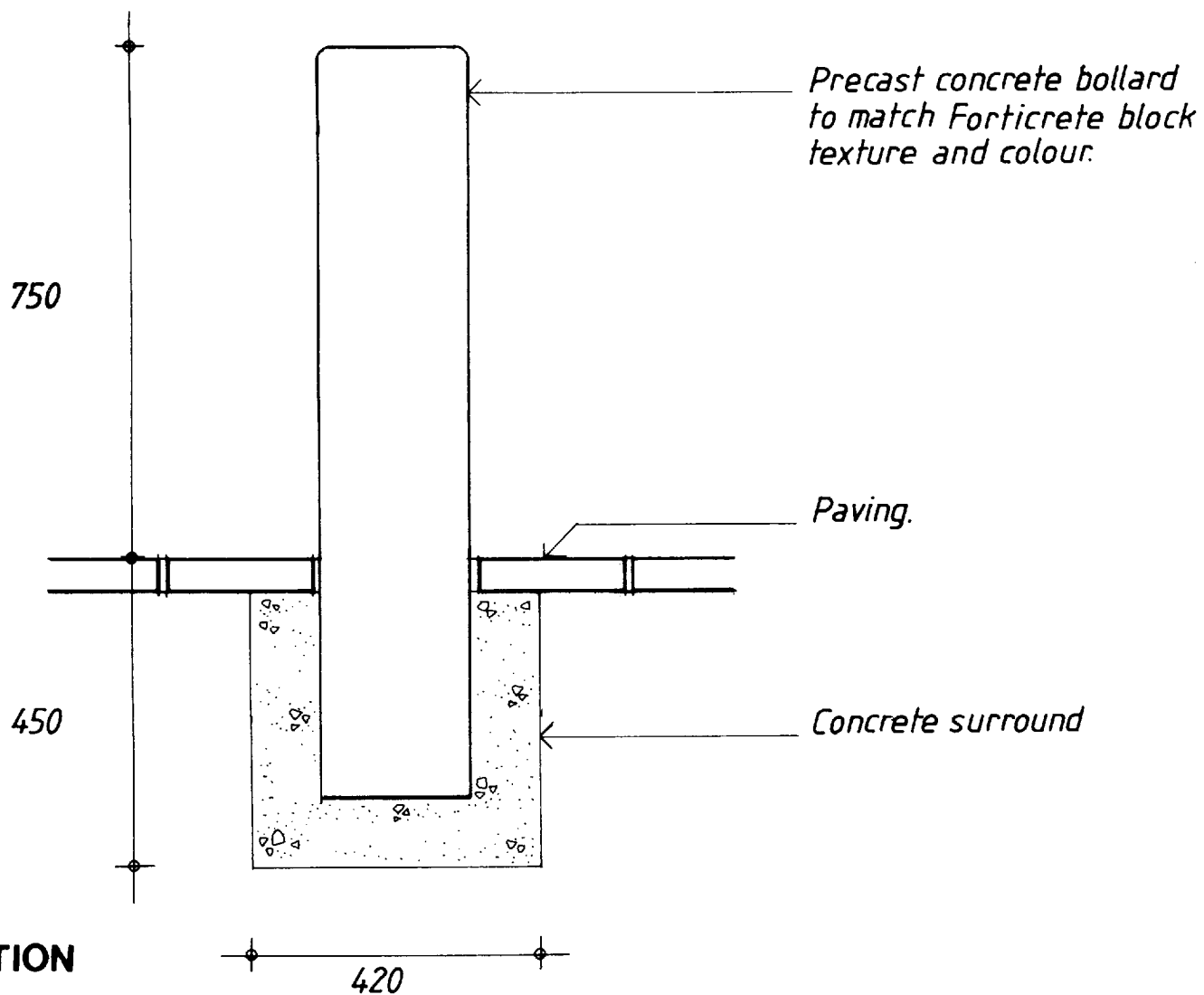




ELEVATION

Scale 1:10

LOW RAIL
timber & steel tube

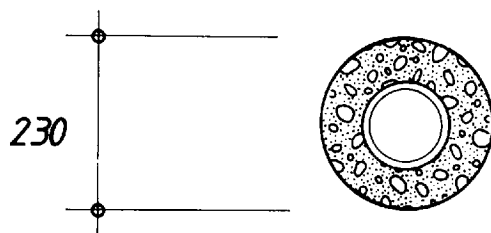
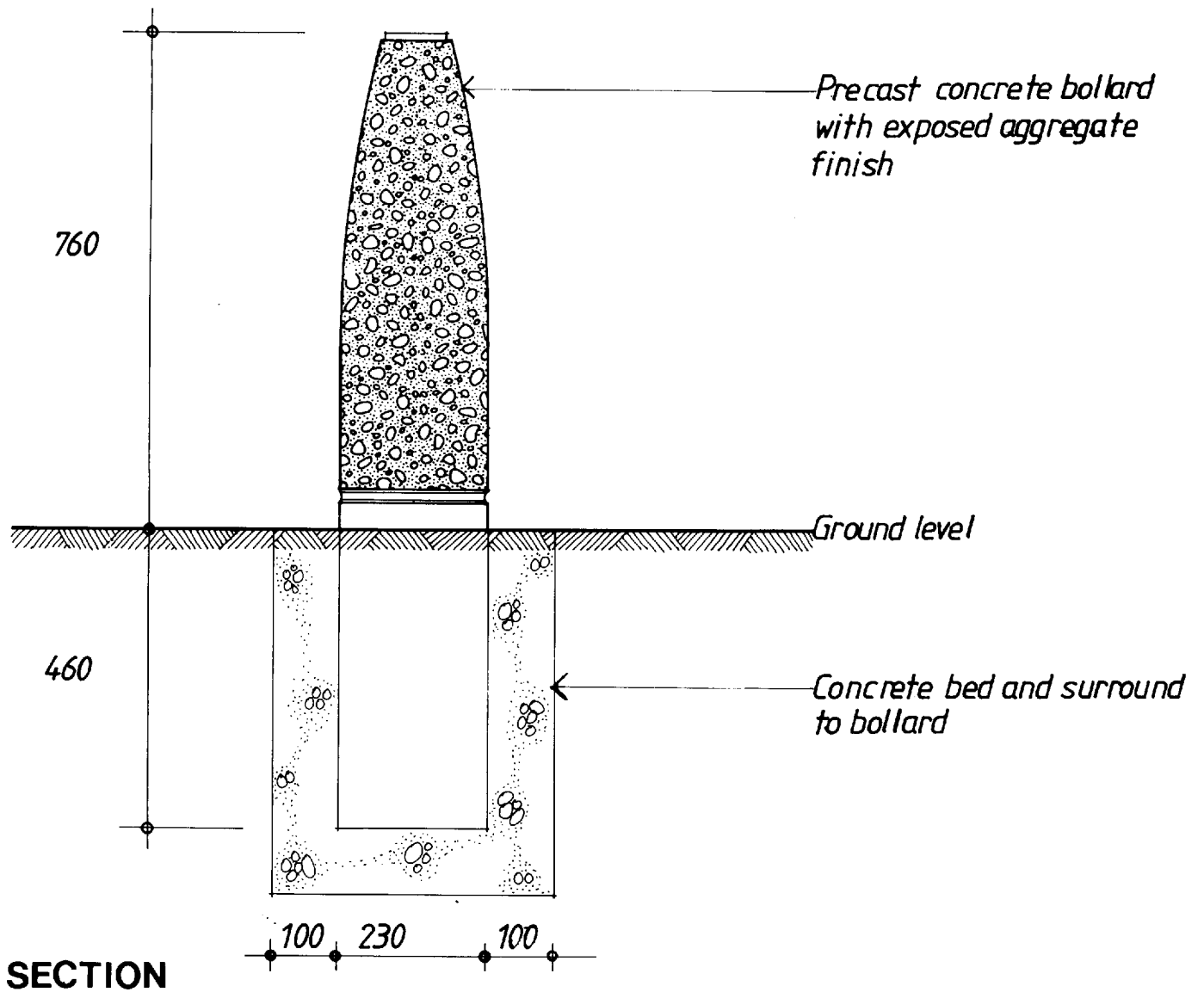


Scale 1:10



BOLLARD
concrete





Scale 1:10



BOLLARD
exposed aggregate



REFERENCES

APPENDIX A

Walls

British Cement Association (BCA)

formerly C&CA

A.K. Tovey, *Concrete Masonry for the Designer*, 1981

Brick Development Association (BDA)

Practical Notes:

5 *Overhand bricklaying*

6 *Damp proof courses and flashings with brickwork and blockwork*

7 *Repointing of brickwork*

Design Notes:

6 *Brick cladding to timber frame construction*

10 *Designing for movement in brickwork*

Design Guide:

9 *Brickwork domestic fireplaces and chimneys*

Publications:

Brickwork Retaining Walls, August 1981

External Walls: Design of Wind Loads

The Design of Free-standing Brick Walls, February 1983

Mortars for Brickwork, September 1973

J.O.A. Korff, *Design of Freestanding Walls*, 1985

W.G. Curtin, G. Shaw, J.K. Beck and W.A. Bray

Design of Brick Diaphragm Walls, 1982

British Precast Concrete Federation

Design of Urban Space: a GLC manual, R. Cartwright (ed.), Architectural Press, 1980

National House Building Council (NHBC)

Practice Note 5 *Timber frame dwellings*

British Ceramic Research Association

Design Guide for Reinforced and Pre-stressed Clay Brickwork,

SP 91:1977

Architects' Journal

Products in practice supplements:

Concrete blocks 27.2.85

Bricks 27.11.85

Salvage and reuse of bricks 6.4.77

Building Research Establishment (BRE)

Digests:

65, 66 *The selection of clay building bricks*

89 *Sulphate attack on brickwork*

77 *Damp proof courses*

160 *Mortars for bricklaying*

165 *Clay brickwork, Part 2*

200 *Repairing brickwork*

214 *Cavity barriers and firestops, Part 1* (withdrawn)

246 *Strength of brickwork and blockwork walls: design for vertical load*

250 *Concrete in sulphate bearing soils and ground water*

277 *Built-in cavity wall insulation for housing*

Defect Action Sheets:

12 *Cavity trays in external cavity walls: preventing water penetration*

17 *External masonry walls insulated with mineral fibre cavity-width batts: resisting rain penetration*

Information Paper:

7/83 *Window to wall jointing*

Current Paper:

16/73 *Long term unrestrained expansion of test bricks*

92/74 *Expansion of unrestrained Fletton brickwork*

Publications:

Thermal insulation: Avoiding risks, 1988

C.W. Newberry and K.J. Eaton,

Wind loading handbook, BRE Report, HMSO

Cement and Concrete Association

Concrete Block Walls, 1966

Screen Walls of Pierced Concrete

A.K. Tovey, *Concrete Masonry for the Designer*, 1981

Fences and gates

Timber Research and Development Association

Timber Fences and Gates

BIBLIOGRAPHY

APPENDIX B

Walls

- J. Ashurst and F.G. Dimes, *Stone in Building*, Architectural Press, 1977
- E. Beazley, *Design and Detail of the Space between Buildings*, Architectural Press, 1960
- Brick in Landscape Architecture Garden Walls*, Technical Notes 29A, Structural Clay Products Institute, 1750 Old Meadow Road, McLean, VA 22101, 1968
- B.A. Heseltine and J.N. Tutt, *Brickwork Retaining Walls*, BDA, 1981
- R.M. Cartwright, *The Design of Urban Space*, Architectural Press, 1980
- M.F. Downing, *Landscape Construction*, E & F N Spon, 1977
- John Duell and Fred Lawson, *Damp proof course Detailing*, Architectural Press, 1983
- Earth retaining structures*, Civil Engineering Code of Practice No. 2 (1951), Institution of Civil Engineers
- M. Gage and T. Kirkbride, *Design in Blockwork*, 2nd Edition, Architectural Press, 1976
- M. Gage and M. Vandenberg, *Hard Landscape in Concrete*, Architectural Press, 1975
- J.O.A. Korff, *Design of freestanding walls*, 1985
- A.E. Munson, *Construction Design for Landscape Architects*, McGraw-Hill, 1974
- C. Tandy, *AJ Handbook of Urban Landscape*, Information Sheet 35, Architectural Press, 1972
- Structural Design of Serpentine Walls*, Technical Notes 33, Brick Institute of America
- Walls and Fences*, Brick Association of North Carolina, PO Box 6305, Greensboro, NC 27405

Fences

Central Electricity Generating

- Board, *Design Memorandum on the Use of Fences*, January 1966
- D. Lovejoy & Partners, *Spon's Landscape Handbook*, E & F N Spon, 1983
- A. Pinder and A. Pinder, *Beazley's Design and Detail of the Space between Buildings*, E & F N Spon, 1990
- Spon's Landscape Handbook*, E & F N Spon
- C. Tandy, *AJ Handbook of Urban Landscape*, Information Sheet 34, 1972
- G.S. Thomas, *Plants for Ground Cover*, Dent, 1970
- A.E. Weddle (ed.), *Techniques of Landscape Architecture*, Heinemann, 1967
- A.E. Weddle, *Landscape Techniques*, Heinemann, 1979

Fences and gates

- T. Blackwell, *Walls and Fences in Japan*, Landscape Australia, 2/1988, pp. 208–210
- British Trust for Conservation Volunteers, *Fencing: A Practical Conservation Handbook*, Wallingford, 1986
- J. Brookes, *The Garden Book* (includes a short section on fences for domestic garden), Dorley Kindersley, 1984
- Central Electricity Generating Board, *Design Memorandum on the Use of Fences*, January 1966
- J. Clarke and R. Boswell, *Tests on Round Timber Fence Posts*, Forestry Commission, HMSO, 1976
- M.F. Downing, *Landscape Construction*, E & F N Spon, 1977
- Forestry Commission, *Forest Fencing*
- Greater London Council, *Finishes for External Softwood Joinery: A Warning*, *Development and Materials Bulletin* 79, Use of Pigmented Water-repellent, October/November 1974
- G. Jaffa, *Fencing Requirements in the Eyes of the Law: Parks and*

Sports Grounds (especially maintenance rights), January 1985, pp. 10–11

T. Jemison, Railings, *Landscape Design*, June 1983, pp. 27–30; August 1983, pp. 33–36; October 1983, pp. 33–36; February 1984, pp. 43–44

A. Lisney and K. Fieldhouse, *Landscape Design Guide*, Vol. 2, *Hard Landscape: The Design of Paved Spaces*, *Landscape Enclosure and Landscape Furniture*, Gower, 1990 (Chapters 12–14 deal with fences)

Derek Lovejoy Partnership, *Spon's Landscape and External Works Price Book: 10th Edition*, E & F N Spon, 1991

Monmouthshire County Council, *Fences in the Countryside*, 1974

National Building Agency, *External Works Detail Sheet*, Architectural Press, 1977, Chapter 3

A. Pinder and A. Pinder, *Beazley's Design and Detail of the Space between Buildings*, E & F N Spon, 1990

Judith Rowe, *Badger Gates*, Forestry Commission, HMSO, 1976

C. Tandy (ed.), *Handbook of Urban Landscape*, Architectural Press, 1972

A.E. Weddle (ed.), *Techniques of Landscape Architecture*, 2nd Edition, Heinemann, 1979

Stone

- J. Ashurst and F.G. Dimes, *Stone in Building*, Architectural Press, 1984
- Building Research Establishment (BRE), *The selection of natural building stone*, BRE Digest 269
- Natural Stone Directory*, 7th Edition, Ealing Publications, April 1987

Specifications

NBS specification

BRITISH STANDARDS

APPENDIX C

Walls generally

- BS 12: 1978 Ordinary and rapid-hardening Portland cement
BS 146: 1973 Part 2: Portland blast furnace cement
BS 743: 1970 Materials for damp-proof courses
BS 890: 1972 Building limes
BS 1178: 1982 Milled lead sheet
BS 1200: 1976 Sands for mortar
BS 1217: 1986 Cast stone
BS 1449: 1983 Part 2: Stainless steel strip
BS 1470: 1987 Aluminium and aluminium alloy strip
BS 2870: 1980 Rolled copper and copper alloy strip
BS 2989: 1982 Zinc and Iron – Zinc alloy coated steel strip
BS 4027: 1980 Sulphate-resisting Portland cement
BS 4729: 1971 Shapes and dimensions of special bricks
BS 5642: 1983 Part 2: Coping units

Brick and block walls

- CP 121: 1973 Part 1: Brick and block masonry
BS 187: 1978 Calcium silicate bricks
BS 3921: 1985 Clay bricks and blocks
BS 4729: 1971 Shapes and dimensions of special bricks
BS 6073: 1981 Precast concrete masonry units

In-situ concrete walls

- BS 1881: 1970 Methods of testing concrete
BS 4482: 1985 Hard drawn mild steel wire for reinforcing concrete
BS 4483: 1983 Steel fabric for reinforcing concrete
BS 5328: 1981 Methods of specifying concrete
BS 5328: 1981 Ready-mixed concrete
BS 5896: 1989 High tensile steel wire for reinforcing concrete
BS 6100: 1984 Glossary of terms for concrete and reinforced concrete and other appropriate standards listed in earlier sections

Rendered walls

- BS 1199: 1976 Sands for external renderings
BS 4764: 1986 Powder cement paints
BS 5262: 1976 Code of practice for external rendered finishes
BS 6213: 1982 Guide to selection of constructional sealants and other appropriate standards listed in previous sections and those defining specific sealant types

Stone walls

- BS 5390: 1976 Code of practice for stone masonry
BS 6100: 1984 Glossary of terms for stone used in building and other appropriate standards already listed above

Retaining walls

- BS 3921: 1974 Clay bricks and blocks
BS 5628: 1978 The structural use of masonry: Part 1
Unreinforced Masonry
Civil Engineering CP 2: 1951 Earth retaining structures
CP 110: 1972 The structural use of concrete
CP 111: 1970 Structural recommendations for loadbearing walls
CP 121: Part 1: 1973 Walling: Brick and block masonry
BS 187: Part 2: 1978 Calcium silicate (sand lime and flint lime) bricks
BS 729: 1971 Hot dip galvanised coatings on iron and steel articles

Timber fences

- BS 144/3051: 1972/7 Coal tar creosote for the preservation of timber
PA 532: 1980 A paint system comprising an undercoat and gloss finish
BS 565: 1972 Glossary of terms relating to timber and woodwork
BS 881/589: 1974: Nomenclature of commercial timbers
BS 913: 1973 Wood preservation by means of pressure creosoting

BS 1201: 1963 Wood screws
 BS 1202: 1974 Nails
 BS 1282: 1975 Guide to the choice, use and application of wood preservatives
 BS 1336: 1971 Knotting
 BS 1722: 1986 Part 4 Cleft Chestnut Pale Fences
 Part 5 Close Boarded Fences
 Part 6 Wooden Palisade Fences
 Part 7 Wooden Post and Rail Fences
 Part 11 Woven Wood and Lap Boarded Panel Fences
 BS 2015: 1985 Glossary of paint terms
 BS 2521/2523: 1966 Lead-based priming paints
 BS 3452/3453: 1962 Waterborne wood preservatives and their application
 BS 4102: 1986 Steel wire for fences
 BS 4261: 1985 Glossary of terms relating to timber preservation
 BS 4471: 1971/73 Dimensions for softwood
 BS 4756: 1971 Aluminium priming paints for woodwork
 BS 5056: 1970 Copper naphthenate wood preservatives
 BS 5082: 1974 Water-thinning priming paints for wood
 BS 5450: 1977 Sizes of hardwoods and methods of measurement
 BS 5358: 1976 Low-lead solvent-thinned priming paint for wood
 BS 5705: 1979/80 Solutions of wood preservatives in organic solvents
 BS 6150: 1982 Code of practice for painting of buildings

Metal fences

BS 4: 1980 Part 1: Hot-rolled steel sections
 BS 405: 1987 Expanded metal (steel)
 BS 481: 1972 Part 2: High tensile steel wire mesh
 BS 729: 1986 Hot-dip galvanized coatings on iron and steel
 BS 1052: 1980 Mild steel wire
 BS 1449: 1983 Steel plate, sheet and strip
 BS 1474: 1988 Wrought aluminium bars, tubes and sections
 BS 1485: 1983 Galvanized wire netting
 BS 1554: 1986 Stainless steel round wire
 BS 1615: 1987 Anodic oxidation coatings on aluminium
 BS 1722: Part 1 1986 Chain Link Fences
 Part 2 1989 Rectangular Wire Mesh and Hexagonal Wire Netting Fences

Part 3 1986 Strained Wire Fences
 Part 8 1978 Mild Steel (low carbon steel)
 Part 9 1979 Mild Steel (low carbon steel) Fences with round or square verticals and flat posts and horizontals
 Part 10 1990 Anti-intruder Fences in Chain Link and Welded Mesh
 Part 12 1990 Steel Palisade Fences
 Part 13 1978 Chain Link Fences for Tennis Court Surroundings
 BS 3830: 1973 Vitreous enamelled steel
 BS 3987: 1974 Anodic oxide coatings on aluminium for external applications
 BS 4102: 1986 Steel wire for fences
 BS 4483: 1985 Steel fabric
 BS 4842: 1984 Storing organic finishes on aluminium
 BS 4848: 1975 Part 2: Hot-rolled hollow steel sections
 BS 4921: 1988 Sheradized coatings on iron and steel
 BS 6323: 1982 Seamless and welded steel tubes
 BS 6722: 1986 Recommendations for metal wire and the standards for paints listed under Timber fencing above

Gates and stiles

BS 1186: 1986 Part 1: Specification for timbers
 BS 1227: 1967 Part 1: Hinges for general building purposes
 BS 3470: 1975 Field gates and posts
 BS 4092: 1966 Domestic front entrance gates
 BS 5707: 1979 Stiles, bridle gates and kissing gates

Changes of level (ramps)

BS 5810: Access for the disabled to buildings
 BSI Education Information, Aid for the disabled

MANUFACTURERS AND SUPPLIERS

(FENCES, RAILINGS, GATES, WALLS)

APPENDIX D

Abacus Municipal Ltd
Sutton-in-Ashfield
Notts NG17 5FT
(0623 511111)

Akzo Coatings
99 Station Road
Didcot
Oxon OX11 7NQ
(0235 815141)

Amstad Systems
Amstad House
Cliftonville Road
Northampton NN1 5BU
(0604 29721)

Anda Crib
Oaklands House
Solartron Road
Farnborough
Hampshire GU14 7QL
(0252 549334)

Ballantine Bo'ness Iron Co. Ltd
Bo'ness
West Lothian
Scotland EH15 9PW
(0506 822721)

Blenheim Estate Sawmills
Combe
Witney
Oxford OX8 8ET
(0993 881206)

Blue Circle Industries Ltd
Church Road
Murston
Sittingbourne
Kent ME10 3TN
(0795 21066)

Bolton Brady Ltd
Turton Street
Bolton
BL1 2SP
(0204 32111)

Braby Group Ltd
Ashton Gate Works
Bristol BS3 2LQ
(0272 664041)

British Gates and Timber Ltd
Castletons Oak Sawmill
Biddenden
Nr Ashford
Kent TN27 8DD
(0580 291555)

Brookbrae Ltd
53 St Leonard's Road
London SW14 7NQ
(081 8769238)

Broxap and Corby Ltd
Walker Street
Radcliffe
Manchester M26 9JH
(061 796 5600)

Bruce and Hyslop Ltd
Well Lane
Park Street
Bootle
Merseyside L20 3BS

Buffalo Fence Ltd
19 Mill Lane
Benson
Oxford OX9 5SA
(0491 38368)

Dorothea Ltd
Pearl House
Hardwick Street
Buxton
Derbyshire SK17 6DH
(0298 79121)

Dimex
Dimex House
116 High Street
Solihull
West Midlands B91 3SD
(021 704 3551)

Durafencing Ltd
37 Nightingale Grove
London SE13
(081-318 1215)

Ferrous Gate Company
Acorn Works
Green-Street-Green Road
Dartford
Kent
(032 72119)

Forest Fencing Ltd
Stanford Court
Stanford Bridge
Nr Worcester
(08865 451)

Graepel-Barrier
BSA Guns Ltd
Armoury Road
Smallheath
Birmingham B11 2PX
(021 772 8543)

Grassphalte-Gaze
Old House Lane
Bisley
Surrey GU24 9DS
(04867 80284)

Hill and Smith
PO Box 4
Canal Street
Brierley Hill
West Midlands DY5 1JL
(0384 480084)

ICI Fibres
Hookstone Road
Harrogate
North Yorkshire
(0423 68021)

H. S. Jackson & Son (Fencing)
Stowting Common
Near Ashford
Kent TN25 6BN
(023-375 393)

Joint Technology Ltd
Unit 12, Claydon Industrial Park
Greater Blakenham
Suffolk IP6 0JD
(0473 831000)

KUFA Plastics Ltd
2 Lyon Close
Chantry Estate
Kempston
Bedford MK42 7SB
(0234 854464)

Lionweld Kennedy Ltd
Marsh Road
Middlesbrough
Cleveland TS1 5JS
(0642 245151)

Marley Building Products
Peasmarsh
Guildford
Surrey GU3 1LS
(0483 69922)

Non-Corrosive Metal Products Ltd
(Street Furniture Ltd)
Horton Road
West Drayton
Middlesex
(0895 42607)

Orsogrill UK Ltd
Southwood Summit Centre
83 Apollo Rise
Farnborough
Hampshire
(0252 521212)

Permafix Products
72-74 Bath Road
Cheltenham
Gloucestershire GL53 7JT
(0242 573202)

PJP Trading Ltd
151 Dixons Hill Road
North Mimms
Hatfield
Herts AL9 7JE
(070 72 66726)

Ranalah Gates
Gloucester Road Trading Estate
Malmesbury
Wiltshire SN16 9JT
(0666 823001)

Reinforced Concrete Construction
Co. Ltd
Delph Road
Brierley Hill
Staffs DY5 2RW
(0384 78079)

River and Sea Gabions (London)
2 Swallow Place
Princes Street
London W1R 85Q
(071 629 8528)

James W. Shenton Ltd
Edison Works
Tinsley Street
Great Bridge
Tipton
West Midlands DY4 7LQ
(021-557 2531)

Sloan Davidson Ltd
Carrick Foundry
Stanningley
Pudsey
Yorkshire LS28 7XE
(0532 571892)

SMP (Playgrounds) Ltd
Pound Road
Chertsey
Surrey KT16 8EJ
(0932 568081)

J. Starkie Gardner Ltd
Lady Lane Industrial Estate
Hadleigh
Ipswich
Suffolk IP7 9DG
(0473 822525)

Steelway-Fensecure Glynwed
Steels & Engineering Ltd
Queensgate Works
Bilston Road
Wolverhampton WV2 2NJ
(0902 451733)

UAC Timber (Wragby)
Wragby
Lincoln LN3 5NE
(067 34 304)

Woodscape Ltd
Upfield
Pike Lowe
Brinsall
Nr Chorley
Lancs PR6 8SP
(0254 830886)

INSTITUTIONS AND ASSOCIATIONS

APPENDIX E

Aluminium Federation

Broadway House
Calthorpe Road
Five Ways
Birmingham B15 1TN
Tel: (021) 455 0311
Technical information and advice.

Association of Safety Fencing Contractors

7 Woodgates Close
North Ferriby
North Humberside HU14 3JS
Tel: (0482) 633569

British Constructional Steelwork Association Limited

4 Whitehall Court
Westminster
London SW1A 2ES
Tel: (071) 839 8566

Fencing Industry Association Ltd

Timbers House
35 Oakwood Avenue
Purley
Surrey CR2 1AR

The Fencing Contractor's Association

St John's House
23 St John's Road
Watford, Herts
Tel: (0923) 248895

National Corrosion Service

National Physical Laboratory
Teddington
Middlesex TW11 0LW
Tel: (081) 977 3222

NBS Services Ltd

Mansion House Chambers
The Close
Newcastle upon Tyne NE1 3RE
Tel: (091) 232 9594
Fax: (091) 232 5714

Timber Research and Development Association (TRADA)

Stocking Lane
Hughenden Valley
High Wycombe
Buckinghamshire HP14 4ND
Tel: (0240) 24 (Naphill) 3091/2771/3956

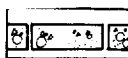
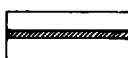
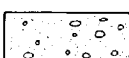
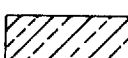
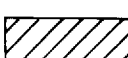
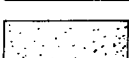
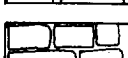
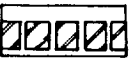
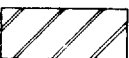
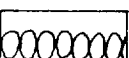
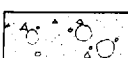
The Welding Institute

Abington Hall
Abington
Cambridge
CB1 6AL

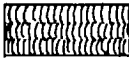
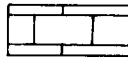
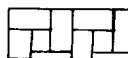
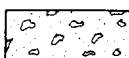
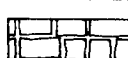
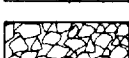
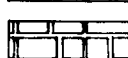
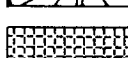
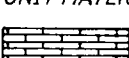
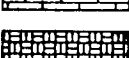
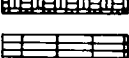
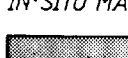
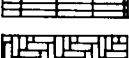
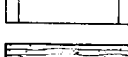
Zinc-Lead Development Association (ZDA)

34 Berkeley Square
London W1X 6AJ
Tel: (071) 499 6636
Technical advice and publications.

APPENDIX F
Standard graphic symbols

	Gravel		Concrete-pc paving units
	Hardcore		Metal
	Hoggin		Setts
	Rock		Stone-natural, cut
	Rubble		Stone-reconstituted
	Sand		Rubble stone-random
	Topsoil		Rubble stone-coursed
	Water		Timber-dressed (wrot)
UNIT MATERIALS			Timber-rough (unwrot)
	Brick paving	IN-SITU MATERIALS	
	Brickwork		Asphalt
	Cobbles		Concrete-in-situ
	Concrete-pc blockwork		Mortar

section

	Grass		Concrete-pc paving slabs
	Gravel		Concrete-pc blocks
	Hoggin		Concrete-pc hexagonal slabs
	Sand		Setts-stack bond
	Soil		Setts-stretcher bond
	Rock		Stone-natural
	Rubble		Stone-reconstituted
	Water		Stone-random paving
UNIT MATERIALS			Tiled paving
	Brick-stretcher bond		Timber
	Brick-basket weave	IN-SITU MATERIALS	
	Brick-stack bond		Asphalt
	Brick-herringbone		Concrete-is broom finish
	Cobbles-random laid		Concrete-is exposed aggregate
	Cobbles-coursed		Concrete-is trowelled finish
	Cobbles-flat, parallel laid		Concrete-is marked finish

plan

APPENDIX G

Conversion tables

Rates

Imperial unit	Metric unit	Conversion factor (×)	Reciprocal (×)
lb/sq yd	kilogram/m ² (kg/m ²)	0.54	1.83
lb/acre	gram/m ² (g/m ²)	0.112	8.922
lb/acre	kilogram/ha (kg/ha)	1.121	0.892
ton/acre	tonne/ha (t/ha)	2.511	0.398
cwt/cu yd	kilogram/m ³	66.5	0.015
lb/cu ft	kilogram/m ³	16.018	0.062
oz/sq yd	gram/m ² (g/m ²)	33.906	0.0295
oz/sq yd	kilogram/ha (kg/ha)	339.057	0.00295
gal/sq yd	litre/m ² (l/m ²)	5.437	0.1839
gal/acre	cubic metre/ha (m ³ /ha)	0.01123	0.089
gal/acre	litre/ha (l/ha)	11.233	0.089
gal per min	litre/sec (l/s)	0.0757	13.158
gal per sec	litre/sec (l/s)	4.5461	0.220
million gal per day	cubic metre/sec (m ³ /s)	0.0526	19.011
miles per hour	metre/sec (m/s)	0.447	2.237
ft/sec	metre/sec (m/s)	0.305	3.281
cub ft/sec	cubic metre/hr (m ³ /h)	102	0.0098
lb/sq in	kilogram/m ² (kg/m ²)	703.07	0.00142
btu/lb	kilojoule/kilogram (kJ/kg)	2.325	0.430

Gradients

Ratio rise: horizontal distance	Angle	Grade (%)
1 in 1	45°	100
1 in 2	26° 34'	50
1 in 3	18° 26'	33(1/3)
1 in 4	14° 02'	25
1 in 5	11° 19'	20
1 in 10	5° 43'	10
1 in 15	3° 49'	6(2/3)
1 in 20	2° 52'	5
1 in 40	1° 25'	2(1/2)
1 in 50	1° 09'	2
1 in 60	0° 57'	1(2/3)
1 in 80	0° 43'	1(1/4)
1 in 100	0° 34'	(2/3)

Landscape Detailing

Volume 1 Enclosures

Michael Littlewood's Landscape Detailing is now well established as a valuable source of reference for architects, landscape architects, other professionals and students designing external works. For this third edition it has been split into two volumes to give a greater depth of coverage than ever before.

Volume 1, Enclosures, covers walls (free-standing, screen and retaining), fences, gates, barriers and bollards. Each section begins with technical guidance notes on design and construction and then provides a list of points against which specifications can be checked. This is followed by a set of drawn-to-scale detail sheets. These details can be traced for direct incorporation into the set of contract drawings.

A list of relevant British Standards, references, bibliography and a list of associations and institutions indicate where further guidance can be obtained.

As a ready reference for landscape designers and as an indispensable time-saving tool, Landscape Detailing is an essential for the design office.

Also available by Michael Littlewood

Landscape Detailing

Volume 2 Surfaces

Covers foundations, paving (vehicular and pedestrian), steps and ramps, margins and edges, kerbs and wheelstops and drainage channels.

Tree Detailing

A companion volume to Landscape Detailing, this is a distilled compendium of practical information on all aspects of tree planting and maintenance. The ten chapters deal with tree stock and sizes, planting techniques, pruning and surgery, protection of new and existing trees, tree surrounds, trees in containers and on roof gardens, maintenance and management, proximity to buildings and drains and survey and evaluation. Each chapter includes a set of scale drawings, ready for use by landscape designers or specifiers.



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